# Protocol v3.1: proofs appendix

Part A: cryptographic and DP guarantees · Part B: sensitivity and statistical guarantees
Part C: composition across runs and the hockey-stick ledger

Draft for review

This appendix collects the proofs for protocol v3.1 (honest-but-curious model). Part A (page 5) covers the encoding sensitivity, smudging, the worst-case error bound $B_{\text{tot}}$, output privacy (Theorem 1), input and cohort privacy, and correctness. Part B (page 45) covers the replace-one sensitivity, operator-norm concentration, the two-stage choice of $(\lambda, \Psi)$, the bound on $\|\boldsymbol{\beta}\|$, coverage of both intervals, and the bootstrap RSS correction. Part C (page 75) proves composition across runs for an adaptive, possibly colluding coalition of researchers (the paper's Proposition `prop:compose`), with a self-contained Rényi filter, and the hockey-stick treatment of the smudging step that the ledger uses (three sums, the reserved share $\xi_L$ of $\varepsilon_L$, and the rule for $\kappa_L$). Its result numbers are those the paper cites as "proofs, composition appendix". Sections 4, 7 and 9 of Part A and S7–S10 of Part B were added after a second reading of the companion book on the protocol, whose proofs of these results were checked step by step (with the corrections noted where they occur), and include new results (Part A, Lemma 7.5 and Proposition 7.7; Part B, Lemma S0(b), Lemma S8′, Propositions S10a and S10b, Theorem S10c). Each part labels every step as exact, first order, asymptotic, numerical or open, and lists its corrections to the specification.

## Status at a glance

| Claim | Status | Notes |
|---|---|---|
| Lemma 1: encoding sensitivity | proven | $\sqrt{2/N}$ and the $\sqrt{N}$ rounding term are correct |
| Lemma 2: smudging | proven | SD $\le \|e\|_2/(\sqrt{2\pi}\sigma_1) + 4\epsilon/(1-\epsilon)$ with $\sigma_1 = \sqrt{\sigma_c^2 - 2.1^2}$; the per-release form $2^{-\kappa}(1+2^{-80}) + 2^{-97}$ needs $\sigma_c \ge 2^{42}$ (A, Corollary 2.4, corrected from $2^{20}$); the hockey-stick form needs no floor (C, Lemma 5.1) |
| Obligation O1: worst-case $B_{\text{tot}}$ | proven (explicit) | $\approx 2^{25.3}$ for the circuit as specified; errors measured in OpenFHE runs stay 6–8 bits below it (no OpenFHE noise estimate is used); assumes (M5) |
| O1 for OpenFHE's gadget, decoupled scales | proven for model (M3′) (A, Lemma 4.1) | $\varsigma_\uparrow = 2\max_j k_j$, $\varsigma_\downarrow = 2k_P$; $B_{\text{tot},2} = 2^{26.59}$ at the evaluation setting; match of OpenFHE 1.5.1 with (M3′) by code reading; (M5) for OpenFHE's sampler numerical |
| Centre sensitivity, decoupled scales | proven (A, Lemma 4.3) | $\Delta_{x,2} = 2^{27.57}$, remainder $\epsilon_x = 2^{-36.5}$ covered by the "+1"; exact rounding in Ecd assumed |
| Decoding offset | proven (A, Proposition 4.5) | $\bar{d}_2$ between $10^{-8}$ and $5 \cdot 10^{-6}$; coefficient-domain $d_{\max} \le 2^{38.6}$ |

| Claim | Status | Notes |
|---|---|---|
| zCDP of the ideal mechanism; tight conversion | proven | $\rho_{\text{run}} = 0.01496$ matches the specification |
| Theorem 1: output privacy | proven (hybrids) | one-sample RLWE for the re-randomizer; plain RLWE for $\mathsf{O}$'s public-key share, the only key material $\mathsf{O}$ sends (circular security only if its rotation-key shares were sent); O1; $\mathsf{R}$'s errors honestly sampled; slack $\gamma$ statistical |
| Release timing (budgeted sampler) | proven under the cost assumption (W) (A, Section 5.5: Lemmas 5.15–5.20, Corollary 5.21) | the sampler runs a fixed budget of $T$ trials with fixed-width work and is exact outside a failure event of probability $\leq \xi_T = 2^{-80}$ per release ($T = 2.39N$ at $\sigma_c = 2^{68.5}$), charged in $\gamma$ and in the ledger's $\Gamma$ (C, Remark 5.9); the timing then depends only on public parameters, the number of accepted trials and machine noise; (W) is not verified for CPython, numpy or OpenFHE's error sampler |
| Composition across runs | proven (C, Theorem 4.2) | any PPT coalition under the filter; runs fixed in advance under rule (N); the ledger charges a run's full cost at admission; RLWE for a finite menu of parameter sets, against non-uniform adversaries; rule (N) with adaptive requests not proved |
| Hockey-stick ledger and $\kappa_L$ | proven (C, Theorem 5.2, Corollary 5.4, Proposition 5.5) | $\kappa_L = 22, 21, 19, 17, 15$ at $(10, 10^{-6}, 0.1, 10^{-3})$ for $\varepsilon_{\text{run}} = 0.5, \ldots, 8$ |
| G1, G2: input and cohort privacy | proven | simulators, up to the cuckoo-hashing failure probability, now bounded (A, Lemma 7.5: $2^{-49.1}$ at $n_R^{\max} = 2{,}000$, $\leq 2^{-40}$ up to load 0.639; needs an insertion that finds an assignment whenever one exists) and $\mathsf{R}$'s observable aborts |
| G4: disclosure from the Link phase | proven, ROM and CDH (A, Theorem 7.3) | first layer, without circuit privacy; second layer (SanitizeBFV) proven for the textbook BFV model (A, Proposition 7.7: evaluation noise $\leq 2^{-31}$ after the switch); OpenFHE's BFVrns not checked against the model |
| Soundness of Calibrate; standard simulation with output; $\mathsf{O}$'s work | proven (A, Section 9) | the run computes its DP functionality against $\mathsf{R}$ when the functionality outputs the centred residue modulo $q_0$; $\Omega(n_{\text{Sc}})$ reads in a cell-probe model |
| Correctness (no wraparound) | proven | failure probability $\leq \delta_B + 2Ne^{-50}$ per release; no condition at the earlier levels (A, Proposition 8.1) |
| S1: replace-one sensitivity | proven, exact for $d_O \geq 2$ | triu and Frobenius suprema are equal; regime 3 reduces to three cubics; $d_O = 1$ is valid but conservative |
| When $\mathbf{y}^\top\mathbf{y}$ is free | partly proven | exact criterion; free if $w \leq 2(r+o)$, never free if $w > 3(r+o)$; boundary in between only numerical |
| S2: operator-norm radius | proven (one-sided) | $r_1$ valid for the event Algorithm 6 uses; about $1.55\times$ conservative |
| S7: exact radius | proven | exact level with fresh draws; transfer to the lattice noise with the TV constant now proved (S0) |
| S8: switch-off | bracket proven; size of the bias partly | bracket tight among Gram matrices with a given $\lambda_{\min}$; spectral bound on the bias by Sudakov–Fernique; empirical factors 1.2/1.4 not proved |
| S9: mechanism bias | proven, explicit constants | leading term, ridge, decay in $\lambda$ |

| Claim | Status | Notes |
|---|---|---|
| S10: FWS $\hat{s}^2$; clipping; full-record $\Delta$ | proven | plug-in RSS biased by $-\sigma^2(\operatorname{tr}\mathbf{P} + \hat{\beta}^\top M_\chi \hat{\beta})$; interval valid under binding clipping with a public variance bound; full-record $\Delta$ exact: closed form for blocks of dimension $\geq 2$, enumeration over five faces for a block of dimension 1; full-record circuit's $B_{\text{tot}}$, $\Delta_x$ and offset re-derived (S10e) |
| S3: two-stage $(\lambda, \Psi)$ | proven | independent of the stage-2 noise; closed form for $\lambda_O$ |
| S4: bound on $\|\boldsymbol{\beta}\|$ | proven | holds for the projection coefficient with no model assumed |
| S5: coverage | partly first order | Wald remainder bounded exactly; plug-in variance first order; normality of the sampling term asymptotic |
| S6: bootstrap RSS correction | proven to first order | removes the $O(\sigma^2)$ bias up to $O(\sigma^4)$; constants not explicit |

## Corrections to the specification

Both parts found claims in `protocol_v31.tex` that need to change. The most consequential:

1. **The CKKS scale is undersized.** Flooding the whole error in $\ell_2$ needs $\log_2 \Omega \approx 61.5$ for the circuit as specified, not 52.5. A restructured circuit brings this down to $\approx 57.2$; adding encoding at scale $2^{60}$ by $\mathsf{R}$ brings it to $\approx 53.7$. Calibrate must use the data-independent worst-case bound, never OpenFHE's noise estimate.
2. **The centre of the mechanism.** The ideal mechanism must be centred at the noiseless circuit output, not at $\mathsf{Ecd}(\mathbf{z})$. Its sensitivity rises by a relative $2^{-33}$. $\Omega$ must be the exact final scale of the circuit.
3. $\delta$ **accounting.** The per-release statistical distance is $2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$, for $\sigma_c \geq \max\{2^\kappa B_{\text{tot}}/\sqrt{2\pi}, 2^{42}\}$ (the floor $2^{20}$ stated earlier is too small for the factor $1 + 2^{-80}$). For a single $(\varepsilon, \delta)$ statement, the slack enters with a factor $(1 + e^\varepsilon)$. Across runs, the ledger charges a run's full cost at admission and accounts the smudging step in hockey-stick divergence (Part C).
4. **Assumptions.** RLWE is needed in two forms. Disclosure G4 needs circuit privacy of $\mathsf{O}$'s BFV replies; the modulus switch alone is heuristic. G1 and G2 need explicit assumptions and the hashing slack.
5. **Algorithm 6.**
   - The radius comment should be one-sided, or use $\ln(2/\alpha_L)$.
   - $U_{yy}$ spends $\alpha_U$ twice, and $\sigma_{yy}$ is undefined.
   - $\boldsymbol{\beta}$ must be defined as a working-model or projection coefficient on the clipped data, and the homoskedastic variance fails when clipping binds.
   - The bootstrap needs truncation, since the moments do not exist otherwise.
   - The bias bound $\bar{b}_j$ is first order only.
   - The private $\|\boldsymbol{\beta}\|$ bound is usually infinite while the ridge is active.
   - The specification should state whether coverage is conditional on the gate.
6. **Smaller points.**
   - The sampler needs a rational $\sigma_c^2$ and a defined notion of constant time.
   - The modulus switch to $q_0$ is a no-op after two rescales.
   - The printed $\Delta = 86.18$ comes from unrounded bounds.
   - The earlier paper's "$\mathbf{y}^\top\mathbf{y}$ is free" and "Frobenius is conservative" are false in general.

## Open items

- Full proofs for the two improved circuit variants; a verification of the cost assumption (W) of the budgeted sampler (A, Assumption 5.18) for the implementation's CPython and numpy operations, and of (CT) for OpenFHE's samplers of $(u, e_0, e_1)$. (Adaptive composition across runs is proved in Part C; rule (N) with adaptively chosen requests is not. No wraparound needs no condition at the earlier levels; the chain is sized by CALIBRATE, which is proved sound.) Whether OpenFHE's BFVrns satisfies the BFV model of the $B_\psi$ bound. (The cuckoo insertion now falls back to augmenting paths, so Lemma 7.5 of Part A covers it.)
- The free-set boundary for $\mathbf{y}^\top\mathbf{y}$; a closed form for $\Delta$ with $\mathbf{y}^\top\mathbf{y}$ when $d_O = 1$; explicit constants for S6; a guarantee for the data-driven variance when clipping binds (a public bound gives a valid interval, S10); a spectrum-sharp bound on the downward bias of $\lambda_{\min}$ (S8).
- Sixteen citation details are flagged in the two parts for verification, among them theorem numbers in Canonne et al. (2020), Vershynin, and Boucheron, Lugosi and Massart.

# Protocol v3.1: proofs of the cryptographic and differential-privacy claims

Technical appendix: encoding sensitivity, smudging, a worst-case error bound, output privacy (SIM-CDP), input and cohort privacy, correctness

Draft for review

**Abstract**

This appendix proves, corrects, or marks as open each cryptographic and differential-privacy claim of the specification *Protocol v3.1* (`protocol_v31.tex`). We work in the honest-but-curious model of that specification and reuse its notation. Section 1 proves the encoding-sensitivity lemma. Section 2 proves the smudging lemma for the exact discrete Gaussian, with explicit constants. Section 3 discharges Obligation O1: an explicit bound $B_{\text{tot}}$ on the $\ell_2$ norm of the whole error ring element, worst case over the clipped data domain, with probability only over keys and fresh randomness. Section 4 extends O1 to decoupled encoding scales and to OpenFHE's uncentred key switching, proves the sensitivity of the centre with decoupled scales, and bounds the decoding offset. Section 5 proves output privacy (Theorem 1) by hybrids. Section 6 proves input and cohort privacy (G1, G2). Section 7 proves the first layer of the disclosure claim G4 (random-oracle model, CDH), bounds the cuckoo-hashing slack, and bounds the reply noise of the PSI in a model of BFV. Section 8 treats correctness, at every level of the circuit. Section 9 proves that Calibrate meets the hypotheses of Theorem 1, that the run computes its DP functionality against $\mathsf{R}$ in the standard sense, and that $\mathsf{O}$'s work is linear in its scope. Section 10 lists the corrections to the specification.

**Summary of results.**

- **Lemma 1** holds with the constant $\sqrt{2/N}$ for the stated convention. One precision: $\Omega$ must be the *exact* final scale $\Omega_{\text{fin}}$ of the circuit, not the nominal scale (Correction 2).
- **Lemma 2** holds with the constant $1/\sqrt{2\pi}$ up to an explicit additive smoothing term ($\leq 2^{-98}$ here). An elementary proof gives the constant $1/2$ with no side condition.
- **O1.** The mechanism must be centred at the noiseless circuit output rather than at $\mathsf{Ecd}(\mathbf{z})$ (Correction 1). With that centre, and with every rounding treated adversarially, the circuit exactly as specified gives $B_{\text{tot}} \approx 2^{25.3}$ for stage 2 of the example, with $\delta_B = 2^{-50}$. This needs $\log_2 \Omega \approx 61.5$, not 52.5. The specification's 52.5 floods only the fresh-encryption error, in mean. With the specification's centre, the deterministic plaintext-arithmetic terms alone reach $2^{30.7}$ to $2^{34.5}$, which would need $\log_2 \Omega \approx 67$ to 71. Changes to the circuit bring $B_{\text{tot}}$ down to $2^{17.6}$, so that $\log_2 \Omega \approx 53.7$ suffices (Section 3, Table 1):
  - evaluate EvalSum before the first rescale;
  - use stochastic rounding and a single final rescale;
  - make the special modulus larger;
  - give $\mathsf{R}$'s encoding a larger scale.
- **Theorem 1** holds under decision-RLWE for two secret distributions, a circular-security assumption for $\mathsf{O}$'s rotation-key shares, and O1, with the corrected centring. It holds up to an

explicit statistical slack $\gamma$. PSI security is *not* needed for G3 under the specification's adjacency. The $(\varepsilon, \delta)$ accounting needs the factor $(1+e^{\varepsilon})$ on $\gamma$ when it is read as indistinguishability-based CDP (Correction 5).

- **G1 and G2** hold under RLWE in $s_R$ (with circular security for R's key material), the random-oracle model for the seed expansion, IND-CPA of the PSI's BFV, and obliviousness of the OPRF. G2 holds up to the cuckoo-hashing failure probability.
- **Correctness.** The $q_0$ condition in CALIBRATE is sufficient, with a correctness-failure probability of at most $\delta_B + 2Ne^{-50}$ per release, which the specification should state. The exact condition is derived below; no condition is needed at the earlier levels.
- **Added after a second reading of the companion book** (Sections 4, 7 and 9): O1 for OpenFHE's uncentred gadget under model (M3′), with $\varsigma_{\uparrow} = 2\max_j k_j$ and $\varsigma_{\downarrow} = 2k_P$ (Lemma 4.1; the match of OpenFHE 1.5.1 with (M3′) is a code reading); the centre sensitivity with decoupled scales (Lemma 4.3); a public bound on the decoding offset (Proposition 4.5); G4's first layer under CDH in the random-oracle model, without circuit privacy (Theorem 7.3); the soundness of CALIBRATE (Proposition 9.1); standard simulation of R's view jointly with its output (Corollary 9.3); and O's $\Omega(n_{\mathrm{Sc}})$ work (Proposition 9.5). New here: a proved bound on the cuckoo-hashing slack (Lemma 7.5) and a worst-case bound on the PSI's evaluation noise in the textbook BFV model (Proposition 7.7), which covers it by the "+1" of the implementation's $B_{\mathrm{err}}$ in that model.

Steps that are not fully rigorous are marked **[Not fully rigorous: ...]**. Citation details we could not verify are marked *[citation detail unverified: ...]*.

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# 1 Conventions and Lemma 1 (encoding sensitivity)

**Ring and embeddings.** Let $N \geq 8$ be a power of two, $\mathcal{R} = \mathbb{Z}[X]/(X^N+1)$, $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$, and $\zeta = e^{i\pi/N}$, a primitive $2N$-th root of unity. The roots of $X^N+1$ are $\zeta^j$ for $j \in \mathbb{Z}_{2N}^*$, the odd residues. Let $\mathcal{H} = \langle 5 \rangle \leq \mathbb{Z}_{2N}^*$. Then:

- $\mathcal{H}$ has order $N/2$ and $\mathcal{H} = \{g \in \mathbb{Z}_{2N} : g \equiv 1 \bmod 4\}$;
- $\mathbb{Z}_{2N}^* = \mathcal{H} \sqcup (-\mathcal{H})$;
- the slots are ordered by $h_k = 5^k \bmod 2N$, for $k = 0, \ldots, N/2-1$.

For $f \in \mathbb{R}[X]/(X^N+1)$, the *canonical embedding* is $\sigma(f) = (f(\zeta^j))_{j \in \mathbb{Z}_{2N}^*} \in \mathbb{C}^N$, and the *slot map* is $\tau(f) = (f(\zeta^{h_k}))_{k<N/2} \in \mathbb{C}^{N/2}$. Since $f$ is real, $f(\zeta^{-h}) = \overline{f(\zeta^h)}$. So $\sigma(f)$ consists of $\tau(f)$ and its complex conjugate. The decoding of the specification is $\mathsf{Dcd}(v) = \tau(v)/\Omega$.

We write $\|f\| = \|f\|_2$, $\|f\|_\infty$ and $\|f\|_1$ for the norms of the coefficient vector, and $\|f\|_\infty^{\mathrm{can}} = \|\sigma(f)\|_\infty$. Norms of vectors in $\mathbb{C}^{N/2}$ are complex $\ell_2$ norms. For $g \in \mathbb{Z}_{2N}^*$, $\psi_g$ is the automorphism $X \mapsto X^g$.

**Lemma 1.1** (Parseval for the canonical embedding)**.** *For every $f \in \mathbb{R}[X]/(X^N+1)$, $\|\sigma(f)\|_2^2 = N\|f\|_2^2$ and $\|\tau(f)\|_2^2 = \frac{N}{2}\|f\|_2^2$. The map $\tau$ is an $\mathbb{R}$-linear bijection onto $\mathbb{C}^{N/2}$.*

*Proof.* We have $\sigma(f) = Wf$ with $W_{j,k} = \zeta^{jk}$, for $j \in \mathbb{Z}_{2N}^*$ and $0 \leq k < N$.

*Orthogonality.* For $k \neq l$, put $d = l-k$, so that $0 < |d| < N$. Then

$$(W^*W)_{k,l} = \sum_{j \text{ odd}} \zeta^{jd} = \zeta^d \sum_{t=0}^{N-1} \zeta^{2td} = \zeta^d \frac{1-\zeta^{2Nd}}{1-\zeta^{2d}} = 0,$$

because $\zeta^{2Nd} = 1 \neq \zeta^{2d}$. Also $(W^*W)_{k,k} = N$, so $W^*W = N\mathbf{I}$.

*Norms.* The first identity follows from $W^*W = N\mathbf{I}$. The second follows because $\sigma(f)$ is $\tau(f)$ together with its conjugate.

*Bijectivity.* Let $y \in \mathbb{C}^N$ satisfy $y_{-j} = \overline{y_j}$. Then $f = N^{-1}W^*y$ has coefficients

$$f_k = N^{-1}\sum_j \zeta^{-jk} y_j, \qquad \overline{f_k} = N^{-1}\sum_j \zeta^{jk}\,\overline{y_j} = N^{-1}\sum_j \zeta^{jk} y_{-j} = f_k,$$

so $f$ is real. Given $z \in \mathbb{C}^{N/2}$, take $y = (z, \bar{z})$ arranged by $\mathcal{H} \sqcup (-\mathcal{H})$. This $f$ satisfies $\tau(f) = z$, and it is unique because $W$ is injective. □

**Lemma 1.2** (Encoding sensitivity; Lemma 1 of the specification)**.** *Let $\tau^{-1} : \mathbb{C}^{N/2} \to \mathbb{R}[X]/(X^N+1)$ be the inverse of the slot map.*

*(a)* $\|\tau^{-1}(\mathbf{z})\|_2 = \sqrt{2/N}\,\|\mathbf{z}\|_2$ *for every* $\mathbf{z} \in \mathbb{C}^{N/2}$.

*(b) Let* $\mathsf{Ecd}_\Omega(\mathbf{z}) = \lfloor \Omega\,\tau^{-1}(\mathbf{z}) \rceil$, *rounded coefficient-wise with any fixed tie rule. For all* $\mathbf{z}, \mathbf{z}'$,

$$\|\mathsf{Ecd}_\Omega(\mathbf{z}) - \mathsf{Ecd}_\Omega(\mathbf{z}')\|_2 \le \sqrt{2/N}\,\Omega\,\|\mathbf{z}-\mathbf{z}'\|_2 + \sqrt{N}.$$

*(c) Let* $\Delta_s = \sup_{D\sim D'} \|\mathbf{z}_s(D) - \mathbf{z}_s(D')\|_2$, *where each released entry occupies exactly one slot and all other slots are zero. Then* $\sup_{D\sim D'} \|\mathsf{Ecd}_\Omega(\mathbf{z}_s(D)) - \mathsf{Ecd}_\Omega(\mathbf{z}_s(D'))\|_2 \le \Delta_{\mathrm{enc},s} = \sqrt{2/N}\,\Omega\,\Delta_s + \sqrt{N}$.

*Proof.* (a) Apply Lemma 1.1 to $f = \tau^{-1}(\mathbf{z})$.

(b) Write $\mathsf{Ecd}_\Omega(\mathbf{z}) = \Omega\tau^{-1}(\mathbf{z}) + \mathbf{r}$ and $\mathsf{Ecd}_\Omega(\mathbf{z}') = \Omega\tau^{-1}(\mathbf{z}') + \mathbf{r}'$, with $\mathbf{r}, \mathbf{r}' \in [-\frac{1}{2}, \frac{1}{2}]^N$. Then $\|\mathbf{r}-\mathbf{r}'\|_\infty \le 1$, so $\|\mathbf{r}-\mathbf{r}'\|_2 \le \sqrt{N}$. Apply (a) to $\mathbf{z}-\mathbf{z}'$, using the $\mathbb{R}$-linearity of $\tau^{-1}$, and the triangle inequality.

(c) Follows from (b). □

**Remark 1.3** (Scope of Lemma 1.2)**.**

(i) The constant $\sqrt{2/N}$ belongs to the convention of $N/2$ complex slots measured in complex $\ell_2$. Under the full $N$-dimensional embedding the constant would be $1/\sqrt{N}$. With real data the imaginary parts are zero, so the specification's $\Delta_s$, a real $\ell_2$ sensitivity, is the right input.

(ii) The rounding term $\sqrt{N}$ cannot be improved in the worst case. At $N = 2^{14}$ it is $2^7$, against $\sqrt{2/N}\,\Omega\Delta_s \approx 2^{52}$.

(iii) The lemma is about $\mathsf{Ecd}_\Omega(\mathbf{z}_s)$. The protocol's decrypted value is centred instead at the output of a noiseless evaluation of the circuit. Section 3 shows that the two differ by a deterministic term of size up to about $2^{30}$, and Corollary 3.13 gives the sensitivity of the correct centre. Here $\Omega$ must be the *exact* final scale $\Omega_{\mathrm{fin}}$ (Correction 2).

**Facts used in Section 3.**

**Lemma 1.4** (Norms and automorphisms)**.** *For all* $f, g \in \mathbb{R}[X]/(X^N+1)$*:*

- $\|fg\|_2 \le \|f\|_\infty^{\mathrm{can}}\,\|g\|_2$ *and* $\|fg\|_2 \le \|f\|_1\|g\|_2$*;*
- $\|f\|_2 \le \|f\|_\infty^{\mathrm{can}} \le \|f\|_1 \le \sqrt{N}\|f\|_2$.

*Every $\psi_g$ acts as a signed permutation on coefficients and as a permutation on $\sigma$. It therefore preserves $\|\cdot\|_2$, $\|\cdot\|_\infty$, $\|\cdot\|_1$ and $\|\cdot\|_\infty^{\mathrm{can}}$.*

*Proof. Products.* We have $\sigma(fg) = \sigma(f) \odot \sigma(g)$. By Lemma 1.1,

$$\|fg\|_2 = N^{-1/2}\|\sigma(f)\odot\sigma(g)\|_2 \le \|f\|_\infty^{\mathrm{can}}\,N^{-1/2}\|\sigma(g)\|_2 = \|f\|_\infty^{\mathrm{can}}\,\|g\|_2.$$

The bound $\|f\|_\infty^{\mathrm{can}} \le \|f\|_1$ holds because $|f(\zeta^j)| \le \sum_k |f_k|$. Also $\|f\|_2 = N^{-1/2}\|\sigma(f)\|_2 \le \|f\|_\infty^{\mathrm{can}}$, and $\|f\|_1 \le \sqrt{N}\|f\|_2$ by Cauchy–Schwarz.

*Automorphisms.* Take $g$ odd. Then $X^k \mapsto X^{gk \bmod 2N} = \pm X^{gk \bmod N}$, and $k \mapsto gk \bmod N$ is a bijection of $\mathbb{Z}_N$. Moreover $\sigma(\psi_g f)_j = f(\zeta^{jg}) = \sigma(f)_{jg}$. □

**Lemma 1.5** (The slot sum is a trace)**.** *Let* $L = \log_2(N/2)$ *and* $g_i = 5^{2^i}$. *For* $0 \le i \le L$, *let* $T_i = \prod_{l=i}^{L-1}(1+\psi_{g_l})$, *so that* $T_i = \sum_{g\in\mathcal{H}_i}\psi_g$ *with* $\mathcal{H}_i = \langle 5^{2^i}\rangle$ *and* $|\mathcal{H}_i| = 2^{L-i}$. *The log-step* EvalSum *evaluates* $\mathrm{A} := T_0 = \sum_{g\in\mathcal{H}}\psi_g$, *and*

$$\mathrm{A}(f) = \tfrac{N}{2}\big(f_0 + f_{N/2}X^{N/2}\big), \qquad \tau(\mathrm{A}f) = \tfrac{N}{2}\big(f_0 + if_{N/2}\big)\mathbf{1} = \Big(\sum_{h\in\mathcal{H}}\sigma(f)_h\Big)\mathbf{1}.$$

*Proof. The product formula.* Every $g \in \mathcal{H}$ has a unique representation $5^{\sum_i b_i 2^i}$ with $b_i \in \{0,1\}$. This gives the formula for $T_i$.

*The trace of a monomial.* By linearity it suffices to treat $f = X^k$. For $j \in \mathcal{H}$ we have $j\mathcal{H} = \mathcal{H}$, so

$$\sigma_j(\mathrm{A}X^k) = S_k := \sum_{g\in\mathcal{H}} \zeta^{gk} = \zeta^k \sum_{t=0}^{N/2-1} \zeta^{4kt}.$$

If $\zeta^{4k} \neq 1$, the geometric sum equals $(1-\zeta^{2Nk})/(1-\zeta^{4k}) = 0$. Now $\zeta^{4k} = 1$ if and only if $k \equiv 0 \pmod{N/2}$. Hence $S_0 = N/2$, $S_{N/2} = \frac{N}{2}\zeta^{N/2} = \frac{N}{2}i$, and $S_k = 0$ otherwise. For $j \in -\mathcal{H}$ the values are the conjugates.

*Identification.* The polynomial $\frac{N}{2}([k{=}0] + [k{=}N/2]X^{N/2})$ has the same embedding, because $\zeta^{jN/2} = i^j$ equals $i$ on $\mathcal{H}$ and $-i$ on $-\mathcal{H}$. The claim follows from the injectivity of $\sigma$. The last expression is $\sigma_j(\mathrm{A}f) = \sum_{g\in\mathcal{H}} \sigma(f)_{jg}$ evaluated at $j = 1$. □

**Lemma 1.6** (Masked slot functionals)**.** *Let $f \in \mathbb{R}[X]/(X^N+1)$, $h \in \mathcal{H}$ and $\mathcal{H}' \subseteq \mathcal{H}$. The real-linear map $\xi \mapsto \sigma_h\big(\sum_{g\in\mathcal{H}'} \psi_g(f\xi)\big)$, for $\xi \in \mathbb{R}^N$, equals $\sum_k \xi_k\beta_k$, with*

$$\|\beta\|_2^2 = N \sum_{g\in\mathcal{H}'} |\sigma(f)_{hg}|^2 \le N \min\big\{N\|f\|_2^2,\ |\mathcal{H}'|\,(\|f\|_\infty^{\mathrm{can}})^2\big\}.$$

*In particular $\|\beta\|_2^2 = N|\mathcal{H}'|$ when $f = 1$. The real and imaginary parts have coefficient vectors $\Re\beta$ and $\Im\beta$, with $\|\Re\beta\|^2 + \|\Im\beta\|^2 = \|\beta\|^2$.*

*Proof.* We have $\sigma_h(\psi_g(f\xi)) = \sigma(f)_{hg}\sum_k \xi_k\zeta^{hgk}$, so $\beta_k = \sum_{g\in\mathcal{H}'} \sigma(f)_{hg}\zeta^{hgk}$. Expanding,

$$\|\beta\|^2 = \sum_{g,g'\in\mathcal{H}'} \sigma(f)_{hg}\overline{\sigma(f)_{hg'}} \sum_{k<N} \zeta^{hk(g-g')}.$$

If $g \neq g'$, then $m = h(g-g')$ is even and $m \not\equiv 0 \pmod{2N}$, so $\zeta^m$ is an $N$-th root of unity other than 1 and the inner sum vanishes. If $g = g'$, the inner sum is $N$. The bounds follow from Lemma 1.1 and from $|\sigma(f)_{hg}| \le \|f\|_\infty^{\mathrm{can}}$. □

**Lemma 1.7** (Masking)**.** *Let $\mathbf{u}_k$ be the one-hot slot vector and $m_{u_k} = \mathsf{Ecd}_{\Omega_u}(\mathbf{u}_k) = \Omega_u\tau^{-1}(\mathbf{u}_k) + r_{u_k}$, with $\|r_{u_k}\|_\infty \le \frac{1}{2}$. For polynomials $y_e$ and distinct slots $k_e$:*

$$\Big\|\sum_e y_e\,\Omega_u\tau^{-1}(\mathbf{u}_{k_e})\Big\|_2 = \Omega_u\sqrt{2/N}\Big(\sum_e |\tau(y_e)_{k_e}|^2\Big)^{1/2}, \qquad \|y_e\,r_{u_k}\|_2 \le \tfrac{N}{2}\|y_e\|_2.$$

*Proof.* The slot map is multiplicative, so $\tau(y_e\tau^{-1}(\mathbf{u}_{k_e})) = \tau(y_e)_{k_e}\mathbf{u}_{k_e}$. Apply Lemma 1.2(a) to the sum, whose slots are disjoint. The second bound is Lemma 1.4 with $\|r_{u_k}\|_1 \le N/2$. □

## 2 Lemma 2 (smudging with the exact discrete Gaussian)

**Conventions.** For $\sigma > 0$ and $x \in \mathbb{R}^N$, let $\rho_\sigma(x) = \exp(-\|x\|^2/(2\sigma^2))$. For a countable set $S$, write $\rho_\sigma(S) = \sum_{x\in S}\rho_\sigma(x)$. For $c \in \mathbb{R}^N$, $D_{\mathbb{Z}^N+c,\sigma}$ is the distribution on $\mathbb{Z}^N + c$ with probability mass function $\rho_\sigma(x)/\rho_\sigma(\mathbb{Z}^N + c)$. The specification's $\mathcal{N}_\mathbb{Z}(0,\sigma^2)^N$ is $D_{\mathbb{Z}^N,\sigma}$, the product of $N$ independent copies of the one-dimensional discrete Gaussian $\mathcal{N}_\mathbb{Z}(0,\sigma^2)$ of Canonne, Kamath and Steinke [4]. SD is statistical (total-variation) distance.

Smoothing parameters are usually stated with the Gaussian width $s = \sqrt{2\pi}\sigma$, where $\rho_\sigma(x) = e^{-\pi\|x\|^2/s^2}$. For a lattice $\Lambda$ and $\epsilon > 0$, $\eta_\epsilon(\Lambda)$ is the least $s$ such that $\sum_{y\in\Lambda^*\setminus\{0\}} e^{-\pi s^2\|y\|^2} \le \epsilon$ [13].

**Lemma 2.1** (Gaussian sums are maximal at the lattice)**.** *For all $\sigma > 0$ and $c \in \mathbb{R}^N$, $\rho_\sigma(\mathbb{Z}^N + c) \le \rho_\sigma(\mathbb{Z}^N)$. Equality holds if $c \in \mathbb{Z}^N$.*

*Proof.* The sum factorizes over coordinates, so it suffices to take $N = 1$. By Poisson summation for the Gaussian,

$$\sum_{z\in\mathbb{Z}} e^{-(z+c)^2/(2\sigma^2)} = \sqrt{2\pi}\,\sigma \sum_{k\in\mathbb{Z}} e^{-2\pi^2\sigma^2k^2}\cos(2\pi kc) \le \sqrt{2\pi}\,\sigma \sum_{k\in\mathbb{Z}} e^{-2\pi^2\sigma^2k^2} = \rho_\sigma(\mathbb{Z}).$$

For integer $c$ the sum is a re-indexing of $\rho_\sigma(\mathbb{Z})$. □

**Lemma 2.2** (Divergences of an integer shift)**.** *Let $\eta \sim D_{\mathbb{Z}^N,\sigma}$, $\mathbf{e} \in \mathbb{Z}^N$, and let $P_{\mathbf{e}}$ be the law of $\eta + \mathbf{e}$. For every $\alpha > 1$,*

$$D_\alpha(P_{\mathbf{e}}\|P_{\mathbf{0}}) \le \frac{\alpha\|\mathbf{e}\|^2}{2\sigma^2} \qquad \textit{and} \qquad \mathrm{KL}(P_{\mathbf{e}}\|P_{\mathbf{0}}) = \frac{\|\mathbf{e}\|^2}{2\sigma^2}.$$

*Proof. Normalizers.* Since $\mathbf{e}$ is integral, $P_{\mathbf{e}}(y) = \rho_\sigma(y-\mathbf{e})/Z$ and $P_{\mathbf{0}}(y) = \rho_\sigma(y)/Z$ with the same $Z = \rho_\sigma(\mathbb{Z}^N)$.

*Rényi divergence.* Use the identity $\alpha\|y-\mathbf{e}\|^2 + (1-\alpha)\|y\|^2 = \|y-\alpha\mathbf{e}\|^2 + \alpha(1-\alpha)\|\mathbf{e}\|^2$. With Lemma 2.1,

$$\sum_y P_{\mathbf{e}}(y)^\alpha P_{\mathbf{0}}(y)^{1-\alpha} = \frac{\rho_\sigma(\mathbb{Z}^N - \alpha\mathbf{e})}{Z}\, e^{\alpha(\alpha-1)\|\mathbf{e}\|^2/(2\sigma^2)} \le e^{\alpha(\alpha-1)\|\mathbf{e}\|^2/(2\sigma^2)}.$$

Take $\frac{1}{\alpha-1}\ln$ of both sides.

*Kullback–Leibler divergence.* By symmetry $\mathbb{E}\eta = \mathbf{0}$, so

$$\mathrm{KL}(P_{\mathbf{e}}\|P_{\mathbf{0}}) = \mathbb{E}\big[\|\eta+\mathbf{e}\|^2 - \|\eta\|^2\big]/(2\sigma^2) = \big(2\mathbb{E}\langle\eta,\mathbf{e}\rangle + \|\mathbf{e}\|^2\big)/(2\sigma^2) = \|\mathbf{e}\|^2/(2\sigma^2).$$

□

**Lemma 2.3** (Smudging; Lemma 2 of the specification, with explicit constants)**.** *Let $\eta \sim D_{\mathbb{Z}^N,\sigma}$ and $\mathbf{e} \in \mathbb{Z}^N$.*

*(a)* Elementary bound, valid for all $\sigma > 0$: $\mathrm{SD}(\eta+\mathbf{e},\eta) \le \|\mathbf{e}\|_2/(2\sigma)$.

*(b)* Sharp constant: *let $\epsilon \in (0,1)$, $\sigma_2 \ge \eta_\epsilon(\mathbb{Z}^N)/\sqrt{2\pi}$, $\sigma > \sigma_2$ and $\sigma_1 = \sqrt{\sigma^2 - \sigma_2^2}$. Then*

$$\mathrm{SD}(\eta+\mathbf{e},\eta) \le \frac{\|\mathbf{e}\|_2}{\sqrt{2\pi}\,\sigma_1} + \frac{4\epsilon}{1-\epsilon}.$$

*By [13, Lemma 3.3], $\eta_\epsilon(\mathbb{Z}^N) \le \sqrt{\ln(2N(1+1/\epsilon))/\pi}$. For $N = 2^{14}$ and $\epsilon = 2^{-100}$ this gives $\eta_\epsilon(\mathbb{Z}^N)/\sqrt{2\pi} \le 2.01$, so $\sigma_2 = 2.1$ is admissible.*

*Proof.* (a) Apply Pinsker's inequality, $\mathrm{SD} \le \sqrt{\mathrm{KL}/2}$, to Lemma 2.2.

(b) The proof constructs a shift-equivariant post-processing $F$ of a continuous Gaussian whose output is close to $\eta$.

*The post-processing.* Let $x \sim \mathcal{N}(0,\sigma_1^2\mathbf{I}_N)$ be continuous. Let $F(x) = x + z'$, where $z' \sim D_{\mathbb{Z}^N - x,\sigma_2}$ is sampled given $x$. Then $F(x) \in \mathbb{Z}^N$. For $\mathbf{e} \in \mathbb{Z}^N$ we have $\mathbb{Z}^N - x - \mathbf{e} = \mathbb{Z}^N - x$, so $F(x+\mathbf{e})$ and $F(x) + \mathbf{e}$ have the same law.

*$F(x)$ is close to $\eta$.* The law of $F(x)$ is

$$P(y) = \int \phi_{\sigma_1}(x)\,\frac{\rho_{\sigma_2}(y-x)}{\rho_{\sigma_2}(\mathbb{Z}^N - x)}\,dx, \qquad y \in \mathbb{Z}^N.$$

Since $\sqrt{2\pi}\sigma_2 \geq \eta_\epsilon(\mathbb{Z}^N)$, the smoothing lemma [13, Lemma 4.4] gives $\rho_{\sigma_2}(\mathbb{Z}^N - x) \in [\frac{1-\epsilon}{1+\epsilon}, 1]\,\rho_{\sigma_2}(\mathbb{Z}^N)$ for every $x$. The Gaussian convolution identity gives $\int \phi_{\sigma_1}(x)\rho_{\sigma_2}(y-x)\,dx = (\sigma_2/\sigma)^N \rho_\sigma(y)$. Hence $P(y) = C\,\kappa(y)\rho_\sigma(y)$, with $\kappa(y) \in [1, r]$ and $r = \frac{1+\epsilon}{1-\epsilon}$. Normalizing, $P(y)/D_{\mathbb{Z}^N,\sigma}(y) \in [1/r, r]$, and therefore $\mathrm{SD}(F(x), \eta) \leq r - 1 = 2\epsilon/(1-\epsilon)$.

*The shift.* By the triangle inequality and the data-processing inequality,

$$\mathrm{SD}(\eta + \mathbf{e}, \eta) \leq \mathrm{SD}(F(x+\mathbf{e}), F(x)) + \frac{4\epsilon}{1-\epsilon} \leq \mathrm{SD}(x+\mathbf{e}, x) + \frac{4\epsilon}{1-\epsilon}.$$

By rotation invariance, $\mathrm{SD}(x+\mathbf{e}, x) = 2\Phi(\|\mathbf{e}\|/(2\sigma_1)) - 1 \leq \|\mathbf{e}\|/(\sqrt{2\pi}\sigma_1)$, since $\Phi' \leq 1/\sqrt{2\pi}$. □

**Corollary 2.4** (Flooding with side information)**.** *Let $(Z, \mathbf{e})$ have any joint law, with $\mathbf{e} \in \mathbb{Z}^N$. Let $\eta \sim D_{\mathbb{Z}^N,\sigma}$ be independent of $(Z, \mathbf{e})$, and let $B > 0$. Then*

$$\mathrm{SD}\big((Z, \mathbf{e}+\eta), (Z, \eta)\big) \leq \Pr[\|\mathbf{e}\|_2 > B] + \sup_{\|\mathbf{e}'\|_2 \leq B} \mathrm{SD}(\mathbf{e}'+\eta, \eta).$$

*The same bound holds after reducing $\mathbf{e}+\eta$ and $\eta$ modulo $q_0$, by data processing. In particular, suppose $\sigma \geq \max\{2^\kappa B/\sqrt{2\pi},\, 2^{42}\}$, with $\epsilon = 2^{-100}$ and $\sigma_2 = 2.1$ as above. Then the right-hand side is at most*

$$\Pr[\|\mathbf{e}\| > B] + 2^{-\kappa}\sigma/\sigma_1 + 2^{-97.9} \leq \Pr[\|\mathbf{e}\| > B] + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}.$$

*Proof.* Condition on $(Z, \mathbf{e}) = (z, \mathbf{e}')$. Because $\eta$ is independent of $(Z, \mathbf{e})$, the conditional distance is $\mathrm{SD}(\mathbf{e}'+\eta, \eta)$. This is at most the supremum when $\|\mathbf{e}'\| \leq B$, and at most 1 otherwise. Averaging over $(z, \mathbf{e}')$ gives the bound. For the numerical form, Lemma 2.3(b) gives $\mathrm{SD}(\mathbf{e}'+\eta, \eta) \leq \|\mathbf{e}'\|/(\sqrt{2\pi}\sigma_1) + 4\epsilon/(1-\epsilon) \leq 2^{-\kappa}\sigma/\sigma_1 + 2^{-97.9}$. Moreover $\sigma/\sigma_1 = (1-\sigma_2^2/\sigma^2)^{-1/2} \leq 1 + \sigma_2^2/\sigma^2$ whenever $\sigma \geq 2\sigma_2$, and for $\sigma \geq 2^{42}$ this is at most $1 + 4.41 \cdot 2^{-84} < 1 + 2^{-81}$. Hence $2^{-\kappa}\sigma/\sigma_1 + 2^{-97.9} \leq 2^{-\kappa}(1+2^{-80}) + 2^{-97}$. □

**Remark 2.5** (The floor $2^{42}$; corrected)**.** An earlier version of this corollary stated the final inequality, in the form $\leq \Pr[\|\mathbf{e}\| > B] + 2^{-\kappa} + 2^{-97}$, "for every $\sigma \geq 2^{20}$". That is false for $\kappa < 60$: at $\sigma = 2^{20}$ the factor $\sigma/\sigma_1$ is $1 + 2^{-38.9}$, and the proof's bound $1 + \sigma_2^2/\sigma^2$ is $1 + 2^{-37.9}$, so the excess $2^{-\kappa} \cdot 2^{-37.9}$ exceeds the room $2^{-97} - 2^{-97.9}$ unless $\kappa \geq 60$. The factor $1 + 2^{-80}$ needs $\sigma \geq 2.1 \cdot 2^{40} \approx 2^{41.1}$; the floor is now $2^{42}$ (composition appendix, Remark 2.6). Nothing numerical changes: at every evaluation setting $\sigma_c \geq 2^\kappa B_{\mathrm{tot}}/\sqrt{2\pi} \geq 2^{55}$. The hockey-stick form of this step (composition appendix, Lemma 5.1) needs no floor at all.

**Remark 2.6** (Comparison with the specification)**.** The specification states Lemma 2 with "$\sigma_c \gg 1$" and "+negl". Lemma 2.3(b) makes both precise:

- the condition is $\sigma_c > \sigma_2 \approx 2.1$;
- the negligible term is $4\epsilon/(1-\epsilon) \leq 2^{-97}$;
- the per-release distance is $2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$ rather than exactly $2^{-\kappa}$.

The difference is immaterial, but the $\delta$ accounting should use the corrected value (Correction 5). An implementation may prefer the elementary bound (a), which needs no smoothing argument. The condition then becomes $\sigma_c \geq 2^\kappa B_{\mathrm{tot}}/2$, which costs $\log_2(\sqrt{2\pi}/2) \approx 0.33$ bits of $\Omega$.

## 3 Obligation O1: a worst-case bound on the whole error

The review requires the bound to be worst case over the data, for two reasons. CKKS decryptions leak the secret through their error [12]. And flooding calibrated to average-case noise estimates has been broken by adversarially chosen inputs [9]. Below, every polynomial determined by the data, or by the public randomness, is treated adversarially. Probabilities are taken only over secrets, errors and fresh coins, which are independent of those polynomials (Lemma 3.6).

### 3.1 Probability tools

A real random variable $X$ is *$v$-subgaussian* if $\mathbb{E}e^{\lambda X} \le e^{v\lambda^2/2}$ for all $\lambda \in \mathbb{R}$.

**Lemma 3.1** (Variance proxies)**.** *(a) The uniform distribution on $\{-1, 0, 1\}$ is $\frac{2}{3}$-subgaussian. Sums of independent variables add proxies, so each coefficient of $s = s_R + s_O$ is $\frac{4}{3}$-subgaussian.*
*(b) $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$ is $\sigma^2$-subgaussian.*
*(c) A mean-zero variable supported on an interval of length 1 is $\frac{1}{4}$-subgaussian.*

*Proof.* (a) Expand in powers of $\lambda$: $\frac{1}{3} + \frac{2}{3}\cosh\lambda = 1 + \sum_{k\ge1} \frac{2}{3(2k)!}\lambda^{2k}$ and $e^{\lambda^2/3} = 1 + \sum_{k\ge1} \frac{\lambda^{2k}}{3^k k!}$. Compare the series term by term, using $\frac{2}{3}\,3^k k! \le (2k)!$ for all $k \ge 1$.

(b) Complete the square:

$$\mathbb{E}e^{\lambda X} = \frac{\sum_z e^{-z^2/(2\sigma^2)+\lambda z}}{\rho_\sigma(\mathbb{Z})} = e^{\lambda^2\sigma^2/2}\,\frac{\rho_\sigma(\mathbb{Z} - \lambda\sigma^2)}{\rho_\sigma(\mathbb{Z})} \le e^{\lambda^2\sigma^2/2},$$

by Lemma 2.1.

(c) This is Hoeffding's lemma. □

**Lemma 3.2** (Hsu, Kakade and Zhang [11], Theorem 1)**.** *Let $x \in \mathbb{R}^n$ satisfy $\mathbb{E}e^{\alpha^\top x} \le e^{\|\alpha\|^2/2}$ for all $\alpha$. This holds, for example, when the coordinates are independent and 1-subgaussian. Let $A \in \mathbb{R}^{m\times n}$ and $\Sigma = A^\top A$. Then, for all $t > 0$,*

$$\Pr\big[\|Ax\|^2 > \operatorname{tr}\Sigma + 2\sqrt{\operatorname{tr}(\Sigma^2)t} + 2\|\Sigma\|t\big] \le e^{-t}.$$

*Since $\operatorname{tr}(\Sigma^2) \le \|\Sigma\|\operatorname{tr}\Sigma$, it follows that $\|Ax\|_2 \le \|A\|_F + \sqrt{2t}\,\|A\|_{\mathrm{op}}$ with probability at least $1 - e^{-t}$.*

**Corollary 3.3** (Products with fixed polynomials)**.** *Let $\xi_1, \dots, \xi_m \in \mathbb{Z}^N$ be independent random vectors with independent coordinates, where those of $\xi_j$ are mean-zero and $v_j$-subgaussian. Let $f_1, \dots, f_m$ be fixed polynomials, or random but independent of the $\xi_j$. Then, with probability at least $1 - e^{-t}$,*

$$\Big\|\sum_j f_j\xi_j\Big\|_2 \le \Big(N\sum_j v_j\|f_j\|_2^2\Big)^{1/2} + \Big(2t\sum_j v_j(\|f_j\|_\infty^{\mathrm{can}})^2\Big)^{1/2}.$$

*More generally, for any fixed real-linear map $\ell : (\mathbb{R}^N)^m \to \mathbb{R}^d$ with matrix $A$ in the scaled variables $\xi_j/\sqrt{v_j}$, $\|\ell(\xi)\| \le \|A\|_F(1 + \sqrt{2t})$ with probability at least $1 - e^{-t}$.*

*Proof.* Multiplication by $f$ has the matrix $M_f = N^{-1}W^*\mathrm{diag}(\sigma(f))W$. This matrix is normal, so $\|M_f\|_F^2 = \|\sigma(f)\|^2 = N\|f\|^2$ and $\|M_f\|_{\mathrm{op}} = \|f\|_\infty^{\mathrm{can}}$. Apply Lemma 3.2 to the block matrix $[\sqrt{v_1}M_{f_1} \mid \cdots \mid \sqrt{v_m}M_{f_m}]$. Its operator norm satisfies $\|A\|_{\mathrm{op}}^2 \le \sum_j v_j\|M_{f_j}\|_{\mathrm{op}}^2$. For the general form, use $\|A\|_{\mathrm{op}} \le \|A\|_F$. When the $f_j$ are random and independent of the $\xi_j$, condition on them. □

**Lemma 3.4** (Norms of random polynomials)**.** *Let $\xi \in \mathbb{Z}^N$ have independent, mean-zero, $v$-subgaussian coordinates. Then:*

- $\|\xi\|_2 \le \sqrt{v}(\sqrt{N} + \sqrt{2t})$ *with probability at least $1 - e^{-t}$;*
- $\|\xi\|_\infty^{\mathrm{can}} \le \sqrt{2vN\ln(2N/\delta)}$ *with probability at least $1 - \delta$.*

*Proof.* The first bound is Lemma 3.2 with $A = \mathbf{I}$.

For the second, take $h$ odd. The real part $\Re\sigma(\xi)_h = \sum_k \xi_k\cos(\pi hk/N)$ is $vN/2$-subgaussian, because $\sum_{k<N}\cos^2(\pi hk/N) = N/2$; the same holds for the imaginary part. So each part exceeds $a$ in absolute value with probability at most $2e^{-a^2/(vN)}$. There are $N/2$ conjugate pairs of slots and two parts each, so a union bound gives $\|\xi\|_\infty^{\mathrm{can}} \le \sqrt{2}\,a$ except with probability $2Ne^{-a^2/(vN)}$. □

### 3.2 The circuit, made explicit

The specification fixes the circuit only up to standard choices for CKKS [7], its RNS variant [8] and hybrid key switching [10]. We make the following choices explicit; Section 10 discusses deviations from them.

(M1) *Moduli.* Ciphertexts start at $Q_2 = q_0q_1q_2$, where $q_0$ may be a product of RNS primes. $P$ is the special modulus. $\textsc{Rescale}_q$ maps $c \mapsto (c - [c]_q)/q$ component-wise, as RNS-CKKS does. Its rounding error on each component lies in $[-\frac{1}{2}, \frac{1}{2}]^N$.

(M2) *Scales.* $\mathsf{R}$ encodes its vectors at scale $\Omega_U$. $\mathsf{O}$ encodes $V_e$ at $\Omega_V$ and the masks at $\Omega_u$. The final scale $\Omega_{\text{fin}} := \Omega_U\Omega_V\Omega_u/(q_1q_2)$ is public, and $\mathsf{R}$ decodes with $\Omega_{\text{fin}}$. The example uses $\Omega_U = q_2$ and $\Omega_V = \Omega_u = q_1 = \Omega_{\text{fin}} = \Omega$.

(M3) *Key switching*, in the hybrid form.

- *Gadget.* The primes of $Q_1 = q_0q_1$ are partitioned into $d_{\text{num}}$ digits with products $\tilde{Q}_j$. Let $\mathsf{g}_j$ be the CRT idempotent of digit $j$: it is $\equiv 1$ modulo the primes of digit $j$ and $\equiv 0$ modulo the others.
- *Decomposition.* $\tilde{D}_j(c) \in \mathcal{R}$ satisfies $\tilde{D}_j(c) \equiv c \pmod{\tilde{Q}_j}$ and $\|\tilde{D}_j(c)\|_\infty \le \varsigma_\uparrow \tilde{Q}_j/2$. Here $\varsigma_\uparrow = 1$ for exact ModUp, and $\varsigma_\uparrow \le k_j$, the number of primes in digit $j$, for fast base extension.
- *Joint rotation keys.* The key for $\psi_g$ is $\text{rk}_j = (b_j, a_j)$, with $b_j = -a_js + P\mathsf{g}_j\psi_g(s) + e_j^{\text{rk}} \bmod PQ_2$. Here $a_j$ is a common random string and $e_j^{\text{rk}} = e_{R,j}^{\text{rk}} + e_{O,j}^{\text{rk}}$. This is the sum of the two parties' shares $-a_js_P + P\mathsf{g}_j\psi_g(s_P) + e_{P,j}^{\text{rk}}$ in the protocol of Mouchet et al. [17]. It is used at level 1 after reduction modulo $PQ_1$.
- *The switch.* $\text{KS}(c_0, c_1) = \big(c_0 + \lfloor P^{-1}\sum_j \tilde{D}_j(c_1)b_j\rceil,\ \lfloor P^{-1}\sum_j \tilde{D}_j(c_1)a_j\rceil\big)$. The ModDown rounding errors $\text{rd}_0, \text{rd}_1$ lie in $[-\frac{1}{2}, \frac{1}{2}]^N$.

(M4) *Release* is Algorithm 5 at level 0, with $p_0 = -as + e^{\text{pk}} \bmod q_0$ and $e^{\text{pk}} = e_R^{\text{pk}} + e_O^{\text{pk}}$. After two rescales the ciphertext is at $q_0$. The "modulus switch to $q_0$" of Algorithm 4 is therefore a no-op, or an exact limb drop.

(M5) *Distributions.* $s_R, s_O, u \leftarrow \chi_{\text{key}}$, uniform ternary. Every error is drawn from $\chi_{\text{err}} = \mathcal{N}_{\mathbb{Z}}(0, \sigma_e^2)^N$, with $\sigma_e = 3.2$.

**Lemma 3.5** (Key switching)**.** *Suppose* $c_0 + c_1\psi_g(s) \equiv \mu \pmod{Q_1}$. *Then* $\text{KS}(c_0, c_1) = (c_0', c_1')$ *satisfies* $c_0' + c_1's \equiv \mu + e_{\text{ks}} \pmod{Q_1}$, *where*

$$e_{\text{ks}} = P^{-1}\sum_j \tilde{D}_j(c_1)\, e_j^{\text{rk}} + \text{rd}_0 + \text{rd}_1 s.$$

*Proof.* Modulo $PQ_1$, $\sum_j \tilde{D}_j(b_j + a_js) = P\sum_j \tilde{D}_j\mathsf{g}_j\psi_g(s) + \sum_j \tilde{D}_j e_j^{\text{rk}}$. By the CRT, $\sum_j \tilde{D}_j(c_1)\mathsf{g}_j \equiv c_1 \pmod{Q_1}$. Divide by $P$, which is exact modulo $Q_1$ for multiples of $P$, and add the rounding errors. □

**Lemma 3.6** (The $c_1$-components are independent of all secrets and errors)**.** *At every step of Algorithm 4, the* $c_1$*-component of every ciphertext is a deterministic function of three things: the seeds' expansions* $a_h$*, the common random strings* $a_j$ *of the rotation keys, and the data. Consequently* $\tilde{D}_j(\cdot)$, $\text{rd}_1$ *and the* $c_1$*-rounding errors of every rescale are independent of* $(s_R, s_O, u,$ *all errors*$)$.

*Proof.* By induction over the operations:

- the conversion leaves $c_1 = a_h$;
- a ct–pt product maps $c_1 \mapsto m_V c_1$, and additions add $c_1$-components;
- a rescale maps $c_1 \mapsto \lfloor c_1/q\rceil$;
- an automorphism maps $c_1 \mapsto \psi_g(c_1)$;
- key switching outputs $\lfloor P^{-1}\sum_j \tilde{D}_j(c_1)a_j\rceil$.

None of these uses $c_0$, a secret or an error. □

### 3.3 The noiseless centre and the error decomposition

For entry $e$, let $U_e$ be R's vector and $V_e$ O's vector. Define the plaintexts $m_{U_e,h} = \mathsf{Ecd}_{\Omega_U}(U_e[h])$, $m_{V_e,h} = \mathsf{Ecd}_{\Omega_V}(V_e[h])$ and $m_{u_e} = \mathsf{Ecd}_{\Omega_u}(\mathbf{u}_{k_e}) = \Omega_u \tau^{-1}(\mathbf{u}_{k_e}) + r_{u_e}$.

**Definition 3.7** (Noiseless evaluation)**.** For release $s$,

$$\widehat{Z}_s(D) = \sum_{e \in \mathcal{E}_s} \frac{1}{q_1 q_2}\, \mathrm{A}\Big(\sum_{h \le m} m_{U_e,h} m_{V_e,h}\Big) m_{u_e} \ \in \mathbb{Q}[X]/(X^N+1), \qquad f_s(D) = \lfloor \widehat{Z}_s(D) \rceil \in \mathbb{Z}^N.$$

This is a deterministic function of O's data, R's data and the public parameters. It does not depend on any key, error or random coin.

**Proposition 3.8** (Error decomposition)**.** *Let $v_s = h_{O,s} + c'_{1,s} s_R \bmod q_0$. Under (M1)–(M5),*

$$v_s \equiv f_s(D) + e_{\mathrm{tot},s} + \eta_s \pmod{q_0}, \qquad e_{\mathrm{tot},s} = \sum_{i=1}^{8} T_i \in \mathbb{Z}^N,$$

*where, writing $\Pi_e(y) := q_1^{-1} y\, \Omega_u \tau^{-1}(\mathbf{u}_{k_e})$ for the main part of masking and rescaling,*

$$\begin{aligned}
T_1 &= \sum_e \Pi_e\big(\mathrm{A}(x_e^R)\big), \quad x_e^R = q_2^{-1} \sum_h e_{R,h}^{(U_e)} m_{V_e,h} && \text{(R's encryption errors)}\\
T_2 &= \sum_e \Pi_e\big(\mathrm{A}(\varepsilon_{0,e} + \varepsilon_{1,e} s)\big) && \text{(first rescale)}\\
T_3 &= \sum_e \Pi_e\Big(\sum_{i<L} T_{i+1}\big(P^{-1} \textstyle\sum_j \tilde{D}_{e,i,j} e_j^{\mathrm{rk},(i)}\big)\Big) && \text{(key-switching gadget)}\\
T_4 &= \sum_e \Pi_e\Big(\sum_{i<L} T_{i+1}\big(\mathrm{rd}_{0,e,i} + \mathrm{rd}_{1,e,i} s\big)\Big) && \text{(ModDown roundings)}\\
T_5 &= \sum_e (\varepsilon'_{0,e} + \varepsilon'_{1,e} s) && \text{(second rescale)}\\
T_6 &= e^{\mathrm{pk}} u + e_0 + e_1 s && \text{(fresh encryption)}\\
T_7 &= \widehat{Z}_s(D) - f_s(D), \quad \|T_7\|_\infty \le \tfrac{1}{2} && \text{(rounding of the centre)}\\
T_8 &= \sum_e q_1^{-1} Y_e\, r_{u_e} && \text{(mask spill of the errors).}
\end{aligned}$$

*Here $Y_e = \mathrm{A}(x_e^R + \varepsilon_{0,e} + \varepsilon_{1,e} s) + \sum_i T_{i+1}(e_{\mathrm{ks},e,i})$. All $\varepsilon$ and* rd *vectors lie in $[-\frac{1}{2}, \frac{1}{2}]^N$, and $e_j^{\mathrm{rk},(i)}$ is the error of digit $j$ of the key for $\psi_{g_i}$.*

*Proof.* Track the decryption invariant $c_0 + c_1 s$, as integer polynomials, through Algorithm 4.

*Conversion and products.* The conversion gives $c_0 + a_h s = m_{U,h} + e_{R,h}$ exactly. After the products and the chunk sum, the value is $\mu_{\mathbb{Z}} + KQ_2$ for some $K \in \mathcal{R}$, with $\mu_{\mathbb{Z}} = \sum_h m_{U,h} m_{V,h} + \sum_h e_{R,h} m_{V,h}$.

*First rescale.* Write the rounding of $c_j/q_2$ as $c_j/q_2 + \varepsilon_j$. Then $c'_0 + c'_1 s = \mu_{\mathbb{Z}}/q_2 + \varepsilon_0 + \varepsilon_1 s + KQ_1$.

EvalSum. By Lemma 3.5 and $\psi_g(c_0) + \psi_g(c_1)\psi_g(s) = \psi_g(c_0 + c_1 s)$, induction over the $L$ steps gives $\mathrm{A}(\mu^{(2)}) + \sum_i T_{i+1}(e_{\mathrm{ks},i}) \pmod{Q_1}$, by Lemma 1.5.

*Masking and second rescale.* Multiplying by $m_{u_e}$ and rescaling by $q_1$ adds $\varepsilon'_0 + \varepsilon'_1 s$ modulo $q_0$.

*Release.* Summing over $e$, the release adds $(-as + e^{\mathrm{pk}})u + e_0 + (au + e_1)s + \eta = e^{\mathrm{pk}} u + e_0 + e_1 s + \eta$.

*Collecting terms.* Split $m_{u_e} = \Omega_u \tau^{-1}(\mathbf{u}_{k_e}) + r_{u_e}$. The terms not involving any key, error or rounding make up exactly $\widehat{Z}_s(D)$; the rest are $T_1, \dots, T_6$ and $T_8$. Since $v_s$, $f_s$ and $\eta_s$ are integral, so is $e_{\mathrm{tot},s}$. □

**Remark 3.9** (Why the centre matters)**.** The specification centres the mechanism at $\mathsf{Ecd}_\Omega(\mathbf{z}_s(D))$ and counts the encoding roundings of $V_e$ and $\mathbf{u}_k$ as error. The deterministic difference $\widehat{Z}_s(D) - \Omega\tau^{-1}(\mathbf{z}_s(D))$ contains the following terms:

- the *mask spill* of the statistic itself, $\sum_e(\Omega_U\Omega_V/(q_1q_2))S_e\, r_{u_e}$, where $S_e$ is the entry's value. Its $\ell_2$ norm is about $\sqrt{N/12}\,(\sum_e S_e^2)^{1/2}$. This is up to $2^{30.7}$ for $\bar{n}_O = 10^6$, and up to $2^{34.5}$ by the triangle inequality;
- the cross terms $\mathrm{A}(r_U \cdot \Omega_V\tau^{-1}(V))$ between $\mathsf{R}$'s rounding and $\mathsf{O}$'s data. These are up to $\frac{\sqrt{N}}{2}\sqrt{m}\,(\bar{n}_O\Gamma_\mathcal{E})^{1/2} \approx 2^{25}$, in the notation of Section 3.4;
- a nominal-versus-exact scale mismatch of relative size about $2^{-30}$ when the implementation decodes with a nominal scale, as OpenFHE's FIXEDMANUAL does. This gives about $2^{23}$; see Section 3.6.

All three are deterministic functions of $D$. They are part of the *query*, not hints about secrets. They need no flooding, provided the mechanism is centred at $f_s(D)$ and the sensitivity of $f_s$ is used (Corollary 3.13). Flooding them instead would require $B_{\mathrm{tot}} \approx 2^{30.7}$ to $2^{34.5}$, and hence $\log_2\Omega \approx 67$ to 71 for stage 2 of the example (script `scratch/ecd_centre.py`).

### 3.4 Term-by-term bounds, worst case over data

Let $|\mathcal{E}|$ be the number of entries in the release and $m = \lceil \bar{n}_O/N_s \rceil$. Fix $\delta_B$ and split it into $n_{\mathrm{ev}} = 12$ equal parts. Put

$$t = \ln(n_{\mathrm{ev}}/\delta_B), \qquad \tau_t = 1 + \sqrt{2t}, \qquad \ell_N = \ln(2Nn_{\mathrm{ev}}/\delta_B), \qquad v_s = \tfrac{4}{3}, \quad v_u = \tfrac{2}{3}.$$

**Lemma 3.10** (Data bounds)**.** *Every dataset in the clipped domain satisfies* $\sum_{e\in\mathcal{E}} V_{e,t}^2 \le \Gamma_\mathcal{E}$ *for every row* $t$*, where*

$$\Gamma_{\mathcal{E}_G} = o^2 + (d_R+1)o, \qquad \Gamma_{\mathcal{E}_2} = o^2 + (d_R+1)(o+w) + ow + w^2.$$

*Consequently* $\sum_e\sum_h \|m_{V_e,h}\|_2^2 \le M_V^2$*, with* $M_V := \Omega_V\sqrt{2/N}\,(\bar{n}_O\Gamma_\mathcal{E})^{1/2} + \frac{1}{2}(|\mathcal{E}|mN)^{1/2}$.

*Proof.* Bound the entries one block at a time:

- for $\mathrm{triu}\,\mathbf{B}$: $\sum_{j\le k}(x_jx_k)^2 = \frac{1}{2}(\|\mathbf{x}\|^4 + \sum_j x_j^4) \le \|\mathbf{x}\|^4 \le o^2$;
- for $\mathbf{C}_{jk}$, where $V = x_{O,k}$ for $d_R+1$ values of $j$: $(d_R+1)\|\mathbf{x}\|^2$;
- for $\mathbf{c}_R$: $(d_R+1)y^2$;
- for $\mathbf{c}_O$: $\|\mathbf{x}\|^2y^2$;
- for $\mathbf{y}^\top\mathbf{y}$: $y^4$.

Dummy and unmatched rows have $V = 0$, or are bounded in the same way. For the second claim, $\|m_{V,h}\| \le \Omega_V\sqrt{2/N}\|V_h\| + \sqrt{N}/2$ by Lemma 1.2. Then apply Minkowski's inequality over the family $(e,h)$, and use $\sum_{e,h}\|V_{e,h}\|^2 \le \bar{n}_O\Gamma_\mathcal{E}$. □

**Proposition 3.11** (Bounds on the terms)**.** *Assume (M1)–(M5), and let* $\tilde{Q}_{\max} = \max_j \tilde{Q}_j$*. Fix all public randomness: the expansions* $a_h$*, the common random strings* $a, a_j$*, and the dataset* $D$ *in the clipped domain. Then each of the following bounds fails with probability at most* $\delta_B/n_{\mathrm{ev}}$ *over*

$(s_R, s_O, u, \textit{all errors})$*; the bound on* $T_6$ *uses five of these events.*

$$\begin{aligned}
\|T_1\| &\le B_1 = \sqrt{\tfrac{2}{N}}\,\tfrac{\Omega_u}{q_1}\,\tfrac{N}{2q_2}\,\sqrt{2}\,\sigma_e\,M_V\,\tau_t,\\
\|T_2\| &\le B_2 = \sqrt{\tfrac{2}{N}}\,\tfrac{\Omega_u}{q_1}\,\tfrac{N}{2}\Big(\sqrt{|\mathcal{E}|/2} + \sqrt{v_s}\sqrt{2|\mathcal{E}|}\,\tfrac{\sqrt{N}}{2}\,\tau_t\Big),\\
\|T_3\| &\le B_3 = \sqrt{\tfrac{2}{N}}\,\tfrac{\Omega_u}{q_1}\,\sqrt{2}\,\sigma_e\,\big(|\mathcal{E}|\,L\,d_{\mathrm{num}}\big)^{1/2}\,\frac{\varsigma_\uparrow N^{3/2}\tilde{Q}_{\max}}{2P}\,\tau_t,\\
\|T_4\| &\le B_4 = \sqrt{\tfrac{2}{N}}\,\tfrac{\Omega_u}{q_1}\,\sqrt{|\mathcal{E}|}\Big(\tfrac{N}{2}\sum_{i<L} 2^{(L-1-i)/2} + \sqrt{v_s}\,L\,\tfrac{N^{3/2}}{2}\,\tau_t\Big),\\
\|T_5\| &\le B_5 = |\mathcal{E}|\tfrac{\sqrt{N}}{2} + \sqrt{v_s}\,|\mathcal{E}|\,\tfrac{N}{2}\,\tau_t,\\
\|T_6\| &\le B_6 = \big(N(v_u\beta_{\mathrm{pk}}^2 + \sigma_e^2 + \sigma_e^2\beta_s^2)\big)^{1/2} + \big(2t(v_u\gamma_{\mathrm{pk}}^2 + \sigma_e^2 + \sigma_e^2\gamma_s^2)\big)^{1/2},\\
\|T_7\| &\le B_7 = \sqrt{N}/2, \qquad \|T_8\| \le B_8 \quad (\textit{defined in the proof}).
\end{aligned}$$

*Here* $\beta_{\mathrm{pk}} = \sqrt{2}\sigma_e(\sqrt{N}+\sqrt{2t})$, $\gamma_{\mathrm{pk}} = (4\sigma_e^2 N\ell_N)^{1/2}$, $\beta_s = \sqrt{v_s}(\sqrt{N}+\sqrt{2t})$ *and* $\gamma_s = (2v_s N\ell_N)^{1/2}$.

*Proof.* Three facts are used throughout.

- By Lemma 1.7, the main part of $\sum_e \Pi_e(y_e)$ has norm $\sqrt{2/N}(\Omega_u/q_1)(\sum_e |\tau(y_e)_{k_e}|^2)^{1/2}$, because the entries occupy distinct slots.
- By Lemma 3.6, the polynomials $\tilde{D}$, $\mathrm{rd}_1$, $\varepsilon_1$ and $\varepsilon_1'$ are independent of the random variables they multiply. Corollary 3.3 therefore applies with them treated as fixed, and with no assumption on their distribution. This is the worst case over data required by the review.
- The deterministic roundings $\varepsilon_0$, $\varepsilon_0'$ and $\mathrm{rd}_0$ are bounded only through $\|\cdot\|_\infty \le \frac{1}{2}$.

$T_1$. By Lemma 1.5, $\tau(\mathrm{A}x)_k = \frac{N}{2}(x_0 + ix_{N/2})$. The coefficients $(fg)_0$ and $(fg)_{N/2}$ are inner products of $f$ with signed permutations of $g$. So $e_R \mapsto (\frac{N}{2}x^R_{e,0}, \frac{N}{2}x^R_{e,N/2})_e$ is linear with Frobenius norm $\frac{N}{2q_2}\sqrt{2}(\sum_{e,h}\|m_{V_e,h}\|^2)^{1/2} \le \frac{N}{2q_2}\sqrt{2}M_V$, by Lemma 3.10. Entries that share R's ciphertext share $e_R$; the general form of Corollary 3.3 allows this. The coordinates of $e_R$ are $\sigma_e^2$-subgaussian (Lemma 3.1).

$T_2$. The deterministic part satisfies $\frac{N}{2}|\varepsilon_{0,0} + i\varepsilon_{0,N/2}| \le \frac{N}{2}\cdot\frac{1}{\sqrt{2}}$ per entry. The random part is linear in $s$. Its $2|\mathcal{E}|$ rows have norm at most $\frac{N}{2}\|\varepsilon_{1,e}\| \le \frac{N}{2}\cdot\frac{\sqrt{N}}{2}$.

$T_3$. Apply Lemma 1.6 with $f = P^{-1}\tilde{D}_{e,i,j}$ and $\mathcal{H}' = \mathcal{H}_{i+1}$. This gives coefficient vectors with $\|\beta\| \le N\|f\|_2 \le N\cdot\sqrt{N}\varsigma_\uparrow\tilde{Q}_j/(2P)$. The coordinates of $e^{\mathrm{rk}} = e_R^{\mathrm{rk}} + e_O^{\mathrm{rk}}$ are $2\sigma_e^2$-subgaussian, and different $(i,j)$ have independent errors. Sum the squared norms over $(e,i,j)$.

$T_4$. For $\mathrm{rd}_0$, Lemma 1.6 with $f=1$ gives $|\sum_{g\in\mathcal{H}'}\sigma(\mathrm{rd}_0)_{hg}| \le \|\mathrm{rd}_0\|_2\sqrt{N|\mathcal{H}'|} \le \frac{N}{2}\sqrt{|\mathcal{H}_{i+1}|}$. For $\mathrm{rd}_1 s$, take $f = \mathrm{rd}_1$: then $\|\beta\| \le N\|\mathrm{rd}_1\| \le N^{3/2}/2$ for each $i$, and at most $LN^{3/2}/2$ after summing over $i$ for a fixed entry.

$T_5$. The deterministic part has $\|\sum_e \varepsilon'_{0,e}\| \le |\mathcal{E}|\sqrt{N}/2$. For the random part, $F = \sum_e \varepsilon'_{1,e}$ is independent of $s$, with $\|F\|_2 \le |\mathcal{E}|\sqrt{N}/2$ and $\|F\|_\infty^{\mathrm{can}} \le \|F\|_1 \le |\mathcal{E}|N/2$. Apply Corollary 3.3.

$T_6$. Condition on $(e^{\mathrm{pk}}, s)$, which is independent of the fresh $(u, e_0, e_1)$. Apply Corollary 3.3 with $(f_j) = (e^{\mathrm{pk}}, 1, s)$. Then bound $\|e^{\mathrm{pk}}\|, \|e^{\mathrm{pk}}\|_\infty^{\mathrm{can}}, \|s\|, \|s\|_\infty^{\mathrm{can}}$ by Lemma 3.4, which costs four further events.

$T_8$. By Lemma 1.7, $\|T_8\| \le \frac{N}{2q_1}\sum_e\|Y_e\|$. We have $\|\mathrm{A}y\|_2 = \frac{N}{2}(y_0^2 + y_{N/2}^2)^{1/2}$, which is at most $\sqrt{N/2}(q_1/\Omega_u)(B_1+B_2)$ for each entry, and $\|T_{i+1}w\| \le |\mathcal{H}_{i+1}|\|w\|$. A union bound over the $|\mathcal{E}|L$ key switchings with $t' = t + \ln(|\mathcal{E}|L)$ bounds each $\|e_{\mathrm{ks}}\|$ by $\epsilon_{\mathrm{ks}} = \sqrt{2}\sigma_e d_{\mathrm{num}}\frac{N\tilde{Q}_{\max}}{2P}\tau_{t'} + \frac{\sqrt{N}}{2} + \sqrt{v_s}\frac{N}{2}\tau_{t'}$. This gives

$$B_8 = \tfrac{N}{2q_1}\big(|\mathcal{E}|\sqrt{N/2}\,(q_1/\Omega_u)(B_1+B_2) + |\mathcal{E}|\tfrac{N}{2}\epsilon_{\mathrm{ks}}\big).$$

□

**Theorem 3.12** (Obligation O1 discharged)**.** *Under (M1)–(M5), for every dataset $D$ in the clipped domain and every value of the public randomness,*

$$\Pr\big[\|e_{\mathrm{tot},s}\|_2 \le B_{\mathrm{tot},s} := B_1 + \cdots + B_8\big] \ge 1 - \delta_B,$$

*with the probability taken over the key shares, all errors and the fresh randomness only. The centre is $f_s(D)$ of Definition 3.7. The bound covers the whole ring element, in the coefficient $\ell_2$ norm.*

*Proof.* Combine Proposition 3.8, Proposition 3.11, the triangle inequality and a union bound over the $n_{\mathrm{ev}}$ events. □

**Corollary 3.13** (Sensitivity of the centre)**.** *Let $U_{\max,e}$ bound $|U_e|$: 1 for $\mathbf{b}$, and $B_R$ for $\dot{\mathbf{x}}_j$. Put $\nu_U = (\sum_e U_{\max,e}^2)^{1/2}$ and $\nu_V = \sup_{D\sim D'}(\sum_e (V_e - V_e')_t^2)^{1/2} \le 2\Gamma_{\mathcal{E}}^{1/2}$. Then, with $\Omega_U = \Omega_V = \Omega_u = q_1 = q_2$,*

$$\sup_{D\sim D'} \|f_s(D) - f_s(D')\|_2 \le \widehat{\Delta}_{\mathrm{enc},s} := \sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_s + \Delta_{x,s} + \sqrt{N},$$

$$\Delta_{x,s} = \sqrt{N/2}\,(\nu_V + \sqrt{N}\nu_U) + \tfrac{\sqrt{N}}{2}\sqrt{|\mathcal{E}|}\,\Delta_s + \tfrac{N^{3/2}}{4q_1}\sqrt{|\mathcal{E}|}(\nu_V + \sqrt{N}\nu_U).$$

*Terms of order $N^{3/2}\sqrt{|\mathcal{E}|}/\Omega$, which come from the $\sqrt{N}/2$ parts of $\|m_U\|$, are omitted; they are below $2^{-20}$ for the example and are covered by adding 1 to $\Delta_{x,s}$. For the example, $\Delta_{x,2} \approx 2^{18.8}$ against $\sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_2 \approx 2^{52.4}$. So $\widehat{\Delta}_{\mathrm{enc}}$ exceeds the specification's $\Delta_{\mathrm{enc}}$ by a relative $2^{-33.7}$, and $\rho_s$ by a relative $2^{-32.7}$.*

*Proof.* Neighbours differ in one row, which sits in chunk $h^*$ and slot $l^*$. Let $\Delta m_e = \mathsf{Ecd}(V_e[h^*]) - \mathsf{Ecd}(V_e'[h^*]) = \Omega_V\tau^{-1}(\Delta V_e \mathbf{u}_{l^*}) + \delta r_e$, with $\|\delta r_e\|_\infty \le 1$.

*The slot value.* By Lemma 1.5, $\mathrm{A}(m_{U_e,h^*}\Delta m_e)$ has all slots equal to

$$c_e = \sum_l \tau(m_U)_l \tau(\Delta m_e)_l = \Omega_U\Omega_V U_{e,l^*}\Delta V_e + \Omega_V\tau(r_U)_{l^*}\Delta V_e + \sum_l \tau(m_U)_l\tau(\delta r_e)_l.$$

The first term is $\Omega_U\Omega_V$ times the change of entry $e$, and these changes have $\ell_2$ norm at most $\Delta_s$. Bound $|\tau(r_U)_{l^*}| \le \|r_U\|_1 \le N/2$. By Cauchy–Schwarz and Lemma 1.1, the last sum is at most $\frac{N}{2}\|m_U\|\|\delta r\| \le \frac{N}{2}(\Omega_U U_{\max} + \frac{\sqrt{N}}{2})\sqrt{N}$.

*Masking.* Apply Lemma 1.7. The main part contributes $\sqrt{2/N}\,(\Omega_u/(q_1q_2))\|(c_e)_e\|$, and the spill contributes $\frac{\sqrt{N}}{2q_1q_2}\sum_e |c_e|$.

*Rounding.* Finally, $\|\lfloor x\rceil - \lfloor x'\rceil\| \le \|x - x'\| + \sqrt{N}$. □

### 3.5 Numerical evaluation

Take the example parameters: $N = 2^{14}$, $L = 13$, $d_R = 4$, $d_O = 6$, $\bar{n}_O = 10^6$ (so $m = 123$), $B_R = 5.59$, $B_O = 6.24$, $B_y = 2.89$, $\sigma_e = 3.2$, ternary $s_R, s_O, u$, $\delta_B = 2^{-50}$ (so $t = 37.1$ and $\tau_t = 9.62$), $d_{\mathrm{num}} = 2$ and $\varsigma_\uparrow = 1$. Then $|\mathcal{E}_G| = 51$, $|\mathcal{E}_2| = 63$ (joint mode), $\Gamma_{\mathcal{E}_G} = 1711$ and $\Gamma_{\mathcal{E}_2} = 2147$.

The budget is $\varepsilon = 1$ and $\delta_{\mathrm{run}} = 10^{-9}$, with three releases, $\kappa = 40$ and $f_1 = 0.1$. The conversion of Proposition 5.4 gives $\rho_{\mathrm{run}} = 0.01496$ and a stage-2 slot s.d. of 525.

The column $\log_2\Omega$ is the least scale meeting CALIBRATE's condition $\sigma_{c,s} \ge 2^\kappa B_{\mathrm{tot},s}/\sqrt{2\pi}$, with $\sigma_{c,s} = \Delta_{\mathrm{enc},s}/\sqrt{2\rho_s}$. The condition is binding for stage 2. Script: `scratch/btot.py`.

| Configuration (stage 2) | $B_1$ | $B_2$ | $B_3$ | $B_4$ | $B_5$ | $B_6$ | $\log_2 B_{\mathrm{tot}}$ | $\log_2 \Omega$ |
|---|---|---|---|---|---|---|---|---|
| (A) Algorithm 4 as specified, $\tilde{Q}_{\max}/P = 1$ | 20.9 | 19.5 | 24.3 | 23.7 | 22.5 | 17.2 | 25.34 | 61.5 |
| (A′) as (A), $P \geq 2^{20}\tilde{Q}_{\max}$ | 20.9 | 19.5 | 4.3 | 23.7 | 22.5 | 17.2 | 24.39 | 60.5 |
| (C) EvalSum before the first rescale; stochastic rounding; one final rescale; $P \geq 2^{10}\tilde{Q}_{\max}$ | 20.9 | 14.1 | $< 0$ | $< 0$ | 14.0 | 17.2 | 21.07 | 57.2 |
| (C′) as (C), with $\Omega_U = q_2 = 2^{60}$ | 13.4 | 14.1 | $< 0$ | $< 0$ | 14.0 | 17.2 | 17.56 | 53.7 |
| Fresh-encryption error only (what v3.1 assumed) | – | – | – | – | – | 17.2 | 17.17 | 53.3 |

Table 1: $\log_2$ of the bounds of Proposition 3.11 for stage 2 of the example. In all rows $B_7 = 2^6$ and $B_8 < 1$. For stage 1 the required scale is 1.4 to 1.6 bits lower; for (A) it is 59.9. Using Lemma 2.3(a) instead of (b) adds 0.33 bits.

**Reading the table.**

- The specification's $\log_2 \Omega \approx 52.5$ floods only the fresh-encryption error, measured by its *mean* $\ell_2$ norm of about $2^{16.4}$. A high-probability bound on that term alone already needs 53.3.
- For the circuit exactly as specified, the worst-case analysis is dominated by three terms:
  - the gadget term $B_3$, unless $P \gg \tilde{Q}_{\max}$;
  - the ModDown roundings $B_4$;
  - the $|\mathcal{E}|$ separate second rescales $B_5$.

  The latter two are large because roundings are treated as adversarially aligned with a slot. That is the price of a bound with no randomness assumption on rounding errors.
- Variant (C) removes these terms in three ways:
  - EvalSum at level 2, so that key-switching errors are divided by $q_2$ and the first-rescale rounding is not traced;
  - stochastic rounding, so that rounding errors are mean-zero and $\frac{1}{4}$-subgaussian given the ciphertext, by Lemma 3.1(c);
  - a single rescale after summing the masked products.

  (C′) also shrinks $T_1$ by encoding $\mathsf{R}$'s vectors at a larger scale.

The bounds for (C) and (C′) use the same lemmas. We give the formulas in the script rather than repeating the proofs. [**Not fully rigorous:** The proofs for variants (C) and (C′) are only sketched. The $T_2$ and $T_5$ formulas for (C) apply Corollary 3.3 conditionally on $s$, to the independent stochastic-rounding coins.]

### 3.6 Empirical sanity check with OpenFHE

**Setup.** We ran the circuit shape of Algorithm 4 in OpenFHE [1], through its Python bindings, (`scratch/ofhe_check.py`). The run used:

- a single key, FIXEDMANUAL, $N = 2^{14}$, scale $2^{50}$, depth 2;
- one entry, with $m \in \{4, 16\}$ chunks;
- 0/1 vectors $U$ and $|V| \leq o/2$.

Two runs on the same data with independent keys and randomness isolate the randomness-dependent error, because the deterministic part cancels in the difference.

**Results.**

- The measured coefficient $\ell_2$ norm of the random error was $2^{14.0}$ for $m = 4$ and $2^{15.7}$ for $m = 16$. Theorem 3.12 at the same setting gives $2^{21.8}$ and $2^{21.9}$. The bound holds, with a margin of 6

to 8 bits, as expected for a worst-case, high-probability bound.

- The deviation of the decryption from $\mathsf{Ecd}(\mathbf{z})$ was $2^{23.0}$. It is dominated by a deterministic relative error of $8.7 \cdot 10^{-10} \approx 2^{-30}$ at the target slot, from decoding with the nominal scale $2^{50}$ rather than the exact $\Omega_{\text{fin}}$. This confirms Remark 3.9 and Correction 2.

# 4 Decoupled scales, OpenFHE's key switching and the decoding offset

The evaluation runs circuit (A) with *decoupled* encoding scales ($\Omega_U = 2^{68}$, $\Omega_u = q_1$, $\Omega_V$ chosen by CALIBRATE) and with the hybrid key switching of OpenFHE 1.5.1. Proposition 3.11 and Theorem 3.12 are already stated for general $\Omega_U, \Omega_V, \Omega_u, q_1, q_2$: no step of their proofs uses $\Omega_U = q_2$ or $\Omega_V = \Omega_u = q_1$. Three things are not covered by Sections 3.2–3.5: OpenFHE's uncentred gadget (Lemma 4.1), the sensitivity of the centre with decoupled scales (Lemma 4.3; Corollary 3.13 assumes equal scales), and a public bound on the decoding offset (Proposition 4.5), which the statistical part needs. This section proves them. The proofs were first written in the companion book on the protocol (its Appendix B, Lemmas `btot.gadget` and `btot.sensZ`, Proposition `btot.offset`, and Chapter 9, Corollary `agg.sensZ` and Lemma `agg.offset`); they were checked here step by step, with the corrections noted, and the numbers recomputed (script `experiments/proofs_check/check_btot_offset.py` of the repository).

## 4.1 OpenFHE's uncentred key switching

(M3′) As (M3), with three differences. (i) The digit $\tilde{D}_j(c)$ is the integer polynomial $\sum_{i \in j} [c_i (\tilde{Q}_j/q_i)^{-1}]_{q_i} (\tilde{Q}_j/q_i)$, with residues $[\cdot]_{q_i} \in [0, q_i)$; so $\tilde{D}_j(c) \equiv c \pmod{\tilde{Q}_j}$ and $\tilde{D}_j(c) \in [0, k_j \tilde{Q}_j)^N$ (fast base extension without centring). (ii) ModDown is $\mathrm{MD}(x) = (x - y_P)/P$, with $y_P = \sum_i [x_i (P/p_i)^{-1}]_{p_i} (P/p_i) \in [0, k_P P)^N$ and $y_P \equiv x \pmod{P}$, where $p_1, \ldots, p_{k_P}$ are the special primes; so $\mathrm{MD}(x) = x/P + \mathrm{rd}$ with $\mathrm{rd} = -y_P/P \in (-k_P, 0]^N$, the division being exact. (iii) The key switch from $s$ to $\psi_g^{-1}(s)$ is applied *before* the automorphism: the key is $b_j = -a_j \psi_g^{-1}(s) + P \mathsf{g}_j s + e_j^{\mathrm{rk}}$, and the rotation outputs $(\psi_g(c_0 + \mathrm{MD}(\sum_j \tilde{D}_j(c_1) b_j)),\ \psi_g(\mathrm{MD}(\sum_j \tilde{D}_j(c_1) a_j)))$.

In the notation of Proposition 3.11, (M3′) gives $\|\tilde{D}_j\|_\infty < k_j \tilde{Q}_j \le \varsigma_\uparrow \tilde{Q}_{\max}/2$ with $\varsigma_\uparrow = 2 \max_j k_j$, and $\|\mathrm{rd}\|_\infty < k_P = \varsigma_\downarrow/2$ with $\varsigma_\downarrow = 2k_P$, against $\varsigma_\uparrow \le \max_j k_j$ and $\varsigma_\downarrow = 1$ for the centred gadget of (M3).

**Lemma 4.1** (Uncentred digits and ModDown rounding)**.** *Assume (M1), (M2), (M3′), (M4), (M5). Then Proposition 3.8 holds verbatim with every $\tilde{D}_{e,i,j}$, $e_j^{\mathrm{rk},(i)}$, $\mathrm{rd}_{0,e,i}$, $\mathrm{rd}_{1,e,i}$ replaced by its image under the automorphism $\psi_{g_i}$ of rotation $i$. Proposition 3.11 and Theorem 3.12 hold with $\varsigma_\uparrow = 2 \max_j k_j$ in $B_3$ and $\epsilon_{\mathrm{ks}}$, and with the ModDown terms (all of $B_4$, and the terms $\sqrt{N}/2$ and $\sqrt{v_s} \frac{N}{2} \tau_{t'}$ of $\epsilon_{\mathrm{ks}}$) multiplied by $\varsigma_\downarrow = 2k_P$. $B_1$, $B_2$, $B_5$, $B_6$, $B_7$ are unchanged. No mean-zero or symmetry property of the digits or of the roundings is used.*

*Proof. The switch.* Let $c_0 + c_1 s \equiv \mu \pmod{Q_1}$. Modulo $PQ_1$, $\sum_j \tilde{D}_j(c_1)(b_j + a_j \psi_g^{-1}(s)) = P \sum_j \tilde{D}_j(c_1) \mathsf{g}_j s + \sum_j \tilde{D}_j(c_1) e_j^{\mathrm{rk}}$, and $\sum_j \tilde{D}_j(c_1) \mathsf{g}_j \equiv c_1 \pmod{Q_1}$ by the CRT, because $\tilde{D}_j(c_1) \equiv c_1 \pmod{\tilde{Q}_j}$ holds for uncentred digits as well. $\mathrm{MD}(x) \bmod Q_1$ depends only on $x \bmod PQ_1$ (adding $PQ_1 K$ to $x$ adds $Q_1 K$ to $\mathrm{MD}(x)$). Dividing by $P$ and adding the roundings, the switched pair

$(c_0', c_1')$ satisfies

$$c_0' + c_1'\psi_g^{-1}(s) \equiv \mu + e'_{\mathrm{ks}} \pmod{Q_1}, \qquad e'_{\mathrm{ks}} = P^{-1}\sum_j \tilde{D}_j(c_1)e_j^{\mathrm{rk}} + \mathrm{rd}_0 + \mathrm{rd}_1\psi_g^{-1}(s).$$

*The automorphism afterwards.* $\psi_g(c_0') + \psi_g(c_1')s = \psi_g(c_0' + c_1'\psi_g^{-1}(s))$, so the rotated pair decrypts to $\psi_g(\mu) + \psi_g(e'_{\mathrm{ks}})$ under $s$, and

$$\psi_g(e'_{\mathrm{ks}}) = P^{-1}\sum_j \tilde{D}_j^\star e_j^\star + \mathrm{rd}_0^\star + \mathrm{rd}_1^\star s, \qquad \tilde{D}_j^\star = \psi_g(\tilde{D}_j(c_1)),\ e_j^\star = \psi_g(e_j^{\mathrm{rk}}),\ \mathrm{rd}_b^\star = \psi_g(\mathrm{rd}_b).$$

This is the form of Lemma 3.5. The proof of Proposition 3.8 only adds such errors and applies the traces $T_{i+1}$ to them, so the decomposition holds with the starred quantities.

*The hypotheses of the bounds.* (a) $\psi_g$ is a signed permutation of coefficients (Lemma 1.4), so the starred digits and roundings have the same $\ell_\infty$, $\ell_2$ and $\ell_1$ norms as the unstarred ones. (b) The coordinates of $e_j^\star$ are those of $e_j^{\mathrm{rk}}$ up to order and sign; a sign flip preserves $v$-subgaussianity, so $e_j^\star$ has independent $2\sigma_e^2$-subgaussian coordinates, independent across $(i, j)$. (c) $\tilde{D}_j^\star$ and $\mathrm{rd}_1^\star$ are functions of the $c_1$-component, which by the induction of Lemma 3.6 (the new step outputs $\psi_g(\mathrm{MD}(\sum_j \tilde{D}_j(c_1)a_j))$ and uses neither $c_0$, nor a secret, nor an error) is a function of the public randomness and the data only. (d) The rescales are those of (M1). The proofs of $B_3$, $B_4$ and $B_8$ use only $\|\tilde{D}\|_\infty$, $\|\mathrm{rd}\|_\infty$, the proxy of the key errors, (c), and Lemma 1.6, which holds for every polynomial $f$; so they hold with the stated factors. □

**Remark 4.2** (Conformance of OpenFHE 1.5.1 to (M3′); a code reading, not a proof)**.** We re-read the relevant code at tag `v1.5.1` (the headers installed with the implementation are identical to the tagged ones):

- `core/include/lattice/hal/default/dcrtpoly-impl.h`, `ApproxSwitchCRTBasis`: residues are stored in $[0, q_i)$; both code paths compute $\sum_i [x_i\hat{Q}_i^{-1}]_{q_i}[\hat{Q}_i]_p \bmod p$, the residue of the integer $\sum_i [x_i\hat{Q}_i^{-1}]_{q_i}\hat{Q}_i \in [0, k\tilde{Q})$. `ApproxModDown`: `ans` $= (x - \mathrm{partPSwitchedToQ}) \cdot P^{-1}$, with no centring; the $t$-multiplications are skipped for CKKS, whose noise scale is 1 (`gen-cryptocontext-ckksrns-internal.h`).
- `pke/lib/keyswitch/keyswitch-hybrid.cpp`: `KeySwitchGenInternal` sets $b = -a\,s_{\mathrm{new}} + e + [P]_{q_i}s_{\mathrm{old}}$ on the primes of digit $j$ and $-a\,s_{\mathrm{new}} + e$ elsewhere, which is $P\mathbf{g}_j s_{\mathrm{old}}$; `EvalKeySwitchPrecomputeCore` keeps the digit's own residues and extends them by `ApproxSwitchCRTBasis`; `EvalFastKeySwitchCore` applies `ApproxModDown` to both components; `KeySwitchInPlace` adds the first to $c_0$ and replaces $c_1$ by the second.
- `pke/lib/schemebase/base-leveledshe.cpp`, `EvalAutomorphism`: `KeySwitchInPlace`, then `AutomorphismTransform` of both components; CKKS does not override it. `base-multiparty.cpp`, `MultiEvalAutomorphismKeyGen`: the new key is $\psi_{g^{-1}}(s_P)$.
- The implementation (`implementation/native/vfr_native.cpp`) builds the rotation-key shares in this layout itself (`rotkey_share`: $b_j = -a_j\psi_g^{-1}(s) + e + P\mathbf{g}_j s$), sums them, calls the scheme's `EvalAutomorphism` inside its own EvalSum, and uses its own centred deterministic rescale (`rescale_poly`, remainder in $(-q/2, q/2]$), its own encodings and its own release. So (M1) holds as stated and only key switching is OpenFHE's.

These are statements about a program and are not proved. (M5) for OpenFHE's table sampler, which draws the errors, remains a numerical check (Section 3.6; the paper's Check (T4)).

### 4.2 The sensitivity of the centre with decoupled scales

For release $s$ let $\nu_U = (\sum_{e\in\mathcal{E}} U_{\max,e}^2)^{1/2}$, with $U_{\max,e} = 1$ for $\mathbf{b}$ and $B_R$ for a column of $\dot{\mathbf{x}}$, and $\nu_V = \sup_{D\sim D'} \max_t (\sum_{e\in\mathcal{E}} (V_{e,t} - V'_{e,t})^2)^{1/2} \le 2\Gamma_{\mathcal{E}}^{1/2}$ (Lemma 3.10).

**Lemma 4.3** (Sensitivity of the centre, decoupled scales)**.** *Assume (M1), (M2) with arbitrary $\Omega_U, \Omega_V, \Omega_u, q_1, q_2$, and that* Ecd *rounds to the nearest integer coefficient-wise. Put*

$$\begin{aligned}
\epsilon_x &= \frac{\Omega_u}{q_1 q_2}\,\frac{N^{3/2}\sqrt{|\mathcal{E}|}}{2\sqrt{2}} + \frac{N^{5/2}|\mathcal{E}|}{8 q_1 q_2},\\
\Delta_{x,s} &= \sqrt{\tfrac{N}{2}}\,\frac{\Omega_u}{q_1 q_2}\big(\Omega_V \nu_V + \sqrt{N}\,\Omega_U \nu_U\big) + \frac{\sqrt{N}}{2}\,\frac{\Omega_U \Omega_V}{q_1 q_2}\sqrt{|\mathcal{E}|}\,\Delta_s\\
&\quad + \frac{N^{3/2}}{4 q_1 q_2}\sqrt{|\mathcal{E}|}\big(\Omega_V \nu_V + \sqrt{N}\,\Omega_U \nu_U\big) + 1.
\end{aligned}$$

*For every pair of neighbours $D \sim D'$ (differing in one row of* O*, matched or not) and every input and alignment of* R*, $\|\widehat{Z}_s(D) - \widehat{Z}_s(D')\|_2 \le \sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_s + \Delta_{x,s} - 1 + \epsilon_x$; hence, if $\epsilon_x \le 1$,*

$$\|f_s(D) - f_s(D')\|_2 \le \widehat{\Delta}_{\mathrm{enc},s} := \sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_s + \Delta_{x,s} + \sqrt{N}.$$

*With equal scales this reduces to Corollary 3.13.*

*Proof.* The changed row sits in chunk $h^*$ and slot $l^*$; only the encodings $m_{V_e,h^*}$ change. Write $\Delta V_e = V_{e,l^*} - V'_{e,l^*}$ and $\Delta m_e = m_{V_e,h^*} - m'_{V_e,h^*} = \Omega_V \tau^{-1}(\Delta V_e \mathbf{u}_{l^*}) + \delta r_e$ with $\|\delta r_e\|_\infty \le 1$. Then $\widehat{Z}_s(D) - \widehat{Z}_s(D') = \sum_e (q_1 q_2)^{-1} y_e m_{u_e}$ with $y_e = \mathrm{A}(m_{U_e,h^*}\Delta m_e)$.

*Slot value.* By Lemma 1.5 and multiplicativity of $\tau$, every slot of $y_e$ equals $c_e = \sum_l \tau(m_{U_e,h^*})_l \tau(\Delta m_e)_l$. With $m_{U_e,h^*} = \Omega_U \tau^{-1}(U_e[h^*]) + r_{U_e}$,

$$c_e = \Omega_U\Omega_V\, U_{e,l^*}\Delta V_e + \Omega_V\, \tau(r_{U_e})_{l^*}\Delta V_e + \sum_l \tau(m_{U_e,h^*})_l \tau(\delta r_e)_l.$$

The vector $(U_{e,l^*}\Delta V_e)_e$ is the change of the statistic, of norm at most $\Delta_s$; it is zero if the row is unmatched, because R's vectors vanish there. $|\tau(r_{U_e})_{l^*}| \le \|r_{U_e}\|_1 \le N/2$, so the second terms have norm at most $\Omega_V \frac{N}{2}\nu_V$. By Cauchy–Schwarz and Lemma 1.1 the third term is at most $\frac{N}{2}\|m_{U_e,h^*}\|\|\delta r_e\| \le \frac{N}{2}(\Omega_U U_{\max,e} + \frac{\sqrt{N}}{2})\sqrt{N}$, using $\|m_U\| \le \Omega_U\sqrt{2/N}\|U_e[h^*]\| + \sqrt{N}/2$ and $\|U_e[h^*]\| \le \sqrt{N/2}\,U_{\max,e}$. So

$$\|(c_e)_e\| \le \Omega_U\Omega_V\Delta_s + \tfrac{N}{2}\big(\Omega_V\nu_V + \sqrt{N}\,\Omega_U\nu_U\big) + \tfrac{N^2}{4}\sqrt{|\mathcal{E}|}.$$

*Masking.* Split $m_{u_e} = \Omega_u \tau^{-1}(\mathbf{u}_{k_e}) + r_{u_e}$. The main part has norm $\sqrt{2/N}\,(\Omega_u/(q_1q_2))\|(c_e)_e\|$ (Lemma 1.7). For the spill, $y_e$ has only the coefficients of 1 and $X^{N/2}$ (Lemma 1.5): $y_e = a + bX^{N/2}$ with $a^2 + b^2 = |c_e|^2$. Multiplication by $X^{N/2}$ is an orthogonal map $J$ with $J^\top = -J$, so $\langle r, Jr\rangle = 0$ and $\|y_e r_{u_e}\|^2 = |c_e|^2\|r_{u_e}\|^2 \le |c_e|^2 N/4$. Hence the spill is at most $\frac{\sqrt{N}}{2q_1q_2}\sum_e |c_e| \le \frac{\sqrt{N}}{2q_1q_2}\sqrt{|\mathcal{E}|}\|(c_e)_e\|$.

*Collecting.* With $\Omega_U\Omega_V\Omega_u/(q_1q_2) = \Omega_{\mathrm{fin}}$, the main part gives $\sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_s$, the first term of $\Delta_{x,s}$, and $\sqrt{2/N}(\Omega_u/(q_1q_2))\frac{N^2}{4}\sqrt{|\mathcal{E}|}$; the spill gives the second and third terms and $\frac{\sqrt{N}}{2q_1q_2}\frac{N^2}{4}|\mathcal{E}|$. The two leftovers sum to $\epsilon_x$. Finally $\|\lfloor x\rceil - \lfloor x'\rceil\| \le \|x - x'\| + \sqrt{N}$. □

**Remark 4.4** (Second reading)**.** The book states the lemma for a changed row that is matched; the proof covers any row, as shown (an unmatched row contributes $\Delta V_e$ only through the rounding

cross terms), and the paper's adjacency (Definition `def:adj`) changes a row "matched or not". A brute-force check at $N = 16$ with decoupled scales $(\Omega_U, \Omega_V, \Omega_u, q_1, q_2) = (2^{10}, 2^9, 2^{10}, 2^{10}, 2^9)$ and exact rational arithmetic gives a largest ratio 0.93 of the change of $f_s$ to the bound over 30 random instances. If the encoder's output differs from exact rounding by at most $\delta$ per coefficient, every rounding bound $\frac{1}{2}$ becomes $\frac{1}{2} + \delta$ and the lemma holds with the first three terms of $\Delta_{x,s}$ multiplied by $(1 + 2\delta)^2$; the implementation's double-double encoder is not analysed here.

### 4.3 The decoding offset

Let $\zeta_{\mathcal{E}}^2 = \sup_{\text{rows}} \sum_e (U_{e,t} V_{e,t})^2$, the largest squared contribution of one row to the statistic: with $r = B_R^2$, $\zeta_{\mathcal{E}_G}^2 \le o^2 + ro$ and $\zeta_{\mathcal{E}_2}^2 \le o^2 + r(o + w) + ow + w^2$ (joint mode), by the block computation of Lemma 3.10. Let $n_{\max}$ bound the number of matched rows ($n_R^{\max}$ in the protocol).

**Proposition 4.5** (A public bound on the decoding offset)**.** *Assume (M1), (M2) and exact rounding in* Ecd*. Put*

$$A_s = \frac{N\sqrt{m}}{2\sqrt{2}} \Big( \frac{\sqrt{n_{\max}}\, \nu_U}{\Omega_V} + \frac{(\bar{n}_O \Gamma_{\mathcal{E}})^{1/2}}{\Omega_U} \Big) + \frac{\sqrt{|\mathcal{E}|}\, mN^2}{8\, \Omega_U \Omega_V}.$$

*For every dataset in the clipped domain with at most $n_{\max}$ matched rows, the decoded centre satisfies, on the slots of the entries,*

$$\Big\| \big( \mathsf{Dcd}(f_s(D))_{k_e} - \mathbf{z}_e(D) \big)_{e \in \mathcal{E}} \Big\|_2 \le \bar{d}_s, \qquad \bar{d}_s := A_s + \frac{N\sqrt{|\mathcal{E}|}}{2\sqrt{2}\, \Omega_u} \big( n_{\max} \zeta_{\mathcal{E}} + A_s \big) + \frac{N}{2\sqrt{2}\, \Omega_{\text{fin}}},$$

*with* $\mathsf{Dcd}(v) = \tau(v)/\Omega_{\text{fin}}$*, and, in the coefficient domain,*

$$\|\widehat{Z}_s(D) - \Omega_{\text{fin}} \tau^{-1}(\mathbf{z}_s(D))\|_2 \le d_{\max},$$

$$d_{\max} := \sqrt{\tfrac{2}{N}}\, \frac{\Omega_u}{q_1 q_2}\, \bar{r}_s + \frac{\sqrt{N}}{2 q_1 q_2} \sqrt{|\mathcal{E}|} \big( \Omega_U \Omega_V n_{\max} \zeta_{\mathcal{E}} + \bar{r}_s \big), \qquad \bar{r}_s = \Omega_U \Omega_V A_s.$$

*Proof.* Write $m_{U_e,h} = \Omega_U \tau^{-1}(U_e[h]) + r_{U,e,h}$ and $m_{V_e,h} = \Omega_V \tau^{-1}(V_e[h]) + r_{V,e,h}$, with roundings of sup-norm at most $\frac{1}{2}$, so $\|\tau(r)\| = \sqrt{N/2}\|r\| \le N/(2\sqrt{2})$. As in the previous proof, every slot of $y_e = \mathrm{A}(\sum_h m_{U_e,h} m_{V_e,h})$ equals

$$c_e = \Omega_U \Omega_V \mathbf{z}_e + \bar{r}_e, \qquad \bar{r}_e = \Omega_U a_e + \Omega_V b_e + d_e,$$

with $a_e = \sum_h \langle U_e[h], \tau(r_{V,e,h}) \rangle$, $b_e = \sum_h \langle \tau(r_{U,e,h}), V_e[h] \rangle$, $d_e = \sum_h \langle \tau(r_{U,e,h}), \tau(r_{V,e,h}) \rangle$ (bilinear sums over the slots). By Cauchy–Schwarz over slots and chunks, $|a_e| \le \frac{N}{2\sqrt{2}} \sqrt{m} \|U_e\|$, $|b_e| \le \frac{N}{2\sqrt{2}} \sqrt{m} \|V_e\|$, $|d_e| \le mN^2/8$. R's vectors vanish off the matched rows, so $\sum_e \|U_e\|^2 \le n_{\max} \nu_U^2$, and $\sum_e \|V_e\|^2 \le \bar{n}_O \Gamma_{\mathcal{E}}$ (Lemma 3.10); hence $\|(\bar{r}_e)_e\| \le \Omega_U \Omega_V A_s = \bar{r}_s$.

*Slots.* Since $\tau(y_{e'} r_{u_{e'}}) = c_{e'} \tau(r_{u_{e'}})$, $\tau(\widehat{Z}_s)_k = (q_1 q_2)^{-1} \sum_{e'} c_{e'} (\Omega_u [k = k_{e'}] + \tau(r_{u_{e'}})_k)$. Dividing by $\Omega_{\text{fin}}$,

$$\mathsf{Dcd}(\widehat{Z}_s)_{k_e} - \mathbf{z}_e = \frac{\bar{r}_e}{\Omega_U \Omega_V} + \sum_{e'} \frac{c_{e'}\, \tau(r_{u_{e'}})_{k_e}}{\Omega_U \Omega_V \Omega_u}.$$

The first term has norm at most $A_s$ over the entries. For the second, $\|(\tau(r_{u_{e'}})_{k_e})_e\| \le \|\tau(r_{u_{e'}})\| \le N/(2\sqrt{2})$, so it is at most

$$\tfrac{N}{2\sqrt{2}\Omega_u} \textstyle\sum_{e'} |c_{e'}|/(\Omega_U \Omega_V) \le \tfrac{N\sqrt{|\mathcal{E}|}}{2\sqrt{2}\Omega_u} (\|\mathbf{z}\| + A_s),$$

and $\|\mathbf{z}\| \le n_{\max}\zeta_{\mathcal{E}}$ by the triangle inequality over the matched rows. Finally $f_s - \widehat{Z}_s$ has sup-norm at most $\frac{1}{2}$, so its slot vector has norm at most $N/(2\sqrt{2})$, which contributes $N/(2\sqrt{2}\,\Omega_{\text{fin}})$. Taking real parts does not increase norms.

*Coefficients.* The main part of $\widehat{Z}_s$ is

$$\textstyle\sum_e (\Omega_u/(q_1q_2))y_e\tau^{-1}(\mathbf{u}_{k_e}) = \Omega_{\text{fin}}\tau^{-1}(\mathbf{z}_s) + (\Omega_u/(q_1q_2))\tau^{-1}(\sum_e \bar{r}_e\mathbf{u}_{k_e}),$$

whose second piece has norm $\sqrt{2/N}(\Omega_u/(q_1q_2))\|(\bar{r}_e)\|$ (Lemma 1.2(a)). The spill is at most $\frac{\sqrt{N}}{2q_1q_2}\sum_e |c_e|$, as in the previous proof. □

**The numbers at the evaluation setting.** Chain $[60, 59, 59, 59]$ with $q_0 = p_0p_1 \approx 2^{119}$, $q_1, q_2 \approx 2^{59}$, two special primes of 60 bits, $d_{\text{num}} = 2$ digits at the EvalSum level ($\max_j k_j = 2$, $\tilde{Q}_{\max}/P \approx 2^{-1}$), $\Omega_U = 2^{68}$, $\Omega_u = q_1$, simulation population ($d_R = 4$, $d_O = 6$, $(B_R, B_O, B_y) = (5.59, 6.24, 2.89)$), $\varepsilon = 1$, $\delta = 10^{-9}$, $\kappa = 40$, previous setting ($f_1 = 0.1$), recomputed with the implementation's Calibrate:

- $\log_2 \Omega_{\text{fin}} = 62.75$; $B_{\text{tot},1} = 2^{26.43}$ and $B_{\text{tot},2} = 2^{26.59}$ by Lemma 4.1 ($\varsigma_\uparrow = 4$, $\varsigma_\downarrow = 4$), the same at $\bar{n}_O = 4{,}096$, 16,384 and $10^6$;
- $\Delta_{x,2} = 2^{27.57}$ and $\Delta_{x,1} = 2^{27.46}$; $\epsilon_x = 2^{-36.5}$ and $2^{-36.7}$, so the "+1" of $\Delta_{x,s}$ covers it;
- $\bar{d}_2 = 9.9 \cdot 10^{-9}$ at $\bar{n}_O = 4{,}096$ with $n_{\max} = n_R^{\max} = 2{,}000$; $2.0 \cdot 10^{-8}$ with $n_{\max} = \bar{n}_O = 4{,}096$; $1.6 \cdot 10^{-8}$ and $4.8 \cdot 10^{-6}$ at $\bar{n}_O = 10^6$ with $n_{\max} = 2{,}000$ and $10^6$; against $\sigma_2 = 525$;
- $d_{\max} = 2^{30.3}$ at $\bar{n}_O = 10^6$ with $n_{\max} = 2{,}000$, and $2^{38.6}$ with $n_{\max} = 10^6$.

A brute-force check at $N = 16$ with decoupled scales gives a largest ratio 0.07 of the offset to $\bar{d}_s$.

# 5 Output privacy (Theorem 1)

## 5.1 Differential-privacy tools

**Definition 5.1** (zCDP, Bun and Steinke [3])**.** A randomized mechanism $\mathcal{M}$ is $\rho$-zCDP with respect to $\sim$ if $D_\alpha(\mathcal{M}(D)\|\mathcal{M}(D')) \le \rho\alpha$ for all $D \sim D'$ and all $\alpha > 1$. Here $D_\alpha$ is the Rényi divergence of order $\alpha$.

**Lemma 5.2** ([3], composition and post-processing)**.**

- Composition. *Suppose $\mathcal{M}_1$ is $\rho_1$-zCDP, and $\mathcal{M}_2(\cdot, y)$ is $\rho_2$-zCDP for every $y$. Then $D \mapsto (y, \mathcal{M}_2(D, y))$ with $y \leftarrow \mathcal{M}_1(D)$ is $(\rho_1 + \rho_2)$-zCDP.*
- Post-processing. *$g \circ \mathcal{M}$ is $\rho$-zCDP for every randomized $g$ that does not depend on $D$.*

**Proposition 5.3** (The ideal mechanism; multivariate discrete Gaussian)**.** *Let $f : \mathcal{D} \to \mathbb{Z}^N$ satisfy $\sup_{D\sim D'}\|f(D) - f(D')\|_2 \le \widehat{\Delta}$, and let $\mathcal{M}(D) = f(D) + \eta$ with $\eta \sim D_{\mathbb{Z}^N,\sigma}$. Then $\mathcal{M}$ is $\widehat{\Delta}^2/(2\sigma^2)$-zCDP.*

*Hence $\mathcal{M}^{\text{id}}(D) = (f_s(D) + \eta_s)_s$, with independent $\eta_s \sim D_{\mathbb{Z}^N,\sigma_{c,s}}$ and $f_s$ from Definition 3.7, is $\sum_s \rho_s$-zCDP, with $\rho_s = \widehat{\Delta}^2_{\text{enc},s}/(2\sigma^2_{c,s})$.*

*Proof. Single release.* $\mathcal{M}(D)$ and $\mathcal{M}(D')$ are integer shifts of one another, by $\mathbf{e} = f(D) - f(D') \in \mathbb{Z}^N$. Lemma 2.2 gives $D_\alpha \le \alpha\|\mathbf{e}\|^2/(2\sigma^2)$. This is the multivariate statement of [4] [*citation detail unverified:* numbered Theorem 14 in the arXiv version], proved here in a self-contained way.

*Several releases.* The releases are computed from fixed ciphertexts $M_1$, $M_2$, $M_y$. R's choices between releases do not change what O releases, so non-adaptive composition (Lemma 5.2) applies.

*The centre.* Corollary 3.13 bounds the sensitivity of $f_s$. The specification's $\Delta_{\text{enc},s}$ is the sensitivity of $\mathsf{Ecd}_\Omega(\mathbf{z}_s)$ (Lemma 1.2); it must be replaced by $\widehat{\Delta}_{\text{enc},s}$, a relative change of $2^{-33}$. □

**Proposition 5.4** (Conversion from zCDP, Canonne, Kamath and Steinke [4])**.** *If $\mathcal{M}$ is $\rho$-zCDP, then it is $(\varepsilon, \delta(\rho,\varepsilon))$-DP for every $\varepsilon \geq 0$, where*

$$\delta(\rho,\varepsilon) = \inf_{\alpha>1} \frac{e^{(\alpha-1)(\alpha\rho-\varepsilon)}}{\alpha-1}\Big(1-\frac{1}{\alpha}\Big)^{\alpha}.$$

*The map $\rho \mapsto \delta(\rho,\varepsilon)$ is continuous and nondecreasing, and $\delta(\rho,\varepsilon) \to 0$ as $\rho \to 0$. Hence $\rho_{\mathrm{run}} = \max\{\rho : \delta(\rho,\varepsilon_{\mathrm{run}}) \leq \delta_{\mathrm{DP}}\}$ in* Calibrate *is well defined, and the $\rho_{\mathrm{run}}$-zCDP mechanism is $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-DP.*

*Proof.* Let $P = \mathcal{M}(D)$ and $Q = \mathcal{M}(D')$, with $D \sim D'$.

*Hockey-stick form.* $(\varepsilon,\delta)$-indistinguishability is equivalent to $\mathbb{E}_Q[(P/Q - e^{\varepsilon})_+] \leq \delta$ (with the usual convention when $P \not\ll Q$; here both have full support on $\mathbb{Z}^N$).

*A pointwise bound.* For $x > 0$ and $\alpha > 1$, $(x - e^{\varepsilon})_+ \leq x^{\alpha} c_{\alpha}$, where

$$c_{\alpha} = \sup_{y>0} \frac{(y-e^{\varepsilon})_+}{y^{\alpha}} = \frac{e^{-(\alpha-1)\varepsilon}}{\alpha-1}\Big(1-\frac{1}{\alpha}\Big)^{\alpha}.$$

The supremum is attained at $y = \alpha e^{\varepsilon}/(\alpha-1)$.

*Conclusion.* Hence $\mathbb{E}_Q[(P/Q - e^{\varepsilon})_+] \leq c_{\alpha}\mathbb{E}_Q[(P/Q)^{\alpha}] = c_{\alpha} e^{(\alpha-1)D_{\alpha}(P\|Q)} \leq c_{\alpha} e^{(\alpha-1)\alpha\rho}$. Take the infimum over $\alpha$. Each term is continuous and increasing in $\rho$, so the infimum is nondecreasing and upper semicontinuous. It is also continuous here, because the infimum may be restricted to a compact set of $\alpha$ for $\rho$ in a compact range. As $\rho \to 0$, taking $\alpha = 1 + \sqrt{\varepsilon/\rho}$ shows $\delta \to 0$ for $\varepsilon > 0$. □

**Remark 5.5.** The conversion is valid for every $\rho$-zCDP mechanism, but it is not the tightest possible for the Gaussian mechanism. For the example, $\varepsilon = 1$ and $\delta_{\mathrm{DP}} \approx 10^{-9}$ give $\rho_{\mathrm{run}} = 0.01496$. This agrees with the specification.

## 5.2 SIM-CDP

Let $\lambda$ be the computational security parameter. $\mathrm{view}_{\mathsf{R}}(D)$ comprises three things: R's input (cohort identifiers and $\mathbf{X}_R$), its random tape, and every message it receives in one run on dataset $D$.

**Definition 5.6** (SIM-CDP for R, after Mironov et al. [16])**.** The protocol is $(\varepsilon,\delta)$-*SIM-CDP for* R *with statistical slack* $\gamma$ if two objects exist:

- a randomized mechanism $\mathcal{M}$ that is $(\varepsilon,\delta)$-DP with respect to $\sim$;
- a probabilistic polynomial-time simulator $\mathcal{S}$.

They must satisfy, for every dataset $D$ and every PPT distinguisher $\mathcal{A}$,

$$\big|\Pr[\mathcal{A}(\mathrm{view}_{\mathsf{R}}(D)) = 1] - \Pr[\mathcal{A}(\mathcal{S}(\mathcal{M}(D))) = 1]\big| \leq \gamma + \mathrm{negl}(\lambda).$$

With $\gamma = 0$ this is the simulation-based notion of [16], which those authors state for pure $\varepsilon$ [*citation detail unverified:* whether MPRV09 also define the $(\varepsilon,\delta)$ variant]. The extension to $(\varepsilon,\delta)$ is the standard one. The slack $\gamma$ makes explicit the statistical terms that are not negligible in $\lambda$ at concrete $\kappa$.

**Lemma 5.7** (From SIM-CDP to indistinguishability)**.** *Suppose the protocol is $(\varepsilon,\delta)$-SIM-CDP with slack $\gamma$. Then for all $D \sim D'$ and every PPT $\mathcal{A}$,*

$$\Pr[\mathcal{A}(\mathrm{view}(D)) = 1] \leq e^{\varepsilon}\Pr[\mathcal{A}(\mathrm{view}(D')) = 1] + \delta + (1+e^{\varepsilon})\gamma + \mathrm{negl}(\lambda).$$

*That is, the protocol is $(\varepsilon, \delta + (1+e^{\varepsilon})\gamma)$-IND-CDP in the sense of [16].*

*Proof.* Chain three steps: $\mathrm{view}(D) \approx \mathcal{S}(\mathcal{M}(D))$; then DP of $\mathcal{S} \circ \mathcal{M}$, which is post-processing; then $\mathcal{S}(\mathcal{M}(D')) \approx \mathrm{view}(D')$. □

**Lemma 5.8** (Post-processing and composition of SIM-CDP)**.** *(a) If the protocol is $(\varepsilon, \delta)$-SIM-CDP with slack $\gamma$, then so is $g(\mathrm{view}_{\mathsf{R}}(D))$ for every PPT $g$, with simulator $g \circ \mathcal{S}$.*

*(b) Consider $K$ runs on the same dataset, with independent keys and randomness and with $\mathsf{R}$'s inputs fixed. Suppose run $k$ is simulated by $\mathcal{S}_k$ from a $\rho_k$-zCDP mechanism $\mathcal{M}_k$ with slack $\gamma_k$. Then the joint view is simulated by $(\mathcal{S}_k)_k$ from $(\mathcal{M}_k)_k$, which is $\sum_k \rho_k$-zCDP, with slack $\sum_k \gamma_k$.*

*In particular, the per-base-register ledger of the specification bounds the privacy loss of all runs, even if researchers collude. This holds when runs use fresh keys, which Algorithm 1 provides.*

*Proof.* (a) is immediate. For (b), use a hybrid over runs: replace run $k$ by $\mathcal{S}_k(\mathcal{M}_k(D))$ one run at a time. This is possible because run $k$ uses keys and randomness independent of the other runs, so the reduction can simulate those runs itself. Then apply Lemma 5.2. If the researchers' inputs and the runs' parameters depend adaptively on earlier outputs, the hybrid is written out in Part C, the composition appendix (Theorem 4.2 there, `thm:main`): all smudging steps are done first in one statistical hybrid, the computational steps follow run by run, and a Rényi filter replaces Lemma 5.2. Part C also requires the ledger to charge a run's full cost at admission, since the $\delta_B$ part of an aborted run's unmade releases must stay charged, and gives the ledger with a hockey-stick slack (Corollary 5.4 there). □

### 5.3 Assumptions

**Assumption 5.9** (RLWE for the re-randomizer)**.** Let $a \leftarrow \mathcal{R}_{q_0}$ be uniform, $u \leftarrow \chi_{\mathrm{key}}$ (uniform ternary) and $e_1 \leftarrow \chi_{\mathrm{err}} = \mathcal{N}_{\mathbb{Z}}(0, \sigma_e^2)^N$. Then $(a, au + e_1) \approx_c (a, \mathrm{U}(\mathcal{R}_{q_0}))$. This is decision-RLWE with one sample, a ternary secret and discrete Gaussian error with $\sigma_e = 3.2$. Let $\mathrm{Adv}_1$ denote the best advantage.

**Assumption 5.10** (RLWE with circular security for $\mathsf{O}$'s key shares)**.** Let $s_O \leftarrow \chi_{\mathrm{key}}$, let every error be drawn from $\chi_{\mathrm{err}}$, and let all $a$'s be uniform common random strings. The tuple

$$\left(a,\ -as_O + e_O^{\mathrm{pk}};\ \{a_{g,j},\ -a_{g,j}s_O + P\mathsf{g}_j\psi_g(s_O) + e_{O,g,j}^{\mathrm{rk}}\}_{g\in\{g_0,\ldots,g_{L-1}\},\, j\le d_{\mathrm{num}}}\right)$$

is computationally indistinguishable from uniform over $\mathcal{R}_{Q_2} \times \mathcal{R}_{PQ_2}^{Ld_{\mathrm{num}}}$; let $\mathrm{Adv}_2$ be the advantage. The first component is plain RLWE in $s_O$. The rotation-key components encrypt key-dependent messages $P\mathsf{g}_j\psi_g(s_O)$, so they need a circular (key-dependent-message) security assumption, as do all CKKS and BFV rotation keys.

These shares are exactly what $\mathsf{R}$ receives from $\mathsf{O}$ in the one-round public-key and rotation-key generation protocols of Mouchet et al. [17]. Under this assumption a semi-honest $\mathsf{R}$'s view of key generation is simulated by uniform elements; this is the "security of threshold key generation" used in the specification. [*citation detail unverified:* Mouchet et al. state that the rotation-key protocol relies on the same circular-security assumption as single-key RLWE schemes; we did not locate the exact statement number.]

The proof also uses three things proved above: Theorem 3.12 (Obligation O1), Lemma 2.3 with Corollary 2.4, and Proposition 5.3 with Proposition 5.4. It also uses $\mathsf{O}$'s honest behaviour in Algorithm 5:

- $(u, e_0, e_1, \eta)$ are fresh and independent for each release;
- $\eta$ is *exactly* $D_{\mathbb{Z}^N, \sigma_{c,s}}$;

- $c'_0$ is neither sent nor reused.

The proof does *not* use PSI security, the random-oracle model, or any assumption on the distribution of rounding errors.

### 5.4 The theorem

**Theorem 5.11** (Output privacy against an honest-but-curious R; Theorem 1 of the specification, corrected)**.** *Assume (M1)–(M5), Assumptions 5.9 and 5.10, and that for every release $s$:*

- $\sigma_{c,s} \geq \max\{2^{\kappa} B_{\mathrm{tot},s}/\sqrt{2\pi},\, 2^{42}\}$, *with $B_{\mathrm{tot},s}$ from Theorem 3.12, or from Lemma 4.1 for OpenFHE's key switching (M3′) (the floor was $2^{20}$ in earlier versions; Remark 2.5);*
- $\rho_s = \widehat{\Delta}^2_{\mathrm{enc},s}/(2\sigma^2_{c,s})$, *with $\widehat{\Delta}_{\mathrm{enc},s}$ from Corollary 3.13 (equal scales) or Lemma 4.3 (decoupled scales, as in the evaluation);*
- $\sum_s \rho_s \leq \rho_{\mathrm{run}}$, *with $\rho_{\mathrm{run}}$ from Proposition 5.4 at $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$.*

*Then R's view of one run is $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-SIM-CDP for $\sim$, with statistical slack*

$$\gamma = n_{\mathrm{rel}}\big(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}\big)$$

*and computational loss $n_{\mathrm{rel}}\mathrm{Adv}_1 + \mathrm{Adv}_2$. The mechanism is*

$$\mathcal{M}(D) = \big(\omega_{\mathsf{R}},\ \Lambda,\ (f_s(D;\pi) + \eta_s)_s\big),$$

*where $\omega_{\mathsf{R}}$ is a fresh random tape for R, and $\Lambda$ is R's view of the Link phase (Algorithm 2) generated with fresh coins for O. Those coins include the OPRF key, the dummy identifiers' labels and the permutation $\tau$, hence $\pi$; the same $\tau$ fixes the arrangement used in $f_s(\cdot;\pi)$. By Lemma 5.7, the view is $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}} + (1 + e^{\varepsilon_{\mathrm{run}}})\gamma)$-IND-CDP.*

*Proof.* *$\mathcal{M}$ is $\rho_{\mathrm{run}}$-zCDP, and hence $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-DP.* The pair $(\omega_{\mathsf{R}}, \Lambda)$ has the same law under $D$ and $D'$. Neighbours differ only in $(\mathbf{x}_O, y)$ of one row, and neither the Link phase nor R's coins depend on $(\mathbf{x}_O, y)$: the labels are positions $\tau(j)$, and the identifiers, including the dummy ones, are part of the fixed part of the dataset. Conditionally on $(\omega_{\mathsf{R}}, \Lambda)$, the last component is the composition of the mechanisms $f_s(\cdot;\pi) + \eta_s$. Their sensitivity is at most $\widehat{\Delta}_{\mathrm{enc},s}$ for every arrangement $\pi$, by Corollary 3.13 or Lemma 4.3. Apply Proposition 5.3, Lemma 5.2 and Proposition 5.4.

*The simulator.* On input $(\omega_{\mathsf{R}}, \Lambda, (w_s)_s)$, $\mathcal{S}$ proceeds as follows:

1. output $\omega_{\mathsf{R}}$ and $\Lambda$;
2. replace O's key-generation shares by uniform elements of the right rings;
3. for each release, sample $c^*_{1,s} \leftarrow \mathrm{U}(\mathcal{R}_{q_0})$ and output $(c^*_{1,s},\ w_s - c^*_{1,s}s_R \bmod q_0)$, reading $s_R$ from $\omega_{\mathsf{R}}$;
4. output the public parameters, including the outputs of Calibrate. These depend on public inputs only, because $B_{\mathrm{tot},s}$ is a worst-case bound.

*Hybrids.*

**H0** The real view.

**H1** Compute $h_{O,s}$ as $v_s - c'_{1,s}s_R$, with $v_s = c'_{0,s} + c'_{1,s}s + \eta_s \bmod q_0$. This is identical to H0, since $h_{O,s} = c'_{0,s} + c'_{1,s}s_O + \eta_s$.

**H2**$^{(s)}$ For $s = 1, \ldots, n_{\mathrm{rel}}$ in turn, replace $v_s$ by $f_s(D;\pi) + \eta_s \bmod q_0$. By Proposition 3.8, $v_s \equiv f_s + e_{\mathrm{tot},s} + \eta_s$. Let $Z$ be everything else in the hybrid. Then $\eta_s$ is independent of $(Z, e_{\mathrm{tot},s})$, because it is fresh and used only in $v_s$. Corollary 2.4 with $B = B_{\mathrm{tot},s}$ and Theorem 3.12, which holds for every $D$ and every value of the public randomness, bound the statistical distance between consecutive hybrids by $\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$. Summing over $s$ gives $\gamma$. Three precisions:

– *The side information includes* R*'s random tape.* $Z$ contains $\omega_{\mathsf{R}}$, hence $s_R$, R's encryption errors and its key-share errors, all of which enter $e_{\mathrm{tot},s}$ (through the term of R's errors times O's data, the shares of $e^{\mathrm{pk}}$ and $e^{\mathrm{rk}}$, and the $s$-dependent terms). Corollary 2.4 allows any joint law of $(Z, e_{\mathrm{tot},s})$, so the simulator needs only $\mathcal{M}(D)$, not the exact $f_s(D)$: this is simulation in the standard sense, not the relaxed sense of Hwang et al. [15] in which the simulator also receives the exact output.

– R*'s errors must be honestly sampled.* Theorem 3.12 bounds $e_{\mathrm{tot},s}$ under (M5), with every error drawn from $\chi_{\mathrm{err}}$. A researcher who inflated its encryption errors would push the term of R's errors times O's data beyond $B_{\mathrm{tot},s}$, and that term is correlated with O's data. This is part of the honest-but-curious model, and it must be stated as such.

– $\gamma$ *is statistical.* The slack $\gamma$ is a concrete statistical distance at the chosen $\kappa$ and $\delta_B$, not a quantity negligible in $\lambda$; it is accounted for explicitly in $\delta$ (Lemma 5.7).

**H3**$^{(s)}$ Replace $c'_{1,s}$ by a uniform element of $\mathcal{R}_{q_0}$. After H2, $u_s$ and $e_{1,s}$ enter the view only through $c'_{1,s} = c_{1,s} + au_s + e_{1,s}$. The reason is that $c'_{0,s}$ is not sent, and neither $v_s$ nor anything else uses $c'_{0,s}$, $e_{0,s}$, $u_s$ or $e_{1,s}$; in particular the hint $e^{\mathrm{pk}}u_s + e_{1,s}s$ was removed with $e_{\mathrm{tot},s}$.

The reduction to Assumption 5.9 works as follows. On a challenge $(a^*, b^*)$, it sets $a$ to a uniform CRT lift of $a^*$, so that $a$ is uniform at the full modulus. It samples $s_O$, all keys, all errors and all other releases itself; it knows $D$. It sets $c'_{1,s} = c_{1,s} + b^*$. Releases $s' < s$ are already uniform and releases $s' > s$ are real, and both are computable because the reduction knows $a$. The loss per release is $\mathrm{Adv}_1$. Fresh $u_s$ per release is essential here. If $u$ were reused, then $c'_{1,s} - c'_{1,s'}$ would equal $c_{1,s} - c_{1,s'} + e_{1,s} - e_{1,s'}$. That is a short perturbation of a deterministic function of the data and the public randomness, and the hybrid would fail.

**H4** Replace O's key-generation shares by uniform elements. After H3, the view depends on $(s_O, e_O^{\mathrm{pk}}, e_O^{\mathrm{rk}})$ only through these shares:

– $h_{O,s}$ no longer contains $s_O$;

– the ciphertexts $M_s$, and therefore the key-switching and conversion steps, no longer influence the view;

– $e_{\mathrm{tot},s}$ has been removed.

The reduction to Assumption 5.10 embeds its challenge as O's shares. It computes the joint public key and R's own shares from the challenge and $\omega_{\mathsf{R}}$, and runs the rest honestly. The loss is $\mathrm{Adv}_2$.

*Only $p_{0,O}$ is sent.* In the protocol as implemented and as stated in the paper, O sends R only its public-key share $p_{0,O} = -as_O + e_O^{\mathrm{pk}}$; O sums the rotation-key shares itself and never sends its own. After H3 the rotation keys enter R's view only through the ciphertexts $M_s$, which no longer enter it. So H4 needs only the first component of Assumption 5.10: plain decision-RLWE with one sample in $s_O$, with no circular-security assumption. The rotation-key components are needed only if O's rotation-key shares are sent.

H4 is distributed exactly as $\mathcal{S}(\mathcal{M}(D))$, with the coins of $\mathcal{M}$ coupled to those of the hybrid: the Link part of H4 is generated exactly as $\Lambda$, and every other component matches the simulator. □

**Remark 5.12** (Comparison with the specification's proof sketch)**.** The sketch's hybrids H0–H3 match ours, with these differences:

(i) the centre is $f_s(D)$, not $\mathsf{Ecd}(\mathbf{z}_s(D))$, which changes $\Delta_{\mathrm{enc}}$ to $\widehat{\Delta}_{\mathrm{enc}}$;

(ii) the per-release distance is $\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$;

(iii) the statistical slack is kept separate from $\delta$, and it enters $\delta$ with the factor $1 + e^{\varepsilon}$ under the indistinguishability reading;

(iv) the sketch's H4 invokes PSI security, which is unnecessary under the specification's adjacency.

If one wants the simulator to use only the leakage G4, that is, $(I, \pi|_I)$, rather than a transcript

generated from the identifiers, then one needs the semi-honest receiver security of the labeled PSI [5]. That in turn needs circuit privacy of O's BFV replies (Correction 7).

**Remark 5.13** (Why exactness of the sampler matters)**.** Proposition 5.3 needs $\eta_s$ to be exactly $D_{\mathbb{Z}^N,\sigma}$, through Lemmas 2.1 and 2.2. A sampler at statistical distance $\nu$ from it adds $\nu$ per coordinate, so $N\nu$ per release, to $\gamma$. That is acceptable only if $N\nu \ll 2^{-\kappa}$. A floating-point sampler at $\sigma \approx 2^{52}$ is not of this kind. Float64 values of magnitude at least $2^{53}$ are spaced at least 2 apart, so the rounded output misses odd integers in that range. More generally, the low-order bits of its output are not smoothed. So $\nu$ is not small, and the Rényi divergence between neighbouring outputs can be infinite. This is the review's point, and it is why SampleDG must be exact (Section 8).

### 5.5 Release timing and the budgeted sampler

The view of Definition 5.6 contains the messages R receives, not the times at which they arrive. An honest-but-curious R can record those times, and the time O takes to answer a release request depends on how long the noise $\eta_s$ took to draw. With the plain CKS20 sampler (Section 8) that time grows with $|\eta_s|$: its discrete Laplace and Bernoulli($e^{-\gamma}$) loops run a value-dependent number of times. This subsection replaces it by a sampler with a *fixed work budget*, proves that its timing depends on its output only on a failure event of probability at most $\xi_T$, and charges $\xi_T$ per release to the statistical slack. Throughout, $\tau_s$ is the time from R's request for release $s$ to the arrival of O's answer (the release or a refusal).

**The sampler.** Fix an integer variance $S = \sigma^2 \geq 1$, a number $n$ of values, $r = 128$ bits per coin, $t = 2^b$ with $b$ a public function of $S$ (the implementation takes the $b$ near $\frac{1}{2}\log_2 S$ that maximizes $p_{\mathrm{lo}}$ below), $C = \lceil 2^r S/(2t^2)\rceil$ and $c = C/2^r \geq S/(2t^2)$. Let $E_i = \lfloor 2^r e^{-i}\rfloor$ for $i \geq 1$ and $J_0 = \min\{i : E_i = 0\}$ ($J_0 = 89$). A *coin* is a uniform $U \in [0,1)$ of which only the first $r$ bits $R = \lfloor 2^r U\rfloor$ are drawn. Public integers $T$, $J_v < J_0$ and $k_{\max}$ are fixed below.

**Definition 5.14** (SampleDGBudget$(S,n)$)**.** For $j = 1,\dots,T$, with fresh coins for each $j$:

(i) a sign bit $s_j$, $u_j$ uniform on $\{0,\dots,t-1\}$, and $v_j = \#\{1 \leq i \leq J_v : R_j^v < E_i\}$ from a coin $U_j^v$;
(ii) $m_j = u_j + tv_j$, and $y_j = m_j$ if $s_j = 0$, $y_j = -(m_j+1)$ if $s_j = 1$;
(iii) $G_j = \lfloor 2^{r-1} y_j^2/S\rfloor - 2^r v_j + C$, computed exactly; $n_j = \lfloor G_j/2^r\rfloor$ and $F_j = G_j \bmod 2^r$;
(iv) $\alpha_j = 1$ iff $[n_j = 0$ or $R_j^a < E_{\min(n_j,J_0)}]$ and $L_j$ is even, where $L_j$ is the number of leading $k \in \{1,\dots,k_{\max}\}$ with $R_j^{(k)} < \lfloor F_j/k\rfloor$, from coins $U_j^a, U_j^{(1)},\dots,U_j^{(k_{\max})}$.

Trial $j$ is *undecided* if $v_j = J_v$; or $R_j^v = E_i$ for some $i \leq J_v$; or $n_j \geq 1$ and $R_j^a = E_{\min(n_j,J_0)}$; or $R_j^{(k)} = \lfloor F_j/k\rfloor$ for some $k \leq k_{\max}$; or $L_j = k_{\max}$. The call *fails* (event F) if some trial is undecided or $A = \sum_{j\leq T}\alpha_j < n$; otherwise it returns the $y_j$ of the first $n$ trials with $\alpha_j = 1$.

All $T$ trials are computed, whatever their outcome, and the surplus is discarded. Step (iv) is Bernoulli($e^{-n_j}$) by one coin and Bernoulli($e^{-f_j}$), $f_j = g_j - n_j$, by Algorithm 1 of CKS20 with exactly $k_{\max}$ coins, where

$$g_j = \frac{y_j^2}{2S} - v_j + c.$$

**The ideal sampler.** On the same probability space, extended by fresh coins, let the *ideal sampler* run an infinite sequence of trials with the same coins for $j \leq T$ and new ones for $j > T$, read every coin to full precision, and use no bound: $v_j^* = \#\{i \geq 1 : U_j^v < e^{-i}\}$, $y_j^*$ from $(s_j, u_j, v_j^*)$ as in (ii), $g_j^*$ as above, and $\alpha_j^* = [U_j^a < e^{-\lfloor g_j^*\rfloor}] \cdot [K_j^* \text{ odd}]$, with $K_j^*$ the first $k \geq 1$ such that $U_j^{(k)} \geq (g_j^* - \lfloor g_j^*\rfloor)/k$ (coins $U_j^{(k)}$ for $k > k_{\max}$ are fresh). Its output is the $y_j^*$ of the first $n$ trials with $\alpha_j^* = 1$.

**Lemma 5.15** (The ideal sampler is exact, and its acceptances carry no information)**.** *The trials* $(y_j^*, \alpha_j^*)_{j\ge1}$ *are i.i.d.,* $\Pr[\alpha_j^* = 1] = p = (1-e^{-1})e^{-c}\Theta/(2t)$ *with* $\Theta = \sum_{y\in\mathbb{Z}} e^{-y^2/(2S)}$*, and*

$$\Pr[y_j^* = y,\ \alpha_j^* = 1] = p \cdot \pi_S(y), \qquad \pi_S(y) = e^{-y^2/(2S)}/\Theta.$$

*Hence the ideal output is exactly* $\mathcal{N}_{\mathbb{Z}}(0,S)^n$ *and is independent of the acceptance sequence* $(\alpha_j^*)_{j\ge1}$*, which is i.i.d.* Bernoulli$(p)$*. Moreover* $p \ge p_{\mathrm{lo}} := (1-e^{-1})e^{-c}\sqrt{2\pi S}/(2t)$*.*

*Proof. Proposal.* $v_j^*$ is geometric, $\Pr[v_j^* \ge k] = \Pr[U_j^v < e^{-k}] = e^{-k}$. The map $(s,u,v) \mapsto y$ of (ii) is a bijection from $\{0,1\} \times \{0,\dots,t-1\} \times \mathbb{N}$ onto $\mathbb{Z}$ (nonnegative $y$ for $s=0$, negative for $s=1$), so $\Pr[y_j^* = y] = q(y) = (1-e^{-1})e^{-v(y)}/(2t)$ with $v(y) = \lfloor m(y)/t \rfloor$, $m(y) = y$ or $-y-1$.

*Acceptance.* $g(y) \ge 0$: in both cases $|y| \ge m(y) \ge tv(y)$, so $g(y) \ge t^2v^2/(2S) - v + S/(2t^2) = \frac{1}{2}(tv/\sigma - \sigma/t)^2 \ge 0$. Given $y$, the two factors of $\alpha_j^*$ use independent coins; the first is 1 with probability $e^{-\lfloor g \rfloor}$, and the second with probability $e^{-\{g\}}$, since $\Pr[K^* > k] = \prod_{i\le k}\{g\}/i = \{g\}^k/k!$ and $\Pr[K^* \text{ odd}] = \sum_{k\ge0}(-1)^k\{g\}^k/k!$ (CKS20, Algorithm 1). Hence $\Pr[\alpha_j^* = 1 \mid y_j^* = y] = e^{-g(y)}$ and

$$\Pr[y_j^* = y, \alpha_j^* = 1] = \frac{1-e^{-1}}{2t}\, e^{-v(y)} e^{-y^2/(2S)+v(y)-c} = \frac{(1-e^{-1})e^{-c}}{2t}\, e^{-y^2/(2S)}.$$

Summing over $y$ gives $p$, and dividing gives $\pi_S$.

*Independence.* The trials use disjoint coins, so they are i.i.d. Since the joint law of $(y_j^*, \alpha_j^*)$ factorizes as above, conditionally on the whole sequence $(\alpha_j^*)_j$ the values $y_j^*$ of the accepted trials are i.i.d. $\pi_S$; so the output, the first $n$ of them, is $\mathcal{N}_{\mathbb{Z}}(0,S)^n$ given every acceptance sequence.

*Lower bound.* By Poisson summation, $\Theta = \sqrt{2\pi S}\sum_{k\in\mathbb{Z}} e^{-2\pi^2 S k^2} \ge \sqrt{2\pi S}$. □

**Lemma 5.16** (Coupling)**.** *Outside* $\mathsf{F}$*, the budgeted sampler returns exactly the ideal sampler's output, and* $A = A^* := \sum_{j\le T}\alpha_j^*$*.*

*Proof.* Let trial $j$ be decided. For a threshold $E = \lfloor 2^r x \rfloor$ and a coin with $R \ne E$: if $R < E$ then $U < (R+1)/2^r \le E/2^r \le x$; if $R > E$ then $U \ge R/2^r \ge (E+1)/2^r > x$. So $[R < E] = [U < x]$. Applied with $x = e^{-i}$, $i \le J_v$: the indicators defining $v_j$ and $v_j^*$ agree for $i \le J_v$, and $v_j < J_v$ gives $U_j^v \ge e^{-J_v}$, so they agree for all $i$, and $v_j = v_j^*$, $y_j = y_j^*$. $G_j = \lfloor 2^r g_j \rfloor$ exactly, because $2^r v_j$ and $C$ are integers; hence $n_j = \lfloor g_j \rfloor$, $F_j = \lfloor 2^r\{g_j\} \rfloor$ and $\lfloor F_j/k \rfloor = \lfloor 2^r\{g_j\}/k \rfloor$. Applied with $x = e^{-n_j}$ (and, for $n_j \ge J_0$, with $E = 0$, where $R \ne 0$ gives $U \ge 2^{-r} > e^{-n_j}$) and with $x = \{g_j\}/k$ for $k \le k_{\max}$, the decisions agree with the ideal ones; $L_j < k_{\max}$ means that $K_j^* = L_j + 1 \le k_{\max}$, so the parity is decided by the coins drawn. Hence $\alpha_j = \alpha_j^*$ for every $j \le T$, $A = A^*$, and $A \ge n$ means that the ideal sampler's first $n$ acceptances lie among the first $T$ trials. □

**Lemma 5.17** (Failure bound)**.**

$$\Pr[\mathsf{F}] \le \xi_{\mathrm{bound}} := \exp\Big(-T\,\mathrm{KL}\Big(\frac{n-1}{T}\ \Big\|\ p_{\mathrm{lo}}\Big)\Big) + T\Big(e^{-J_v} + \frac{1}{k_{\max}!} + (J_v + 1 + k_{\max})2^{-r}\Big) \quad \textit{whenever } n-1 < Tp_{\mathrm{lo}},$$

*with* $\mathrm{KL}(a\|p) = a\ln\frac{a}{p} + (1-a)\ln\frac{1-a}{1-p}$. Calibrate *(implementation:* `sampler.budget`*) takes* $T$ *the smallest integer whose first term is at most* $\xi_T/2$*,* $J_v = \lceil \ln(8T/\xi_T) \rceil$ *and* $k_{\max}$ *the smallest with* $T/k_{\max}! \le \xi_T/8$*, and checks that the tie term is at most* $\xi_T/8$*; then* $\xi_{\mathrm{bound}} \le \xi_T$*.*

*Proof.* If no trial is undecided, $A = A^*$ by Lemma 5.16, so $\mathsf{F} \subseteq \{A^* < n\} \cup \bigcup_j \{j \text{ undecided}\}$. $A^* \sim \mathrm{Binomial}(T,p)$ with $p \ge p_{\mathrm{lo}}$ (Lemma 5.15); the lower tail is nonincreasing in $p$, and the Chernoff bound $\Pr[\mathrm{Bin}(T,p) \le k] \le e^{-T\mathrm{KL}(k/T\|p)}$ for $k < Tp$ gives the first term. For one trial: $v_j = J_v$ requires $R_j^v < E_{J_v}$, of probability $E_{J_v}2^{-r} \le e^{-J_v}$; $L_j = k_{\max}$ requires $U_j^{(k)} < \{g_j\}/k$ for all

$k \le k_{\max}$, of probability at most $1/k_{\max}!$; and each of the $J_v + 1 + k_{\max}$ ties is a coin equal to a threshold computed from other coins, of probability $2^{-r}$. A union bound over the $T$ trials completes the proof. The constants are certified: $E_i$ is computed from rational bounds on $e^{-1}$ (consecutive partial sums of its alternating series) with directed rounding at 768 bits, and both bounds must give the same floor. □

At $\xi_T = 2^{-80}$, $N = 2^{14}$ and the largest $\sigma_c^2$ of the $10^6$-row runs ($\sigma_c^2 = 2^{136.7}$, $\sigma_c \approx 2^{68.4}$, quoted as $2^{68.5}$ elsewhere): $b = 68$, $p_{\text{lo}} = 0.4457$, $T = 39{,}088$ $(2.39N)$, $J_v = 69$, $k_{\max} = 29$, $\xi_{\text{bound}} = 2^{-80.9}$. For the flooding of SanitizeBFV ($\sigma_\psi = 2^{61}$, $N_B = 2^{14}$): $p_{\text{lo}} = 0.4805$, $T = 36{,}189$, the same $J_v$ and $k_{\max}$, $\xi_{\text{bound}} = 2^{-80.9}$.

**The cost assumption.** A side channel needs a model of what the observer measures. We use the following.

**Assumption 5.18** ((W): work, not values, determines time)**.** The time $\mathsf{O}$ takes to execute a sequence of operations is $\Phi(\varpi, Z)$, where the *trace* $\varpi$ lists the operations together with their public shapes (array lengths, numbers of limbs, the native function called and the sizes of its operands, but not the operand values), $Z$ is machine noise (scheduling, other processes, the state of caches and of the allocator) independent of every secret of the run and of $D$, and $\Phi$ does not depend on the operand values.

**Lemma 5.19** (The trace of a call)**.** *In the implementation (`sampler.sample_residues`), the trace of a call that does not fail is a function of $(S, n, \xi_T)$, the moduli and $A$ only.*

*Proof (by inspection of the code).* The call draws $4T(1 + L_y + 8 + 4k_{\max})$ bytes from the CSPRNG ($L_y$ limbs of 32 bits hold $|y| < 2^{b+\lceil \log_2(J_v+2)\rceil+1}$), and then executes a fixed sequence of numpy operations on arrays of shape $(\cdot, T)$ with limb counts that are functions of $(S, T, J_v, k_{\max})$: the comparisons for $v$ against the $J_v$ table values, the placement of $v$ at bit $b$, the limb products $y^2$, $y^2\lfloor 2^P/S\rfloor$ and $\hat{Q} \cdot S$, one correction, the addition of $C$ and subtraction of $v$ at limb 4, a gather from the $(J_0 + 1)$-entry table, and $k_{\max}$ long divisions of $F$ by $k$ (by multiplication with $\lceil 2^{28}/k\rceil$ on 16-bit digits, not by a hardware division) and comparisons. No Python branch depends on a value. The only shape that varies is that of the index array of accepted trials (length $A$), from which the first $n$ are taken; the residues of these $n$ values modulo each $q$ are computed by a fixed reciprocal multiplication and one correction. The failure test is evaluated in every call. □

**Lemma 5.20** (Release timing)**.** *Under (W), for each release $s$ there is a random variable $\tau_s^*$, whose law depends only on the public parameters and the machine, which is independent of $\mathsf{R}$'s view with exact noise (and of $D$), and such that, with $V$ the rest of $\mathsf{R}$'s view (its tape and messages),*

$$\mathrm{SD}\big((V, \tau_s),\ (V^*, \tau_s^*)\big) \le \xi_T,$$

*where $V^*$ is $V$ with $\eta_s$ replaced by the ideal sampler's output on the same coins, which is exactly $D_{\mathbb{Z}^N, \sigma_{c,s}}$ and independent of everything else in the run.*

*Proof.* $\mathsf{O}$'s work for release $s$ after the request is: the sampler call, the re-randomized partial decryption (a native call on fixed-shape polynomials), the deadline test and the hold, and the send. Let $\varpi(A)$ be the trace of a non-failing release with $A$ accepted trials (Lemma 5.19; the other steps have fixed traces). Set $\tau_s^* = \Psi(\Phi(\varpi(A^*), Z), Z')$, where $\Psi$ applies the deadline rule (hold until $D$, or refuse if the busy time exceeds $D$) and $Z'$ is the network and scheduling noise of the send. On $\mathsf{F}^c$, $\eta_s$ equals the ideal output and $A = A^*$ (Lemma 5.16), so $(V, \tau_s) = (V^*, \tau_s^*)$; hence the statistical

distance is at most $\Pr[\mathsf{F}] \le \xi_T$ (Lemma 5.17). $A^*$ is a function of the acceptance sequence, which is independent of the ideal output (Lemma 5.15) and of every other coin of the run; $Z, Z'$ are independent of all secrets by (W); and $D$, the deadline, is fixed before the first release from timed dummy calls on public parameters with independent coins. So $\tau_s^*$ is independent of $V^*$ and of $D$. □

**Corollary 5.21** (Output privacy with release timing; the charge)**.** *Under the hypotheses of Theorem 5.11 and (W), with every release drawn by* SampleDGBudget *at failure bound $\xi_T$, $\mathsf{R}$'s view of one run* including the arrival times $(\tau_s)_s$ *is $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-SIM-CDP with statistical slack*

$$\gamma = n_{\mathrm{rel}}\big(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97} + \xi_T\big),$$

*for the mechanism $\mathcal{M}(D)$ of Theorem 5.11 extended by the independent, data-independent component $(\tau_s^*)_s$. In the hockey-stick ledger of Part C, the run's additive slack becomes $n_{\mathrm{rel}}(\delta_B + \delta' + \xi_T)$.*

*Proof.* Add a hybrid H0′ before H1: replace, for $s = 1, \dots, n_{\mathrm{rel}}$ in turn, $(\eta_s, \tau_s)$ by $(\eta_s^*, \tau_s^*)$. Each step costs at most $\xi_T$ by Lemma 5.20 (the lemma holds conditionally on everything outside release $s$, because the sampler's coins are fresh). After H0′ every $\eta_s$ is exact and independent of the rest, which is what H2 uses, and the $\tau_s^*$ are independent of everything else and have a law that does not depend on $D$. The remaining hybrids are unchanged and carry $(\tau_s^*)_s$ along. The extended mechanism is still $\rho_{\mathrm{run}}$-zCDP: a component independent of $D$ and of the rest does not change any Rényi divergence (Lemma 5.2), and the simulator outputs it from $\mathcal{M}(D)$. The $n_{\mathrm{rel}}\xi_T$ is a statistical distance introduced before any other hybrid, exactly like $\delta_B$; in Part C it enters the purely additive slack sum $\Gamma$ (the proofs of Theorem 5.2 and Corollary 5.4 there treat every such term identically), and by Lemma 5.7 it is converted with the factor $1 + e^{\varepsilon}$. □

**Remark 5.22** (The deadline is defence in depth; the overrun is not charged)**.** The theorem rests on the work budget, not on the padding. Under (W) the busy time of a release is a function of $(A^*, Z)$, so whether it overruns the deadline is independent of $\eta_s$ and of $D$: an overrun refusal is part of $\tau_s^*$ and needs no charge, whatever the machine load. The padding still helps where (W) is only approximately true: residual value-dependent costs are hidden below $D$, and only an overrun could expose them.

**Remark 5.23** (The flooding of SanitizeBFV)**.** The flooding vectors of the PSI replies are drawn by the same sampler, one call per vector of $N_B$ values ($2\alpha F$ calls per cuckoo table, $n_{\mathrm{calls}}$ in total). By Lemmas 5.16–5.20 applied to each call, the reply timing makes the view of the Link phase differ by at most $n_{\mathrm{calls}}\xi_T$ from one in which every flooding vector is exact and independent of the timing, which is what the smudging step of G4's second layer (Section 7) needs; a failed call aborts the run, inside the same bound. This adds $n_{\mathrm{calls}}\xi_T$ to the statistical distance of the second layer, not to $\delta$: neighbouring datasets have the same identifiers and positions, so the Link phase, timing included, has the same law under both (Theorem 5.11). Setup checks $n_{\mathrm{calls}}\xi_T \le \delta_{\mathrm{psi}}$; at $10^6$ rows ($\alpha = 5$, $F = 2$, one table) it is $20 \cdot 2^{-80} = 2^{-75.7}$.

**Remark 5.24** (What (W) asks of the implementation, and what is not verified)**.** (W) is an assumption about the platform, not a theorem. The implementation is written so that only operations whose cost plausibly does not depend on values remain: fixed-shape numpy loops over `uint64` ($+, -, \times$, shifts, bitwise operations, comparisons, boolean reductions), with no hardware division (whose latency depends on its operands on many x86 cores) and no Python branch on a value; ChaCha20 from OpenSSL for the coins; and OpenFHE's NTT-based polynomial arithmetic. Not verified: that numpy's loops have no value-dependent fast paths on this machine; the gather from the 2.9 kB table $(E_i)$, whose addresses depend on $n_j$ (a cache channel, invisible over a network

but not on O's machine); the CPython allocator; and OpenFHE's own samplers for $(u, e_0, e_1)$, which (CT) of Section 8 also covers. The measurements in `reports/v31_timing/` support (W) for the sampler: on coins forced to the extreme values ($v = 0$ everywhere, $v = J_v - 1$ everywhere, $u = 0$ everywhere) one call's trials take the same time as on random coins to within the machine's noise (medians within 6% in two runs of 30 and 100 calls per kind, at a load average near 20 from other jobs, with an ordering of the kinds that did not repeat), while the plain CKS20 sampler's median time per value grows with $|\eta|/\sigma$, from 9.7 $\mu$s below $1\sigma$ to 25.6 $\mu$s beyond $3\sigma$. This is evidence, not a proof.

# 6 Input privacy (G1) and cohort privacy (G2)

**O's view.** Besides its own input $D$ and random tape, O receives the following from R:

(i) *Key-generation shares:* $p_{0,R} = -as_R + e_R^{\mathrm{pk}}$, and the rotation-key shares $-a_{g,j}s_R + P\mathsf{g}_j\psi_g(s_R) + e_{R,g,j}^{\mathrm{rk}}$.

(ii) *Link messages:* the blinded OPRF queries, exactly $n_R^{\max}$ of them; R's BFV public key and any evaluation keys used by the PSI; and the BFV ciphertexts of windowed powers. Their number is fixed by public PSI parameters and $n_R^{\max}$.

(iii) *Uploads:* $(c_{0,h}, \varsigma_h)$ for each of the $(d_R + 2)m$ chunks, with $c_{0,h} = -a_h s_R + e_h + m_{U,h}$ and $a_h = H(\varsigma_h)$.

(iv) *Control messages:* the requests for the two or three releases. Their schedule is fixed by Algorithm 6 and does not depend on R's data. R's refusal at line 7 happens after the last release and is not sent to O.

O receives no decryption share and no ciphertext component that involves $s_R$ other than these.

**Assumption 6.1** (For G1 and G2)**.** (a) *RLWE with circular security in $s_R$.* Let $s_R \leftarrow \chi_{\mathrm{key}}$, let every error be drawn from $\chi_{\mathrm{err}}$, and let all $a$'s be uniform. Then the tuple consisting of

- R's public-key share,
- its $Ld_{\mathrm{num}}$ rotation-key shares, and
- $(d_R + 2)m$ samples $(a_h, -a_h s_R + e_h)$ at modulus $Q_2$

is pseudorandom. This is multi-sample RLWE plus the circular assumption for the rotation-key shares. Samples at modulus $Q_2$ are reductions of samples at modulus $PQ_2$, so a single modulus suffices.

(b) *Random oracle.* The seed expansion $H$ is a random oracle, and seeds are $\lambda$-bit strings drawn uniformly.

(c) *The PSI's encryption.* The PSI's BFV is IND-CPA under RLWE, with circular security for any relinearization key R sends.

(d) *The OPRF.* The blinded OPRF queries of [5] are distributed independently of the queried identifiers. For the Diffie–Hellman-type OPRF this holds perfectly in a prime-order group. [*citation detail unverified:* the concrete OPRF instantiation in [5] and whether its query blinding is perfectly or computationally hiding]

**Theorem 6.2** (G1: R-input privacy)**.** *Under Assumption 6.1 there is a PPT simulator $\mathcal{S}_O$ with the following property. Given O's input and random tape, $(n_R^{\max}, \bar{n}_O)$ and the public parameters, $\mathcal{S}_O$ outputs a view that is indistinguishable from O's real view, for every input of R. The distance is at most*

$$\mathrm{Adv}_{\mathrm{RLWE},R} + \mathrm{Adv}_{\mathrm{BFV}} + q_H^2/2^\lambda + p_{\mathrm{cuckoo}},$$

*where $q_H$ is the number of oracle queries and $p_{\mathrm{cuckoo}}$ is the probability that R's cuckoo hashing fails in a way that O can observe, for example as an abort or a retry (bounded in Lemma 7.5). R's abort*

*on a detected false match is observable too; its probability, at most $p_{\text{false}}$ (the paper's Remark on collisions), is added to the slack in the same way.*

*Proof. The simulator.* $\mathcal{S}_O$ outputs:

- uniform key-generation shares;
- $n_R^{\max}$ uniform group elements as the OPRF queries;
- a fresh BFV public key, any evaluation keys, and encryptions of zero in the public number;
- $(d_R + 2)m$ pairs $(c^*_{0,h}, \varsigma^*_h)$, with $c^*_{0,h}$ uniform in $\mathcal{R}_{Q_2}$ and $\varsigma^*_h$ uniform seeds;
- the fixed control messages.

*Hybrids.*

1. Replace the OPRF queries by uniform group elements. The distance is zero for a perfectly hiding blinding, and the OPRF advantage otherwise.
2. Replace R's BFV ciphertexts of powers by encryptions of zero, at cost $\text{Adv}_{\text{BFV}}$.
3. In the random-oracle model, the $a_h = H(\varsigma_h)$ are uniform and independent unless two seeds collide, which happens with probability at most $q_H^2/2^\lambda$. The reduction programs $H(\varsigma_h)$ to the $a$-components of its RLWE challenge. The tuple of R's key shares and the values $-a_h s_R + e_h$ is then replaced by uniform elements, at cost $\text{Adv}_{\text{RLWE},R}$ by Assumption 6.1(a). Adding the known plaintext $m_{U,h}$ to a uniform element leaves it uniform.
4. The real protocol leaks the cuckoo-hashing outcome. Removing that leak costs at most $p_{\text{cuckoo}}$.

The final hybrid is $\mathcal{S}_O$'s output. O's own messages, such as the OPRF evaluations, PSI replies and releases, are functions of its view, so they need no simulation. □

**Corollary 6.3** (G2: cohort privacy)**.** *$\mathcal{S}_O$ does not use R's cohort, $n_R$ or $\mathbf{X}_R$. Hence, for any two cohorts of size at most $n_R^{\max}$, O's views are indistinguishable, up to twice the distance of Theorem 6.2. Padding the OPRF queries to exactly $n_R^{\max}$ is what makes the number of OPRF calls cohort-independent. The per-run fresh OPRF key and the quota are not needed against an honest-but-curious O; they matter only for linkage across runs.*

**Remark 6.4.** (i) The statement "O never receives a decryption share from R" is what makes G1 a pure encryption argument. If any future version sent R's share $c_1 s_R + e$, G1 would need its own flooding.

(ii) The random-oracle model is used only to treat the expanded $a_h$ as uniform. A standard-model alternative is to send $a_h$ in full, at twice the upload size.

(iii) G1 and G2 hold against an honest-but-curious O only. A deviating O could, for example, choose its PSI replies so as to learn the outcome of R's decryption through R's later behaviour. This is outside the model.

# 7 Disclosure (G4), the cuckoo-hashing slack and the reply noise

This section proves the first layer of G4 (what R learns from the Link phase) in the random-oracle model under CDH, bounds the cuckoo-hashing failure probability that enters the slack of G1 and G2, and bounds the BFV evaluation noise of the PSI replies in an explicit model of BFV. The first result was first written in the companion book (Chapter 8, Lemma `aln.keyed` and Theorem `aln.R`) and was checked here; the other two are new. They replace Correction 7 as far as the first layer is concerned.

**The Link phase.** O draws a fresh OPRF key $k \leftarrow \mathbb{Z}^*_{q_{\mathbb{G}}}$ and a uniform permutation $\tau$ of its $\bar{n}_O$ padded rows. Row $j$ gets $y_j = F_k(\text{id}_j) = H_2(\text{id}_j, H_1(\text{id}_j)^k)$; its item, bins and label key are

domain-separated hashes of the two halves of $y_j$, and its label is $\tau(j)$ padded with the label key modulo $t$. R pads its identifiers to a list $\bar{X}$ of exactly $n_R^{\max}$, obtains $F_k$ on $\bar{X}$ through the blinded protocol ($\alpha = H_1(x)^r$, $\beta = \alpha^k$), cuckoo-hashes its items, and sends BFV encryptions of the windowed powers under its own key. O evaluates the masked membership and label polynomials of each partition and replies with SanitizeBFV of the results. R's output is $(I, \pi|_I)$, or an abort.

**Assumption 7.1** (CDH in the OPRF group)**.** Given $(g, g^k, h)$ with $k$ and $h$ uniform, no PPT algorithm (with non-uniform advice) outputs $h^k$ with probability above $\mathrm{Adv}_{\mathrm{CDH}}$.

**Lemma 7.2** (OPRF-keyed items and labels are pseudorandom outside the match)**.** *Model $H_1$, $H_2$ and the derived hash functions as random oracles, let* R *be semi-honest, and let the distinguisher make at most $q_{H_2}$ queries to $H_2$. Let $J$ be the set of* O*'s rows whose identifiers are not in $\bar{X}$. Then* R*'s view of the OPRF phase together with $(F_k(\mathrm{id}_j))_{j\in J}$ changes by at most $q_{H_2}\mathrm{Adv}_{\mathrm{CDH}} + \epsilon_{\mathrm{coll}}$ in distinguishing advantage when these outputs are replaced by independent uniform strings, for every input; $\epsilon_{\mathrm{coll}} \le \bar{n}_O n_R^{\max} 2^{-\ell_{\mathrm{id}}} + (q_{H_1} + \bar{n}_O)/q_{\mathbb{G}}$ accounts for a collision of a fresh dummy identifier of length $\ell_{\mathrm{id}}$ with the other list and for $H_1$ hitting the identity.*

*Proof. Identical until a bad event.* In the real experiment $F_k(\mathrm{id}_j) = H_2(\mathrm{id}_j, H_1(\mathrm{id}_j)^k)$; in the modified one it is a fresh uniform string, and $H_2$ is answered uniformly elsewhere. An honest R queries $H_2$ only at points $(x, H_1(x)^k)$ with $x \in \bar{X}$, so the two experiments are identical unless the distinguisher queries $H_2$ at $(\mathrm{id}_j, H_1(\mathrm{id}_j)^k)$ for some $j \in J$ (event Bad), which has the same probability in both.

*A CDH solver.* Given $(g, g^k, h)$, run the modified experiment with O's input fixed (non-uniform advice), programming $H_1(x) = g^{a_x}$ at every queried point other than O's identifiers in $J$, and $H_1(\mathrm{id}_j) = h\, g^{b_j}$ for $j \in J$, with fresh uniform exponents. Answer a blinded query $\alpha = g^{a_x r}$ with $(g^k)^{a_x r} = \alpha^k$; compute the outputs of the matched rows as $H_2(x, (g^k)^{a_x})$ and those of the rows in $J$ as fresh strings. Preprocessing, evaluation and sanitization use no other secret, and R's own coins are sampled by the solver. At the end, pick a uniformly random $H_2$ query $(\mathrm{id}_j, Z)$ with $j \in J$ and output $Z\,(g^k)^{-b_j}$. On Bad some query has $Z = h^k (g^k)^{b_j}$, picked with probability at least $1/q_{H_2}$. So $\Pr[\mathsf{Bad}] \le q_{H_2}\mathrm{Adv}_{\mathrm{CDH}}$; the programmed values are uniform except on the events counted in $\epsilon_{\mathrm{coll}}$. □

**Theorem 7.3** (G4, first layer: R learns only the match and its positions)**.** *Under the hypotheses of Lemma 7.2 there is a PPT simulator $\mathcal{S}_\Lambda$ that, given* R*'s input, its output $(I, \pi|_I)$ and the public sizes, outputs* R*'s view of the Link phase up to distinguishing advantage $q_{H_2}\mathrm{Adv}_{\mathrm{CDH}} + \epsilon_{\mathrm{coll}}$, for every input. Jointly with the output, the distance grows by the probability of the correctness failures on which* R*'s output differs from $(I, \pi|_I)$ (overflow of a bin, a cuckoo failure, a false match; Lemma 7.5 and the paper's Remark on collisions). The statement needs neither circuit privacy of the BFV evaluation nor* SanitizeBFV*, nor a bound on the reply noise.*

*Proof.* $\mathcal{S}_\Lambda$ samples R's tape, runs R's code to produce the padded list, the blinded elements, the BFV keys and the encrypted windowed powers; draws a key $k^*$ and answers the blinded elements with it; creates one row per $i \in I$ with output $F_{k^*}(i)$ (computed by R's code) and position $\pi(i)$, and $\bar{n}_O - |I|$ rows with uniform outputs and positions forming a uniform bijection onto $[\bar{n}_O] \setminus \pi(I)$; and runs O's preprocessing (partition placement included), evaluation, masking and sanitization with fresh coins. By Lemma 7.2 the real view is within the stated distance of the experiment in which O's rows outside the match carry uniform outputs. In that experiment the key enters only through R's own OPRF answers, so $k^*$ may replace $k$; positions outside the match are a uniform bijection onto the complement of $\pi(I)$ because $\tau$ is uniform; the labels outside the match are padded by uniform keys. Every reply, its noise included, is a fixed randomized function of R's ciphertexts and

of these inputs, so the two experiments coincide. Overflow aborts, cuckoo failures, false matches and the partition index of a match arise from identically distributed items and bins and are reproduced exactly. □

**Remark 7.4** (Second reading; what remains of Correction 7)**.** The book's lemma omits the collision term $\epsilon_{\text{coll}}$, which is negligible; otherwise its proof is correct as written. The reduction is non-uniform (it embeds $\mathsf{O}$'s input), so $\text{Adv}_{\text{CDH}}$ is against non-uniform adversaries. Circuit privacy is not needed for the first layer: whatever the noise of a reply reveals is a function of pseudorandom items and labels. The second layer, SanitizeBFV, makes the reply noise independent of $\mathsf{O}$'s data up to a statistical distance, given a worst-case bound $B_\psi$ (Proposition 7.7 below bounds its evaluation part in a model of BFV).

**Lemma 7.5** (The cuckoo-hashing slack)**.** *Let $n$ items each have $h$ candidate bins drawn independently and uniformly from $m$ bins (in the random-oracle model for the bin hashes; with $m = 2^{14}$ dividing $2^{256}$ the reduction modulo $m$ of a uniform hash is exactly uniform). The probability that no assignment of the items to distinct candidate bins exists is at most*

$$p_{\text{noassign}}(n, m, h) \le \sum_{s=2}^{n} \binom{n}{s} \binom{m}{s-1} \left(\frac{s-1}{m}\right)^{hs}.$$

*An insertion procedure that finds an assignment whenever one exists (augmenting paths, or the implementation's random walk followed, on failure, by an augmenting-path search) fails with exactly this probability, so $p_{\text{cuckoo}} \le p_{\text{noassign}}$. At $m = 2^{14}$ and $h = 3$ the bound is $1.7 \cdot 10^{-15} = 2^{-49.1}$ for $n = 2{,}000$ (the evaluation's $n_R^{\max}$, one table), $2^{-47.0}$ for $n = 4{,}096$, $2^{-45.0}$ for $n = 8{,}192$, and at most $2^{-40}$ for every $n \le 10{,}472$ (load $0.639$).*

*Proof.* By Hall's theorem an assignment exists unless some set $S$ of $s \ge 2$ items has fewer than $s$ candidate bins in total (a single item always has a bin). Fix $S$ and a set $T$ of $s-1$ bins; all $hs$ candidates of $S$ lie in $T$ with probability $((s-1)/m)^{hs}$. A union bound over $S$ and $T$ gives the sum (a set with fewer than $s-1$ bins is contained in some $T$). The numbers are from `experiments/proofs_check/tv1_cuckoo.py`, which sums every term. □

**Remark 7.6** (What is not covered)**.** The implementation's insertion is a random walk with at most 2,000 evictions per item. Until 2026-10-25 it had no fallback, and its failure probability when an assignment exists was not bounded. It now completes the walk's partial assignment by augmenting paths when the walk exceeds its step limit (`implementation/vfr/psi.py`, `cuckoo` and `_augment`), so it finds an assignment whenever one exists and $p_{\text{cuckoo}} = p_{\text{noassign}}$. The union bound is useless above load about 0.64 at $m = 2^{14}$ (the implementation allows tables up to load 0.9, used only when $n_R^{\max} > 10{,}472$).

## 7.1 The BFV evaluation noise of the replies

We bound the evaluation part $w_{\text{eval}} = (q'/q_B)v$ of the switched reply error in the following model of BFV (with $\mathsf{R}$'s ternary key $s_B$, plaintext modulus $t$, $\Omega_{\text{B}} = \lfloor q_B/t \rfloor$, $r_t = q_B - t\Omega_{\text{B}} \in [0, t)$, ring dimension $N = N_B$, and every integer polynomial represented with centred coefficients).

(B1) A ciphertext $(c_0, c_1)$ with plaintext $m$, $\|m\|_\infty \le t/2$, satisfies $c_0 + c_1 s_B = \Omega_{\text{B}} m + v + q_B k$ as integer polynomials; $\|c_i\|_\infty \le q_B/2$ and $\|v\|_\infty \le q_B/2$ give $\|k\|_\infty \le \kappa_k := (N+3)/2$.

(B2) Every error coefficient has absolute value at most $B_e$ (OpenFHE's table sampler at $\sigma = 3.19$ saturates at $B_e = 27$), so $\mathsf{R}$'s fresh encryptions have $\|v\|_\infty \le E_0 := (2N+1)B_e$: $v = e_B^{\text{pk}} u' + e_0 + e_1 s_B$ for a public-key encryption, and $v = e$, with $\|v\|_\infty \le B_e$, for the secret-key encryption that the implementation uses.

(B3) A ciphertext product computes $\tilde{d}_i = \lfloor t\, d_i/q_B \rceil$ from the tensor $(d_0, d_1, d_2) = (c_0c_0',\ c_0c_1' + c_1c_0',\ c_1c_1')$, with rounding errors of sup-norm at most $\frac{1}{2}$ up to an additive error of sup-norm at most $E_{\mathrm{rns}}$ (an allowance for an RNS implementation), and relinearizes $\tilde{d}_2$, adding an error of sup-norm at most $E_{\mathrm{rl}}$.

(B4) A plaintext multiplication multiplies both components by a plaintext polynomial $a$ with $\|a\|_\infty \le t/2$; a plaintext addition adds $\Omega_{\mathrm{B}} a$ to $c_0$.

**Proposition 7.7** (The reply's evaluation noise in the BFV model)**.** *Under (B1)–(B4), let* $F(E,E') = \frac{Nt^2}{4} + t(\frac{Nt}{4} + \frac{1}{2}) + \frac{Nt}{2}(E+E') + \frac{tN}{q_B}EE' + Nt^2\kappa_k + tN\kappa_k(E+E') + \frac{1+N+N^2}{2} + E_{\mathrm{rl}} + E_{\mathrm{rns}}$*. A product of two ciphertexts with noise sup-norms at most* $E$ *and* $E'$ *(both below* $q_B/(2t)$*) has noise sup-norm at most* $F(E,E')$*. Hence, if every power* $\mathbf{w}^d$*,* $d \le D$*, is a product tree of depth at most* $L_\times$ *over the fresh windows, its noise is at most* $E_{L_\times}$ *with* $E_{l+1} = F(E_l, E_l)$*, and a reply* $\sum_{d=0}^{D} a_d \odot \mathrm{ct}(\mathbf{w}^d)$ *(with* $\mathrm{ct}(\mathbf{w}^0)$ *the plaintext* $1$*) has noise at most*

$$E_{\mathrm{eval}} = D\Big(\tfrac{Nt}{2}E_{L_\times} + t\big(\tfrac{Nt}{4} + \tfrac{1}{2}\big)\Big) + \tfrac{t(D+1)}{2},$$

*so that* $\|w_{\mathrm{eval}}\|_2 \le \sqrt{N}\,(q'/q_B)\,E_{\mathrm{eval}}$*, provided* $E_{\mathrm{eval}} < q_B/(2t)$ *(which also gives correct decryption before the switch).*

*Proof. Product.* Multiply the two invariants of (B1) and scale by $t/q_B$. With $t\Omega_{\mathrm{B}} = q_B - r_t$ and $mm' = [mm']_t + t\mu$, $\|\mu\|_\infty \le Nt/4 + \frac{1}{2}$, the identity

$$\begin{aligned}\tfrac{t}{q_B}(\Omega_{\mathrm{B}}m + v + q_Bk)(\Omega_{\mathrm{B}}m' + v' + q_Bk') \equiv \Omega_{\mathrm{B}}[mm']_t - \tfrac{r_t}{q_B}\Omega_{\mathrm{B}}mm' - r_t\mu \\ + \tfrac{t\Omega_{\mathrm{B}}}{q_B}(mv' + m'v) + \tfrac{t}{q_B}vv' - r_t(mk' + m'k) + t(vk' + v'k)\end{aligned}$$

holds modulo $q_B$ (the terms $t\Omega_{\mathrm{B}}(mk' + m'k)$ and $tq_Bkk'$ reduce to $-r_t(mk' + m'k)$ and $0$). The rounded tensor adds $r_0 + r_1s_B + r_2s_B^2$, of sup-norm at most $\frac{1}{2}(1 + \|s_B\|_1 + \|s_B^2\|_1) \le \frac{1}{2}(1 + N + N^2)$, plus $E_{\mathrm{rns}}$; relinearization adds $E_{\mathrm{rl}}$. Bound each term with $\|ab\|_\infty \le N\|a\|_\infty\|b\|_\infty$: $\frac{r_t\Omega_{\mathrm{B}}}{q_B} \le 1$ and $\|mm'\|_\infty \le Nt^2/4$; $r_t < t$; $\|mv' + m'v\|_\infty \le \frac{Nt}{2}(E+E')$; $\|vv'\|_\infty \le NEE'$; $\|mk' + m'k\|_\infty \le Nt\kappa_k$; $\|vk' + v'k\|_\infty \le N\kappa_k(E+E')$. The sum is $F(E,E')$. The bound is increasing in $E$ and $E'$, so a tree of depth $l$ has noise at most $E_l$.

*Evaluation.* $a \odot (c_0, c_1)$ has invariant $a(\Omega_{\mathrm{B}}m + v + q_Bk) \equiv \Omega_{\mathrm{B}}[am]_t - r_t\mu_a + av$, with $\|\mu_a\|_\infty \le Nt/4 + \frac{1}{2}$ and $\|av\|_\infty \le \frac{Nt}{2}\|v\|_\infty$. Summing $D$ such terms and the plaintext term $a_0$, the messages add up to at most $(D+1)t/2$ in sup-norm; reducing the sum modulo $t$ changes the noise by $r_t\nu$ with $\|\nu\|_\infty \le (D+1)/2$, at most $t(D+1)/2$. The switch multiplies the evaluation part of the noise by $q'/q_B$, and $\|\cdot\|_2 \le \sqrt{N}\|\cdot\|_\infty$. □

**Corollary 7.8** (A proved $B_\psi$ in the model)**.** *At the evaluation's PSI parameters (*$N_B = 2^{14}$*,* $t = 4{,}293{,}918{,}721$*,* $D = 64$*,* $W = 7$ *windows and product trees of depth* $L_\times = 3$*, as in the implementation's balanced split of* $d$ *by popcount,* $q_B$ *of seven* 60*-bit limbs and* $q'$ *of two), Proposition 7.7 gives* $E_0 = 2^{19.8}$*,* $E_3 = 2^{211}$*,* $E_{\mathrm{eval}} = 2^{262} < q_B/(2t) \approx 2^{387}$ *and* $\|w_{\mathrm{eval}}\|_2 \le 2^{-31}$ *for any* $E_{\mathrm{rl}}, E_{\mathrm{rns}} \le 2^{26}$*, and* $2^{-30}$ *for any* $E_{\mathrm{rl}}, E_{\mathrm{rns}} \le 2^{90}$ *(script* `experiments/proofs_check/bpsi_model.py`*). So, in the model, the evaluation part is covered by the "+1" of the implementation's* $B_{\mathrm{err}} = 2(\sqrt{N_B}/2 + N_B^{3/2}/2) + 1$*, and* $B_\psi := \sqrt{2\pi}B_{\mathrm{err}}$ *bounds* $\|w\|_2$ *with the rounding part (exact) and the fresh-encryption part (whose high-probability bound* $(q'/q_B)\beta_{\mathrm{fresh}}$ *is below* $2^{-250}$*).*

**Remark 7.9** (Status of $B_\psi$)**.** Proposition 7.7 is a theorem about the textbook BFV model (B1)–(B4); it is worst case over $\mathsf{O}$'s inputs and needs only the honest sampling of $\mathsf{R}$'s errors. That

OpenFHE 1.5.1's BFVrns (its RNS multiplication and scaling, and its relinearization) satisfies (B3) with small $E_{\mathrm{rns}}$ and $E_{\mathrm{rl}}$ was not checked; the measured evaluation noise after the switch is $2^{-72}$, far below the model's $2^{-31}$. G4's second layer is therefore proved for the model and remains conditional on this conformance for the implementation.

## 8 Correctness: no wraparound, and the exact sampler

**Proposition 8.1** (Decryption without wraparound)**.** *Let $S_{\max} = \max_e \sup_D |\mathbf{z}_e(D)|$. Each entry is a sum over at most $\bar{n}_O$ rows of a product bounded by one of $o$, $\sqrt{ro}$, $\sqrt{rw}$, $\sqrt{ow}$ or $w$; so $S_{\max} \le \bar{n}_O \max(o, \sqrt{ro}, \sqrt{rw}, \sqrt{ow}, w)$. Let $d_{\max}$ bound $\|\widehat{Z}_s(D) - \Omega_{\mathrm{fin}}\tau^{-1}(\mathbf{z}_s(D))\|_2$ over all $D$; Proposition 4.5 gives an explicit, proved $d_{\max}$ ($2^{30.3}$ at the evaluation setting with $\bar{n}_O = 10^6$ and $n_R^{\max} = 2{,}000$, and $2^{38.6}$ if every row may match). Suppose*

$$q_0 > 2\Big(\Omega_{\mathrm{fin}}\,\tfrac{2|\mathcal{E}_s|}{N}\,S_{\max} + d_{\max} + \tfrac{1}{2} + B_{\mathrm{tot},s} + t_\eta\,\sigma_{c,s}\Big). \tag{Q}$$

*Then $v_s$, reduced to its centred representative modulo $q_0$, equals the integer polynomial $f_s(D) + e_{\mathrm{tot},s} + \eta_s$. This holds except with probability at most*

$$\delta_B + 2Ne^{-t_\eta^2/2}$$

*over the keys and the fresh randomness. Moreover $\mathsf{Dcd}(v_s) = \tau(\widehat{Z}_s)/\Omega_{\mathrm{fin}} + \tau(e_{\mathrm{tot},s} + T_7' + \eta_s)/\Omega_{\mathrm{fin}}$, where $T_7' = f_s - \widehat{Z}_s$.*

*Proof. Size of the centre.* From the inversion formula in the proof of Lemma 1.1, $\|\tau^{-1}(\mathbf{z})\|_\infty \le \frac{2}{N}\|\mathbf{z}\|_1 \le \frac{2|\mathcal{E}|}{N}S_{\max}$. So $\|f_s\|_\infty \le \Omega_{\mathrm{fin}}\frac{2|\mathcal{E}|}{N}S_{\max} + d_{\max} + \frac{1}{2}$.

*Size of the error.* $\|e_{\mathrm{tot},s}\|_\infty \le \|e_{\mathrm{tot},s}\|_2 \le B_{\mathrm{tot},s}$, except with probability $\delta_B$ (Theorem 3.12).

*Size of the noise.* Each $\eta_k$ is $\sigma^2$-subgaussian (Lemma 3.1(b)), so $\Pr[|\eta_k| > t_\eta\sigma] \le 2e^{-t_\eta^2/2}$. A union bound over the $N$ coefficients gives the second term.

Under (Q), the integer polynomial $f_s + e_{\mathrm{tot},s} + \eta_s$ lies strictly inside $(-q_0/2, q_0/2)^N$. So the centred reduction of $v_s$ returns it exactly, and the decoding identity follows.

*Earlier levels: no condition is needed.* Proposition 3.8 is an identity of congruences that holds whatever the size of the intermediate plaintexts. A rescale satisfies $c_0' + c_1's = (c_0 + c_1 s)/q + \varepsilon_0 + \varepsilon_1 s$ exactly in $\mathbb{Q}[X]/(X^N + 1)$; if $c_0 + c_1 s = \mu + KQ$ with $K$ integral, whatever the size of $\mu$, then $(c_0 + c_1 s)/q = \mu/q + K(Q/q)$, so the invariant modulo $Q/q$ is $\mu/q + \varepsilon_0 + \varepsilon_1 s$. Plaintext products, additions and automorphisms are congruences modulo the current modulus, and so is key switching (Lemmas 3.5 and 4.1), again with no size condition; the rounding errors are bounded through their sup-norms only. Hence only the final reduction modulo $q_0$ can wrap around, and (Q) suffices. (This replaces the argued step of an earlier version, which bounded the intermediate plaintexts; the argument is that of the companion book, Chapter 9, Proposition `agg.wrap`, checked here.) □

**Remark 8.2** (The specification's condition)**.** CALIBRATE requires

$$q_0 > 2\big(\Omega S_{\max} + 2^\kappa B_{\mathrm{tot}} + 10\,\sigma_{c,\max}\big).$$

This implies (Q) with $t_\eta = 10$ when $\Omega = \Omega_{\mathrm{fin}}$. The reasons:

- $2|\mathcal{E}|/N \le 1$, and the slack $\Omega_{\mathrm{fin}}S_{\max}(1 - 2|\mathcal{E}|/N)$ is far larger than $d_{\max} + \frac{1}{2}$;
- $2^\kappa B_{\mathrm{tot}} \ge B_{\mathrm{tot}}$.

The specification should still state the correctness-failure probability $2Ne^{-50} \approx 2^{-57.1}$ per release. This is a correctness event, not a privacy event, because the view is the same whether or not wraparound occurs. The term $\Omega S_{\max}$ overestimates by a factor $N/(2|\mathcal{E}|) \approx 2^7$, so $q_0$ could be about 7 bits smaller.

**The exact sampler.** SampleDG is Algorithm 3 of Canonne, Kamath and Steinke [4]. It draws a discrete Laplace candidate $Y$ with integer scale $t = \lfloor \sigma \rfloor + 1$, accepts it with probability $\exp(-(|Y| - \sigma^2/t)^2/(2\sigma^2))$, and repeats otherwise. The acceptance coins come from an exact Bernoulli($e^{-\gamma}$) sampler for rational $\gamma$. CKS20 prove the following, for rational $\sigma^2$:

- the output is *exactly* $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$;
- only integer arithmetic and uniform random bits are used;
- the expected number of iterations is bounded by a constant.

[*citation detail unverified:* the exact constant bounding the expected iterations in CKS20]

Two implementation requirements follow:

(i) $\sigma^2_{c,s}$ *must be rational.* Set $\sigma^2_{c,s} := \lceil \widehat{\Delta}^2_{\mathrm{enc},s}/(2\rho_s) \rceil$. Increasing $\sigma$ lowers $\rho_s$ and only helps flooding.

(ii) *Big integers must be arbitrary-precision.* At $\sigma \approx 2^{55}$, $t$ and $\sigma^2$ exceed 64 bits, so all arithmetic must be exact arbitrary-precision integer or rational arithmetic, and the coins must come from a CSPRNG.

**What "constant-time" must mean.** The model of the specification abstracts side channels away. Its remaining obligation O2 nevertheless asks for a constant-time sampler, and R can at least observe O's response time. The requirement is therefore:

> (CT) The joint distribution of everything R can observe about the execution of the release, in particular its duration, is independent of $(\eta, s_O, u, e_0, e_1)$ given the public parameters.

The rejection structure helps only in part:

- In rejection sampling, the number of *rejected* trials is independent of the accepted value.
- But the time of the accepting trial depends on the accepted $Y$. The discrete Laplace and Bernoulli($e^{-\gamma}$) subroutines loop for value-dependent numbers of iterations, and big-integer operations take time that depends on operand length.

So a direct implementation of CKS20 does not satisfy (CT).

A sufficient implementation has four parts:

(a) every inner loop runs for a fixed public number of iterations $T_{\max}$, with the result selected branch-free;

(b) the probability that some loop would have needed more than $T_{\max}$ iterations is at most $2^{-\kappa'}$ per sample, and the total $Nn_{\mathrm{rel}}2^{-\kappa'}$ is added to the slack $\gamma$;

(c) all arithmetic is done in fixed-width limbs;

(d) the $N$ samples of a release are produced in one batch whose duration is padded to a public bound.

**A concrete variant (Section 5.5).** The implementation now uses SampleDGBudget (Definition 5.14), which realizes (a)–(c) in a stronger form: the whole batch, not each loop, gets a fixed budget of $T$ candidate trials; every trial executes the same fixed-width operations (limb arithmetic, every coin of the bounded loops drawn and compared), the surplus is discarded, and the call fails with probability at most $\xi_T$ (Lemma 5.17) instead of $Nn_{\mathrm{rel}}2^{-\kappa'}$; $n_{\mathrm{rel}}\xi_T$ is added to $\gamma$ (Corollary 5.21). It is exact on the complement of the failure event (Lemma 5.16), so no Rényi or statistical distance of the output enters, unlike generic constant-time samplers by convolution such as that of Micciancio and Walter [14]. (d) is kept as defence in depth. (CT) then holds for $\eta$ up to $\xi_T$ per release under the cost assumption (W) (Assumption 5.18). **[Not fully rigorous:** (W) is not verified for the implementation's CPython and numpy operations (Remark 5.24), and (CT) for $(u, e_0, e_1)$ depends on OpenFHE's samplers, which were not analysed.]

# 9 Calibrate, the run as a secure computation, and O's work

The three results of this section were first written in the companion book (Chapter 19, Proposition `par.calibrate`; Chapter 4, Proposition `sc.secureDP` with Chapter 17, Theorem `run.secure`; Chapter 20, Proposition `cost.lower`). They were checked here, with one correction to the second.

## 9.1 Soundness of Calibrate

CALIBRATE (the paper's Section on calibration) takes the parameter vector $\mathcal{P}$ and the public ledger parameters and runs seven steps: the security test $\log_2(QP) \le L_{\max}(N, \lambda_{\sec})$; the sensitivities and the mode; $\kappa = \max\{\kappa_{\min}, \kappa_{\run}, \kappa_L\}$ (unless $\mathcal{P}$ fixes $\kappa$), with the test of the run's share of the ledger's slack capacity; $\gamma$ and $\delta_{\DP}$; $\rho_{\run}$ and its split, the stage-2 share rounded down; the smallest grid scale at which $\sigma_{c,s} = (\lceil \widehat{\Delta}^2_{\enc,s}/(2\rho_s) \rceil)^{1/2} \ge \max\{2^{\kappa} B_{\tot,s}/\sqrt{2\pi}, 2^{42}\}$ for every release; and the test on $q_0$. It refuses when a step has no solution.

**Proposition 9.1** (CALIBRATE is sound)**.** *Suppose* CALIBRATE *does not refuse, and that it evaluates $B_{\tot,s}$ and $\Delta_{x,s}$ by the formulas of Theorem 3.12 with Lemma 4.1, and of Lemma 4.3, for the circuit model in use. Then:*

*(a) every output is a deterministic function of $\mathcal{P}$ and the public ledger parameters;*

*(b) for every release, $\sigma_{c,s} \ge \max\{2^{\kappa} B_{\tot,s}/\sqrt{2\pi}, 2^{42}\}$, $\widehat{\Delta}^2_{\enc,s}/(2\sigma^2_{c,s}) \le \rho_s$ with $\sigma^2_{c,s}$ an integer, $\sum_s \rho_s \le \rho_{\run}$, $\rho_{\run} = \max\{\rho : \delta(\rho, \varepsilon_{\run}) \le \delta_{\DP}\}$ with $\delta_{\DP} = \delta_{\run} - (1 + e^{\varepsilon_{\run}})\gamma > 0$; these are the hypotheses of Theorem 5.11;*

*(c) condition (Q) of Proposition 8.1 holds for every release, with $d_{\max}$ of Proposition 4.5, and the chain meets the security bound;*

*(d) if $\kappa$ is derived, $(1 + e^{\varepsilon_{\run}})\gamma \le \zeta \delta_{\run}$ and $\rho_{\run} \ge \tilde{\rho}_{\run}$; with a ledger cap, $n_{\rel}\varepsilon'(\kappa) \le (\rho_{\run}/\rho_L)E'_L$ and $n_{\rel}(\delta_B + \delta') \le (\rho_{\run}/\rho_L)\Gamma_L$, and the costs it returns are the per-run costs of the composition appendix (Part C, Section 5).*

*Proof.* (a) No step reads data, keys, errors or coins: $B_{\tot,s}$ and $\Delta_{x,s}$ are functions of public parameters only (Theorem 3.12, Lemmas 4.1 and 4.3), the grid and its order are fixed, and every step is deterministic. (b) The scale loop stops only where the flooding condition holds for every release. $\sigma^2_{c,s} = \lceil \widehat{\Delta}^2_{\enc,s}/(2\rho_s) \rceil \ge \widehat{\Delta}^2_{\enc,s}/(2\rho_s)$. The split sums to at most $\rho_{\run}$ because the stage-2 share is rounded down. $\delta_{\DP} > 0$ by the test, and $\rho_{\run}$ exists and is positive because $\delta(\cdot, \varepsilon)$ is continuous, increasing and 0 at 0. (c) The $q_0$ test is $q_0 > 2(\Omega_{\fin} S_{\max} + 2^{\kappa} B_{\tot,\max} + t_\eta \sigma_{c,\max})$ with $S_{\max} = \bar{n}_O \max(r, o, w, \sqrt{ro}, \sqrt{rw}, \sqrt{ow})$. Since $\Omega_{\fin} S_{\max} = \Omega_{\fin} \frac{2|\mathcal{E}_s|}{N} S_{\max} + (1 - \frac{2|\mathcal{E}_s|}{N})\Omega_{\fin} S_{\max}$ and the second part exceeds $d_{\max} + \frac{1}{2}$ whenever $d_{\max} + \frac{1}{2} \le (1 - 2|\mathcal{E}_s|/N)\Omega_{\fin} S_{\max}$ (at the evaluation setting $2^{38.6}$ against more than $2^{74}$), and $2^{\kappa} B_{\tot,\max} \ge B_{\tot,s}$, $\sigma_{c,\max} \ge \sigma_{c,s}$, the test implies (Q). (d) $\gamma(k)$ is strictly decreasing, so $\kappa \ge \kappa_{\run}$ gives the share and $\delta_{\DP} \ge (1 - \zeta)\delta_{\run}$, hence $\rho_{\run} \ge \tilde{\rho}_{\run}$ by monotonicity; $\varepsilon'(k)$ is decreasing in $k$, so $\kappa \ge \kappa_L$ and $\tilde{\rho}_{\run} \le \rho_{\run}$ give the smudging inequality, and the share test gives the slack inequality. □

**Remark 9.2** (Second reading)**.** The proof is in exact arithmetic. The implementation computes $\sigma^2_{c,s}$ as an exact fraction, rounds the split and the ledger's capacities in the safe direction, and stores the ledger's sums exactly; it evaluates $B_{\tot,s}$ in double precision (relative error about $10^{-15}$) and, since 2026-10-25, makes the flooding test conservative: $B_{\tot,s}$ and the needed $\sigma_{c,s}$ are rounded up by a relative $2^{-40}$, and the integer $\sigma^2_{c,s}$ is compared with the square of the rounded value in exact rational arithmetic, so floating-point error cannot pass a case that exact arithmetic would refuse. Its formula for the term $T_8$ uses $d_{\num}$ where $\sqrt{d_{\num}}$ suffices, which is conservative. Its diagnostic value `q0_need_Q` used $d_{\max} = 2^{35}$ until 2026-10-25, which Proposition 4.5 proves when at most

$n_R^{\max} = 2{,}000$ rows match ($2^{30.3}$) but not when every row of a register of $10^6$ may match ($2^{38.6}$). It now uses the $d_{\max}$ of Proposition 4.5 for $n_{\max} = \min(n_R^{\max}, \bar{n}_O)$; the condition (Q) still needs 74.1 bits at $\bar{n}_O = 4{,}096$ and 81.9 bits at $10^6$ ($\varepsilon = 1$), because its dominant term is about $2^{81}$. The refusal test itself uses the condition of (c), which needs no $d_{\max}$.

### 9.2 The run computes the DP functionality against R

Let $\mathcal{M}(D) = (\omega_{\mathsf{R}}, \Lambda, ([f_s(D;\pi) + \eta_s]_{q_0})_s)$ be the mechanism of Theorem 5.11, where $[\cdot]_{q_0}$ is the centred residue modulo $q_0$ (a post-processing, so $\mathcal{M}$ is still $\rho_{\text{run}}$-zCDP), and let Out be R's output map: its decryption of the Link replies and the centred residues $v_s = h_{O,s} + c'_{1,s} s_R \bmod q_0$.

**Corollary 9.3** (Standard simulation for R, jointly with its output)**.** *Under the hypotheses of Theorem 5.11, the pair (R's view, R's output) is within statistical distance $\gamma$ plus computational loss $n_{\text{rel}}\text{Adv}_1 + \text{Adv}_2$ of $(\mathcal{S}(\mathcal{M}(D)), \mathcal{M}(D))$: the run computes the functionality that hands $\mathcal{M}(D)$ to R, against a semi-honest R, in the standard (not the liberal) sense. With Theorem 7.3 in place of the transcript $\Lambda$, the functionality may output $(I, \pi|_I)$ instead of $\Lambda$, at the additional cost of that theorem.*

*Proof.* Out is an efficient deterministic function of the view, and the simulator of Theorem 5.11 is output-consistent: on input $w_s$ it outputs $(c^*_{1,s}, w_s - c^*_{1,s} s_R \bmod q_0)$, whose reconstruction $h^* + c^* s_R \bmod q_0$, taken as a centred residue, is $w_s$ exactly because $w_s = [f_s + \eta_s]_{q_0}$ is already reduced. Applying $v \mapsto (v, \text{Out}(v))$ to the two views of Theorem 5.11 preserves both the statistical and the computational distance, and maps the simulated pair to $(\mathcal{S}(\mathcal{M}(D)), \mathcal{M}(D))$. □

**Remark 9.4** (Correction to the book)**.** The book's mechanism outputs $f_s + \eta_s$ unreduced, and its claim that the reconstruction returns it "exactly" fails on the wraparound event $\|f_s + \eta_s\|_\infty \geq q_0/2$, of probability at most $2Ne^{-t_\eta^2/2}$ per release under (Q). Either the functionality outputs the centred residue, as above (nothing changes for privacy), or that probability is added to the slack. The book's general statement (SIM-CDP with an output-consistent simulator implies security jointly with the outputs, and the converse fails without output-consistency) is correct as written.

### 9.3 O's work is linear in its scope

Model O's computation as reading cells of its register (one cell per row), its final state as a function of its coins and the cells read, and the decrypted statistic as a function of the final states; the protocol is $(\tau_{\text{dec}}, p_{\text{err}})$-correct if, for every input in the clipped domain, the decrypted $\mathbf{z}$ is within $\tau_{\text{dec}}$ of the true one in every entry except with probability $p_{\text{err}}$.

**Proposition 9.5** (Work linear in the scope)**.** *Let the protocol be $(\tau_{\text{dec}}, p_{\text{err}})$-correct with $B_O^2 > 2\tau_{\text{dec}}$, and let every efficient distinguisher of O's views for two cohorts of the same size have advantage at most* adv*. If $n_{\text{Sc}}$ rows of the scope register are non-dummy, each matchable by some cohort, then O reads at least $(1 - 2p_{\text{err}} - \text{adv})\, n_{\text{Sc}}$ rows in expectation, for every input.*

*Proof.* Fix a non-dummy row $t$ and let $E_t$ be the event that cell $t$ is never read; it is a function of O's view. If $t$ is matched, couple two executions with the same coins and inputs except $\mathbf{x}_{O,t} = \mathbf{0}$ in one and $B_O \mathbf{e}_1$ in the other. They coincide until cell $t$ is read, so $E_t$ has the same probability in both, and on $E_t$ the decrypted statistics coincide while the true entries $\mathbf{B}_{11}$ differ by $B_O^2 > 2\tau_{\text{dec}}$; so on $E_t$ at least one execution errs, and $\Pr[E_t] \leq 2p_{\text{err}}$, for every content of the other cells. If $t$ is unmatched, replace one identifier of the cohort by that of row $t$; the indicator of $E_t$ is an efficient function of O's view, so its probability changes by at most adv. Summing $\Pr[t \text{ read}] \geq 1 - 2p_{\text{err}} - \text{adv}$ over the $n_{\text{Sc}}$ rows gives the bound. □

In v3.1, adv is the slack of G2 (Corollary 6.3), so O's work is $\Omega(n_{\mathrm{Sc}})$; O may skip its dummy rows, which it knows, so the bound is in $n_{\mathrm{Sc}}$, not $\bar{n}_O$. The argument is that of the classical lower bound on the server's work in single-server private information retrieval.

## 10 Corrections to the specification

Items are ordered by importance. "Spec" refers to `protocol_v31.tex`.

**Correction 1** (The centre of the mechanism; affects Lemma 1, Theorem 1 and Calibrate)**.** *Spec:* the ideal mechanism is $\bar{\mathbf{z}}_s(D) + \eta_s$ with $\bar{\mathbf{z}}_s = \mathsf{Ecd}(\mathbf{z}_s(D))$, and $e_{M,s}$ includes O's encoding roundings of $V_e$ and $\mathbf{u}_k$.

*Problem:* the decrypted value differs from $\mathsf{Ecd}(\mathbf{z}_s(D))$ by deterministic, data-dependent terms. Chief among them is the mask spill $\sum_e S_e\, r_{u_e}$, of norm about $2^{30.7}$ to $2^{34.5}$ for $\bar{n}_O = 10^6$ (Remark 3.9). Flooding them would require $\log_2 \Omega \approx 67$ to 71.

*Corrected statement:* centre the mechanism at the noiseless circuit output $f_s(D) = \lfloor \widehat{Z}_s(D) \rceil$ (Definition 3.7), and use the sensitivity

$$\widehat{\Delta}_{\mathrm{enc},s} = \sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_s + \Delta_{x,s} + \sqrt{N}$$

(Corollary 3.13), where $\Delta_{x,2} \approx 2^{18.8}$. Then $e_{\mathrm{tot},s}$ consists only of randomness-dependent terms (Proposition 3.8), and Theorem 5.11 holds with $\rho_s = \widehat{\Delta}^2_{\mathrm{enc},s}/(2\sigma^2_{c,s})$. Numerically this changes $\rho_s$ by a relative $2^{-32.7}$. Lemma 1.2 itself is correct as stated.

**Correction 2** (Exact final scale)**.** Throughout Lemma 1, Calibrate, $\sigma_s$ and $\mathsf{Dcd}$, $\Omega$ must be the exact final scale $\Omega_{\mathrm{fin}} = \Omega_U\Omega_V\Omega_u/(q_1q_2)$. It must be tracked exactly, as in FLEXIBLEAUTO, or computed from the primes. Decoding with a nominal power-of-two scale, as FIXEDMANUAL does, causes a deterministic relative bias of about $2^{-30}$; we measured $8.7 \cdot 10^{-10}$ (Section 3.6). This bias is not a privacy problem once the mechanism is centred at $f_s$, but it is an avoidable estimation bias.

**Correction 3** (Obligation O1: the value of $B_{\mathrm{tot}}$ and the required scale)**.** *Spec:* $\log_2 \Omega \approx 52.5$ at $\varepsilon = 1$.

*Problem:* that figure floods only the fresh-encryption error, and only at its mean $\ell_2$ norm. For the circuit of Algorithm 4 as written, Theorem 3.12 gives a rigorous worst-case bound of $B_{\mathrm{tot}} \approx 2^{25.3}$ for stage 2, with $\delta_B = 2^{-50}$. That requires $\log_2 \Omega \approx 61.5$, or 60.5 if $P \geq 2^{20}\tilde{Q}_{\max}$ (Table 1).

*Recommended changes:* the following give $B_{\mathrm{tot}} \approx 2^{17.6}$ and $\log_2 \Omega \approx 53.7$:

(i) evaluate EvalSum before the first rescale, at level 2;
(ii) sum the masked products of a release *before* a single second rescale, instead of rescaling each $P_e$;
(iii) use stochastic rounding in Rescale and in ModDown;
(iv) choose $P \geq 2^{10}\tilde{Q}_{\max}$;
(v) encode R's vectors at $\Omega_U = q_2 \approx 2^{60}$.

By a rough count, $\log_2(QP) \approx 320$ for (A) and $\approx 285$ for (C′). Both are below the 128-bit limit of about 438 for $N = 2^{14}$ [2]. [**Not fully rigorous:** This count is approximate; the modulus chain should be re-derived with the chosen $d_{\mathrm{num}}$.]

*Further requirement:* Calibrate must use the *worst-case*, data-independent $B_{\mathrm{tot},s}(\Omega)$ of Theorem 3.12. A value estimated from the actual ciphertexts, for example by OpenFHE's noise estimation, would depend on the data. It would then leak through $\sigma_{c,s}$, and it would not be a worst-case bound. Such an estimate may serve as a sanity check only.

**Correction 4** (Lemma 2: constants)**.** *Corrected statement:* for $e \in \mathbb{Z}^N$ and $\eta \sim D_{\mathbb{Z}^N,\sigma_c}$ with $\sigma_c > \sigma_2 \geq \eta_\epsilon(\mathbb{Z}^N)/\sqrt{2\pi}$,

$$\mathrm{SD}(\eta + e, \eta) \leq \frac{\|e\|_2}{\sqrt{2\pi}\sqrt{\sigma_c^2 - \sigma_2^2}} + \frac{4\epsilon}{1-\epsilon}.$$

Alternatively, $\mathrm{SD}(\eta + e, \eta) \leq \|e\|_2/(2\sigma_c)$ for every $\sigma_c > 0$. With CALIBRATE's condition, the per-release distance is at most $2^{-\kappa}(1 + 2^{-80}) + 2^{-97}$ (Lemma 2.3, Corollary 2.4).

**Correction 5** ($\delta$ accounting)**.** *Spec:* $\delta_{\mathrm{DP}} = \delta_{\mathrm{run}} - n_{\mathrm{rel}}(\delta_B + 2^{-\kappa})$.

*Corrected statement:* Theorem 5.11 gives $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{DP}})$-SIM-CDP with statistical slack

$$\gamma = n_{\mathrm{rel}}\big(\delta_B + 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}\big),$$

plus the sampler-cap failures of Section 8. A single indistinguishability guarantee $(\varepsilon_{\mathrm{run}}, \delta_{\mathrm{run}})$ requires

$$\delta_{\mathrm{DP}} = \delta_{\mathrm{run}} - (1 + e^{\varepsilon_{\mathrm{run}}})\gamma$$

(Lemma 5.7). The factor $1 + e^{\varepsilon}$ is missing in the spec. For the example ($\gamma \approx 2.7 \cdot 10^{-12}$) the effect on $\rho_{\mathrm{run}}$ is negligible.

**Correction 6** (Assumptions of Theorem 1)**.**

- *RLWE in two forms.* RLWE is needed both for the re-randomizer (one sample per release, at modulus $q_0$, with ternary $u$ and $\mathcal{N}_{\mathbb{Z}}(0, 3.2^2)$ errors) and for O's key shares (multi-sample, at modulus $PQ_2$, with circular security for the rotation-key shares). These are Assumptions 5.9 and 5.10.
- *What "threshold key-generation security" means here.* For the one-round protocols of [17] it is exactly Assumption 5.10.
- *Unneeded assumptions.* PSI security is *not* needed for G3 under the spec's adjacency, because the identifiers are fixed and the PSI transcript is independent of $(\mathbf{x}_O, y)$. Nor is the random-oracle model.
- *An implicit requirement.* CALIBRATE's outputs must be data-independent.

**Correction 7** (PSI and the disclosure claim G4)**.** G4 says that R learns only $I$ and $\pi|_I$ from the Link phase. This requires the labeled PSI to be secure against a semi-honest receiver. For the FHE-based PSI of [6, 5], that requires *circuit privacy* of O's BFV replies, for example by noise flooding or re-randomization before the reply. Without it, the reply noise depends on O's identifier set. A modulus switch alone (Algorithm 2, line 4) is a heuristic measure. *Update:* with OPRF-keyed items and labels, the first layer of G4 holds without circuit privacy, in the random-oracle model under CDH (Section 7, Theorem 7.3); SANITIZEBFV adds a statistical second layer, proved for a model of BFV (Proposition 7.7). [*citation detail unverified:* how CLR17 and CHLR18 phrase the noise-flooding requirement]

**Correction 8** (G1 and G2: assumptions and slack)**.** *Missing assumptions:* G1 needs Assumption 6.1:

- RLWE with circular security in $s_R$, covering R's rotation-key shares;
- the random oracle for the seed expansion;
- IND-CPA of the PSI's BFV, with circular security for any relinearization key;
- obliviousness of the OPRF.

*Simulator inputs:* the simulator also takes the public parameters and O's own input.

*Slack:* G1 and G2 hold up to the probability $p_{\mathrm{cuckoo}}$ that R's cuckoo hashing fails observably.

**Correction 9** (Correctness)**.** The $q_0$ condition in CALIBRATE is sufficient once $\Omega = \Omega_{\mathrm{fin}}$ (Proposition 8.1). The spec should state the per-release correctness-failure probability, $\delta_B + 2Ne^{-50} \approx 2^{-50}$.

The sharper condition (Q) saves about 7 bits of $q_0$. The "modulus switch to $q_0$" in Algorithm 4 is a no-op, or an exact limb drop, for a chain $q_0 q_1 q_2$. A scaling modulus switch would add a rounding term bounded like $T_5$.

**Correction 10** (Sampler)**.**

- Set $\sigma_{c,s}^2$ to a rational, for example $\lceil \widehat{\Delta}_{\mathrm{enc},s}^2/(2\rho_s) \rceil$, as CKS20's exact sampler requires.
- "Constant-time" must mean (CT) of Section 8. A direct implementation of CKS20 does not satisfy (CT). The budgeted sampler of Section 5.5 does for $\eta$, under the cost assumption (W), up to a failure probability $\xi_T$ per release that is added to $\gamma$.
- The slot-domain noise has variance at most $\sigma_{c,s}^2 N/(2\Omega_{\mathrm{fin}}^2)$ per real slot component. The variance of $\mathcal{N}_{\mathbb{Z}}(0, \sigma^2)$ is below $\sigma^2$, and at $\sigma \gg 1$ the difference is negligible. So $\sigma_s = \sigma_{c,s}\sqrt{N/2}/\Omega_{\mathrm{fin}}$ is an upper bound on the noise s.d., not an exact value.

**Remark 10.1** (Statements verified as correct)**.** The following claims of the spec were checked and need no correction:

- the identity $\|\tau^{-1}(\mathbf{z})\|_2 = \sqrt{2/N}\|\mathbf{z}\|_2$ and the rounding term $\sqrt{N}$ of Lemma 1;
- the zCDP cost $\rho_s = \Delta_{\mathrm{enc},s}^2/(2\sigma_{c,s}^2)$ of the lattice discrete Gaussian, once $\Delta_{\mathrm{enc}}$ is replaced by $\widehat{\Delta}_{\mathrm{enc}}$;
- the validity of the conversion used in CALIBRATE;
- the release format $(c_1', h_O)$ with $c_0'$ withheld, which removes the C3 leak;
- the need for fresh $(u, e_0, e_1, \eta)$ in every release;
- the absence of any decryption share from R.

The spec's remark that stage 1's error "cannot depend on the entries released only in stage 2" is true but not needed: flooding covers any correlation.

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# Protocol v3.1 — statistical appendix

Proofs of the sensitivity, concentration and inference claims (S0–S6)

Technical appendix to `protocol_v31.tex`

**Abstract**

We prove the statistical claims of Protocol v3.1. Proposition S1 gives the exact replace-one $\ell_2$ sensitivity of the released vector $(\mathrm{triu}\,\mathbf{B}, \mathbf{C}, \mathbf{c}_R, \mathbf{c}_O, \mathbf{y}^\top\mathbf{y})$ in all three regimes, including a proof that the norms sit at their bounds (and where they do not), the triu-versus-Frobenius question, the case $d_O = 1$, the sensitivities $\Delta_{\neg yy}$, $\Delta_G$, $\Delta_y$, and the set of bounds on which $\mathbf{y}^\top\mathbf{y}$ is free. Lemma S2 bounds the operator norm of the masked symmetric noise matrix; Lemma S3 analyses the two-stage choice of the ridge; Lemma S4 bounds $\|\boldsymbol{\beta}\|$; Proposition S5 states the coverage of both intervals of Algorithm 6, separating exact, first-order and asymptotic parts; Lemma S6 analyses the parametric-bootstrap RSS correction. Sections S7–S10, added after a second reading of the companion book on the protocol, prove the exact stage-1 radius and its transfer to the lattice noise, a bracket for the switch-off of the ridge and a bound on the downward bias of $\lambda_{\min}$, the mechanism bias with explicit constants, the bias of the plug-in residual variance used in the FWS comparison, a valid bias-aware interval under binding clipping, and a closed form for the full-record sensitivity. Section 14 lists every place where the specification is false, imprecise or needs an extra condition, with corrected statements. Every step that is not fully rigorous is marked [first order], [asymptotic], [numerical] or [open].

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# 1 Summary

# 2 Setting, notation and the noise model

**Notation.** We follow `protocol_v31.tex`. R's block has $d_A := d_R + 1$ coordinates (intercept included) and O's block $d_O$, so $p = d_A + d_O$. For a matched record, $\mathbf{a} := \dot{\mathbf{x}} \in \mathbb{R}^{d_A}$ is R's full row ($\|\mathbf{a}\| \le B_R$, $a_1 = 1$), $\mathbf{x} := \mathbf{x}_O \in \mathbb{R}^{d_O}$ ($\|\mathbf{x}\| \le B_O$) and $y$ ($|y| \le B_y$). We write

$$r = B_R^2, \quad o = B_O^2, \quad w = B_y^2, \quad b = B_y.$$

$\Pi_O$ is the coordinate projection on O's block. The mask $\chi \in \{0,1\}^{p\times p}$ has $\chi_{ij} = 0$ iff $i, j \le d_A$. For the ridge chosen by Algorithm 6 we write

$$\mathbf{G}_\lambda = \mathbf{G} + \lambda\Psi, \qquad \mathbf{P}_\lambda = \mathbf{G}_\lambda^{-1}, \qquad \boldsymbol{\theta}_\lambda = \mathbf{P}_\lambda \mathbf{c}, \qquad \vartheta_0 = \frac{\lambda_{\min}(\mathbf{G}_\lambda)}{2\sigma_2\sqrt{p}}, \qquad \hat{\vartheta} = \frac{\lambda_{\min}(\hat{\mathbf{G}}_2 + \lambda\Psi)}{2\sigma_2\sqrt{p}}.$$

$\vartheta_0$ is the (unobservable) validity ratio of the ridge at the true Gram matrix; $\hat{\vartheta}$ is the gate statistic of Algorithm 6, line 7. The notation table of the specification calls $\hat{\vartheta}$ "the validity ratio"; the guarantee of Lemma S3 is about $\vartheta_0$. $\sigma_{yy}$ is the noise s.d. of $\widehat{yy}$ ($\sigma_2$ in joint mode, $\sigma_y$ in separate mode). $z_a = \Phi^{-1}(a)$, $\phi$ and $\Phi$ are the standard normal density and distribution function, and $Q = 1 - \Phi$.

**The masked noise model.** A *masked symmetric noise matrix* with scale $\sigma$ is $\mathbf{E} = \mathbf{E}^\top \in \mathbb{R}^{p\times p}$ with $E_{ij} = 0$ when $\chi_{ij} = 0$ and $\{E_{ij} : i \le j,\ \chi_{ij} = 1\}$ independent, each $\mathcal{N}(0, \sigma^2)$ (the diagonal of the O block included, since triu $\mathbf{B}$ is released entry by entry). Its covariance is

$$\mathrm{Cov}(E_{ij}, E_{kl}) = \sigma^2\chi_{ij}\big(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} - \delta_{ij}\delta_{ik}\delta_{il}\big). \tag{1}$$

Stage $s$ releases $\hat{\mathbf{G}}_s = \mathbf{G} + \mathbf{E}_s$ and (stage 2) $\hat{\mathbf{c}}_2 = \mathbf{c} + \mathbf{f}_2$ with $\mathbf{f}_2 \sim \mathcal{N}(0, \sigma_2^2 \mathbf{I}_p)$ (every entry of $\mathbf{c}_R$ and $\mathbf{c}_O$ depends on $\mathbf{y}$), and $\widehat{yy} = \mathbf{y}^\top \mathbf{y} + g$ with $g \sim \mathcal{N}(0, \sigma_{yy}^2)$; the noises of different releases are independent and independent of the data. For $\boldsymbol{\theta} \in \mathbb{R}^p$ we use

$$\mathbf{K}(\boldsymbol{\theta}) := \sigma^{-2} \operatorname{Cov}(\mathbf{E}\boldsymbol{\theta}), \qquad K(\boldsymbol{\theta})_{ik} = \delta_{ik}\Big(\sum_j \chi_{ij}\theta_j^2 - \chi_{ii}\theta_i^2\Big) + \chi_{ik}\theta_i\theta_k, \tag{2}$$

and $\mathbf{M}_\chi(\mathbf{P}) := \sigma^{-2}\mathbb{E}[\mathbf{EPE}] = \chi \circ \mathbf{P} + \operatorname{diag}_i\big(\sum_j \chi_{ij} P_{jj} - \chi_{ii} P_{ii}\big)$, both direct consequences of (1). $0 \preceq \mathbf{K}(\boldsymbol{\theta})$ and $\|\mathbf{K}(\boldsymbol{\theta})\| \le 2\|\boldsymbol{\theta}\|^2$.

**Assumption 2.1** (Gaussian slot model, N)**.** Every released entry equals its exact value plus an independent $\mathcal{N}(0, \sigma_s^2)$ variable, independently across releases and of the data.

All statistical statements below are proved under Assumption 2.1. Lemma S0 shows how they transfer to the actual mechanism, whose noise is the linear image of a lattice discrete Gaussian.

**Lemma 2.2** (S0: the slot noise)**.** *Let $\eta \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma_c^2)^N$ and let $\boldsymbol{\xi}^{\mathrm{raw}} \in \mathbb{R}^m$ be the noise in $m$ released slot values (real parts of distinct slots, or any $m$ distinct coordinates of $(\operatorname{Re}\tau(\eta), \operatorname{Im}\tau(\eta))/\Omega$). Write $\sigma_s = \sigma_c\sqrt{N/2}/\Omega$.*

*(a) $\boldsymbol{\xi}^{\mathrm{raw}} = (\sigma_s/\sigma_c)\,\mathbf{U}\eta$ for a matrix $\mathbf{U} \in \mathbb{R}^{m\times N}$ with orthonormal rows. Hence the coordinates of $\boldsymbol{\xi}^{\mathrm{raw}}$ are* exactly *uncorrelated, with common variance $\sigma_s^2\, v(\sigma_c)/\sigma_c^2$, where $v(\sigma) = \operatorname{Var}\mathcal{N}_{\mathbb{Z}}(0,\sigma^2)$ satisfies $\sigma^2\big(1 - O(\sigma^2 e^{-2\pi^2\sigma^2})\big) \le v(\sigma) \le \sigma^2$; and $\boldsymbol{\xi}^{\mathrm{raw}}$ is $\sigma_s^2$-subgaussian: $\mathbb{E}e^{\langle \mathbf{u}, \boldsymbol{\xi}^{\mathrm{raw}}\rangle} \le e^{\sigma_s^2\|\mathbf{u}\|^2/2}$ for all $\mathbf{u}$.*

*(b) There is a coupling of $\eta$ with $\mathbf{Z} \sim \mathcal{N}(0, \sigma_c^2\mathbf{I}_N)$ such that $\Pr[\eta \ne \lfloor \mathbf{Z} \rceil] \le N\,\mathrm{TV}_1(\sigma_c)$, where $\mathrm{TV}_1(\sigma) = d_{\mathrm{TV}}\big(\mathcal{N}_{\mathbb{Z}}(0,\sigma^2), \lfloor \mathcal{N}(0,\sigma^2)\rceil\big)$ satisfies, for every $\sigma > 0$,*

$$\mathrm{TV}_1(\sigma) \le \frac{\phi(1)}{12\,\sigma^2} + \frac{V}{48\,\sigma^3} + \sum_{j\ge 1} e^{-2\pi^2\sigma^2 j^2}, \qquad V = 2\phi(0) + 8\phi(\sqrt{3}) = 1.5100,$$

*that is, $\mathrm{TV}_1(\sigma) \le 0.020165\,\sigma^{-2} + 0.031459\,\sigma^{-3} + 1.0001\,e^{-2\pi^2\sigma^2}$ for $\sigma \ge 1$. On that event $\boldsymbol{\xi}^{\mathrm{raw}} = \boldsymbol{\xi}^G + \boldsymbol{\nu}$ with $\boldsymbol{\xi}^G = (\sigma_s/\sigma_c)\mathbf{U}\mathbf{Z} \sim \mathcal{N}(0, \sigma_s^2\mathbf{I}_m)$ exactly and $\|\boldsymbol{\nu}\| \le N/(2\sqrt{2}\,\Omega)$.*

*(c) The decrypted value also carries the flooded error $e_{\mathrm{tot},s}$ and the encoding rounding. On $\{\|e_{\mathrm{tot},s}\| \le B_{\mathrm{tot},s}\}$ (probability $\ge 1 - \delta_B$) their slot-domain contribution has $\ell_2$ norm at most $N/(2\sqrt{2}\,\Omega) + \sqrt{2\pi}\,2^{-\kappa}\sigma_s$, using $\sigma_c \ge 2^\kappa B_{\mathrm{tot}}/\sqrt{2\pi}$.*

*Consequently every statement proved under Assumption 2.1 holds for Protocol v3.1 after adding $\delta_B + N\,\mathrm{TV}_1(\sigma_{c,s})$ (and the wraparound probability excluded by* Calibrate*) per release to the failure probabilities, and enlarging every radius, in operator or Frobenius norm, by $\sqrt{2}\,\bar{\nu}$ with $\bar{\nu} := N/(\sqrt{2}\,\Omega) + \sqrt{2\pi}\,2^{-\kappa}\sigma_s$ (the factor $\sqrt{2}$ accounts for the mirrored off-diagonal entries of $\hat{\mathbf{B}}$). With the indicative parameters ($N = 2^{14}$, $\log_2\Omega \approx 52.5$, $\kappa = 40$, $\sigma_c \approx 2^{50}$) both additions are below $10^{-11}$ in relative terms.*

*Proof.* (a) For $m(X) = \sum_k \eta_k X^k$, slot $j$ is $\tau(m)_j = \sum_k \eta_k \zeta^{e_j k}$ with $\zeta = e^{i\pi/N}$ and $e_j = 5^j \bmod 2N$ odd. For $j \ne l$, $\sum_{k=0}^{N-1}\cos(\pi k e_j/N)\cos(\pi k e_l/N) = \frac{1}{2}\sum_k[\cos(\pi k(e_j - e_l)/N) + \cos(\pi k(e_j + e_l)/N)] = 0$, because $e_j \pm e_l$ is even and non-zero modulo $2N$ (the subgroup generated by 5 does not contain $-1$), and $\sum_{k<N} e^{i\pi k m/N} = (1 - e^{i\pi m})/(1 - e^{i\pi m/N}) = 0$ for even $m \not\equiv 0$. The same computation gives $\sum_k \cos^2 = \sum_k \sin^2 = N/2$ and $\sum_k \cos\cdot\sin = 0$. Hence the $N$ real numbers $(\operatorname{Re}\tau(m), \operatorname{Im}\tau(m))$ equal $\sqrt{N/2}\,\mathbf{U}_0\eta$ for an orthogonal $\mathbf{U}_0$ (this is the identity $\|\tau^{-1}(\mathbf{z})\| = \sqrt{2/N}\|\mathbf{z}\|$ of Lemma 1 of the specification), and $\mathbf{U}$ is a set of $m$ rows of $\mathbf{U}_0$. The variance and subgaussian bounds for $\mathcal{N}_{\mathbb{Z}}(0,\sigma^2)$ are in Canonne, Kamath and Steinke [6] (Section 2); the lower bound on $v$ follows from

Poisson summation, $v(\sigma) = \sigma^2 \frac{1+\sum_{m\neq 0}(1-4\pi^2\sigma^2 m^2)e^{-2\pi^2\sigma^2 m^2}}{1+\sum_{m\neq 0} e^{-2\pi^2\sigma^2 m^2}}$. Subgaussianity of $\boldsymbol{\xi}^{\mathrm{raw}}$ follows because $\langle \mathbf{u}, \boldsymbol{\xi}^{\mathrm{raw}} \rangle = (\sigma_s/\sigma_c)\langle \mathbf{U}^\top \mathbf{u}, \eta \rangle$ with $\|\mathbf{U}^\top \mathbf{u}\| = \|\mathbf{u}\|$ and independent coordinates.

(b) Couple coordinatewise with a maximal coupling; a union bound gives $N\,\mathrm{TV}_1$. Write $p_k \propto e^{-k^2/2\sigma^2}$ and $q_k = \int_{k-1/2}^{k+1/2} \phi_\sigma$, so $\mathrm{TV}_1 = \frac{1}{2}\sum_k |p_k - q_k|$. First, $p_k = \phi_\sigma(k)/\sum_l \phi_\sigma(l)$ and, by Poisson summation, $\sum_l \phi_\sigma(l) = 1+2\epsilon$ with $\epsilon = \sum_{j\geq 1} e^{-2\pi^2\sigma^2 j^2}$; so $\sum_k |p_k - \phi_\sigma(k)| = |1 - \sum_l \phi_\sigma(l)| = 2\epsilon$. Second, $q_k - \phi_\sigma(k) = \int_{-1/2}^{1/2} (\phi_\sigma(k+u) - \phi_\sigma(k) - u\phi'_\sigma(k))\,du$ (the linear term integrates to zero), so $|q_k - \phi_\sigma(k)| \leq \frac{1}{24}\sup_{I_k} |\phi''_\sigma|$ with $I_k = [k - \frac{1}{2}, k + \frac{1}{2}]$. For any function $f$ of bounded variation, $\sup_{I_k} |f| \leq \int_{I_k} |f| + \mathrm{TV}_{I_k}(|f|)$, so $\sum_k \sup_{I_k} |\phi''_\sigma| \leq \int |\phi''_\sigma| + \mathrm{TV}(|\phi''_\sigma|) = 4\phi(1)\sigma^{-2} + V\sigma^{-3}$: $|\phi''(x)| = \phi(x)|x^2 - 1|$ for the standard density has zeros at $\pm 1$ and local maxima $\phi(0)$ at 0 and $2\phi(\sqrt{3})$ at $\pm\sqrt{3}$, so its total variation is $2\phi(0) + 8\phi(\sqrt{3})$, and $\int |\phi''| = 4\phi(1)$. Halving the sum of the two parts gives the bound. It is within a factor 1.02 of the exact value at $\sigma = 100$ and within a factor 3.2 at $\sigma = 1$ (exact high-precision evaluation, `experiments/proofs_check/tv1_cuckoo.py`); at the evaluation setting ($N = 2^{14}$, $\sigma_c \approx 2^{65.3}$) $N\,\mathrm{TV}_1(\sigma_c) \leq 2^{-122.2}$. (An earlier version gave only the leading constant, with the $O(\sigma^{-3})$ term checked numerically; so does the companion book's Lemma `rel.slot`.) Finally $\|\lfloor \mathbf{Z} \rceil - \mathbf{Z}\| \leq \sqrt{N}/2$ and $\|(\sigma_s/\sigma_c)\mathbf{U}\|_{\mathrm{op}} = \sqrt{N/2}/\Omega$.

(c) The rounding of $\mathsf{Ecd}$ has sup-norm $\leq 1/2$; the slot image of a ring element $e$ has norm $\sqrt{N/2}\|e\|/\Omega$, and $\sqrt{N/2}B_{\mathrm{tot}}/\Omega \leq \sqrt{N/2}\sqrt{2\pi}\sigma_c 2^{-\kappa}/\Omega = \sqrt{2\pi}2^{-\kappa}\sigma_s$. □

# 3 Proposition S1: replace-one sensitivity

## 3.1 The change vector

Replacing $(\mathbf{x}, y)$ by $(\mathbf{x}', y')$ in a matched record with R-row $\mathbf{a}$ changes $(\mathrm{triu}\,\mathbf{B}, \mathbf{C}, \mathbf{c}_R, \mathbf{c}_O, \mathbf{y}^\top\mathbf{y})$ by

$$\big(\mathrm{triu}\,\mathbf{D},\ \mathbf{a}(\mathbf{x} - \mathbf{x}')^\top,\ \mathbf{a}(y - y'),\ \mathbf{x}y - \mathbf{x}'y',\ y^2 - y'^2\big), \qquad \mathbf{D} := \mathbf{x}\mathbf{x}^\top - \mathbf{x}'\mathbf{x}'^\top.$$

A change in an unmatched (or dummy) row changes nothing, because its O-side entries are multiplied by $b_t = 0$ or by $\dot{\mathbf{x}}_t = \mathbf{0}$ (Algorithm 4). Hence $\Delta^2 = \max_{\mathcal{D}} \Phi_{\mathrm{triu}}$ with

$$\Phi_{\mathrm{triu}}(\mathbf{a}, \mathbf{x}, \mathbf{x}', y, y') = \|\,\mathrm{triu}\,\mathbf{D}\|^2 + \|\mathbf{a}\|^2\big(\|\mathbf{x} - \mathbf{x}'\|^2 + (y - y')^2\big) + \|\mathbf{x}y - \mathbf{x}'y'\|^2 + (y^2 - y'^2)^2,$$
$$\mathcal{D} = \{\|\mathbf{a}\| \leq B_R,\ \|\mathbf{x}\|, \|\mathbf{x}'\| \leq B_O,\ |y|, |y'| \leq B_y\}$$

(compact, so the supremum is a maximum). $\Phi_F$ denotes the same expression with $\|\,\mathrm{triu}\,\mathbf{D}\|^2$ replaced by $\|\mathbf{D}\|_F^2$, and $\Phi_{\mathrm{triu}}^{\neg yy}$, $\Phi_F^{\neg yy}$ the expressions without the last term. The intercept constraint $a_1 = 1$ is irrelevant: the maximum has $\|\mathbf{a}\| = B_R \geq 1$, which is attained by $\mathbf{a} = (1, \mathbf{z})$ with $\|\mathbf{z}\|^2 = r - 1$ whenever $d_R \geq 1$ (if $d_R = 0$ then $B_R = 1$). $\Delta$ is the supremum over all cohorts; for a given cohort the sensitivity can only be smaller.

Define on $[-o, o] \times [-b, b]$

$$F(s,t) = 2o^2 - 2s^2 + 2r(o - s) + r(b - t)^2 + o(w + t^2) - 2bts + (w - t^2)^2, \tag{3}$$
$$F^{\neg yy}(s,t) = F(s,t) - (w - t^2)^2, \qquad s^*(t) = \mathrm{clip}\big(-(r + bt)/2;\ -o,\ o\big), \qquad G(t) = F(s^*(t), t).$$

$F$ is the function of the previous paper (with $s = oc$), so the reduction below is the missing justification of its use.

### 3.2 Statement

**Theorem 3.1** (S1)**.** *Let $d_O \geq 2$ and $b > 0$ ($b = 0$ is covered by setting $w = 0$).*

*(i)* ***Reduction.*** $\Delta^2 = \max_{|s|\leq o, |t|\leq b} F(s,t) = \max_{|t|\leq b} G(t)$.

*(ii)* ***Regime 1****, $|r-w| \leq 2o$:* $\Delta^2 = \frac{1}{2}(2o+r+w)^2 + 2rw$*, attained exactly at $t = -b$, $s = (w-r)/2$.*

*(iii)* ***Regime 2****, $r - w \geq 2o$:* $\Delta^2 = 4r(o+w)$*, attained exactly at $t = -b$, $s = -o$.*

*(iv)* ***Regime 3****, $w - r > 2o$: with $t_1 = -(r+2o)/b$ and $t_2 = (2o-r)/b$ one has $-b < t_1 < t_2 < b$ and $G$ is $C^1$ and piecewise quartic:*

$$
\begin{aligned}
G_A(t) &= t^4 + (r+o-2w)t^2 - 2b(r+o)t + (r+o)w + w^2 && t \in [-b, t_1] \ (s^* = o),\\
G_B(t) &= t^4 + (r+o-\tfrac{3}{2}w)t^2 - rbt + 2o^2 + 2ro + (r+o)w + w^2 + \tfrac{1}{2}r^2 && t \in [t_1, t_2] \ (|s^*| < o),\\
G_C(t) &= t^4 + (r+o-2w)t^2 + 2b(o-r)t + 4ro + (r+o)w + w^2 && t \in [t_2, b] \ (s^* = -o).
\end{aligned}
$$

*$\Delta^2 = \max_{t\in\mathcal{T}} G(t)$ for the finite set $\mathcal{T} = \{-b, t_1, t_2, b\} \cup \{$real roots of $G'_A, G'_B, G'_C$ in their intervals$\}$; this is exact (three cubics). $t = -b$ is a strict local maximizer with $G(-b) = 4w(r+o)$; if $\Delta^2 > 4w(r+o)$, every maximizer has $|t^*| < b$.*

*(v)* ***Norms.*** *At every maximizer $\|\mathbf{a}\| = B_R$, $\|\mathbf{x}\| = \|\mathbf{x}'\| = B_O$ and $\max(|y|, |y'|) = B_y$. Both $|y| = |y'| = B_y$ in regimes 1–2; in regime 3 this fails whenever $\mathbf{y}^\top\mathbf{y}$ is not free ($|y'| = |t^*| < B_y$).*

*(vi)* triu ***versus Frobenius.*** $\max_{\mathcal{D}} \Phi_{\text{triu}} = \max_{\mathcal{D}} \Phi_F$ *(and the same without $\mathbf{y}^\top\mathbf{y}$); a configuration attains the* triu *maximum iff it attains the Frobenius maximum and $\mathbf{D}$ is diagonal.*

*(vii)* ***Worst case.*** *Let $s^\dagger$, $t^\dagger$ be a maximizer of $F$ and $c = s^\dagger/o$, $\alpha_\pm = \frac{1}{2}(\sqrt{1+c} \pm \sqrt{1-c})$. A worst-case pair is $\|\mathbf{a}\| = B_R$, $y = B_y$, $y' = t^\dagger$,*

$$\mathbf{x} = B_O(\alpha_+\mathbf{e}_1 + \alpha_-\mathbf{e}_2), \qquad \mathbf{x}' = B_O(\alpha_-\mathbf{e}_1 + \alpha_+\mathbf{e}_2),$$

*for which $\mathbf{D} = o\sqrt{1-c^2}\,(\mathbf{e}_1\mathbf{e}_1^\top - \mathbf{e}_2\mathbf{e}_2^\top)$. In regime 1: $y' = -y$, $\cos\angle(\mathbf{x}, \mathbf{x}') = -(r-w)/(2o)$; in regime 2: $y' = -y$, $\mathbf{x}' = -\mathbf{x}$ ($\mathbf{D} = 0$).*

*(viii)* ***Other sensitivities.*** *$\Delta^2_{\neg yy} = \frac{1}{2}(2o+r+w)^2 + 2rw$ if $|r-w| \leq 2o$ and $\Delta^2_{\neg yy} = 4M(o+m)$ otherwise, with $M = \max(r,w)$ and $m = \min(r,w)$. $\Delta_G = \Delta(B_R, B_O, 0)$ equals $(2o+r)/\sqrt{2}$ if $r \leq 2o$ and $2\sqrt{ro}$ otherwise. $\Delta_y = w$.*

*(ix)* ***When $\mathbf{y}^\top\mathbf{y}$ is free.*** *$\Delta = \Delta_{\neg yy}$ iff $\max_{|t|\leq b} G(t) \leq 4w(r+o)$ or $w \leq r + 2o$. In particular it is free whenever $w \leq 2(r+o)$ (this covers regimes 1, 2 and part of regime 3), and costly whenever $w > 3(r+o)$. For $o = 0$ the exact threshold is $w \leq \frac{11+5\sqrt{5}}{8}\, r \approx 2.7725\, r$.*

The specification's DeltaRep ("exact for $d_O \geq 2$") is therefore correct in regimes 1–2 and, after the reduction (i), in regime 3, where (iv) replaces the grid search by an exact finite computation. For $d_O = 1$ see Proposition 3.5.

### 3.3 Two lemmas

**Lemma 3.2** (triu versus Frobenius)**.** *For all $\mathbf{x}, \mathbf{x}'$: $\|\operatorname{triu}\mathbf{D}\|^2 = \frac{1}{2}\big(\|\mathbf{D}\|_F^2 + \sum_j D_{jj}^2\big) \leq \|\mathbf{D}\|_F^2$, with equality iff $\mathbf{D}$ is diagonal. If $d_O \geq 2$, there is an orthogonal $\mathbf{Q}$ such that $\mathbf{Q}\mathbf{D}\mathbf{Q}^\top$ is diagonal; replacing $(\mathbf{x}, \mathbf{x}')$ by $(\mathbf{Q}\mathbf{x}, \mathbf{Q}\mathbf{x}')$ changes $\mathbf{D}$ into $\mathbf{Q}\mathbf{D}\mathbf{Q}^\top$ and leaves every other term of $\Phi$ and the constraints unchanged. Hence $\max\Phi_{\text{triu}} = \max\Phi_F$. For $d_O = 1$, $\operatorname{triu}\mathbf{D} = \mathbf{D}$.*

*Proof.* The identity holds because $\mathbf{D}$ is symmetric. $\mathbf{D}$ has rank $\leq 2$ and range in $\operatorname{span}\{\mathbf{x}, \mathbf{x}'\}$, so $\mathbf{D} = \mu_1\mathbf{q}_1\mathbf{q}_1^\top + \mu_2\mathbf{q}_2\mathbf{q}_2^\top$ with orthonormal $\mathbf{q}_1, \mathbf{q}_2$; any orthogonal $\mathbf{Q}$ with $\mathbf{Q}\mathbf{q}_i = \mathbf{e}_i$ diagonalizes $\mathbf{D}$ (this needs $d_O \geq 2$). The other terms depend on $(\mathbf{x}, \mathbf{x}')$ only through $\|\mathbf{x}\|, \|\mathbf{x}'\|, \mathbf{x}^\top\mathbf{x}'$, which are invariant. Thus $\Phi_{\text{triu}}(\cdot, \mathbf{Q}\mathbf{x}, \mathbf{Q}\mathbf{x}', \cdot) = \Phi_F(\cdot, \mathbf{x}, \mathbf{x}', \cdot) \geq \Phi_{\text{triu}}(\cdot, \mathbf{x}, \mathbf{x}', \cdot)$. □

**Lemma 3.3** (Reduction)**.** *Let $d_O \geq 2$. With $\rho = \|\mathbf{a}\|^2$, $u = \|\mathbf{x}\|^2$, $v = \|\mathbf{x}'\|^2$, $s = \mathbf{x}^\top \mathbf{x}'$,*

$$\Phi_F = \Upsilon := u^2 + v^2 - 2s^2 + \rho\big(u + v - 2s + (y - y')^2\big) + y^2 u + y'^2 v - 2yy's + (y^2 - y'^2)^2,$$

*and the image of $\mathcal{D}$ is $\{0 \leq \rho \leq r,\ 0 \leq u, v \leq o,\ s^2 \leq uv,\ |y|, |y'| \leq b\}$.*

*(a) $\Upsilon$ is non-decreasing in $\rho$, strictly unless $(\mathbf{x}, y) = (\mathbf{x}', y')$.*

*(b) For fixed $(\rho, v, s, y, y')$, $\Upsilon$ is strictly increasing in $u$ on the feasible set $\{u \in [0, o] : s^2 \leq uv\}$, which is an interval with right end $o$. Hence every maximizer has $u = o$, and likewise $v = o$.*

*(c) For fixed $(\rho, u, v, s)$, $\Upsilon = \mathcal{Q}(y, y') + (y^2 - y'^2)^2 + \text{const}$ with $\mathcal{Q}$ positive semidefinite; hence $\Upsilon(\kappa y, \kappa y') \geq \Upsilon(y, y')$ for $\kappa \geq 1$, and every maximizer has $\max(|y|, |y'|) = b$.*

*(d) $\Upsilon$ is invariant under $(u, y) \leftrightarrow (v, y')$ and under $(y, y') \mapsto (-y, -y')$.*

*Consequently $\Delta^2 = \max F$ over $[-o, o] \times [-b, b]$: set $\rho = r$, $u = v = o$, $y = b$, $y' = t$.*

*Proof.* For $d_O \geq 2$, every $(u, v, s)$ with $s^2 \leq uv$ is realized by some $(\mathbf{x}, \mathbf{x}')$. (a) The coefficient of $\rho$ is $\|\mathbf{x} - \mathbf{x}'\|^2 + (y - y')^2$. (b) $\partial_u \Upsilon = 2u + \rho + y^2 > 0$ (as $\rho \geq 1$ on $\mathcal{D}$); the feasible set in $u$ is $[s^2/v, o]$ (or $[0, o]$ if $v = 0$), which is upward closed. This is the step the earlier proof sketch asserted without proof: monotonicity holds at fixed inner product $s$, *not* at fixed cosine (at fixed cosine $c$, $\partial_u(-2s^2) = -2vc^2$ can dominate). (c) $\mathcal{Q}(y, y') = \rho(y - y')^2 + uy^2 + vy'^2 - 2syy'$ has matrix $\left(\begin{smallmatrix} \rho+u & -(\rho+s) \\ -(\rho+s) & \rho+v \end{smallmatrix}\right)$ with non-negative diagonal and determinant $\rho(u + v - 2s) + uv - s^2 \geq 0$. So $\Upsilon(\kappa y, \kappa y') - \Upsilon(y, y') = (\kappa^2 - 1)\mathcal{Q} + (\kappa^4 - 1)(y^2 - y'^2)^2 \geq 0$. If a maximizer had $\max(|y|, |y'|) < b$, scaling would not increase $\Upsilon$ only if $\mathcal{Q}(y, y') = 0$ and $y^2 = y'^2$. Either $y = y' = 0$, and then moving to $(b, -b)$ increases $\Upsilon$ by $\mathcal{Q}(b, -b) = w(4\rho + u + v + 2s) > 0$; or $\det = 0$, which with $\rho > 0$ forces $u = v = s$, i.e. $\mathbf{x} = \mathbf{x}'$, and then $\mathcal{Q} = (\rho + u)(y - y')^2 = 0$ forces $y = y'$ and $\Upsilon = 0$, not a maximum. (d) is immediate. By (d) we may take $|y| = b$ and then $y = b$. □

### 3.4 Proof of Theorem 3.1

*(i)* Lemmas 3.2 and 3.3. For fixed $t$, $F$ is a strictly concave quadratic in $s$, $F(s, t) = -2s^2 - 2s(r + bt) + H(t)$ with $H(t) = 2o^2 + 2ro + r(b - t)^2 + o(w + t^2) + (w - t^2)^2$, maximized over $[-o, o]$ at $s^*(t)$.

*(ii)* Relax $s \in \mathbb{R}$: $G_u(t) := \max_{s \in \mathbb{R}} F = H(t) + \frac{1}{2}(r + bt)^2 = t^4 + (r + o - \frac{3}{2}w)t^2 - rbt + K$, with $K$ the constant term of $G_B$. Using $w - t^2 = (b - t)(b + t)$ and $w^2 - t^4 = (w - t^2)(w + t^2)$,

$$G_u(-b) - G_u(t) = (b + t)\big[rb + (b - t)\big(r + o - \tfrac{w}{2} + t^2\big)\big]. \tag{4}$$

If $w \leq r + 2o$ then $r + o - w/2 \geq r/2 > 0$, so (4) is $> 0$ for $t > -b$. If also $r - w \leq 2o$, the relaxed maximizer at $t = -b$, $s = -(r - w)/2$, is feasible. Hence $\Delta^2 = G_u(-b) = 2o^2 + 2o(r + w) + 4rw + \frac{1}{2}(r - w)^2 = \frac{1}{2}(2o + r + w)^2 + 2rw$, with a unique maximizer (strict concavity in $s$).

*(iii)* If $r - w \geq 2o$ then $-(r + bt)/2 \leq -(r - w)/2 \leq -o$ for all $|t| \leq b$, so $s^* \equiv -o$ and $G(t) = F(-o, t) = 4ro + r(b - t)^2 + o(b + t)^2 + (w - t^2)^2$. Then

$$G(-b) - G(t) = (b + t)\big[r(3b - t) - o(b + t) - (b - t)^2(b + t)\big].$$

With $r \geq 2o + w$ and $t = b\tau$, the bracket is at least $o(5b - 3t) + b^3\big[(3 - \tau) - (1 - \tau)^2(1 + \tau)\big] = o(5b - 3t) + b^3(2 + \tau^2 - \tau^3) \geq 2ob + 2b^3 > 0$. So the maximum is $G(-b) = 4r(o + w)$, attained only at $t = -b$.

*(iv)* $s^*(t) = o$ iff $r + bt \leq -2o$ iff $t \leq t_1$, and $s^*(t) = -o$ iff $t \geq t_2$; in regime 3, $t_1 > -b$ and $t_2 < b$. Substituting $s = \pm o$ or $s = -(r + bt)/2$ in (3) gives $G_A, G_B, G_C$. By Danskin's theorem [8] (unique maximizer $s^*(t)$, continuous in $t$), $G'(t) = \partial_t F(s^*(t), t)$ is continuous, so a maximizer of $G$ on $[-b, b]$ is an endpoint or a zero of $G'$, i.e. a root of the cubic $G'_k$ of the piece that contains it.

Further $G_A(-b) = 4w(r+o)$, $G'_A(-b) = -4b(r+o) < 0$, and $G(b) = G_C(b) = 4o(r+w) < 4w(r+o)$ because $w > r$ in regime 3. So if $\max G > 4w(r+o)$ every maximizer lies in $(-b, b)$.

*(v)* From Lemma 3.3 and the uniqueness statements in (ii)–(iv).

*(vi)* Lemma 3.2. If $\mathbf{D}$ is not diagonal at a Frobenius maximizer, the triu value is strictly smaller.

*(vii)* With $\|\mathbf{x}\| = \|\mathbf{x}'\| = B_O$ and $\mathbf{x}^\top \mathbf{x}' = oc$, $\mathbf{D}$ has eigenvalues $\pm o\sqrt{1-c^2}$ and triu attains $\|\mathbf{D}\|_F$ iff the eigenvectors are coordinate vectors; the stated pair has $\alpha_+^2 + \alpha_-^2 = 1$, $2\alpha_+\alpha_- = c$ and $\alpha_+^2 - \alpha_-^2 = \sqrt{1-c^2}$, which gives the displayed diagonal $\mathbf{D}$. A pair with the right norms and angle but non-diagonal $\mathbf{D}$ is *not* worst case: for $\mathbf{x} = B_O\mathbf{e}_1$, $\mathbf{x}' = B_O(c\mathbf{e}_1 + \sqrt{1-c^2}\mathbf{e}_2)$ one loses $o^2c^2(1-c^2)$ in $\|\operatorname{triu}\mathbf{D}\|^2$.

*(viii)* Lemma 3.3(a),(b) hold verbatim for $\Phi^{\neg yy}$. Without the quartic term, $\Upsilon^{\neg yy}$ is a positive semidefinite quadratic form in $(y, y')$ plus a constant, hence convex, and its maximum over the square is at a vertex; by (d), $y = b$, $t = y' \in \{b, -b\}$. Now $F^{\neg yy}(s, -b) = 2o^2 + 2ro + 2ow + 4rw - 2s^2 - 2(r-w)s$ and $F^{\neg yy}(s, b) = 2o^2 - 2s^2 + 2(r+w)(o-s)$. Maximizing each over $|s| \le o$:

- $t = -b$: $\frac{1}{2}(2o + r + w)^2 + 2rw$ if $|r - w| \le 2o$; $4r(o+w)$ if $r - w > 2o$; $4w(o+r)$ if $w - r > 2o$.
- $t = b$: $\frac{1}{2}(2o + r + w)^2$ if $r + w \le 2o$, otherwise $4o(r+w)$.

Since $\frac{1}{2}(2o + r + w)^2 \ge 4o(r+w)$ (AM–GM), $4r(o+w) - 4o(r+w) = 4w(r-o) \ge 0$ in regime 2, and $4w(o+r) - 4o(r+w) = 4r(w-o) > 0$ in regime 3, the $t = -b$ values win, which gives the stated $\Delta_{\neg yy}$. $\Delta_G$ is the case $w = 0$: regime 1 if $r \le 2o$, regime 2 otherwise. $\Delta_y = \sup|y^2 - y'^2| = w$, attained at $(y, y') = (b, 0)$.

*(ix)* In regimes 1–2 the maximizer has $y' = -y$, so the $\mathbf{y}^\top\mathbf{y}$ term vanishes there and $\Delta = \Delta_{\neg yy}$ by the explicit formulas. In regime 3, $\Delta^2_{\neg yy} = 4w(r+o) = G(-b) \le \Delta^2$, so freeness is equivalent to $\max G \le 4w(r+o)$. By homogeneity, $F(s,t;r,o,w) = w^2 F(s/w, t/b; r/w, o/w, 1)$, so take $w = b = 1$. Put $s = o - \varsigma$ with $\varsigma \in [0, 2o]$, $t = u - 1$ with $u \in [0, 2]$, $S = r + o$ and $\delta = 1 - r - 2o > 0$ (regime 3). A direct expansion (verified exactly in rational arithmetic by `s1_symbolic.py`, as are the other polynomial identities of this proof) gives

$$F(s,t) - 4S = 2\varsigma(u - \delta) - 2\varsigma^2 - S\,u(4-u) + u^2(2-u)^2. \tag{5}$$

Maximizing over $\varsigma \ge 0$ gives at most $\frac{1}{2}(u-\delta)_+^2 \le \frac{1}{2}u^2$. If $w \le 2(r+o)$, i.e. $S \ge \frac{1}{2}$, then $-Su(4-u) \le -\frac{1}{2}u(4-u)$, and

$$F(s,t) - 4S \le \tfrac{1}{2}u^2 - \tfrac{1}{2}u(4-u) + u^2(2-u)^2 = u(u-2)(u-1)^2 \le 0 \qquad (0 \le u \le 2).$$

Hence $\Delta^2 \le 4w(r+o) = \Delta^2_{\neg yy}$. If $w > 3(r+o)$, then $G(0) \ge F(o, 0) = w(r + o + w) > 4w(r+o)$. For $o = 0$, $G(t) = r(b-t)^2 + (w - t^2)^2$, which is the function $h$ of Proposition 3.5(d) with $o$ replaced by $w$; the threshold follows from that computation. □

**Remark 3.4** (The free set, numerically)**.** For fixed $(r, o)$ the free set in regime 3 is an interval $r + 2o < w \le w^*(r, o)$ [numerical: checked on a fine grid in $w$ for $o/r \in [0, 100]$]; Theorem 3.1(ix) proves $2(r+o) \le w^* \le 3(r+o)$ and $w^*(r, 0) = \frac{11+5\sqrt{5}}{8}r$. The ratio $w^*/(r+o)$ decreases from 2.7725 at $o = 0$ to 2 as $o/r \to \infty$:

| $o/r$ | 0 | 0.01 | 0.1 | 0.25 | 0.5 | 1 | 2 | 5 | 10 | 100 |
|---|---|---|---|---|---|---|---|---|---|---|
| $w^*/(r+o)$ | 2.773 | 2.767 | 2.725 | 2.677 | 2.623 | 2.534 | 2.414 | 2.250 | 2.153 | 2.019 |

Example: $(B_R, B_O, B_y) = (1, 1, 5)$ has $w = 25 > 3(r+o) = 6$, so $\mathbf{y}^\top\mathbf{y}$ costs: $\Delta = 26.0706$ (maximizer $t^* = -0.0704$, $s^* = -0.324$), against $\Delta_{\neg yy} = \sqrt{200} = 14.142$. For $(\sqrt{5}, \sqrt{5}, 5)$, $w = 25 \le w^* = 25.33$, so $\mathbf{y}^\top\mathbf{y}$ is free ($\Delta = \Delta_{\neg yy} = 31.62$), which is why the earlier paper saw no difference there. The indicative bounds $(5.59, 6.24, 2.89)$ are in regime 1 (free), with $\Delta = 86.15$ and $\Delta_G = 77.16$.[1]

[1] The specification prints $\Delta = 86.18$ and $\Delta_G = 77.20$, presumably computed from unrounded bounds; with the bounds as printed the formulas give 86.152 and 77.162.

### 3.5 The case $d_O = 1$

**Proposition 3.5** ($d_O = 1$)**.** *Let $d_O = 1$ and $\Phi_1(x, x', y, y') = (x^2 - x'^2)^2 + r(x - x')^2 + r(y - y')^2 + (xy - x'y')^2 + (y^2 - y'^2)^2$ (here* triu $\mathbf{D} = \mathbf{D}$*, so there is no* triu *issue).*

*(a)* $\Delta^{(1)} \le \Delta$, $\Delta^{(1)}_{\neg yy} \le \Delta_{\neg yy}$ *and* $\Delta^{(1)}_G \le \Delta_G$*; so the $d_O \ge 2$ formulas are valid (conservative) for $d_O = 1$.*

*(b)* $\Delta^{(1)2} = \max\{\max_{x',t} \Phi_1(B_O, x', b, t),\ \max_{x',t} \Phi_1(B_O, x', t, b)\}$ *over* $(x', t) \in [-B_O, B_O] \times [-b, b]$.

*(c)* $\Delta^{(1)} = \Delta$ *in regime 2 and whenever some maximizer of Theorem 3.1 has $|s^\dagger| = o$ (for instance in the free part of regime 3, where the maximizer is $\mathbf{x}' = \mathbf{x}$, $y' = -y$). $\Delta^{(1)} < \Delta$ in the interior of regime 1.*

*(d)* $\Delta^{(1)2}_G = 4ro$ *if* $o \le \frac{11+5\sqrt{5}}{8} r$*; otherwise* $\Delta^{(1)2}_G = \frac{o^2}{16}(3 - q)^3(1 + q)$ *with* $q = \sqrt{1 - 2r/o}$.

*Proof.* (a) A $d_O = 1$ configuration embedded in $\mathbb{R}^2$ (second coordinates zero) has diagonal $\mathbf{D}$ and the same value. (b) $\Phi_1$ is non-decreasing under $(x, x') \mapsto \kappa(x, x')$, $\kappa \ge 1$ (each $x$-dependent term is a non-negative form of degree 4 or 2 in $(x, x')$), and under $(y, y') \mapsto \kappa(y, y')$ (as in Lemma 3.3(c), with $\mathcal{Q} = r(y - y')^2 + (xy - x'y')^2$, a sum of squares). So $\max(|x|, |x'|) = B_O$ and $\max(|y|, |y'|) = b$. Using the symmetries $(x, y) \leftrightarrow (x', y')$, $(x, x') \mapsto -(x, x')$ and $(y, y') \mapsto -(y, y')$, take $x = B_O$; then either $y = b$ or $y' = b$. (c) In regime 2 the maximizer $\mathbf{x}' = -\mathbf{x}$ is collinear. In the interior of regime 1 every maximizer of the $d_O \ge 2$ problem has $u = v = o$ and $s = (w - r)/2 \in (-o, o)$, so it is not collinear. The $d_O = 1$ feasible set is a compact subset of the $d_O = 2$ set that contains no maximizer, so its maximum is strictly smaller. (d) With $w = 0$, $x = B_O$, $x' = B_O\tau$: $\Phi_1 = o\, h(\tau)$ with $h(\tau) = o(1 - \tau^2)^2 + r(1 - \tau)^2$, and $h'(\tau) = -2(1 - \tau)\big[2o\tau(1 + \tau) + r\big]$. If $o \le 2r$ the bracket is $\ge r - o/2 \ge 0$, so $h$ decreases and the maximum is $h(-1) = 4r$. If $o > 2r$, $h$ has a local maximum at $\tau_+ = (q - 1)/2$, where $h(\tau_+) = (1 - \tau_+)^2[o(1 + \tau_+)^2 + r] = \frac{o}{16}(3 - q)^3(1 + q)$, while $h(-1) = 4r = 2o(1 - q^2)$. The interior value wins iff $(3 - q)^3 > 32(1 - q)$, i.e. $q > q_0 = 5 - 2\sqrt{5}$, i.e. $o/r > 2/(1 - q_0^2) = \frac{11+5\sqrt{5}}{8}$ (and $o \le 2r$ falls in the first case). □

For $d_O = 1$ we give no closed form for $\Delta^{(1)}$ in regimes 1 and 3. (b) is an exact two-variable polynomial maximization over a rectangle, solvable to machine precision; for calibration the $d_O \ge 2$ formula is a valid upper bound. The losses are substantial in regime 1: $\Delta^{(1)}/\Delta = 0.894$ at $(1, 1, 1)$, 0.784 at $(1, 2, 1)$ and 0.912 at the indicative bounds (Table 2).

### 3.6 Numerical verification

Script `scratch_stats/s1_check.py` (output `s1_check.out`) compares, for 14 bound settings covering all regimes: the exact value of Theorem 3.1 (closed form, or candidate enumeration in regime 3); `sim_core.delta_rep` (read only); a brute-force multistart BFGS maximization of the triu objective $\Phi_{\text{triu}}$ over raw configurations $(\mathbf{x}, \mathbf{x}', y, y')$ for $d_O = 2$ and $d_O = 3$; and the explicit worst-case configuration of (vii) evaluated with the triu norm. All agree to $10^{-6}$ (relative difference to `delta_rep` is $\le 10^{-9}$, which is its deliberate rounding up). $\Delta_{\neg yy}$ agrees with brute force, and the $d_O = 1$ reduction (b) agrees with brute force over $d_O = 1$ configurations. Script `s1_freeset.py` produced the table of Remark 3.4, and `s1_band.py` confirms $\max_t G - 4(r + o) \le 4.4 \times 10^{-16}$ over a $300 \times 120$ grid of the band $r + 2o < w \le 2(r + o)$.

## 4 Lemma S2: operator norm of the masked noise

**Lemma 4.1** (S2)**.** *Let* $\mathbf{E}$ *be a masked symmetric noise matrix with scale $\sigma$ (Section 2); the proof allows any mask and any variances $\le \sigma^2$ off the diagonal and $\le 2\sigma^2$ on it, so it covers the full*

*GOE-type matrix too. Let* $\mathbf{g} \sim \mathcal{N}(0, \mathbf{I}_p)$.

*(a)* $\mathbb{E}\lambda_{\max}(\mathbf{E}) \le 2\sigma\,\mathbb{E}\|\mathbf{g}\| < 2\sigma\sqrt{p}$, *and* $\lambda_{\max}(-\mathbf{E})$ *has the same law as* $\lambda_{\max}(\mathbf{E})$.

*(b)* $\mathbf{E} \mapsto \lambda_{\max}(\mathbf{E})$ *and* $\mathbf{E} \mapsto \|\mathbf{E}\|_{\mathrm{op}}$ *are* $\sqrt{2}\sigma$*-Lipschitz functions of the standard Gaussian vector of independent entries. Hence, for* $t \ge 0$,

$$\Pr[\lambda_{\max}(\mathbf{E}) \ge \mathbb{E}\lambda_{\max}(\mathbf{E}) + t] \le e^{-t^2/(4\sigma^2)}, \qquad \Pr[\|\mathbf{E}\|_{\mathrm{op}} \ge \mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} + t] \le e^{-t^2/(4\sigma^2)}.$$

*(c)* $\Pr\big[\lambda_{\max}(\mathbf{E}) \ge \sigma\big(2\sqrt{p} + 2\sqrt{\ln(1/\alpha)}\big)\big] \le \alpha$ *and* $\Pr\big[\|\mathbf{E}\|_{\mathrm{op}} \ge \sigma\big(2\sqrt{p} + 2\sqrt{\ln(2/\alpha)}\big)\big] \le \alpha$.

*(d)* $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} \le 2\sigma\sqrt{p} + c_0\sigma$, *with* $c_0 = 2\sqrt{\ln 2} + 4\sqrt{\pi}\,Q(\sqrt{2\ln 2}) \approx 2.512$, *and* $(\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}}^4)^{1/4} \le \sigma(2\sqrt{p} + c_0 + 2\sqrt{2})$.

*In Algorithm 6,* $r_1 = \sigma_1(2\sqrt{p} + 2\sqrt{\ln(1/\alpha_L)})$ *therefore satisfies* $\Pr[\lambda_{\max}(\hat{\mathbf{G}}_1 - \mathbf{G}) > r_1] \le \alpha_L$, *and this one-sided event is all the algorithm uses (Lemma S3).*

*Proof.* (a) For a unit vector $\mathbf{u}$ let $X_{\mathbf{u}} = \mathbf{u}^\top\mathbf{E}\mathbf{u} = \sum_{i\le j,\chi_{ij}=1} E_{ij}\,w_{ij}(\mathbf{u})$, with $w_{ii} = u_i^2$ and $w_{ij} = 2u_iu_j$. Then

$$\begin{aligned}\mathbb{E}(X_{\mathbf{u}} - X_{\mathbf{v}})^2 &\le \sigma^2\Big[2\sum_i (u_i^2 - v_i^2)^2 + 4\sum_{i<j}(u_iu_j - v_iv_j)^2\Big] = 2\sigma^2\|\mathbf{u}\mathbf{u}^\top - \mathbf{v}\mathbf{v}^\top\|_F^2 \\ &= 4\sigma^2\big(1 - (\mathbf{u}^\top\mathbf{v})^2\big) \le 4\sigma^2\|\mathbf{u} - \mathbf{v}\|^2,\end{aligned}$$

using $1 - (\mathbf{u}^\top\mathbf{v})^2 = (1 - \mathbf{u}^\top\mathbf{v})(1 + \mathbf{u}^\top\mathbf{v}) \le 2(1 - \mathbf{u}^\top\mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|^2$. The process $Y_{\mathbf{u}} = 2\sigma\,\mathbf{g}^\top\mathbf{u}$ has $\mathbb{E}(Y_{\mathbf{u}} - Y_{\mathbf{v}})^2 = 4\sigma^2\|\mathbf{u} - \mathbf{v}\|^2$. By the Sudakov–Fernique inequality [12, 9] (see [14, Thm. 7.2.11]; both processes are centred, Gaussian and continuous on the sphere), $\mathbb{E}\lambda_{\max}(\mathbf{E}) = \mathbb{E}\sup_{\mathbf{u}} X_{\mathbf{u}} \le \mathbb{E}\sup_{\mathbf{u}} Y_{\mathbf{u}} = 2\sigma\mathbb{E}\|\mathbf{g}\| \le 2\sigma\sqrt{p}$. $\mathbf{E}$ and $-\mathbf{E}$ have the same law.

(b) Write $E_{ij} = \sigma_{ij}\zeta_{ij}$ with independent standard $\zeta_{ij}$. For a unit $\mathbf{u}$, $|\mathbf{u}^\top(\mathbf{E}(\zeta) - \mathbf{E}(\zeta'))\mathbf{u}| \le \|\zeta - \zeta'\|\big(\sum \sigma_{ij}^2 w_{ij}(\mathbf{u})^2\big)^{1/2} \le \sqrt{2}\sigma\|\zeta - \zeta'\|$, since $\sum_i u_i^4 \cdot 2 + \sum_{i<j} 4u_i^2u_j^2 = 2$ (and with diagonal variance $\sigma^2$ the bound is $\sigma^2(2 - \sum_i u_i^4)$). Both $\lambda_{\max}$ and $\|\cdot\|_{\mathrm{op}} = \sup_{\mathbf{u}}|\mathbf{u}^\top\cdot\mathbf{u}|$ are therefore $\sqrt{2}\sigma$-Lipschitz, and Gaussian concentration [13] (see [5, Thm. 5.6]) gives $\Pr[f - \mathbb{E}f \ge t] \le e^{-t^2/(2L^2)}$ with $L = \sqrt{2}\sigma$.

(c) Combine (a) and (b) with $t = 2\sigma\sqrt{\ln(1/\alpha)}$. For the norm, $\|\mathbf{E}\|_{\mathrm{op}} = \max(\lambda_{\max}(\mathbf{E}), \lambda_{\max}(-\mathbf{E}))$ and a union bound gives the factor 2.

(d) With $m = 2\sigma\sqrt{p} \ge \mathbb{E}\lambda_{\max}(\pm\mathbf{E})$, $\|\mathbf{E}\|_{\mathrm{op}} \le m + \max\big((\lambda_{\max}(\mathbf{E}) - m)_+, (\lambda_{\max}(-\mathbf{E}) - m)_+\big)$, and the last maximum exceeds $t$ with probability at most $\min(1, 2e^{-t^2/4\sigma^2})$; integrating gives $c_0$. For the fourth moment, $\|\mathbf{E}\|_{\mathrm{op}} = \mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} + X$ with $\Pr[|X| \ge t] \le 2e^{-t^2/(4\sigma^2)}$ by (b), so $\mathbb{E}X^4 \le 8\int_0^\infty t^3 e^{-t^2/(4\sigma^2)}dt = 64\sigma^4$, and Minkowski's inequality gives the claim. □

**Remark 4.2** (What is not proved)**.** The claim $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} \le 2\sigma\sqrt{p}$, which would give the two-sided bound with $\ln(1/\alpha)$, is [open] here. Sudakov–Fernique applied to $\sup_{\mathbf{u},\pm} \pm\mathbf{u}^\top\mathbf{E}\mathbf{u}$ fails, because the cross-sign increments $\mathbb{E}(\mathbf{u}^\top\mathbf{E}\mathbf{u} + \mathbf{v}^\top\mathbf{E}\mathbf{v})^2$ reach $8\sigma^2$ at $\mathbf{u} = \mathbf{v}$. The suggested Gaussian comparison does reduce the masked case to the full one: if $\mathbf{W} = \mathbf{E} + \mathbf{E}'$ with $\mathbf{E}'$ an independent centred Gaussian symmetric matrix supported on the masked entries (and, if desired, extra diagonal variance up to $2\sigma^2$), then $\mathbb{E}\|\mathbf{W}\|_{\mathrm{op}} = \mathbb{E}_{\mathbf{E}}\mathbb{E}_{\mathbf{E}'}\|\mathbf{E} + \mathbf{E}'\|_{\mathrm{op}} \ge \mathbb{E}\|\mathbf{E}\|_{\mathrm{op}}$ by Jensen's inequality, since $\|\cdot\|_{\mathrm{op}}$ is convex and $\mathbb{E}\mathbf{E}' = 0$. So the mask can only help, and it would suffice to prove $\mathbb{E}\|\mathbf{W}\|_{\mathrm{op}} \le 2\sigma\sqrt{p}$ for the full GOE-type matrix. For the GOE this holds asymptotically (the largest eigenvalue has mean $2\sigma\sqrt{p} - O(\sigma p^{-1/6})$ by Tracy–Widom asymptotics), but I know of no non-asymptotic proof with the constant exactly 2. Known sharp non-asymptotic bounds, such as that of Bandeira and van Handel [4], carry an additional $O(\sigma\sqrt{\log p})$ term. Monte Carlo (`s2_goe_ratio.py`) gives $\mathbb{E}\|\mathbf{W}\|_{\mathrm{op}}/(2\sigma\sqrt{p}) = 0.567, 0.724, 0.799, 0.864, 0.926, 0.954, 0.966$ for $p = 1, 2, 3, 5, 11, 20, 30$ (GOE normalization), increasing towards 1 from below, and 0.81 for the protocol's masked $p = 11$ (Table 3). Algorithm 6 does not need the claim.

**Subgaussian (discrete-Gaussian) noise.** By Lemma 2.2(b),(c), (c) holds for the protocol's noise with $\alpha$ replaced by $\alpha + N\,\mathrm{TV}_1(\sigma_{c,1}) + \delta_B$ and $r_1$ by $r_1 + \sqrt{2}\,\bar{\nu}$. Without the coupling, using only that the noise vector is $\sigma^2$-subgaussian (Lemma 2.2(a)), a $\frac{1}{4}$-net $\mathcal{N}$ of the sphere ($|\mathcal{N}| \le 9^p$) and $\|\mathbf{E}\|_{\mathrm{op}} \le 2\max_{\mathbf{v}\in\mathcal{N}} |\mathbf{v}^\top \mathbf{E}\mathbf{v}|$ (see [14, Cor. 4.2.13, Ex. 4.4.3]), together with the fact that $\mathbf{v}^\top\mathbf{E}\mathbf{v} = \langle \mathbf{w}(\mathbf{v}), \boldsymbol{\xi}\rangle$ with $\|\mathbf{w}(\mathbf{v})\|^2 \le 2$ is $2\sigma^2$-subgaussian, give

$$\Pr\Big[\|\mathbf{E}\|_{\mathrm{op}} \ge 4\sigma\sqrt{p\ln 9 + \ln(2/\alpha)}\Big] \le \alpha.$$

For $p = 11$ and $\alpha = 0.0025$ this is $22.2\sigma$, against $11.5\sigma$ for the Gaussian $r_1$. The coupling route is therefore the one to use.

**Monte Carlo** (`s2_check.py`, $2 \times 10^5$ draws). For the protocol shape ($p = 11$, $5 \times 5$ exact block): $\mathbb{E}\lambda_{\max} = 4.98\sigma$, $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} = 5.38\sigma$, against $2\sigma\sqrt{p} = 6.63\sigma$, a ratio of 0.81. The 0.9975-quantile of $\lambda_{\max}$ is $7.43\sigma$ (of $\|\mathbf{E}\|_{\mathrm{op}}$, $7.62\sigma$), while $r_1/\sigma_1 = 11.53$ at $\alpha_L = 0.0025$, so $r_1$ is conservative by a factor of about 1.55. Table 3 gives other shapes. The discrete-Gaussian version (ring $N = 128$, canonical embedding, $\sigma_c \in \{0.8, 3\}$) has exactly uncorrelated slots and the same $\lambda_{\max}$ distribution to Monte Carlo accuracy (mean 4.97, 0.9975-quantile 7.38).

## 5 Lemma S3: the two-stage choice of the ridge

Let $\mathbf{E}_1 = \hat{\mathbf{G}}_1 - \mathbf{G}$ (scale $\sigma_1$), $\gamma := g^\star + r_1$ with $g^\star = 2\vartheta^\star\sigma_2\sqrt{p}$, and

$$\Omega_L := \{\lambda_{\max}(\mathbf{E}_1) \le r_1\}.$$

**Lemma 5.1** (S3)**.** *Under Assumption 2.1:*

*(a)* ***Measurability and independence.*** $(\lambda, \Psi) = \Lambda(\mathbf{A}, \hat{\mathbf{B}}_1, \hat{\mathbf{C}}_1)$ *for a Borel map* $\Lambda$ *that depends otherwise only on public constants* $(\sigma_1, \sigma_2, p, \vartheta^\star, \alpha_L, \kappa_\Psi)$*. Hence, conditionally on the data* $(\mathbf{X}, \mathbf{y})$*,* $(\lambda, \Psi, \hat{L})$ *is independent of the stage-2 noise* $(\mathbf{E}_2, \mathbf{f}_2, g)$ *and of the separate* $\mathbf{y}^\top\mathbf{y}$ *noise.*

*(b)* ***Radius event.*** $\Pr[\Omega_L \mid \mathbf{X}, \mathbf{y}] \ge 1 - \alpha_L$*. On* $\Omega_L$*:* $\lambda_{\min}(\mathbf{G} + \lambda\Psi) \ge g^\star$*, i.e.* $\vartheta_0 \ge \vartheta^\star$*; and* $\hat{L} \le \lambda_{\min}(\mathbf{G})$*.*

*(c)* ***The*** $\Pi_O$ ***ridge.*** $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O) \le \lambda_{\min}(\mathbf{A})$ *for every* $\lambda \ge 0$ *(Cauchy interlacing), and* $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O) \uparrow \lambda_{\min}(\mathbf{A})$ *as* $\lambda \to \infty$*. If* $\lambda_{\min}(\mathbf{A}) > \gamma$*,*

$$\lambda_O = \max\big\{0,\ \gamma - \lambda_{\min}\big(\hat{\mathbf{B}}_1 - \hat{\mathbf{C}}_1^\top(\mathbf{A} - \gamma\mathbf{I})^{-1}\hat{\mathbf{C}}_1\big)\big\} < \infty;$$

*if* $\lambda_{\min}(\mathbf{A}) < \gamma$*,* $\lambda_O = \infty$*; the case* $\lambda_{\min}(\mathbf{A}) = \gamma$ *has probability zero (set* $\lambda_O = \infty$*). No bisection is needed.*

*(d)* ***Gate.*** *Conditionally on the data and stage 1,* $\Pr[\hat{\vartheta} < 1] \le \Pr\big[\lambda_{\max}(\mathbf{E}_2) > 2(\vartheta_0 - 1)\sigma_2\sqrt{p}\big] \le \exp\big(-p(\vartheta_0 - 2)^2\big)$ *for* $\vartheta_0 > 2$*. Hence* $\Pr[\textsf{refused} \mid \mathbf{X}, \mathbf{y}] \le \alpha_L + \exp(-p(\vartheta^\star - 2)^2)$ *when* $\vartheta^\star > 2$*.*

*Proof.* (a) $\lambda_{\min}$ is continuous; $\hat{L}$, $\lambda_I$ and the closed form in (c) are continuous on their domains, and the comparison $\lambda_O \le \kappa_\Psi\lambda_I$ is a Borel event. Equivalently, $\{\lambda_O \le a\} = \{\lambda_{\min}(\hat{\mathbf{G}}_1 + a\Pi_O) \ge \gamma\}$ is closed, because $\lambda \mapsto \lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O)$ is continuous and non-decreasing; so $\lambda_O$ is lower semicontinuous. A bisection with a fixed number of steps is also Borel. The stage-2 noise is fresh: $\eta$ and $(u, e_0, e_1)$ are drawn anew in Algorithm 5, and $(\lambda, \Psi)$ is a function of the data and $\mathbf{E}_1$ only. For the real protocol this independence holds exactly for the ideal mechanism $\mathcal{M}$ and up to the perturbation of norm $\le \sqrt{2}\,\bar{\nu}$ of Lemma 2.2.

(b) $\Pr[\Omega_L] \ge 1 - \alpha_L$ is Lemma 4.1(c). For any symmetric $\mathbf{M}$, $\lambda_{\min}(\mathbf{G} + \mathbf{M}) = \lambda_{\min}(\hat{\mathbf{G}}_1 + \mathbf{M} - \mathbf{E}_1) \ge \lambda_{\min}(\hat{\mathbf{G}}_1 + \mathbf{M}) - \lambda_{\max}(\mathbf{E}_1)$ (Weyl). With $\mathbf{M} = \lambda\Psi$: if $\Psi = \mathbf{I}$, $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda_I\mathbf{I}) = \lambda_{\min}(\hat{\mathbf{G}}_1) + \lambda_I \ge \gamma$;

if $\Psi = \Pi_O$ (then $\lambda_O < \infty$), $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda_O \Pi_O) \geq \gamma$ by definition. So on $\Omega_L$, $\lambda_{\min}(\mathbf{G}_\lambda) \geq \gamma - r_1 = g^\star$. With $\mathbf{M} = 0$: $\lambda_{\min}(\mathbf{G}) \geq \lambda_{\min}(\hat{\mathbf{G}}_1) - r_1$, and $\lambda_{\min}(\mathbf{G}) \geq 0$, so $\lambda_{\min}(\mathbf{G}) \geq \hat{L}$.

(c) Interlacing: $\mathbf{A}$ is a principal submatrix of $\hat{\mathbf{G}}_1 + \lambda\Pi_O$ (the $\mathbf{A}$ block is exact and unaffected by $\lambda\Pi_O$). For $\gamma < \lambda_{\min}(\mathbf{A})$, $\lambda_{\min}(\hat{\mathbf{G}}_1 + \lambda\Pi_O) \geq \gamma$ iff $\left(\begin{smallmatrix} \mathbf{A}-\gamma\mathbf{I} & \hat{\mathbf{C}}_1 \\ \hat{\mathbf{C}}_1^\top & \hat{\mathbf{B}}_1+(\lambda-\gamma)\mathbf{I} \end{smallmatrix}\right) \succeq 0$ iff (Schur complement, $\mathbf{A} - \gamma\mathbf{I} \succ 0$) $\hat{\mathbf{B}}_1 - \hat{\mathbf{C}}_1^\top(\mathbf{A}-\gamma\mathbf{I})^{-1}\hat{\mathbf{C}}_1 + (\lambda-\gamma)\mathbf{I} \succeq 0$. The limit follows from the Schur complement as $\lambda \to \infty$. If $\gamma = \lambda_{\min}(\mathbf{A})$, feasibility requires $\hat{\mathbf{C}}_1^\top \mathbf{v} = 0$ for an eigenvector $\mathbf{v}$ of $\mathbf{A}$, a null event under continuous noise.

(d) $\lambda_{\min}(\hat{\mathbf{G}}_2 + \lambda\Psi) \geq \lambda_{\min}(\mathbf{G}_\lambda) - \lambda_{\max}(-\mathbf{E}_2)$, so $\hat{\vartheta} < 1$ implies $\lambda_{\max}(-\mathbf{E}_2) > 2(\vartheta_0 - 1)\sigma_2\sqrt{p} = 2\sigma_2\sqrt{p} + 2(\vartheta_0 - 2)\sigma_2\sqrt{p}$. Apply Lemma 4.1(a),(b) with $t = 2(\vartheta_0 - 2)\sigma_2\sqrt{p}$ and (b) of this lemma. □

**Remark 5.2** (Size of the guarantees)**.** With $\vartheta^\star = 2.5$ and $p = 11$ the rigorous refusal bound is $\alpha_L + e^{-2.75} = \alpha_L + 0.064$. Monte Carlo at the worst case $\vartheta_0 = \vartheta^\star$ found no exceedance of $\lambda_{\max}(\mathbf{E}) > 9.95\sigma$ in $2 \times 10^5$ draws, and Algorithm 6 never refused in our simulations (Table 5). Because $r_1$ is conservative (Lemma S2), the realized $\vartheta_0$ is typically far above $\vartheta^\star$: it was 7.1–8.2 in the ridge scenarios of Table 4. The private bound $\sqrt{U_{yy}/\hat{L}}$ is finite only if $\lambda_{\min}(\hat{\mathbf{G}}_1) > r_1$; when the ridge is active this often fails, and the $\beta$ interval then rests on $B_\beta^{\text{pub}}$ (Correction 10).

## 6 Lemma S4: the bound on $\|\boldsymbol{\beta}\|$

**Lemma 6.1** (S4)**.** *Let* $\mathbf{X}$ *be the matched (clipped) design,* $\mathbf{G} = \mathbf{X}^\top\mathbf{X}$ *with* $\lambda_{\min}(\mathbf{G}) > 0$*, and* $\mathbf{y}$ *the clipped response,* $|y_i| \leq B_y$*.*

*(a) If* $\mathbb{E}[\mathbf{y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$*, then* $\|\boldsymbol{\beta}\|^2 \leq \boldsymbol{\beta}^\top\mathbf{G}\boldsymbol{\beta}/\lambda_{\min}(\mathbf{G}) \leq \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}]/\lambda_{\min}(\mathbf{G})$*. Without any model, the projection coefficient* $\boldsymbol{\beta}^* := \mathbf{G}^{-1}\mathbf{X}^\top\mathbb{E}[\mathbf{y} \mid \mathbf{X}]$ *satisfies the same bound.*

*(b) If the* $y_i$*,* $i \in I$*, are conditionally independent given* $\mathbf{X}$*, then for* $s \geq 0$

$$\Pr\big[\widehat{yy} \leq \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}] - s \;\big|\; \mathbf{X}\big] \leq \exp\Big(-\frac{s^2}{2(nB_y^4/4 + \sigma_{yy}^2)}\Big).$$

*Hence* $U_{yy}^{\text{c}} := \widehat{yy} + \sqrt{2\ln(1/\alpha_U)\,(nB_y^4/4 + \sigma_{yy}^2)}$ *satisfies* $\Pr[U_{yy}^{\text{c}} \geq \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}] \mid \mathbf{X}] \geq 1 - \alpha_U$*. The specification's* $U_{yy} = \widehat{yy} + z_{1-\alpha_U}\sigma_{yy} + B_y^2\sqrt{n\ln(1/\alpha_U)/2}$ *satisfies this with probability* $\geq 1 - 2\alpha_U$ *only; with* $\alpha_U/2$ *in both terms it satisfies it with probability* $\geq 1 - \alpha_U$*. In all cases one may replace* $U$ *by* $\min\{U, nB_y^2\}$*, since* $\mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}] \leq nB_y^2$ *deterministically.*

*(c) Let* $\Omega_U = \{U \geq \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}]\}$*. On* $\Omega_L \cap \Omega_U$*, and if* $\|\boldsymbol{\beta}\| \leq B_\beta^{\text{pub}}$*, we have* $\|\boldsymbol{\beta}\| \leq \hat{B}_\beta = \min\{B_\beta^{\text{pub}}, \sqrt{U/\hat{L}}\}$*.*

*Proof.* (a) $\mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}] = \|\mathbb{E}[\mathbf{y} \mid \mathbf{X}]\|^2 + \sum_i \mathrm{Var}(y_i \mid \mathbf{X}) \geq \|\mathbb{E}[\mathbf{y} \mid \mathbf{X}]\|^2$. Under the model, $\|\mathbb{E}[\mathbf{y} \mid \mathbf{X}]\|^2 = \boldsymbol{\beta}^\top\mathbf{G}\boldsymbol{\beta} \geq \lambda_{\min}(\mathbf{G})\|\boldsymbol{\beta}\|^2$. In general, $\boldsymbol{\beta}^{*\top}\mathbf{G}\boldsymbol{\beta}^* = \|\mathbf{H}\mathbb{E}[\mathbf{y} \mid \mathbf{X}]\|^2 \leq \|\mathbb{E}[\mathbf{y} \mid \mathbf{X}]\|^2$ with $\mathbf{H}$ the hat matrix.

(b) $y_i^2 \in [0, w] = [0, B_y^2]$, so by Hoeffding's lemma [10] each centred $y_i^2$ is $(w^2/4) = (B_y^4/4)$-subgaussian given $\mathbf{X}$; independence gives $nw^2/4$ for the sum, and adding the independent $\mathcal{N}(0, \sigma_{yy}^2)$ release noise gives $nw^2/4 + \sigma_{yy}^2$. The Chernoff bound gives the display. For the specification's form, Hoeffding gives $\Pr[\mathbf{y}^\top\mathbf{y} \leq \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}] - w\sqrt{n\ln(1/\alpha)/2}] \leq \alpha$ (this term is correct as stated) and $\Pr[\widehat{yy} \leq \mathbf{y}^\top\mathbf{y} - z_{1-\alpha}\sigma_{yy}] = \alpha$; the union bound gives $2\alpha$, and $\alpha/2$ in each restores $\alpha$. Without conditional independence, Hoeffding does not apply; a martingale version controls $\sum_i (y_i^2 - \mathbb{E}[y_i^2 \mid \mathcal{F}_{i-1}])$, which is not the target, so only the deterministic bound $nB_y^2$ remains.

(c) On $\Omega_L$, $\hat{L} \le \lambda_{\min}(\mathbf{G})$ (Lemma 5.1(b)); on $\Omega_U$, $U \ge \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}]$; combine with (a). If $\hat{L} = 0$ the private bound is $+\infty$ and the claim reduces to the public one. □

**Clipping and the estimand.** R clips $\mathbf{x}_R$, O clips $\mathbf{x}_O$ and $y$. The clipped data need not follow a linear model even if the raw data do. Clipping $y$ is a non-linear map of $\mathbf{x}^\top\boldsymbol{\beta}+\varepsilon$, and clipping $\mathbf{x}$ breaks linearity in the original features. Clipping also makes the errors heteroskedastic: $\mathrm{Var}(\mathrm{clip}(y) \mid \mathbf{x})$ depends on $\mathbf{x}$ through the probability of clipping. Precisely:

- The Wald interval targets $\boldsymbol{\theta}_\lambda = (\mathbf{G} + \lambda\Psi)^{-1}\mathbf{c}$ computed on the clipped matched data. It is well defined without any model; its randomness is the mechanism only, given the data and stage 1.
- The bias-aware interval targets $\boldsymbol{\beta}$ in the *working model on the clipped scale*: $\mathbb{E}[\tilde{y}_i \mid \tilde{\mathbf{X}}] = \tilde{\mathbf{x}}_i^\top\boldsymbol{\beta}$ with conditionally independent, homoskedastic errors, where tildes denote clipped values. Without that model the natural target is the conditional projection $\boldsymbol{\beta}^* = \mathbf{G}^{-1}\tilde{\mathbf{X}}^\top\mathbb{E}[\tilde{\mathbf{y}} \mid \tilde{\mathbf{X}}]$. For $\boldsymbol{\beta}^*$ the bias identity $\mathbb{E}[\boldsymbol{\theta}_\lambda \mid \mathbf{X}] - \boldsymbol{\beta}^* = -\lambda\mathbf{P}_\lambda\Psi\boldsymbol{\beta}^*$ (because $\tilde{\mathbf{X}}^\top\mathbb{E}[\tilde{\mathbf{y}} \mid \tilde{\mathbf{X}}] = \mathbf{G}\boldsymbol{\beta}^*$) and Lemma 6.1(a) still hold, but the homoskedastic variance formula does not. The interval is then approximate, and a sandwich variance (a further release) would be needed for validity.
- The raw-scale coefficient equals the clipped-scale one only if clipping never binds on the support, i.e. if the bounds are true bounds. Exactly Gaussian errors are incompatible with $|y| \le B_y$, so normality of the sampling term can only be asymptotic (Section 7).

# 7 Proposition S5: coverage

Condition on the data and stage 1, $\mathcal{F}_1 = \sigma(\mathbf{X}, \mathbf{y}, \mathbf{E}_1)$, so $\lambda, \Psi, \mathbf{G}_\lambda, \mathbf{P}_\lambda, \boldsymbol{\theta}_\lambda, \vartheta_0$ are fixed. Let

$$\boldsymbol{\xi} := \mathbf{P}_\lambda(\mathbf{f}_2 - \mathbf{E}_2\boldsymbol{\theta}_\lambda), \qquad \mathbf{V}_0 := \mathrm{Var}(\boldsymbol{\xi} \mid \mathcal{F}_1) = \sigma_2^2\,\mathbf{P}_\lambda\big(\mathbf{I} + \mathbf{K}(\boldsymbol{\theta}_\lambda)\big)\mathbf{P}_\lambda, \qquad q := \|\mathbf{G}_\lambda^{-1/2}\mathbf{E}_2\mathbf{G}_\lambda^{-1/2}\|_{\mathrm{op}}.$$

**Theorem 7.1** (S5(a): Wald interval for $\boldsymbol{\theta}_\lambda$)**.** *Under Assumption 2.1:*

*(i)* *[exact]* $\boldsymbol{\xi} \mid \mathcal{F}_1 \sim \mathcal{N}(0, \mathbf{V}_0)$*; and on* $\{\mathbf{G}_\lambda + \mathbf{E}_2$ *invertible*$\}$, $\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_\lambda = (\mathbf{I} + \mathbf{P}_\lambda\mathbf{E}_2)^{-1}\boldsymbol{\xi}$.

*(ii)* *[exact] If* $q < 1$*, then* $|\tilde{\theta}_j - \theta_{\lambda,j} - \xi_j| \le \frac{q}{1-q}\sqrt{(\mathbf{P}_\lambda)_{jj}}\,\|\mathbf{G}_\lambda^{1/2}\boldsymbol{\xi}\|$.

*(iii)* *[exact, finite sample] Let* $s_j^2 = (\mathbf{V}_0)_{jj}$, $z = z_{1-\alpha/2}$, $\varpi \in (0, z)$ *and* $\eta \in (0,1)$ *with* $\eta\vartheta_0 \ge 1$*. The oracle interval* $\tilde{\theta}_j \pm z\,s_j$ *satisfies*

$$\Pr\big[|\tilde{\theta}_j - \theta_{\lambda,j}| > z s_j \;\big|\; \mathcal{F}_1\big] \le \alpha + 2\varpi\,\phi(z - \varpi) + 2e^{-p(\eta\vartheta_0 - 1)^2} + \Big(\frac{\eta}{1-\eta}\Big)^2\frac{\omega_j}{\varpi^2},$$

*with* $\omega_j := (\mathbf{P}_\lambda)_{jj}\,\sigma_2^2\,\mathrm{tr}\big((\mathbf{I} + \mathbf{K}(\boldsymbol{\theta}_\lambda))\mathbf{P}_\lambda\big)/s_j^2 \le (1 + 2\|\boldsymbol{\theta}_\lambda\|^2)\,\mathrm{tr}(\mathbf{P}_\lambda)/(\mathbf{P}_\lambda)_{jj}$.

*(iv)* *[first order] The reported interval uses* $\mathbf{V}_{\mathrm{mech}}$ *in place of* $\mathbf{V}_0$*. By Lemma S6,* $(\mathbf{V}_{\mathrm{mech}})_{jj} = s_j^2(1 + O_p(\vartheta_0^{-1}) + O(\vartheta_0^{-2}) + O_p(B^{-1/2}))$.

*(v)* *[asymptotic] Unconditionally,* $\Pr[\theta_{\lambda,j} \notin \mathrm{CI}_j$*, not refused*$] \le \alpha_{\mathrm{CI}} + \alpha_L + \epsilon_j$*, where* $\epsilon_j \to 0$ *as* $\vartheta^\star \to \infty$ *and* $B \to \infty$ *with* $p$*,* $\|\boldsymbol{\theta}_\lambda\|$ *and* $\mathrm{tr}(\mathbf{P}_\lambda)/(\mathbf{P}_\lambda)_{jj}$ *bounded (choose* $\eta = c/\vartheta_0$ *and* $\varpi \to 0$ *slowly in (iii)).*

*Proof.* (i) Given $\mathcal{F}_1$, $(\mathbf{E}_2, \mathbf{f}_2)$ is Gaussian, centred and independent of $(\lambda, \Psi)$ (Lemma 5.1(a)), and $\mathbf{E}_2 \perp \mathbf{f}_2$; so $\boldsymbol{\xi}$ is a centred Gaussian linear image with covariance $\mathbf{P}_\lambda(\sigma_2^2\mathbf{I} + \sigma_2^2\mathbf{K})\mathbf{P}_\lambda$. From $(\mathbf{G}_\lambda+\mathbf{E}_2)\tilde{\boldsymbol{\theta}} = \mathbf{c}+\mathbf{f}_2$ and $\mathbf{G}_\lambda\boldsymbol{\theta}_\lambda = \mathbf{c}$ we get $(\mathbf{G}_\lambda+\mathbf{E}_2)(\tilde{\boldsymbol{\theta}}-\boldsymbol{\theta}_\lambda) = \mathbf{f}_2-\mathbf{E}_2\boldsymbol{\theta}_\lambda$. (ii) Let $\mathbf{M} = \mathbf{G}_\lambda^{-1/2}\mathbf{E}_2\mathbf{G}_\lambda^{-1/2}$. Then $\mathbf{I} + \mathbf{P}_\lambda\mathbf{E}_2 = \mathbf{G}_\lambda^{-1/2}(\mathbf{I} + \mathbf{M})\mathbf{G}_\lambda^{1/2}$, so $\mathbf{G}_\lambda^{1/2}(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_\lambda - \boldsymbol{\xi}) = -(\mathbf{I} + \mathbf{M})^{-1}\mathbf{M}\mathbf{G}_\lambda^{1/2}\boldsymbol{\xi}$, whose norm is at most $\frac{q}{1-q}\|\mathbf{G}_\lambda^{1/2}\boldsymbol{\xi}\|$. Finally $|\mathbf{e}_j^\top\mathbf{v}| \le \|\mathbf{G}_\lambda^{-1/2}\mathbf{e}_j\|\|\mathbf{G}_\lambda^{1/2}\mathbf{v}\|$. (iii) $\Pr[|\xi_j + R_j| > z s_j] \le \Pr[|\xi_j| > (z - \varpi)s_j] + \Pr[|R_j| > \varpi s_j]$. The first term is $2Q(z - \varpi) \le \alpha + 2\varpi\phi(z - \varpi)$. For the second,

$q \le \|\mathbf{E}_2\|_{\mathrm{op}}/\lambda_{\min}(\mathbf{G}_\lambda)$ and Lemma 4.1(c) give $\Pr[q > \eta] \le 2e^{-p(\eta\vartheta_0-1)^2}$. On $\{q \le \eta\}$, (ii) and Markov's inequality with $\mathbb{E}\|\mathbf{G}_\lambda^{1/2}\boldsymbol{\xi}\|^2 = \operatorname{tr}(\mathbf{G}_\lambda\mathbf{V}_0) = \sigma_2^2\operatorname{tr}((\mathbf{I}+\mathbf{K})\mathbf{P}_\lambda)$ give the last term. The bound on $\omega_j$ uses $s_j^2 \ge \sigma_2^2\|\mathbf{P}_\lambda\mathbf{e}_j\|^2 \ge \sigma_2^2(\mathbf{P}_\lambda)_{jj}^2$ (as $\mathbf{K} \succeq 0$) and $\|\mathbf{K}\| \le 2\|\boldsymbol{\theta}\|^2$. (v) On $\Omega_L$, $\vartheta_0 \ge \vartheta^\star$ (Lemma 5.1(b)); integrate (iii)–(iv) over $\mathcal{F}_1$ and add $\Pr[\Omega_L^c] \le \alpha_L$. □

The bound in (iii) is a certificate that the error vanishes as $\vartheta_0 \to \infty$, but it is loose (it uses the worst direction of $\mathbf{M}$ and Markov's inequality). The *typical* size of the remainder relative to $s_j$ is governed by

$$\sigma_2 \operatorname{tr}(\mathbf{P}_\lambda) \;\le\; \frac{p\,\sigma_2}{\lambda_{\min}(\mathbf{G}_\lambda)} = \frac{\sqrt{p}}{2\vartheta_0},$$

the ratio of the $O(\sigma_2^2)$ mechanism bias $\sigma_2^2\mathbf{P}_\lambda\mathbf{M}_\chi(\mathbf{P}_\lambda)\boldsymbol{\theta}_\lambda$ to the $O(\sigma_2)$ standard deviation. At $\vartheta_0 = \vartheta^\star = 2.5$ and $p = 11$ this can reach 0.66 when all eigenvalues of $\mathbf{G}_\lambda$ sit at the floor. "First order in $1/\vartheta$" therefore hides a factor $\sqrt{p}$ (Correction 7). In the simulations (Table 4) $\sigma_2\operatorname{tr}(\mathbf{P}_\lambda) \le 0.18$ in the realistic scenarios and 0.59 in the stress scenario.

**Theorem 7.2** (S5(b): bias-aware interval for $\beta_j$)**.** *Assume (M1) the working model on the clipped scale: $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $(\varepsilon_i)$ conditionally independent given $\mathbf{X}$, $\mathbb{E}[\varepsilon_i \mid \mathbf{X}] = 0$ and $\operatorname{Var}(\varepsilon_i \mid \mathbf{X}) = s^2$; (M2) $\|\boldsymbol{\beta}\| \le B_\beta^{\mathrm{pub}}$ (possibly $+\infty$); (M3) $n > p$. Decompose*

$$\tilde\theta_j - \beta_j = b_j + S_j + \xi_j + R_j, \qquad b_j = -\lambda\,\mathbf{e}_j^\top\mathbf{P}_\lambda\Psi\boldsymbol{\beta}, \quad S_j = \mathbf{e}_j^\top\mathbf{P}_\lambda\mathbf{X}^\top\boldsymbol{\varepsilon}, \quad R_j = \tilde\theta_j - \theta_{\lambda,j} - \xi_j.$$

*(i) [exact] $\mathbb{E}[\boldsymbol{\theta}_\lambda \mid \mathbf{X}, \mathbf{E}_1] = \boldsymbol{\beta} - \lambda\mathbf{P}_\lambda\Psi\boldsymbol{\beta}$; $|b_j| \le \lambda\|\Psi\mathbf{P}_\lambda\mathbf{e}_j\|\,\|\boldsymbol{\beta}\|$, with equality for $\boldsymbol{\beta} \parallel \Psi\mathbf{P}_\lambda\mathbf{e}_j$. $\operatorname{Var}(S_j \mid \mathbf{X}, \mathbf{E}_1) = s^2(\mathbf{P}_\lambda\mathbf{G}\mathbf{P}_\lambda)_{jj}$, and $\operatorname{Cov}(S_j, \xi_j) = 0$.*

*(ii) [exact] If $T \sim \mathcal{N}(b, v^2)$ and $\bar B$ is a random variable with $|b| \le \bar B$ on an event $\Omega$, then $\Pr[|T| > v\,\mathrm{cv}_\alpha(\bar B/v)] \le \alpha + \Pr[\Omega^c]$, where $\mathrm{cv}_\alpha(t)$ is the $(1-\alpha)$-quantile of $|Z + t|$.*

*(iii) [exact on events] On $\Omega_L \cap \Omega_U$, $|b_j| \le \bar b_j^0 := \lambda\|\Psi\mathbf{P}_\lambda\mathbf{e}_j\|\hat B_\beta$.*

*(iv) [exact for the oracle interval] Let $v_j^2 = s^2(\mathbf{P}_\lambda\mathbf{G}\mathbf{P}_\lambda)_{jj} + (\mathbf{V}_0)_{jj}$. If $S_j + \xi_j \mid (\mathbf{X}, \mathbf{E}_1) \sim \mathcal{N}(0, v_j^2)$ and $R_j = 0$, then $\tilde\theta_j \pm v_j\,\mathrm{cv}_{\alpha_{\mathrm{CI}}}(\bar b_j^0/v_j)$ misses $\beta_j$ with probability at most $\alpha_{\mathrm{CI}} + \alpha_L + \alpha_U'$, where $\alpha_U' = \alpha_U$ for the corrected $U_{yy}^{\mathrm{c}}$ and $2\alpha_U$ for the specification's $U_{yy}$.*

*(v) [first order] The reported interval replaces $\mathbf{P}_\lambda$ by $\tilde{\mathbf{P}}$ in $\bar b_j$ (relative error $\le \|\tilde{\mathbf{P}}\|\|\mathbf{E}_2\| = O_p(\vartheta_0^{-1})$, since $\mathbf{P}_\lambda - \tilde{\mathbf{P}} = \tilde{\mathbf{P}}\mathbf{E}_2\mathbf{P}_\lambda$), and $v_j^2$ by $V_{jj} = (\mathbf{V}_{\mathrm{mech}})_{jj} + \tilde s^2(\tilde{\mathbf{P}}\hat{\mathbf{G}}_2\tilde{\mathbf{P}})_{jj}$. It also drops $R_j$ (Theorem 7.1(ii)) and ignores the dependence of $\mathbf{V}_0$ on $\boldsymbol{\theta}_\lambda$, hence on $\boldsymbol{\varepsilon}$, which is a relative $O(\|\boldsymbol{\theta}_\lambda - \mathbb{E}\boldsymbol{\theta}_\lambda\|/\|\boldsymbol{\theta}_\lambda\|)$ effect.*

*(vi) [asymptotic] Under (M1) the errors are bounded (clipping), so $S_j$ is normal only asymptotically: by the Berry–Esseen bound for independent summands [11], $\sup_x |\Pr[S_j \le x\,\mathrm{sd}(S_j) \mid \mathbf{X}] - \Phi(x)| \le C_{\mathrm{BE}}\sum_i |a_{ij}|^3\mathbb{E}|\varepsilon_i|^3/(s^3\|\mathbf{a}_j\|^3)$ with $\mathbf{a}_j = \mathbf{X}\mathbf{P}_\lambda\mathbf{e}_j$. Together with (v): $\Pr[\beta_j \notin \mathrm{CI}_j,\ \text{not refused}] \le \alpha_{\mathrm{CI}} + \alpha_L + \alpha_U' + \epsilon_j'$, with $\epsilon_j' \to 0$ when the Lyapunov ratio $\to 0$, $\vartheta^\star \to \infty$ and $B \to \infty$.*

*Proof.* (i) $\mathbb{E}[\mathbf{c} \mid \mathbf{X}] = \mathbf{G}\boldsymbol{\beta}$ and $\mathbf{P}_\lambda\mathbf{G} = \mathbf{I} - \lambda\mathbf{P}_\lambda\Psi$; $(\lambda, \Psi)$ is independent of $\boldsymbol{\varepsilon}$ given $\mathbf{X}$, because it depends on $\mathbf{G}$ and $\mathbf{E}_1$ only. Cauchy–Schwarz with $\Psi, \mathbf{P}_\lambda$ symmetric. $\mathbb{E}[\boldsymbol{\xi} \mid \mathbf{X}, \boldsymbol{\varepsilon}, \mathbf{E}_1] = 0$ gives the zero covariance. (ii) $t \mapsto \Pr[|Z + t| \le c]$ is even and non-increasing in $|t|$ (Anderson's lemma [1]), so $\mathrm{cv}_\alpha$ is non-decreasing in $|t|$. On $\Omega$, $\mathrm{cv}_\alpha(\bar B/v) \ge \mathrm{cv}_\alpha(|b|/v)$, and $\Pr[|T| > v\,\mathrm{cv}_\alpha(|b|/v)] = \Pr[|Z + b/v| > \mathrm{cv}_\alpha(|b|/v)] = \alpha$. This is the construction of Armstrong and Kolesár [2], as used for regularized regression by Armstrong, Kolesár and Kwon [3]. (iii) Lemma 6.1(c) and (i). (iv) Apply (ii) with $\Omega = \Omega_L \cap \Omega_U$ and $\bar B = \bar b_j^0$. The dependence between $\bar b_j^0$ and $T$ (through $\boldsymbol{\varepsilon}$ in $U$, and through $\mathbf{E}_1$) is harmless, because (ii) does not need $\bar B$ independent of $T$. (v), (vi) are as stated. □

**Variance estimate.** $\tilde{s}^2$ targets $\mathrm{RSS}_\lambda/(n-p)$, and $\mathbb{E}[\mathrm{RSS}_\lambda \mid \mathbf{X}] = s^2\sum_i(1-h_i)^2 + \|\lambda\mathbf{X}\mathbf{P}_\lambda\Psi\boldsymbol{\beta}\|^2 \geq (n-p)s^2$, where $h_i \in [0,1)$ are the eigenvalues of $\mathbf{X}\mathbf{P}_\lambda\mathbf{X}^\top$ (at most $p$ of them non-zero). So $\tilde{s}^2$ is conservative under the ridge (Table 5: 1.05, 1.48 and 1.79 against $s^2 = 1$ in three ridge scenarios). This conservativeness, and the worst-case nature of $\bar{b}_j$, explain the over-coverage of the $\beta$ intervals.

**Refusals and conditioning on the gate.** All statements are unconditional, with a refusal producing no interval (equivalently, the trivial interval $\mathbb{R}$). Conditionally on passing, $\Pr[\text{miss} \mid \text{pass}] \leq \Pr[\text{miss}, \text{pass}]/(1 - \Pr[\mathsf{refused}])$, with $\Pr[\mathsf{refused}]$ bounded in Lemma 5.1(d). The gate does not make the bootstrap moments finite (Lemma S6).

## 8 Lemma S6: the bootstrap RSS correction and covariance

Fix $\mathcal{F}_1$. Write $\mathrm{RSS}(\mathbf{G}', \mathbf{c}', yy') = yy' - \mathbf{c}'^\top\boldsymbol{\theta}' - \lambda\boldsymbol{\theta}'^\top\Psi\boldsymbol{\theta}'$ with $\boldsymbol{\theta}' = (\mathbf{G}' + \lambda\Psi)^{-1}\mathbf{c}'$; this equals $yy' - 2\mathbf{c}'^\top\boldsymbol{\theta}' + \boldsymbol{\theta}'^\top\mathbf{G}'\boldsymbol{\theta}'$, so $\mathrm{RSS}_\lambda := \mathrm{RSS}(\mathbf{G}, \mathbf{c}, \mathbf{y}^\top\mathbf{y}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}_\lambda\|^2$ and $\widetilde{\mathrm{RSS}} = \mathrm{RSS}(\hat{\mathbf{G}}_2, \hat{\mathbf{c}}_2, \widehat{yy})$. Define the normalized quantities

$$\mathbf{M} = \mathbf{G}_\lambda^{-1/2}\mathbf{E}_2\mathbf{G}_\lambda^{-1/2}, \quad \mathbf{h} = \mathbf{G}_\lambda^{-1/2}\mathbf{f}_2, \quad \boldsymbol{\phi} = \mathbf{G}_\lambda^{1/2}\boldsymbol{\theta}_\lambda, \quad \mathbf{N} = \lambda\mathbf{G}_\lambda^{-1/2}\Psi\mathbf{G}_\lambda^{-1/2} \ (0 \preceq \mathbf{N} \preceq \mathbf{I}),$$

so that, exactly, $\widetilde{\mathrm{RSS}} = \mathbf{y}^\top\mathbf{y} + g - (\boldsymbol{\phi} + \mathbf{h})^\top(\mathbf{I} + \mathbf{M})^{-1}\big[\mathbf{I} + \mathbf{N}(\mathbf{I} + \mathbf{M})^{-1}\big](\boldsymbol{\phi} + \mathbf{h})$. For a symmetric truncation level $\eta \in (0,1)$ let $T_\eta = \{\|\mathbf{M}\|_{\mathrm{op}} \leq \eta\}$. In the *truncated bootstrap* a draw is kept iff $\|(\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1/2}\mathbf{E}^*(\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1/2}\|_{\mathrm{op}} \leq \eta$; this event is invariant under $\mathbf{E}^* \mapsto -\mathbf{E}^*$.

**Lemma 8.1** (S6)**.** *Under Assumption 2.1, conditionally on $\mathcal{F}_1$:*

*(a)* ***Expansion.*** $\widetilde{\mathrm{RSS}} = \mathrm{RSS}_\lambda + R_1 + R_2 + R_3 + \ldots$, *converging on $T_\eta$, where $R_k$ is homogeneous of degree $k$ in $(\mathbf{M}, \mathbf{h}, g)$:*

$$R_1 = g - 2\boldsymbol{\theta}_\lambda^\top\mathbf{f}_2 + \boldsymbol{\theta}_\lambda^\top\mathbf{E}_2\boldsymbol{\theta}_\lambda - 2\lambda\boldsymbol{\theta}_\lambda^\top\Psi\boldsymbol{\xi}, \qquad R_2 = -\boldsymbol{\xi}^\top(\mathbf{G} + 2\lambda\Psi)\boldsymbol{\xi} + 2\lambda\boldsymbol{\theta}_\lambda^\top\Psi\mathbf{P}_\lambda\mathbf{E}_2\boldsymbol{\xi},$$

*with $\boldsymbol{\xi} = \mathbf{P}_\lambda(\mathbf{f}_2 - \mathbf{E}_2\boldsymbol{\theta}_\lambda)$, and $|R_k| \leq c_k\|\mathbf{M}\|^{k-2}\big(\|\mathbf{M}\boldsymbol{\phi}\| + \|\mathbf{h}\| + \|\mathbf{N}\boldsymbol{\phi}\|\big)\big(\|\mathbf{M}\boldsymbol{\phi}\| + \|\mathbf{h}\|\big)$ for $k \geq 2$, with combinatorial constants $c_k \leq 4(k+1)$.*

*(b)* ***Second-order bias.*** $\mathbb{E}[R_1] = \mathbb{E}[R_3 \mid T_\eta] = 0$ *(odd degree, symmetric event), and*

$$\mathbb{E}[R_2] = b_{\mathrm{RSS}}(\mathbf{G}, \boldsymbol{\theta}_\lambda) := -\sigma_2^2\Big[\mathrm{tr}\big((\mathbf{G} + 2\lambda\Psi)\mathbf{P}_\lambda(\mathbf{I} + \mathbf{K}(\boldsymbol{\theta}_\lambda))\mathbf{P}_\lambda\big) + 2\lambda\,\boldsymbol{\theta}_\lambda^\top\Psi\mathbf{P}_\lambda\mathbf{M}_\chi(\mathbf{P}_\lambda)\boldsymbol{\theta}_\lambda\Big].$$

*Hence $\mathbb{E}[\widetilde{\mathrm{RSS}} \mid T_\eta] - \mathrm{RSS}_\lambda = b_{\mathrm{RSS}}(\mathbf{G}, \boldsymbol{\theta}_\lambda) + \rho_1$, with $|\rho_1| \leq C_\eta\,\vartheta_0^{-2}\,\mathcal{S}_2 + O(e^{-p(\eta\vartheta_0 - 1)^2})$, where $\mathcal{S}_2$ is the scale of the second-order terms, $\mathcal{S}_2 \asymp \sigma_2^2(1 + \|\boldsymbol{\theta}_\lambda\|^2)\,\mathrm{tr}(\mathbf{P}_\lambda) + \|\mathbf{N}\boldsymbol{\phi}\|\,\sigma_2(1 + \|\boldsymbol{\theta}_\lambda\|)\sqrt{\mathrm{tr}\,\mathbf{P}_\lambda}/\vartheta_0$.*

*(c)* ***RSS correction.*** *Let $\mathrm{Corr}_\infty = \mathbb{E}^*[\mathrm{RSS}^* \mid T_\eta^*] - \widetilde{\mathrm{RSS}}$ be the $B \to \infty$ limit of* Corr *for the truncated bootstrap. Then $\mathrm{Corr}_\infty = b_{\mathrm{RSS}}(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}}) + \hat{\rho}_1$, and*

$$\mathbb{E}\big[\widetilde{\mathrm{RSS}} - \mathrm{Corr}_\infty \;\big|\; T_\eta\big] - \mathrm{RSS}_\lambda = O\big(\vartheta_0^{-2}\mathcal{S}_2\big).$$

*The uncorrected bias $b_{\mathrm{RSS}} = O(\mathcal{S}_2)$ is removed up to a remainder smaller by a factor $\vartheta_0^{-2}$: $O(\sigma_2^2)$ becomes $O(\sigma_2^4)$. The Monte Carlo error of* Corr *with $B$ draws is $O_p(\mathcal{S}_1/\sqrt{B})$ with $\mathcal{S}_1^2 = \mathrm{Var}(R_1) = \sigma_{yy}^2 + \sigma_2^2\,O\big((1 + \|\boldsymbol{\theta}_\lambda\|^2)^2\big)$. Antithetic pairs $(\mathbf{E}^*, \mathbf{f}^*, g^*), (-\mathbf{E}^*, -\mathbf{f}^*, -g^*)$ cancel $R_1$ and $R_3$ exactly and reduce it to $O_p(\mathcal{S}_2/\sqrt{B})$.*

*(d)* ***Covariance.*** $\mathrm{Var}(\tilde{\boldsymbol{\theta}} \mid T_\eta) = \mathbf{V}_0\,(1 + O(\vartheta_0^{-2}))$ *and* $\mathrm{Cov}^*(\boldsymbol{\theta}^* \mid T_\eta^*) = \mathbf{V}_0(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}})(1 + O(\hat{\vartheta}^{-2}))$, *with $\mathbf{V}_0(\mathbf{G}', \boldsymbol{\theta}') = \sigma_2^2(\mathbf{G}' + \lambda\Psi)^{-1}(\mathbf{I} + \mathbf{K}(\boldsymbol{\theta}'))(\mathbf{G}' + \lambda\Psi)^{-1}$; and $\mathbf{V}_0(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}}) = \mathbf{V}_0(\mathbf{G}, \boldsymbol{\theta}_\lambda)(1 + O_p(\vartheta_0^{-1}))$, in the sense of the diagonal entries relative to themselves. So $\mathbf{V}_{\mathrm{mech}}$ estimates the mechanism covariance to first order, with relative error $O_p(\vartheta_0^{-1}) + O(\vartheta_0^{-2}) + O_p(B^{-1/2})$.*

(e) ***Untruncated bootstrap.*** *Without truncation,* $\mathbb{E}^*\|\boldsymbol{\theta}^*\| = \infty$ *(already for* $p = 1$*,* $\mathbb{E}|1/(a+e)| = \infty$ *for Gaussian* $e$*), so* $\mathrm{Cov}_b(\boldsymbol{\theta}_b^*)$ *and* $\overline{\mathrm{RSS}^*}$ *do not converge as* $B \to \infty$*. For fixed* $B$ *they coincide with the truncated versions except on an event of probability* $\le B \cdot 2e^{-p(\eta\hat{\vartheta}-1)^2}$*; the gate* $\hat{\vartheta} \ge 1$ *alone does not make this small.*

*Proof [first order: constants* $c_k$*,* $C_\eta$ *and the scale* $\mathcal{S}_2$ *are not made fully explicit].* (a) Expand $(\mathbf{I}+\mathbf{M})^{-1} = \sum_k (-\mathbf{M})^k$, which converges on $T_\eta$. Every term of degree $k \ge 2$ in $(\mathbf{M}, \mathbf{h})$ is a bilinear form whose two outer vectors are $\mathbf{h}$, $\mathbf{M}\boldsymbol{\phi}$ or $\mathbf{N}\boldsymbol{\phi}$, with at most one of them $\mathbf{N}\boldsymbol{\phi}$ and at most $k-2$ further factors $\mathbf{M}$ or $\mathbf{N}$ between them ($\|\mathbf{N}\| \le 1$); the number of such terms is at most $4(k+1)$ (outer pairs $(\boldsymbol{\phi},\boldsymbol{\phi})$: $k+2$ terms; $(\mathbf{h},\boldsymbol{\phi})$ and $(\boldsymbol{\phi},\mathbf{h})$: $k+1$ each; $(\mathbf{h},\mathbf{h})$: $k$), each with coefficient $\pm 1$. The explicit $R_1, R_2$ follow by differentiation: with $\boldsymbol{\theta}' = \boldsymbol{\theta}_\lambda + \boldsymbol{\xi} + \boldsymbol{\zeta} + \ldots$, $\boldsymbol{\zeta} = -\mathbf{P}_\lambda \mathbf{E}_2 \boldsymbol{\xi}$, use $\mathbf{c} - \mathbf{G}\boldsymbol{\theta}_\lambda = \lambda\Psi\boldsymbol{\theta}_\lambda$ and $\mathbf{f}_2 - \mathbf{E}_2\boldsymbol{\theta}_\lambda = \mathbf{G}_\lambda\boldsymbol{\xi}$.

(b) Odd-degree terms have zero mean under the symmetric Gaussian law and on the symmetric event $T_\eta$. $\mathbb{E}[\boldsymbol{\xi}^\top \mathbf{A}\boldsymbol{\xi}] = \mathrm{tr}(\mathbf{A}\mathbf{V}_0)$ and $\mathbb{E}[\mathbf{E}_2\mathbf{P}_\lambda\mathbf{E}_2] = \sigma_2^2\mathbf{M}_\chi(\mathbf{P}_\lambda)$ give $b_{\mathrm{RSS}}$; $\mathbb{E}[\mathbf{E}_2\mathbf{P}_\lambda\mathbf{f}_2] = 0$. The degree-4 terms are bounded via (a) by $c_4\mathbb{E}[\|\mathbf{M}\|^2(\cdot)(\cdot)]$, and $\mathbb{E}\|\mathbf{M}\|^2 \le (\mathbb{E}\|\mathbf{E}_2\|^2)/\lambda_{\min}(\mathbf{G}_\lambda)^2 = \vartheta_0^{-2}(1 + O(p^{-1/2}))$ by Lemma 4.1(d) and (b); also $\mathbb{E}\|\mathbf{h}\|^2 = \sigma_2^2 \,\mathrm{tr}\,\mathbf{P}_\lambda$ and $\mathbb{E}\|\mathbf{M}\boldsymbol{\phi}\|^2 = \sigma_2^2\,\mathrm{tr}(\mathbf{P}_\lambda\mathbf{K}(\boldsymbol{\theta}_\lambda)) \le 2\sigma_2^2\|\boldsymbol{\theta}_\lambda\|^2\,\mathrm{tr}\,\mathbf{P}_\lambda$. Degrees $\ge 5$ are summed geometrically on $T_\eta$. Removing the conditioning costs $\Pr[T_\eta^c] \le 2e^{-p(\eta\vartheta_0-1)^2}$ (Lemma 4.1(c)) times bounded conditional moments.

(c) Apply (b) in the bootstrap world centred at $(\hat{\mathbf{G}}_2, \hat{\mathbf{c}}_2, \widehat{yy})$, where the bootstrap law is exactly the mechanism law. Then $b_{\mathrm{RSS}}(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}}) - b_{\mathrm{RSS}}(\mathbf{G}, \boldsymbol{\theta}_\lambda)$ is a smooth function of $(\mathbf{E}_2, \mathbf{f}_2)$ whose linear part has zero mean; its quadratic part is $O(\vartheta_0^{-2}\mathcal{S}_2)$ in mean, since $b_{\mathrm{RSS}}$ is homogeneous of degree 2 in the noise scale and every derivative brings a factor of order $\vartheta_0^{-1}$. The Monte Carlo statements follow from $\mathrm{Var}(\mathrm{RSS}^*) = \mathrm{Var}(R_1^*) + O(\mathcal{S}_2^2)$, and the antithetic average is even in the noise.

(d) $\mathrm{Var}(\tilde{\boldsymbol{\theta}}) = \mathrm{Var}(\boldsymbol{\xi}) + 2\,\mathrm{Cov}(\boldsymbol{\xi}, \boldsymbol{\zeta}) + \ldots$; $\mathrm{Cov}(\boldsymbol{\xi}, \boldsymbol{\zeta})$ is an odd (third) moment and vanishes, and the remaining terms are $O(\vartheta_0^{-2})$ relative by (a). The plug-in error comes from $\tilde{\mathbf{P}} - \mathbf{P}_\lambda = -\tilde{\mathbf{P}}\mathbf{E}_2\mathbf{P}_\lambda$ ($\|\cdot\| \le \|\tilde{\mathbf{P}}\|\|\mathbf{E}_2\|\|\mathbf{P}_\lambda\|$, relative $O_p(\vartheta_0^{-1})$) and $\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}_\lambda = O_p(\|\boldsymbol{\xi}\|)$.

(e) For $p = 1$, $\int |a+e|^{-1}\phi_\sigma(e)\,de = \infty$ because the density is positive at $e = -a$. For general $p$, $\|(\hat{\mathbf{G}}_2 + \mathbf{E}^* + \lambda\Psi)^{-1}\| \ge 1/|\lambda_{\min}(\cdot)|$, and the law of $\lambda_{\min}$ has a positive density at 0. Moreover $\hat{\vartheta} \ge 1$ only gives $\lambda_{\min}(\hat{\mathbf{G}}_2 + \lambda\Psi) \ge 2\sigma_2\sqrt{p}$, which for $p = 11$ is only $1.33\,\mathbb{E}\lambda_{\max}(\mathbf{E}^*)$, so near-singular draws are not rare when $\hat{\vartheta}$ is close to 1. □

**Closed-form alternative.** Because $b_{\mathrm{RSS}}$ and $\mathbf{V}_0$ are explicit, R may replace the Monte Carlo by $\mathrm{Corr}^{\mathrm{cf}} = b_{\mathrm{RSS}}(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}})$ and $\mathbf{V}_{\mathrm{mech}}^{\mathrm{cf}} = \mathbf{V}_0(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}})$. These have the same first-order guarantees as the truncated bootstrap, no Monte Carlo error, and no moment problem. They are post-processing, so they cost no privacy.

**Numerical checks** (`s3_s6_sim.py`, fixed design and data, $2 \times 10^4$ stage-2 releases for the "truth", 200 independent stage-2 releases each with $B = 400$ bootstrap draws). Table 4 shows that $b_{\mathrm{RSS}}$ matches the Monte Carlo bias within 2% when $\sigma_2\,\mathrm{tr}\,\mathbf{P}_\lambda \le 0.18$ and within 5% at 0.59. $\mathbf{V}_0$ and $\mathbf{V}_{\mathrm{mech}}$ match the true mechanism standard deviations within 2% in the realistic scenarios. The spread of the bootstrap standard errors across releases (3.5–4%) is essentially the Monte Carlo error $1/\sqrt{2B} = 3.5\%$. Antithetic pairs reduce the Monte Carlo s.d. of Corr by a factor of 5–100 at the same cost. In the stress scenario ($\vartheta_0 = 2.52$, outside the default design) $\mathbf{V}_0$ is 2–6% too small, the bootstrap is 3–10% too large, and Corr overshoots the bias by 12%. The corrected RSS still removes 87% of the bias.

# 9 S7: the exact stage-1 radius and the transfer to the protocol

The exact radius of settings I and P replaces the union bound of Lemma S2(c) by a Monte Carlo quantile of $\lambda_{\max}$ of the unit stage-1 noise matrix. Let $F_{d_A,d_O}$ be the distribution function of $\lambda_{\max}(\mathbf{E})$ for a masked symmetric noise matrix of unit scale with blocks $(d_A, d_O)$, and $q_{1-\alpha} = F^{-1}_{d_A,d_O}(1-\alpha)$. Draw $M$ independent copies, sort their largest eigenvalues $L_{(1)} \le \cdots \le L_{(M)}$, let $j$ be the smallest integer with $\Pr[\mathrm{Bin}(M, 1-\alpha) \le j] \ge c$, and set $\hat{q} = L_{(j+1)}$, $r_1^{\mathrm{ex}} = \sigma_1 \hat{q}$ (the implementation's rule, with $c = 0.99$ and $M = \max\{2 \cdot 10^5, \lceil 2500/\alpha \rceil\}$). For release $s$ let

$$\bar{w}_s = \frac{N}{2\sqrt{2}\,\Omega_f} + \sqrt{2\pi}\,2^{-\kappa}\sigma_s + \bar{d}_s, \qquad \pi_s = \delta_B + N\,\mathrm{TV}_1(\sigma_{c,s}) + 2Ne^{-t_\eta^2/2},$$

with $\bar{d}_s$ the proved bound on the decoding offset (crypto proofs, Part A, Proposition on the decoding offset; it includes the rounding of the centre) and $\mathrm{TV}_1$ bounded by Lemma S0(b).

**Proposition 9.1** (S7: exact stage-1 radius)**.** *(a) The law of $\lambda_{\max}(\mathbf{E}_1)/\sigma_1$ is $F_{d_A,d_O}$, which depends only on $(d_A, d_O)$ and is continuous.*

*(b) Over the Monte Carlo draws, $\Pr[\hat{q} \ge q_{1-\alpha}] \ge c$.*

*(c) If the draws are independent of the release, $\Pr[\lambda_{\max}(\mathbf{E}_1) > r_1^{\mathrm{ex}}] = (M-j)/(M+1)$, at most $\alpha$ whenever $j+1 \ge (1-\alpha)(M+1)$.*

*(d) Under the protocol, (c) holds with $r_1^{\mathrm{ex}}$ enlarged by $\sqrt{2}\,\bar{w}_1$ and the probability enlarged by $\pi_1$; so do Lemma S2(c) and the radius event of Lemma S3.*

*Proof.* (a) $\mathbf{E}_1 = \sigma_1 \mathbf{E}$ with $\mathbf{E}$ of unit scale, whose law is fixed by the mask. $\lambda_{\max}(\mathbf{E}) \ge 0$ by interlacing (the R block is zero). If $\mathbf{C} \ne 0$, pick $\mathbf{a}, \mathbf{v}$ with $\mathbf{a}^\top \mathbf{C} \mathbf{v} > 0$; the vector $(\mathbf{a}, t\mathbf{v})$ gives $2t\mathbf{a}^\top \mathbf{C}\mathbf{v} + t^2 \mathbf{v}^\top \mathbf{B}\mathbf{v} > 0$ for small $t > 0$; so $\Pr[\lambda_{\max}(\mathbf{E}) = 0] \le \Pr[\mathbf{C} = 0] = 0$. For $t > 0$, $\{\lambda_{\max} \le t\}$ is a convex set of the Gaussian coordinates with non-empty interior (it contains a neighbourhood of $\mathbf{C} = 0$, $\mathbf{B} = -\mathbf{I}$), and $\{\lambda_{\max} = t\}$ lies in its boundary, which has Lebesgue measure zero. (b) $F(q_{1-\alpha}) = 1-\alpha$ by continuity, the number $K$ of draws below $q_{1-\alpha}$ is $\mathrm{Bin}(M, 1-\alpha)$, and $\hat{q} < q_{1-\alpha}$ iff $K \ge j+1$, of probability $1 - \Pr[K \le j] \le 1-c$. (c) $F(L_i)$ are independent uniforms, so $F(\hat{q})$ is $\mathrm{Beta}(j+1, M-j)$ with mean $(j+1)/(M+1)$; conditionally on the draws the exceedance probability is $1 - F(\hat{q})$. (d) Lemma S0(b),(c) couple each release with Gaussian slot noise; on the coupling event (probability $\ge 1 - \pi_s$: coupling, certificate, no wraparound) the decoded entries differ from $\mathbf{z}_s + \boldsymbol{\xi}_s^G$ by a vector of norm at most $\bar{w}_s$ (the rounding of $\mathbf{Z}$, the flooded error, and the decoding offset), and the symmetric matrix assembled from it has operator norm at most $\sqrt{2}\,\bar{w}_s$; Weyl's inequality gives the enlargement. □

At $p = 11$, $d_A = 5$, $\alpha = 0.0025$: $M = 10^6$, $j = 997{,}615$, $\hat{q} = 7.415$ against 11.529 for the union bound, and the exact level is $(M-j)/(M+1) = 0.002385$. At the evaluation setting $N\,\mathrm{TV}_1(\sigma_{c,s}) \le 2^{-122.2}$ ($\approx 10^{-37}$) and $\bar{w}_s$ is between $2 \cdot 10^{-8}$ ($\bar{n}_O = 4{,}096$) and $5 \cdot 10^{-6}$ ($\bar{n}_O = 10^6$, every row matchable), against $\sigma_1 \ge 128$. With the fixed public seed used in practice $\hat{q}$ is one number; (b) says that the procedure producing it errs with probability at most $1-c$, and (c) is a statement about the procedure with fresh draws. Parts (a)–(c) and the structure of (d) follow the companion book (Chapter 13, Proposition `inf.r1exact`); the book's caveat on (d), a numerically checked total-variation constant, is removed by Lemma S0(b) below, and its offset bound is proved in Part A.

# 10 S8: where the ridge switches off

Write $g_1^\star = g^\star + r_1$. The stage-1 rule returns $\lambda = 0$ iff $\lambda_{\min}(\hat{\mathbf{G}}_1) \ge g_1^\star$: then $\lambda_I = \lambda_O = 0$; conversely $\lambda_I > 0$ forces $\lambda_O > 0$, since $\lambda_O = 0$ means $\lambda_{\min}(\hat{\mathbf{G}}_1) \ge g_1^\star$.

**Proposition 10.1** (S8: a bracket for the switch-off)**.** *Condition on the data, let $\alpha' \in (0,1)$, and let $q'_{1-\alpha'}$ be the $(1-\alpha')$ quantile of $\lambda_{\max}$ of the unit stage-1 noise matrix (or its union bound $2\sqrt{p} + 2\sqrt{\ln(1/\alpha')}$). Under Assumption 2.1:*

*(a) if $\lambda_{\min}(\mathbf{G}) \geq g_1^\star + \sigma_1 q'_{1-\alpha'}$, the rule returns $\lambda = 0$ with probability at least $1-\alpha'$;*

*(b) if $\lambda_{\min}(\mathbf{G}) \leq g_1^\star - \sqrt{2}\,\sigma_1 z_{1-\alpha'}$, it returns $\lambda > 0$ with probability at least $1-\alpha'$;*

*(c) the threshold in (a) cannot be lowered in terms of $\lambda_{\min}(\mathbf{G})$ alone: for $\mathbf{G} = \lambda_{\min}(\mathbf{G})\mathbf{I}$ the rule returns $\lambda = 0$ with probability exactly $F_{d_A,d_O}((\lambda_{\min}(\mathbf{G}) - g_1^\star)/\sigma_1)$.*

*Under the protocol the thresholds move by $\sqrt{2}\,\bar{w}_1$ and the probabilities by $\pi_1$ (Proposition 9.1(d)).*

*Proof.* (a) Weyl: $\lambda_{\min}(\hat{\mathbf{G}}_1) \geq \lambda_{\min}(\mathbf{G}) - \lambda_{\max}(-\mathbf{E}_1)$, and $\lambda_{\max}(-\mathbf{E}_1)$ has the law of $\sigma_1\lambda_{\max}(\mathbf{E})$. (b) For a unit bottom eigenvector $\mathbf{u}$ of $\mathbf{G}$, $\lambda_{\min}(\hat{\mathbf{G}}_1) \leq \lambda_{\min}(\mathbf{G}) + \mathbf{u}^\top\mathbf{E}_1\mathbf{u}$, a centred Gaussian with variance $\sigma_1^2(\sum_{i\in O} u_i^4 + 4\sum_{i<j,\chi_{ij}=1} u_i^2u_j^2) \leq 2\sigma_1^2$. (c) $\lambda_{\min}(c\mathbf{I} + \mathbf{E}_1) = c - \lambda_{\max}(-\mathbf{E}_1)$. □

With $\lambda_{\min}(\mathbf{G}) \approx n\ell$ the switch-off lies between $(1 - \sqrt{2}\sigma_1 z_{1-\alpha'}/g_1^\star)n_{\text{off}}$ and $(1 + \sigma_1 q'_{1-\alpha'}/g_1^\star)n_{\text{off}}$; on the simulation population at $\alpha' = 0.05$ ($q'_{0.95} = 6.303$, $\sigma_2/\sigma_1 = 0.645$) this is $[0.87, 1.35]\,n_{\text{off}}$ under setting I and $[0.79, 1.58]\,n_{\text{off}}$ under setting P, at every $\varepsilon$.

**Lemma 10.2** (S8′: the size of the downward bias of $\lambda_{\min}$)**.** *For fixed symmetric $\mathbf{G}$ and a masked symmetric noise matrix $\mathbf{E}$ of scale $\sigma$ (any variances $\leq \sigma^2$ off the diagonal and $\leq 2\sigma^2$ on it), with $\mathbf{g} \sim \mathcal{N}(0, \mathbf{I}_p)$:*

*(a) $\mathbb{E}\lambda_{\min}(\mathbf{G}+\mathbf{E}) \leq \lambda_{\min}(\mathbf{G})$;*

*(b) $\mathbb{E}\lambda_{\min}(\mathbf{G}+\mathbf{E}) \geq m(\mathbf{G},\sigma) := \mathbb{E}\min_{\|\mathbf{u}\|=1}(\mathbf{u}^\top\mathbf{G}\mathbf{u} - 2\sigma\mathbf{g}^\top\mathbf{u}) \geq \lambda_{\min}(\mathbf{G}) - 2\sigma\mathbb{E}\|\mathbf{g}\|$; $m(\mathbf{G},\sigma)$ depends only on the eigenvalues $\mu_1 \leq \cdots \leq \mu_p$ of $\mathbf{G}$ and equals $\mathbb{E}[\theta - \sum_k \sigma^2 g_k^2/(\mu_k - \theta)]$, with $\theta < \mu_1$ the root of $\sum_k \sigma^2 g_k^2/(\mu_k - \theta)^2 = 1$;*

*(c) $\Pr[\lambda_{\min}(\mathbf{G}+\mathbf{E}) \leq \mathbb{E}\lambda_{\min}(\mathbf{G}+\mathbf{E}) - t] \leq e^{-t^2/(4\sigma^2)}$ for $t \geq 0$.*

*Proof.* (a) $\lambda_{\min}(\mathbf{G}+\mathbf{E}) \leq \mathbf{u}^\top\mathbf{G}\mathbf{u} + \mathbf{u}^\top\mathbf{E}\mathbf{u}$ for a bottom eigenvector $\mathbf{u}$ of $\mathbf{G}$, and $\mathbb{E}\mathbf{u}^\top\mathbf{E}\mathbf{u} = 0$. (b) $-\lambda_{\min}(\mathbf{G}+\mathbf{E}) = \sup_{\mathbf{u}}(-\mathbf{u}^\top\mathbf{G}\mathbf{u} - X_{\mathbf{u}})$ with $X_{\mathbf{u}} = \mathbf{u}^\top\mathbf{E}\mathbf{u}$. The processes $(-\mathbf{u}^\top\mathbf{G}\mathbf{u} - X_{\mathbf{u}})$ and $(-\mathbf{u}^\top\mathbf{G}\mathbf{u} + Y_{\mathbf{u}})$, $Y_{\mathbf{u}} = 2\sigma\mathbf{g}^\top\mathbf{u}$, are Gaussian with equal means, and by the proof of Lemma S2(a) $\mathbb{E}(X_{\mathbf{u}} - X_{\mathbf{v}})^2 \leq \mathbb{E}(Y_{\mathbf{u}} - Y_{\mathbf{v}})^2$. The Sudakov–Fernique inequality holds for Gaussian vectors with equal means (its interpolation proof uses only the covariances; Chatterjee [7]), and it extends to the sphere by continuity; so $\mathbb{E}[-\lambda_{\min}(\mathbf{G}+\mathbf{E})] \leq \mathbb{E}\sup_{\mathbf{u}}(-\mathbf{u}^\top\mathbf{G}\mathbf{u} + 2\sigma\mathbf{g}^\top\mathbf{u})$, which is (b). The minimum over the sphere of $\mathbf{u}^\top\mathbf{G}\mathbf{u} - 2\mathbf{h}^\top\mathbf{u}$ is a trust-region problem whose value is $\theta - \mathbf{h}^\top(\mathbf{G} - \theta\mathbf{I})^{-1}\mathbf{h}$ at the root $\theta < \mu_1$ of $\|(\mathbf{G} - \theta\mathbf{I})^{-1}\mathbf{h}\| = 1$ (the hard case has probability zero), and $\mathbf{g}$ is rotation invariant. The last inequality takes $\mathbf{u}^\top\mathbf{G}\mathbf{u} \geq \lambda_{\min}(\mathbf{G})$. (c) $\lambda_{\min}(\mathbf{G}+\mathbf{E})$ is an infimum of affine functions of the Gaussian coordinates with Lipschitz constants at most $\sqrt{2}\sigma$ (proof of Lemma S2(b)); Gaussian concentration. □

**Remark 10.3** (What the lemma gives, and what remains open)**.** On the simulation population ($\Sigma$ of the clipped, standardized design, eigenvalues $0.599, 0.621, 0.680, 0.754, \ldots$, $p = 11$), at $n = n_{\text{off}}$ under setting I and $\varepsilon = 1$, (b) bounds the downward bias by $3.19\,\sigma_1$, against $1.70\,\sigma_1$ in simulation, and $2\sigma_1\mathbb{E}\|\mathbf{g}\| \approx 6.5\,\sigma_1$ for $\mathbf{G} \propto \mathbf{I}$ (script `experiments/proofs_check/lmin_bias_bound.py`). Combined with (c), it guarantees the switch-off with probability 0.95 from $1.4\,n_{\text{off}}$ under setting I (above $1.5\,n_{\text{off}}$ under P), which is weaker than the bracket's 1.35 (and 1.58). By Proposition 10.1(c) no bound in terms of $\lambda_{\min}(\mathbf{G})$ alone can do better than the bracket; the empirical factors 1.2 (I) and 1.4 (P) reflect the spread of the population's spectrum. Turning them into bounds needs a tail bound for $\lambda_{\min}(\mathbf{G}+\mathbf{E})$ that uses the spectrum and is sharp to within much less than the factor 1.9 that the comparison of (b) loses: second-order perturbation, $\lambda_{\min}(\mathbf{G}+\mathbf{E}) \approx \mu_1 + \mathbf{u}_1^\top\mathbf{E}\mathbf{u}_1 - \sum_{k\geq 2}(\mathbf{u}_k^\top\mathbf{E}\mathbf{u}_1)^2/(\mu_k - \mu_1)$, needs gaps $\mu_k - \mu_1$ above about $2\|\mathbf{E}\|$, and at $n_{\text{off}}$ the three smallest eigenvalues of $n\Sigma$ lie within $0.1\,n\ell \approx 2.4\,\sigma_1$ of each other. This remains [open].

# 11 S9: mechanism bias with explicit constants

Fix $(\lambda, \Psi)$ independently of the stage-2 noise, write $\mathbf{G}$ for $\mathbf{G}_\lambda$, $\mathbf{P} = \mathbf{G}^{-1}$, $\sigma = \sigma_2$, $\vartheta = \lambda_{\min}(\mathbf{G})/(2\sigma\sqrt{p})$, $\mathbf{H} = \mathbf{PE}$, $\mathbf{M} = \mathbf{G}^{-1/2}\mathbf{E}\mathbf{G}^{-1/2}$, $q = \|\mathbf{M}\|_{\mathrm{op}}$, and for $\bar\eta \in (0,1)$ let $T = \{q \le \bar\eta\}$; $T$ is invariant under $\mathbf{E} \mapsto -\mathbf{E}$. Let $\kappa_p = (2\sqrt{p} + c_0 + 2\sqrt{2})/(2\sqrt{p})$ ($\kappa_p = 1.81$ at $p = 11$), so that $(\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}}^4)^{1/4} \le 2\sigma\sqrt{p}\,\kappa_p$ (Lemma S2(d)) and $\mathbb{E}q^4 \le \kappa_p^4\vartheta^{-4}$. The bias matrix of the paper is $\mathbf{PM}_\chi(\mathbf{P})$, with $\mathbf{M}_\chi$ the contraction of Section 2.

**Theorem 11.1** (S9: leading-order conditional bias, explicit remainder)**.** *If $\bar\eta\vartheta > 1$, then $\mathbb{E}[\tilde{\boldsymbol\theta} \mid T]$ exists and $\mathbb{E}[\tilde{\boldsymbol\theta} \mid T] - \boldsymbol\theta_\lambda = \sigma^2\mathbf{PM}_\chi(\mathbf{P})\boldsymbol\theta_\lambda + \boldsymbol\rho$ with*

$$\|\boldsymbol\rho\|_{\mathbf{G}} \le \frac{1}{\Pr(T)}\Big[\frac{\kappa_p^4}{(1-\bar\eta)\vartheta^4}\big(\|\boldsymbol\theta_\lambda\|_{\mathbf{G}} + \sigma\sqrt{\operatorname{tr}\mathbf{P}}\big) + \frac{2\sqrt{2}\,\kappa_p^2}{\vartheta^2}e^{-p(\bar\eta\vartheta-1)^2/2}\|\boldsymbol\theta_\lambda\|_{\mathbf{G}}\Big],$$

*where $\Pr(T^c) \le 2e^{-p(\bar\eta\vartheta-1)^2}$, and $\|\sigma^2\mathbf{PM}_\chi(\mathbf{P})\boldsymbol\theta_\lambda\|_{\mathbf{G}} \le \kappa_p^2\vartheta^{-2}\|\boldsymbol\theta_\lambda\|_{\mathbf{G}}$. The moment noise $\mathbf{f}$ contributes nothing at order $\sigma^2$. For $\Psi = \mathbf{I}$, $\|\sigma^2\mathbf{PM}_\chi(\mathbf{P})\boldsymbol\theta_\lambda\| \le (3 + p + d_O)\sigma^2\|\mathbf{c}\|/(\lambda_{\min}(\mathbf{G}) + \lambda)^3$ with $\mathbf{G}$ the unpenalized Gram; for $\Psi = \Pi_O$ and $\mathbf{A}$ invertible it is $O(\sigma^2/\lambda)$.*

*Proof. Expansion.* On $T$, $(\mathbf{I}+\mathbf{H})^{-1} = \mathbf{I} - \mathbf{H} + \mathbf{H}^2 - \mathbf{H}^3 + \mathbf{H}^4(\mathbf{I}+\mathbf{H})^{-1}$, so $\tilde{\boldsymbol\theta} = (\mathbf{I} - \mathbf{H} + \mathbf{H}^2 - \mathbf{H}^3)(\boldsymbol\theta_\lambda + \mathbf{Pf}) + \mathbf{R}_4$ with $\|\mathbf{R}_4\|_{\mathbf{G}} \le \frac{q^4}{1-q}(\|\boldsymbol\theta_\lambda\|_{\mathbf{G}} + \|\mathbf{G}^{-1/2}\mathbf{f}\|)$, because $\mathbf{G}^{1/2}\mathbf{H}^k\mathbf{G}^{-1/2} = \mathbf{M}^k$; and $\|\tilde{\boldsymbol\theta}\|_{\mathbf{G}} \le (1-\bar\eta)^{-1}\|\mathbf{G}^{-1/2}(\mathbf{c}+\mathbf{f})\|$ on $T$, which is integrable. *Odd terms.* $\mathbb{E}[\mathbf{H}^m \mid T] = 0$ for odd $m$ by symmetry, and every term containing $\mathbf{Pf}$ has zero conditional mean ($\mathbf{f}$ is centred and independent of $\mathbf{E}$). *Remainder.* $\mathbb{E}[q^4\mathbf{1}_T] \le \kappa_p^4\vartheta^{-4}$, and $\mathbb{E}\|\mathbf{G}^{-1/2}\mathbf{f}\| \le \sigma\sqrt{\operatorname{tr}\mathbf{P}}$, independently of $q$. *Conditioning.* $\mathbb{E}[\mathbf{H}^2 \mid T] - \mathbb{E}\mathbf{H}^2 = (\mathbb{E}[\mathbf{H}^2]\Pr(T^c) - \mathbb{E}[\mathbf{H}^2\mathbf{1}_{T^c}])/\Pr(T)$, of $\mathbf{G}$-operator norm at most $2(\mathbb{E}q^4)^{1/2}\Pr(T^c)^{1/2}/\Pr(T)$ by Cauchy–Schwarz; $\Pr(T^c) \le \Pr[\|\mathbf{E}\|_{\mathrm{op}} > 2\bar\eta\vartheta\sigma\sqrt{p}] \le 2e^{-p(\bar\eta\vartheta-1)^2}$ by Lemma S2(c). *Contraction.* $\mathbb{E}\mathbf{H}^2 = \mathbf{P}\mathbb{E}[\mathbf{EPE}] = \sigma^2\mathbf{PM}_\chi(\mathbf{P})$, of $\mathbf{G}$-norm at most $\mathbb{E}q^2 \le (\mathbb{E}q^4)^{1/2}$. *Decay.* $\mathbf{M}_\chi(\mathbf{P}) = \Pi_O\mathbf{P} + \Pi_R\mathbf{P}\Pi_O + \operatorname{tr}(\mathbf{P})\Pi_O + \operatorname{tr}(\Pi_O\mathbf{P})\Pi_R - \Pi_O\operatorname{diag}(\mathbf{P})$, with term norms at most $\|\mathbf{P}\|$, $\|\mathbf{P}\|$, $p\|\mathbf{P}\|$, $d_O\|\mathbf{P}\|$, $\|\mathbf{P}\|$; with $\|\mathbf{P}_\lambda\| = 1/(\lambda_{\min}(\mathbf{G}) + \lambda)$ and $\|\boldsymbol\theta_\lambda\| \le \|\mathbf{P}_\lambda\|\|\mathbf{c}\|$ for $\Psi = \mathbf{I}$. For $\Psi = \Pi_O$ the $\mathsf{O}$ rows of $\mathbf{P}_\lambda$ are $(\mathbf{S}+\lambda\mathbf{I})^{-1}[-\mathbf{C}^\top\mathbf{A}^{-1},\ \mathbf{I}]$, of norm at most $(1 + \|\mathbf{A}^{-1}\mathbf{C}\|)/(\lambda_{\min}(\mathbf{S}) + \lambda)$, and every term of $\mathbf{M}_\chi(\mathbf{P}_\lambda)\boldsymbol\theta_\lambda$ contains a factor $\Pi_O\mathbf{P}_\lambda$ or $\mathbf{P}_\lambda\Pi_O$. □

This is the paper's Theorem on the leading-order bias and its ridge version, with the constants made explicit; it follows the companion book (Chapter 14, Theorems `bias.main` and `bias.ridge`, Proposition `bias.shrink`), whose proofs were checked step by step, including $c_0 = 2.512$ and the fourth moment $64\sigma^4$ of the centred norm. At $p = 11$, $\vartheta = 5$ and $\bar\eta = \frac{1}{2}$ the leading term is at most $0.13\|\boldsymbol\theta_\lambda\|_{\mathbf{G}}$ and the remainder at most $0.034(\|\boldsymbol\theta_\lambda\|_{\mathbf{G}} + \sigma\sqrt{\operatorname{tr}\mathbf{P}})$ (worst direction); in simulation the leading term matches the bias to within 2%. The correction $\tilde{\boldsymbol\theta} - \sigma^2\tilde{\mathbf{P}}\mathbf{M}_\chi(\tilde{\mathbf{P}})\tilde{\boldsymbol\theta}$ leaves a bias $O(\vartheta^{-4})$ and a fluctuation $O_p(\vartheta^{-3})$ relative to $\|\boldsymbol\theta_\lambda\|_{\mathbf{G}}$ (book, Proposition `bias.corr`; checked, with its constant not explicit).

# 12 S10: the FWS residual variance, binding clipping, and the full-record sensitivity

## 12.1 The plug-in residual variance of the FWS comparison

The paper's FWS comparison uses $\hat s^2 = \widehat{\mathrm{RSS}}/(n-p)$ with $\widehat{\mathrm{RSS}} = \widehat{yy} - \hat{\mathbf{c}}_2^\top\hat{\mathbf{G}}_2^{-1}\hat{\mathbf{c}}_2$, the case $\lambda = 0$ of Lemma S6. Let $\vartheta, T, \kappa_p$ be as in Section 11 with $\lambda = 0$, and $\hat{\boldsymbol\beta} = \mathbf{Pc}$.

**Proposition 12.1** (S10a: bias of the plug-in RSS)**.** *If $\bar\eta\vartheta > 1$, then* $\mathbb{E}[\widehat{\mathrm{RSS}} \mid T] - \mathrm{RSS} = -\sigma^2\big(\operatorname{tr}\mathbf{P} + \hat{\boldsymbol\beta}^\top \mathbf{M}_\chi(\mathbf{P})\hat{\boldsymbol\beta}\big) + \rho$ *with*

$$|\rho| \le \frac{1}{\Pr(T)}\Big[\frac{\kappa_p^4}{(1-\bar\eta)\vartheta^4}\big(\|\hat{\boldsymbol\beta}\|_{\mathbf{G}}^2 + \sigma^2 \operatorname{tr}\mathbf{P}\big) + \frac{\kappa_p^2}{\vartheta^2}\sigma^2 \operatorname{tr}\mathbf{P} + \frac{2\sqrt{2}\,\kappa_p^2}{\vartheta^2} e^{-p(\bar\eta\vartheta-1)^2/2}\|\hat{\boldsymbol\beta}\|_{\mathbf{G}}^2\Big].$$

*The leading term is negative: $\hat s^2$ underestimates the residual variance at order $\sigma_2^2$.*

*Proof.* With $\boldsymbol\phi = \mathbf{G}^{1/2}\hat{\boldsymbol\beta}$ and $\mathbf{h} = \mathbf{G}^{-1/2}\mathbf{f}$, $\widehat{\mathrm{RSS}} - \mathrm{RSS} = g - [(\boldsymbol\phi+\mathbf{h})^\top(\mathbf{I}+\mathbf{M})^{-1}(\boldsymbol\phi+\mathbf{h}) - \|\boldsymbol\phi\|^2]$. Expand $(\mathbf{I}+\mathbf{M})^{-1}$ to $\mathbf{M}^4(\mathbf{I}+\mathbf{M})^{-1}$; on the symmetric event $T$ the terms odd in $\mathbf{M}$ vanish, and so do the terms linear in $\mathbf{h}$; $\mathbb{E}\|\boldsymbol\phi+\mathbf{h}\|^2 = \|\boldsymbol\phi\|^2 + \sigma^2\operatorname{tr}\mathbf{P}$; $\boldsymbol\phi^\top\mathbb{E}[\mathbf{M}^2]\boldsymbol\phi = \sigma^2\hat{\boldsymbol\beta}^\top\mathbf{M}_\chi(\mathbf{P})\hat{\boldsymbol\beta}$. The remainder collects $\boldsymbol\phi^\top(\mathbb{E}[\mathbf{M}^2 \mid T] - \mathbb{E}\mathbf{M}^2)\boldsymbol\phi$ (bounded as in Theorem 11.1), $\mathbb{E}[\mathbf{h}^\top\mathbf{M}^2\mathbf{h} \mid T] \le \mathbb{E}[q^2]\sigma^2\operatorname{tr}\mathbf{P}/\Pr(T)$, and the quartic term, at most $\mathbb{E}[q^4\mathbf{1}_T](\|\boldsymbol\phi\|^2 + \sigma^2\operatorname{tr}\mathbf{P})/((1-\bar\eta)\Pr(T))$. $\mathbf{M}_\chi(\mathbf{P}) \succeq 0$ because it is $\sigma^{-2}\mathbb{E}[\mathbf{EPE}]$ with $\mathbf{P} \succ 0$. □

A Monte Carlo check with antithetic pairs ($p = 11$, $\vartheta = 7.0$, 3.5, 1.8) matches the leading term to 0.05–0.65% (`experiments/proofs_check/fws_s2_bias.py`). Rerunning the paper's FWS comparison on the synthetic population (setting I, the same cohorts, releases and FWS random numbers, 200 cohorts per cell) with $\widehat{\mathrm{RSS}}$ corrected by the leading term changes the coverage of the ten slopes by at most 0.0005 in each of the 41 ridge-on cells, and the median half-width by at most 3%: where the ridge is on, the mechanism noise dominates FWS's width, and $\hat s^2$ is floored at 0.1 in up to 49% of releases (`experiments/proofs_check/fws_s2_corrected.py`).

## 12.2 Binding clipping and the bias-aware interval

When clipping binds, the clipped response is heteroskedastic and $\mathbb{E}[\tilde{\mathbf{y}} \mid \mathbf{X}]$ need not be linear. Take as estimand the projection coefficient $\boldsymbol\beta^* = \mathbf{G}^{-1}\mathbf{X}^\top\mathbb{E}[\tilde{\mathbf{y}} \mid \mathbf{X}]$ of the clipped data (fixed design), with the $\tilde y_i$ conditionally independent given $\mathbf{X}$, and let $\bar\sigma^2 \ge \max_i \operatorname{Var}(\tilde y_i \mid \mathbf{X})$.

**Proposition 12.2** (S10b: a valid interval under binding clipping)**.** $\mathbb{E}[\boldsymbol\theta_\lambda \mid \mathbf{X}] - \boldsymbol\beta^* = -\lambda\mathbf{P}_\lambda\Psi\boldsymbol\beta^*$, $\operatorname{Cov}(\boldsymbol\theta_\lambda \mid \mathbf{X}) \preceq \bar\sigma^2\mathbf{P}_\lambda\mathbf{G}\mathbf{P}_\lambda$*, and the norm bound of Lemma S4 holds for $\boldsymbol\beta^*$. Consequently Theorem S5(b) holds for $\boldsymbol\beta^*$, with the same exact and first-order parts, when $\tilde s^2$ is replaced by $\bar\sigma^2$ in $\mathbf{V} = \mathbf{V}_{\mathrm{mech}} + \tilde s^2\tilde{\mathbf{P}}\hat{\mathbf{G}}_2\tilde{\mathbf{P}}$. Admissible public choices are: $\bar\sigma^2 = B_y^2$, with no model; and, when only the response is clipped and $y_i = \mathbf{x}_i^\top\boldsymbol\beta + \varepsilon_i$ before clipping with $\operatorname{Var}(\varepsilon_i \mid \mathbf{X}) \le \sigma_0^2$, any public $\bar\sigma^2 \ge \sigma_0^2$.*

*Proof.* $\mathbb{E}[\mathbf{c} \mid \mathbf{X}] = \mathbf{X}^\top\mathbb{E}[\tilde{\mathbf{y}} \mid \mathbf{X}] = \mathbf{G}\boldsymbol\beta^*$, so $\mathbb{E}[\boldsymbol\theta_\lambda \mid \mathbf{X}] = \mathbf{P}_\lambda\mathbf{G}\boldsymbol\beta^* = \boldsymbol\beta^* - \lambda\mathbf{P}_\lambda\Psi\boldsymbol\beta^*$; $\operatorname{Cov}(\boldsymbol\theta_\lambda \mid \mathbf{X}) = \mathbf{P}_\lambda\mathbf{X}^\top\boldsymbol\Omega\mathbf{X}\mathbf{P}_\lambda$ with $\boldsymbol\Omega = \operatorname{diag}(\operatorname{Var}(\tilde y_i \mid \mathbf{X})) \preceq \bar\sigma^2\mathbf{I}$. Lemma S4 uses no model. The interval is $\tilde\theta_j \pm \operatorname{cv}_\alpha(\bar b_j/\sqrt{V_{jj}})\sqrt{V_{jj}}$; if the true standard deviation is $s \le s' = \sqrt{V_{jj}}$ and $|b| \le \bar b_j$, then $c = \operatorname{cv}_\alpha(\bar b_j/s')s' > \bar b_j \ge |b|$ (since $\operatorname{cv}_\alpha(t) > t$ for $\alpha < \frac12$), and $s \mapsto \Pr[|sZ+b| \le c] = \Phi((c-b)/s) - \Phi((-c-b)/s)$ has derivative $-[(c-b)\phi((c-b)/s) + (c+b)\phi((c+b)/s)]/s^2 < 0$; so coverage at $s$ is at least coverage at $s'$, which is the guarantee of Theorem S5(b). $|\tilde y_i| \le B_y$ gives $\operatorname{Var}(\tilde y_i \mid \mathbf{X}) \le B_y^2$. Clipping is 1-Lipschitz, and $\operatorname{Var} f(Y) = \frac12\mathbb{E}(f(Y) - f(Y'))^2 \le \frac12\mathbb{E}(Y-Y')^2 = \operatorname{Var} Y$ for 1-Lipschitz $f$ and an independent copy $Y'$, so $\operatorname{Var}(\operatorname{clip}(y_i) \mid \mathbf{X}) \le \sigma_0^2$. □

What remains [open] is a guarantee with the data-driven $\tilde s^2$: it estimates an average of the $\operatorname{Var}(\tilde y_i \mid \mathbf{X})$ plus a misspecification term, not their maximum, and a sandwich variance would need per-record residuals of O's data, that is, a further release. The public bound $B_y^2$ is conservative (on the synthetic population $B_y^2 = 8.35$ against a clipped residual variance of about 0.5, which widens the sampling part by about 4.1).

## 12.3 The full-record sensitivity in closed form

The paper's full-record baseline releases the upper triangle of $\sum_i \mathbf{w}_i\mathbf{w}_i^\top$, $\mathbf{w} = (\dot{\mathbf{x}}, \mathbf{x}_O, y)$, $\dot{\mathbf{x}} = (1, \mathbf{u})$ with an intercept ($\iota = 1$; $\dot{\mathbf{x}} = \mathbf{u}$ and $\iota = 0$ without), every entry with the same noise. Let $R_u = B_R^2 - \iota$, $o = B_O^2$, $w = B_y^2$, and $m_1 = \min(R_u, o)$, $m_2 = \max(R_u, o)$. Define

$$H(a) = \begin{cases} \frac{2}{3}a^2, & |a| \le 3m_1, \\ \frac{1}{2}a^2 + |a|m_1 - \frac{3}{2}m_1^2, & 3m_1 < |a| \le m_1 + 2m_2, \\ 2|a|(m_1+m_2) - 2(m_1^2 + m_2^2 + m_1m_2), & |a| > m_1 + 2m_2, \end{cases}$$

$K = \iota(2R_u + 2o) + 2R_u^2 + 2o^2 + 2R_uo$, and, for $t \in [-w, w]$,

$$\Psi_2(t) = K + (\iota + R_u + o)\Big(w + \frac{t^2}{w}\Big) - 2\iota t + \Big(w - \frac{t^2}{w}\Big)^2 + H(\iota + t),$$

and $\Psi_2^{\neg yy}$ the same without the term $(w - t^2/w)^2$.

**Theorem 12.3** (S10c: full-record sensitivity)**.** *If* $\mathbf{u}$ *and* $\mathbf{x}_O$ *have dimension at least* 2*, the replace-one* $\ell_2$ *sensitivities of the full-record release are* $\Delta^2_{\text{full}} = \max_{|t|\le w} \Psi_2(t)$ *(joint mode),* $\Delta^2_{\text{full},\neg yy} = \max_{|t|\le w} \Psi_2^{\neg yy}(t)$ *(separate mode) and* $\Delta^2_{G,\text{full}} = K + H(\iota)$ *(stage 1).* $\Psi_2$ *is a piecewise quartic in* $t$*, so the maximum is attained at* $t = \pm w$*, at a point where* $|\iota + t| \in \{3m_1, m_1 + 2m_2\}$*, or at a real root of the cubic* $\Psi_2'$ *on a piece: a finite enumeration, as for Proposition S1. If a block has dimension* 1*, the same values are valid, conservative upper bounds.*

*Proof. The bracket.* For a record $\mathbf{w}$ and its replacement $\mathbf{w}'$, write $n_k, n_k'$ for the squared norms of the blocks $k \in \{\mathbf{u}, \mathbf{x}_O, y\}$ and $g_k$ for their inner products. The change of the first moments has squared norm $\iota\sum_k(n_k + n_k' - 2g_k)$; that of the off-diagonal blocks of $\mathbf{w}\mathbf{w}^\top - \mathbf{w}'\mathbf{w}'^\top$ has squared Frobenius norm $\sum_{k<l}(n_kn_l + n_k'n_l' - 2g_kg_l)$; and $\|\operatorname{triu}\mathbf{S}_k\|^2 \le \|\mathbf{S}_k\|_F^2 = n_k^2 + n_k'^2 - 2g_k^2$ for each diagonal block. The supremum of the sum (the paper's expression for $\Delta^2_{\text{full}}$) is attained: for a block of dimension at least 2 the two vectors can be chosen with any norms and inner product so that $\mathbf{S}_k$ is diagonal, and for $y$ the bound is exact with $g_y = yy'$.

*Norms at the bounds.* For fixed $g$ and $(y, y')$ every coefficient of $n_k, n_k'$ ($k = \mathbf{u}, \mathbf{x}_O$) is non-negative, and the admissible set $|g_k| \le \sqrt{n_kn_k'}$ grows with the norms; so the supremum has $n_{\mathbf{u}} = n_{\mathbf{u}}' = R_u$ and $n_O = n_O' = o$. Then, with $s = y^2 + y'^2$ and $t = yy'$, the objective is

$$K + \iota(s - 2t) + (R_u + o)s + (s^2 - 4t^2) + h(g_{\mathbf{u}}, g_O), \qquad h = -2(\iota + t)(g_{\mathbf{u}} + g_O) - 2(g_{\mathbf{u}}^2 + g_O^2 + g_{\mathbf{u}}g_O),$$

over $|g_{\mathbf{u}}| \le R_u$, $|g_O| \le o$ (the term $s^2 - 4t^2 = (y^2 - y'^2)^2$ is absent in separate mode, and every term in $y$ at stage 1).

*The inner maximum.* $h$ is a concave quadratic; its unconstrained maximizer is $g_{\mathbf{u}} = g_O = -a/3$, $a = \iota + t$, with value $\frac{2}{3}a^2$. If $|a| > 3m_1$, the coordinate with the smaller box is clipped at $-\operatorname{sgn}(a)m_1$ and the other is $-\operatorname{sgn}(a)(|a| - m_1)/2$, feasible while $|a| \le m_1 + 2m_2$, with value $\frac{1}{2}a^2 + |a|m_1 - \frac{3}{2}m_1^2$; beyond, both are clipped, with value $2|a|(m_1 + m_2) - 2(m_1^2 + m_2^2 + m_1m_2)$. Each candidate satisfies the KKT conditions (for the clipped coordinate the partial derivative $-2a - 4g_1 - 2g_2$ has the sign of $-a$ once $|a| > 3m_1$), so it is the maximum of the concave problem; this is $H(a)$, continuous at the breakpoints.

*The outer maximum.* For fixed $t$ the objective is strictly increasing in $s$ (its derivative is $\iota + R_u + o + 2s > 0$). Given $t$, the pairs $(y, y')$ with $yy' = t$ and $y^2, y'^2 \le w$ have $s \le w + t^2/w$, with equality iff one of $|y|, |y'|$ equals $B_y$; and $(w + t^2/w)^2 - 4t^2 = (w - t^2/w)^2$. So the supremum is $\max_{|t|\le w} \Psi_2(t)$. A configuration with a block of dimension 1 embeds in dimension 2 (collinear vectors), so its supremum is at most the value for dimension 2. □

At unit bounds without an intercept, $\Psi_2(t) = 9 + t^4 + \frac{2}{3}t^2$, so $\Delta^2_{\text{full}} = 32/3$ (at $t = \pm 1$) and $\Delta^2_{G,\text{full}} = 6$, the paper's values. On the simulation population the enumeration gives $\Delta_{\text{full}} = 92.670$ and $\Delta_{G,\text{full}} = 85.816$ (the paper's 92.7 and 85.8). Over 19 bound vectors, with and without an intercept, it agrees with the paper's grid-and-local-search computation (`experiments/baselines/sens_full.py`) to the relative $10^{-12}$ of that computation's safeguard, in all three modes; with a block of dimension 1 it is above the grid value, as it must be (`experiments/proofs_check/sens_full_closed.py`). Of the 29 benchmark datasets two (3droad and tamielectric, both with $d = 3$ and $d_R = 1$, so $\mathbf{u}$ has dimension 1) use the dimension-1 case, for which the theorem gives only a valid conservative bound. Lemma S10d below gives the exact value there.

### 12.4 The full-record sensitivity for a block of dimension one

Let $\mathbf{u}$ have dimension 1 and $\mathbf{x}_O$ dimension at least 2 (for the converse exchange $R_u$ and $o$ below: the bracket of Theorem 12.3 is symmetric in the two blocks). Write $r = \sqrt{R_u}$, $B = B_y$, $p = u$ and $q = u'$ for the two values of R's scalar feature, and put

$$G(a) = \begin{cases} \frac{1}{2}a^2, & |a| \le 2o, \\ 2o|a| - 2o^2, & |a| > 2o, \end{cases} \qquad a = \iota + pq + yy',$$

$$U(p,q,y,y') = \iota\big[(p-q)^2 + (y-y')^2\big] + 2\iota o + 2o^2 + o\big(p^2+q^2+y^2+y'^2\big) + (p^2-q^2)^2 + (py-qy')^2 + (y^2-y'^2)^2 + G(a)$$

on the box $\mathcal{B} = [-r,r]^2 \times [-B,B]^2$; in separate mode the term $(y^2 - y'^2)^2$ is dropped, and at stage 1 every term in $y, y'$ (with $a = \iota + pq$).

**Lemma 12.4** (S10d: exact full-record sensitivity, one block of dimension one)**.** *In each of the three modes $\Delta^2_{\text{full}} = \max_{\mathcal{B}} U$, and the maximum is attained on one of the faces*

$$F_1 = \{p = r,\ y = B\}, \qquad F_2 = \{q = r,\ y = B\}, \qquad F_3 = \{p = q = 0,\ y = B\},$$
$$F_4 = \{p = r,\ y = y' = 0\}, \qquad F_5 = \{0\}$$

*($F_4$ and $F_5$ at stage 1). On each face $a$ is affine in the free coordinates (at most two), the lines $a = \pm 2o$ cut the face into convex polygons on each of which $U$ is a polynomial, and $\max_{\mathcal{B}} U$ is the largest value of $U$ over the following finite set: the vertices of these polygons; the zeros of the derivative of $U$ along each of their edges in its relative interior (roots of a univariate polynomial); and the interior stationary points of each of the three polynomials, which are common zeros of two bivariate polynomials and are found from the real roots of their resultant. The set is finite whenever no resultant vanishes identically. The baseline code (`experiments/baselines/sens_full.py, sup2_dim1`) enumerates it with exact rational coefficients and exact real-root isolation, and refuses if a resultant vanishes identically.*

*Proof. Reduction to $U$.* For a block of dimension 1, $g = \pm\sqrt{nn'}$ and $\|\operatorname{triu}\mathbf{S}\|^2 = (n-n')^2 = n^2 + n'^2 - 2g^2$, so the bracket of Theorem 12.3 is exact for $\mathbf{u}$ and for $y$, with $g_{\mathbf{u}} = pq$, $g_y = yy'$, and the cross block of $\mathbf{u}$ and $y$ contributes $p^2y^2 + q^2y'^2 - 2pqyy' = (py - qy')^2$. For $\mathbf{x}_O$, of dimension at least 2, the bracket is attained (proof of Theorem 12.3); every coefficient of $n_O, n'_O$ is non-negative and the admissible set $|g_O| \le \sqrt{n_O n'_O}$ grows with them, so $n_O = n'_O = o$. The terms in $g_O$ are $-2g_O^2 - 2ag_O$, whose maximum over $|g_O| \le o$ is $G(a)$ (at $g_O = -a/2$ if $|a| \le 2o$, else at $g_O = -o\operatorname{sgn} a$). Collecting the remaining terms gives $U$, so $\Delta^2_{\text{full}} = \max_{\mathcal{B}} U$.

*Rays.* Fix $(y, y')$ and $(p_0, q_0) \neq 0$, and put $\phi(\mu) = U(\sqrt{\mu}p_0, \sqrt{\mu}q_0, y, y')$. The terms $\iota(p-q)^2$, $o(p^2+q^2)$ and $(py - qy')^2$ are linear in $\mu$, $(p^2-q^2)^2$ is $\mu^2(p_0^2-q_0^2)^2$, and $G(\iota + \mu p_0 q_0 + yy')$ is convex in $\mu$ because $G$ is convex and its argument affine. So $\phi$ is convex, and its maximum over the admissible $\mu$ is at $\mu = 0$ or where $\max(|p|, |q|) = r$. The same holds for $(y, y')$ with $(p, q)$ fixed: $(y^2 - y'^2)^2$ is quadratic in $\mu$ with a non-negative leading coefficient and every other term is linear in $\mu$ or $G$ of an affine function. Applying the two arguments in turn to a maximizer gives a maximizer with $[(p, q) = 0$ or $\max(|p|, |q|) = r]$ and $[(y, y') = 0$ or $\max(|y|, |y'|) = B]$.

*Symmetries.* $U$ is invariant under exchanging the two records, $(p, q, y, y') \mapsto (q, p, y', y)$, and under $(p, q) \mapsto (-p, -q)$ and $(y, y') \mapsto (-y, -y')$: each term, and $a$, is. If $(y, y') \neq 0$, exchange the records so that $|y| = B$ and flip $(y, y')$ so that $y = B$; then either $(p, q) = 0$ (face $F_3$) or one of $|p|, |q|$ is $r$, and flipping $(p, q)$ gives $F_1$ or $F_2$. If $(y, y') = 0$ and $(p, q) \neq 0$, exchange and flip to $p = r$ ($F_4$). Otherwise the point is $F_5$.

*Enumeration.* On $F_1$ and $F_2$, $a = \iota + rx + Bv$ with $x$ the free one of $p, q$ and $v = y'$; on $F_3$ and $F_4$, $a = \iota + Bv$ and $a = \iota + rq$. So the lines $a = \pm 2o$ cut each face into closed convex polygons $P$ on each of which $U = U_k$ for one $k \in \{\mathrm{mid}, \mathrm{hi}, \mathrm{lo}\}$, a polynomial. A maximizer of $U$ over the face lies in some $P$ and maximizes the polynomial $U_k$ over $P$. Then it is a vertex of $P$, or lies in the relative interior of an edge of $P$, where the derivative of $U_k$ along the edge vanishes (a univariate polynomial on a parametrization of the edge: a side of the box, or a segment of a line $a = \pm 2o$, along which $U_{\mathrm{mid}} = U_{\mathrm{hi}}$ or $U_{\mathrm{mid}} = U_{\mathrm{lo}}$ identically because $G$ is continuous), or lies in the interior of $P$, where $\partial_x U_k = \partial_v U_k = 0$. The resultant $\mathrm{Res}_v(\partial_x U_k, \partial_v U_k)$ is a combination of the two partials with polynomial coefficients, so it vanishes at the $x$-coordinate of every common zero; if it is not identically zero, it has finitely many real roots $x^*$, and for each the common zeros are among the roots of $\partial_v U_k(x^*, \cdot)$ (or of $\partial_x U_k(x^*, \cdot)$ if the first vanishes identically; if both do, $U_k$ is constant on the segment $x = x^*$ and any one of its points represents it). So the candidate set is finite and contains a maximizer. Every candidate is a point of $\mathcal{B}$, and the code evaluates the true $U$ (with the true $G$) there, so spurious candidates cannot raise the value: the maximum over the candidates is $\max_{\mathcal{B}} U$. □

*Arithmetic.* The code converts the bounds exactly to rationals ($R_u = B_R^2 - \iota$, $o = B_O^2$, $B = B_y$ as given in double precision), replaces $r$ by the smallest double $\tilde{r} \geq \sqrt{R_u}$ (a box containing $\mathcal{B}$, so the result is an upper bound that exceeds $\max_{\mathcal{B}} U$ by a relative $O(2^{-52})$), isolates the real roots of the univariate polynomials exactly, refines them to 60 digits, evaluates $U$ in 60-digit arithmetic and rounds the result up; $\Delta$ is then rounded up by the relative $10^{-12}$ of Proposition S1's code. *Checks* (`python experiments/baselines/sens_full.py check`, `reports/baselines_v2/sens_full_check.json`): over 33 bound vectors the value is never below a dense four-variable grid refined by local search, which does not use the reduction to faces, and never above the conservative value of Theorem 12.3 (strictly below it for 32 of the 33); a direct maximization over actual vectors with $d_{\mathbf{u}} = 1$ attains it. On the bounds of the benchmark (3droad and tamielectric, every fold, both preprocessings) it agrees with the earlier grid computation to a relative $9 \cdot 10^{-16}$, and on all 29 datasets, with Theorem 12.3 for the other 27, to $1.6 \cdot 10^{-15}$ (`python experiments/baselines/real_full.py sens`, `reports/baselines_v2/sens_full_datasets.json`): the reported sensitivities, noise scales and results do not change. For example, under block normalization ($B_R^2 = 2$ with the intercept, $B_O = 1$) the stage-1 value is $\Delta_{G,\mathrm{full}} = \sqrt{10} = 3.1623$ (at $u = r$, $u' = -r$), against 3.2660 from Theorem 12.3.

## 12.5 The full-record circuit: error bound and centre

The cost of full-record SSP was measured inside the encrypted architecture of v3.1 (the paper's "Full-record SSP" paragraph): its releases also contain the $n_A = d_R(d_R+3)/2$ entries of triu $\mathbf{A}$ without the constant $n$, namely the $d_R$ first moments $\sum_t b_t x_{j,t}$, aggregated from R's existing vectors $U = \mathbf{b} \circ \dot{\mathbf{x}}_j$, and the $d_R(d_R+1)/2$ products $\sum_t b_t x_{j,t} x_{k,t}$, from extra vectors $U = \mathbf{b} \circ \mathbf{x}_j \circ \mathbf{x}_k$ that R uploads; for all of them O's vector is $V = \mathbf{1}$ (every row). Here $\mathbf{b}$ is R's match indicator in O's order. The results of Part A are stated for the v3.1 circuit; the following lemma re-derives the ones that depend on the circuit, in Part A's notation.

**Lemma 12.5** (S10e: the full-record circuit)**.** *Under the circuit model of Part A ((M1), (M2), (M3′), (M4), (M5)), for the full-record circuit:*

*(a) the error decomposition (Part A, Proposition 3.8), the term bounds (Proposition 3.11, with Lemma 4.1) and Theorem 3.12 hold with $|\mathcal{E}|$ replaced by $|\mathcal{E}| + n_A$ and $\Gamma_{\mathcal{E}}$ by $\Gamma_{\mathcal{E}} + n_A$, for stages 1 and 2; no bound on R's vectors enters $B_{\mathrm{tot}}$;*

*(b) for neighbours that replace a whole record (the full-record adjacency, R's row included), Lemma 4.3 of Part A holds with $\Delta_s$ the full-record sensitivity, with $\nu_U^2$ increased by $d_R B_R^2 + \frac{1}{2} d_R(d_R+1) B_R^4$, with the bracket $\Omega_V \nu_V + \sqrt{N}\,\Omega_U \nu_V$ replaced by*

$$K_s = \Omega_V \nu_V + \sqrt{N}\,\Omega_U \nu_U + 2\,\Omega_U \nu_U + \sqrt{N}\,\Omega_V \bar{\nu}_V, \qquad \bar{\nu}_V = \Big(\sum_e V_{\max,e}^2\Big)^{1/2},$$

*where $V_{\max,e} \geq \max_t |V_{e,t}|$ (o for* triu $\mathbf{B}$*, $B_O$ for $\mathbf{C}$, $B_y$ for $\mathbf{c}_R$, $B_O B_y$ for $\mathbf{c}_O$, $w$ for $\mathbf{y}^\top \mathbf{y}$, 1 for the new entries), and with $\epsilon_x$ doubled ($\nu_V \leq 2\Gamma_{\mathcal{E}}^{1/2}$ with the v3.1 $\Gamma_{\mathcal{E}}$, since $V = \mathbf{1}$ never changes);*

*(c) the decoding offset (Proposition 4.5) holds with $|\mathcal{E}|$, $\Gamma_{\mathcal{E}}$ and $\nu_U$ of the full-record circuit and $\zeta_{\mathcal{E}}^2$ increased by $r^2 = B_R^4$; the bound $S_{\max}$ of Proposition 8.1 is unchanged.*

*Proof.* (a) In Proposition 3.8 R's plaintexts $m_{U_e,h}$ enter only the centre $\widehat{Z}_s(D)$; the error terms involve R's encryption errors times O's plaintexts ($T_1$), roundings, key-switching errors and fresh randomness, and Lemma 3.6 is unchanged (the $c_1$-components depend on the data only through O's plaintexts). In Proposition 3.11 the circuit enters $B_1$ through $M_V$, that is through Lemma 3.10's row bound $\sum_e V_{e,t}^2 \leq \Gamma_{\mathcal{E}}$, which gains $\sum_{e \in \mathcal{E}_A} 1^2 = n_A$, and $B_2, \ldots, B_5, B_8$ only through $|\mathcal{E}|$. No step uses a bound on $U$.

(b) Let the changed record sit in chunk $h^*$ and slot $l^*$. Now both $m_{U_e,h^*}$ and $m_{V_e,h^*}$ change, and $m_U m_V - m'_U m'_V = m_U \Delta m_V + \Delta m_U\, m'_V$, with $\Delta m_U = \Omega_U \tau^{-1}(\Delta U_e \mathbf{u}_{l^*}) + \delta r_e^U$, $\|\delta r_e^U\|_\infty \leq 1$. The first product is the one of Lemma 4.3. The slot value of the second is

$$\Omega_U \Omega_V\, \Delta U_e\, V'_{e,l^*} + \Omega_U\, \Delta U_e\, \tau(r_{V'_e})_{l^*} + \sum_l \tau(\delta r_e^U)_l\, \tau(m'_{V_e,h^*})_l.$$

The first terms of the two products add to $\Omega_U \Omega_V (U_{e,l^*} V_{e,l^*} - U'_{e,l^*} V'_{e,l^*})$, $\Omega_U \Omega_V$ times the change of entry $e$, of norm at most $\Omega_U \Omega_V \Delta_s$. $|\tau(r_{V'_e})_{l^*}| \leq N/2$ and $\|(\Delta U_e)_e\| \leq 2\nu_U$, so the second terms have norm at most $\frac{N}{2}\, 2\Omega_U \nu_U$. By Cauchy–Schwarz and Parseval the third is at most $\frac{N}{2}\sqrt{N}\|m'_{V_e,h^*}\| \leq \frac{N}{2}\sqrt{N}(\Omega_V V_{\max,e} + \frac{\sqrt{N}}{2})$, using $\|V_e[h^*]\| \leq \sqrt{N/2}\, V_{\max,e}$. So $\|(c_e)_e\| \leq \Omega_U \Omega_V \Delta_s + \frac{N}{2} K_s + \frac{N^2}{2}\sqrt{|\mathcal{E}|}$, and the masking and rounding steps of Lemma 4.3 give the stated $\Delta_{x,s}$, with leftover $2\epsilon_x$. The $U_{\max,e}$ of the new entries are $B_R$ ($|x_j| \leq \|\dot{\mathbf{x}}\|$) and $B_R^2$ ($|x_j x_k| \leq \|\dot{\mathbf{x}}\|^2$).

(c) The proof of Proposition 4.5 uses $\sum_e \|U_e\|^2 \leq n_{\max} \nu_U^2$ (the new vectors vanish off the matched rows), $\sum_e \|V_e\|^2 \leq \bar{n}_O \Gamma_{\mathcal{E}}$, and $\|\mathbf{z}\| \leq n_{\max} \zeta_{\mathcal{E}}$, where a row adds $\sum_{(j,k) \neq (0,0),\, j \leq k} (x_j x_k)^2 \leq \|\dot{\mathbf{x}}\|^4 - 1 < r^2$. The new entries satisfy $|z_e| \leq \bar{n}_O r$, within $S_{\max}$. □

*What changes.* Numbers from `experiments/eng/fullrecord_btot.py` (in `reports/eng/fullrecord_btot.json`), for the simulation population, $d_R = 4$, $d_O = 6$, $n_A = 14$, $\bar{n}_O = 65{,}536$, $\varepsilon = 1$, $\delta = 10^{-9}$ and setting I. $B_{\mathrm{tot},2}$ rises from $2^{26.592}$ to $2^{26.745}$, essentially all from $|\mathcal{E}|$: the earlier re-evaluation with the entry counts increased only gives the same value to 0.001 bits, because $\Gamma_{\mathcal{E}_2}$ rises only from 2,147 to 2,161. $\Delta_{x,2}$ rises from $2^{27.57}$ to $2^{29.24}$, still about $2^{-33}$ of $\sqrt{2/N}\,\Omega_{\mathrm{fin}}\Delta_2$, and $\epsilon_x$ stays below $2^{-35}$. The v3.1 stage-2 noise floods the old bound with a margin of only 0.03 bits, so it does not flood the new one; CALIBRATE for the full-record circuit (with its own sensitivities) chooses $\log_2 \Omega_{\mathrm{fin}} = 62.656$ instead of 62.625 (one grid step), keeps $\kappa = 40$, the chain and $N$, and needs $q_0 \geq 2^{84.9}$ of $2^{119}$. The operations are those of the v3.1 circuit at different plaintext scales, so the cost is unchanged; it was re-measured with the full-record calibration (`experiments/eng/fullrecord_bench.py`).

## 13 Simulation of Algorithm 6 end to end

`alg6.py` implements Algorithm 6 line by line (stage 1, $r_1$, $\hat{L}$, $\lambda_I$, closed-form $\lambda_O$ checked against bisection, the $\kappa_\Psi$ rule, stage 2, gate, ridge solve, bootstrap with $B = 300$, Corr, $\tilde{s}^2$, both intervals, $U_{yy}$ as specified and corrected). `s3_s6_sim.py` runs 2000 replications per scenario: fixed clipped design ($d_R = 4$, $d_O = 6$, $p = 11$, correlated features clipped at the 99% norm quantile), bounded uniform errors ($s = 1$), $B_y$ a true bound so that the working model holds exactly, $\alpha = 0.05$, $\alpha_L = \alpha_U = 0.0025$, $\vartheta^\star = 2.5$, $\kappa_\Psi = 2$, $\sigma_1 = 2.5\sigma_2$, joint mode.

The Wald interval for $\boldsymbol{\theta}_\lambda$ attains its nominal level in every realistic scenario and over-covers slightly in the stress scenario, because the bootstrap variance there is too large. The $\beta$ interval over-covers whenever the ridge is active: $\bar{b}_j$ is a worst-case bound and $\tilde{s}^2$ is conservative. It is uninformative ($\hat{L} = 0$) without a public bound in two of the three ridge scenarios.

## 14 Corrections to the specification

Each item states what `protocol_v31.tex` says, what is wrong or imprecise, and the corrected statement.

**Correction 1** (Sensitivity: $d_O = 1$, the regime-3 algorithm, and the $\mathbf{y}^\top\mathbf{y}$ mode)**.** DELTAREP's comment "for $d_O \geq 2$ these are exact suprema" is correct (Theorem 3.1). For $d_O = 1$ the same numbers are valid but conservative upper bounds; the exact values are in Proposition 3.5, e.g. $\Delta_G^{(1)} = 2\sqrt{ro}$ for $o \leq \frac{11+5\sqrt{5}}{8} r$. In regime 3, replace "one-dimensional maximization, rounded up" by the exact candidate enumeration of Theorem 3.1(iv) (three cubics). Rounding up may be kept as a safeguard. The joint/separate rule of Algorithm 1, line 5, is consistent with Theorem 3.1(ix): $\mathbf{y}^\top\mathbf{y}$ is free iff $\max_t G \leq 4w(r+o)$ in regime 3, always free if $w \leq 2(r+o)$, never free if $w > 3(r+o)$.

**Correction 2** (Statements inherited from the earlier paper)**.** (i) "$\mathbf{y}^\top\mathbf{y}$ is free" is false in general: $(1, 1, 5)$ gives $\Delta = 26.07$ against 14.14. (ii) "Using the full Frobenius norm of $\Delta\mathbf{B}$ would be conservative" is false for $d_O \geq 2$: the supremum is the same (Lemma 3.2); the triu maximum is attained only when $\mathbf{D}$ is diagonal. (iii) The worst case is not "$(\mathbf{x}, y) \mapsto (-\mathbf{x}, -y)$" in general; that is regime 2 only. In regime 1 it is $y' = -y$ with $\cos\angle(\mathbf{x}, \mathbf{x}') = -(r-w)/(2o)$ and $\mathbf{D}$ diagonal, and in regime 3 $|y'| < B_y$ when $\mathbf{y}^\top\mathbf{y}$ costs. (iv) "At the optimum the norms are at their bounds" is true for $\|\dot{\mathbf{x}}\|, \|\mathbf{x}\|, \|\mathbf{x}'\|$ (Lemma 3.3), false for $|y'|$ in costly regime 3, and false for $d_O = 1$, where only $\max(|x|, |x'|) = B_O$.

**Correction 3** (The radius $r_1$)**.** The comment "$\Pr[\|\mathbf{G}-\hat{\mathbf{G}}_1\|_{\mathrm{op}} > r_1] \le \alpha_L$" (Algorithm 6, line 2) is not proved and would need $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} \le 2\sigma\sqrt{p}$ (Remark 4.2). Replace it by

$$\Pr\big[\lambda_{\max}(\hat{\mathbf{G}}_1 - \mathbf{G}) > r_1\big] \le \alpha_L,$$

which is proved (Lemma 4.1(c)) and is the only event Algorithm 6 uses. For a two-sided statement use $r_1 = \sigma_1(2\sqrt{p} + 2\sqrt{\ln(2/\alpha_L)})$. Optional improvement: since the law of $\lambda_{\max}(\mathbf{E}_1)/\sigma_1$ depends only on $(p, d_A)$, $r_1$ may be set to its exact $(1-\alpha_L)$-quantile, computed offline by Monte Carlo with a certified error. For $p = 11$ this is $7.43\sigma_1$ instead of $11.53\sigma_1$, and it costs no privacy.

**Correction 4** ($\lambda_O$ and the $\Pi_O$ ridge)**.** Replace the bisection by the closed form of Lemma 5.1(c). State explicitly that $\lambda_O < \infty$ iff $\lambda_{\min}(\mathbf{A}) > g^\star + r_1$ (almost surely), and that the $\Pi_O$ ridge can never certify $\vartheta_0 > \lambda_{\min}(\mathbf{A})/(2\sigma_2\sqrt{p})$.

**Correction 5** ($U_{yy}$)**.** As written, $U_{yy}$ bounds $\mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}]$ with probability $\ge 1 - 2\alpha_U$, not $1-\alpha_U$ (two one-sided events at level $\alpha_U$ each). The Hoeffding term $B_y^2\sqrt{n\ln(1/\alpha_U)/2}$ is itself correct, but it requires the $y_i$ to be conditionally independent given $\mathbf{X}$. Corrected:

$$U_{yy} = \min\Big\{\widehat{yy} + z_{1-\alpha_U/2}\,\sigma_{yy} + B_y^2\sqrt{n\ln(2/\alpha_U)/2},\ nB_y^2\Big\} \qquad \text{or}$$
$$U_{yy} = \min\Big\{\widehat{yy} + \sqrt{2\ln(1/\alpha_U)(nB_y^4/4 + \sigma_{yy}^2)},\ nB_y^2\Big\}.$$

**Correction 6** (Estimand under clipping)**.** "CI for $\beta_j$" must name its target. Either (A) assume the working linear model, conditionally independent and homoskedastic, *for the clipped data*; this holds if the bounds are true bounds, so that clipping never binds. Or (B) target the projection coefficient $\boldsymbol{\beta}^* = \mathbf{G}^{-1}\tilde{\mathbf{X}}^\top\mathbb{E}[\tilde{\mathbf{y}} \mid \tilde{\mathbf{X}}]$, for which the bias bound remains valid but the homoskedastic variance is not, because clipping induces heteroskedasticity, so coverage is not guaranteed. Exactly Gaussian errors cannot hold with $|y| \le B_y$; normality is asymptotic (Theorem 7.2(vi)).

**Correction 7** (First-order accuracy is governed by $\sigma_2\,\mathrm{tr}\,\mathbf{P}_\lambda$, not by $1/\vartheta$ alone)**.** The validity ratio $\vartheta_0$ controls convergence of the Neumann series. The relative size of the neglected mechanism bias in the Wald interval is of order $\sigma_2\,\mathrm{tr}(\mathbf{P}_\lambda) \le \sqrt{p}/(2\vartheta_0)$, which at $\vartheta^\star = 2.5$, $p = 11$ may be 0.66. Coverage claims are asymptotic as $\vartheta_0 \to \infty$ at fixed $p$. Report $\sigma_2\,\mathrm{tr}(\tilde{\mathbf{P}})$ as a diagnostic, and consider centring at the bootstrap bias-corrected $2\tilde{\boldsymbol{\theta}} - \overline{\boldsymbol{\theta}^*}$.

**Correction 8** (Bootstrap)**.** Without truncation the bootstrap mean and covariance of $\boldsymbol{\theta}^*$ do not exist (Lemma 8.1(e)); the gate $\hat{\vartheta} \ge 1$ does not fix this. Use the symmetric truncation of Lemma 8.1 (keep a draw iff $\|(\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1/2}\mathbf{E}^*(\hat{\mathbf{G}}_2 + \lambda\Psi)^{-1/2}\|_{\mathrm{op}} \le \eta$, e.g. $\eta = 1/2$), or the closed forms $b_{\mathrm{RSS}}(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}})$ and $\mathbf{V}_0(\hat{\mathbf{G}}_2, \tilde{\boldsymbol{\theta}})$. Use antithetic pairs if Monte Carlo is kept. "$\mathbf{V}_{\mathrm{mech}}$ is the mechanism covariance" holds to first order only (relative $O_p(\vartheta_0^{-1})$).

**Correction 9** (The bias bound $\bar{b}_j$)**.** $\bar{b}_j = \lambda\|\Psi\tilde{\mathbf{P}}\mathbf{e}_j\|\hat{B}_\beta$ uses $\tilde{\mathbf{P}}$; the rigorous bound uses $\mathbf{P}_\lambda$. Either state the interval as first order, or inflate: on $\{\|\mathbf{E}_2\|_{\mathrm{op}} \le r_2\}$ with $r_2 = \sigma_2(2\sqrt{p} + 2\sqrt{\ln(2/\alpha_2)})$ and $\|\tilde{\mathbf{P}}\|r_2 < 1$, $\|\Psi\mathbf{P}_\lambda\mathbf{e}_j\| \le \|\Psi\tilde{\mathbf{P}}\mathbf{e}_j\| + \|\tilde{\mathbf{P}}\|r_2\|\tilde{\mathbf{P}}\mathbf{e}_j\|/(1 - \|\tilde{\mathbf{P}}\|r_2)$, at the price of a further $\alpha_2$ in the split.

**Correction 10** (The private bound on $\|\boldsymbol{\beta}\|$)**.** $\sqrt{U_{yy}/\hat{L}} < \infty$ iff $\lambda_{\min}(\hat{\mathbf{G}}_1) > r_1$. When the ridge is active, $\lambda_{\min}(\hat{\mathbf{G}}_1) < g^\star + r_1$, and in our scenarios $\hat{L} = 0$ in every replication of two of three ridge settings. Even when finite, the bound is of order $\sqrt{(s^2 + \boldsymbol{\beta}^\top\Sigma\boldsymbol{\beta})/\lambda_{\min}(\Sigma)}$, far above $\|\boldsymbol{\beta}\|$. The specification should say that the $\beta$ interval is informative essentially only with a public $B_\beta^{\mathrm{pub}}$ when the ridge binds.

**Correction 11** (Gate and refusals)**.** State whether coverage is unconditional (refusal = no claim) or conditional on passing; the latter loses a factor $1/(1-\Pr[\mathsf{refused}])$. The rigorous refusal bound at the defaults is weak, $\alpha_L + e^{-p(\vartheta^\star-2)^2} = \alpha_L + 0.064$ for $p = 11$, although Monte Carlo gives essentially zero. The notation table should distinguish $\vartheta_0$ (guaranteed $\geq \vartheta^\star$ on $\Omega_L$) from the gate statistic $\hat{\vartheta}$.

**Correction 12** (Minor)**.** (i) The reported "shrinkage $-\lambda\tilde{\mathbf{P}}\Psi\tilde{\boldsymbol{\theta}}$" is not $\boldsymbol{\theta}_\lambda - \hat{\boldsymbol{\beta}}_{\mathrm{OLS}}$, which equals $-\lambda\mathbf{G}^{-1}\Psi\boldsymbol{\theta}_\lambda$ exactly; it agrees with it only to first order in $\lambda\|\mathbf{P}_\lambda\Psi\|$, which is not small when the ridge binds. Rename it or report the exact form when $\mathbf{G}$ is well conditioned. (ii) $\tilde{s}^2$ is conservative under the ridge (Section 7); this is a feature, but it should be stated. (iii) The printed $\Delta = 86.18$, $\Delta_G = 77.20$ correspond to unrounded bounds; with $(5.59, 6.24, 2.89)$ the values are 86.15, 77.16. (iv) $\sigma_{yy}$ is used in Algorithm 6 but not defined: it is $\sigma_2$ in joint mode and $\sigma_y$ in separate mode.

## 15 Steps not made fully rigorous

(1) $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} \leq 2\sigma\sqrt{p}$ (two-sided) is not proved (Remark 4.2); it is not needed by the algorithm.
(2) (Resolved.) The constant of $\mathrm{TV}_1$ is now proved, with an explicit $\sigma^{-3}$ term (Lemma 2.2(b)).
(3) The boundary $w^*(r, o)$ of the free set in regime 3 is characterized exactly as $\max_t G = 4w(r+o)$, but its interval structure and the table of Remark 3.4 are numerical. Only $2(r+o) \leq w^* \leq 3(r+o)$ and $w^*(r, 0) = \frac{11+5\sqrt{5}}{8}r$ are proved.
(4) For $d_O = 1$, $\Delta^{(1)}$ (with $\mathbf{y}$) has no closed form; it is an exact two-variable maximization.
(5) Lemma S6: the constants $c_k$, $C_\eta$ and the scale $\mathcal{S}_2$ are given in order of magnitude only; the remainder orders are rigorous in form but not in constants.
(6) Proposition S5: replacing the oracle quantities ($\mathbf{V}_0$, $\mathbf{P}_\lambda$ in $\bar{b}_j$, $v_j$) by plug-ins is first order. Normality of the sampling term is asymptotic. The variation of $\mathbf{V}_0$ with $\boldsymbol{\theta}_\lambda$ is neglected.
(7) The homoskedastic variance is not valid under binding clipping (Correction 6). A public variance bound gives a valid interval (Proposition 12.2); a guarantee with the data-driven $\tilde{s}^2$ is open.
(8) The size of the downward bias of $\lambda_{\min}(\hat{\mathbf{G}}_1)$ is bounded only by the comparison of Lemma 10.2, which is about a factor 1.9 loose on the simulation population; the empirical switch-off factors 1.2 and 1.4 are not bounds (Remark 10.3).

## Scripts

All in `v31/proofs/scratch_stats/` (Python 3, NumPy, SciPy; `sim_core.py` imported read-only): `s1_common.py` (exact $\Delta$, pieces, worst-case configuration), `s1_symbolic.py` (exact verification of the polynomial identities of the S1 proof), `s1_check.py` (Table 2), `s1_freeset.py` and `s1_band.py` (free set), `s2_check.py` (Table 3, discrete-Gaussian and TV checks), `s2_goe_ratio.py` (Remark 4.2), `alg6.py` (Algorithm 6), `s3_s6_sim.py` (Tables 4–5); outputs in `*.out`.

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| Item | What is proved | Status |
|---|---|---|
| S0 | The slot noise is the image of the lattice discrete Gaussian under a scaled orthogonal map: exactly uncorrelated, $\sigma_s^2$-subgaussian, and within total variation $N(0.0202\,\sigma_c^{-2} + 0.0315\,\sigma_c^{-3} + e^{-2\pi^2\sigma_c^2})$ of i.i.d. $\mathcal{N}(0, \sigma_s^2)$ plus a perturbation of norm $\leq N/(2\sqrt{2}\,\Omega)$. | exact (TV constant now proved) |
| S1 | Exact $\Delta$ in regimes 1–2 (closed form), exact finite algorithm in regime 3; norms at bounds; triu = Frobenius supremum for $d_O \geq 2$; $d_O = 1$ exact reduction and closed-form $\Delta_G$; $\Delta_{\neg yy}$ in closed form in all regimes; $\mathbf{y}^\top\mathbf{y}$ free if $w \leq 2(r+o)$, costly if $w > 3(r+o)$. | exact; boundary of the free set [numerical] |
| S2 | $\mathbb{E}\lambda_{\max}(\mathbf{E}) \leq 2\sigma\sqrt{p}$; $\sqrt{2}\sigma$-Lipschitz concentration; $\Pr[\lambda_{\max}(\mathbf{E}) \geq r_1/\sigma_1 \cdot \sigma] \leq \alpha_L$. | exact (one-sided); $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}} \leq 2\sigma\sqrt{p}$ [open] |
| S3 | Measurability and independence; on $\Omega_L$: $\lambda_{\min}(\mathbf{G} + \lambda\Psi) \geq 2\vartheta^\star\sigma_2\sqrt{p}$, $\hat{L} \leq \lambda_{\min}(\mathbf{G})$; closed-form $\lambda_O$; gate-failure bound. | exact |
| S4 | $\|\boldsymbol{\beta}\|^2 \leq \mathbb{E}[\mathbf{y}^\top\mathbf{y} \mid \mathbf{X}]/\lambda_{\min}(\mathbf{G})$, also for the projection estimand; $U_{yy}$ valid at level $2\alpha_U$ as written, $\alpha_U$ after correction. | exact (needs conditional independence) |
| S5 | Exact Gaussian first-order term, exact resolvent remainder identity, rigorous finite-sample coverage inequality for the oracle Wald interval; AKK inequality and union bound for the $\beta_j$ interval. | plug-in [first order]; normality of sampling term [asymptotic] |
| S6 | Second-order RSS bias in closed form; Corr removes it up to a relative $O(\vartheta_0^{-2})$ remainder; bootstrap covariance consistent to relative $O_p(\vartheta_0^{-1})$. | [first order]; constants not explicit |
| S7 | Exact stage-1 radius: exact level $(M-j)/(M+1)$ with fresh Monte Carlo draws, confidence $c$ over the draws; transfer to the protocol with the proved TV constant and offset bound. | exact |
| S8 | Bracket for the switch-off, tight among Gram matrices with a given $\lambda_{\min}$; spectral lower bound on $\mathbb{E}\lambda_{\min}(\mathbf{G} + \mathbf{E})$ by Sudakov–Fernique. | exact; a spectrum-sharp tail bound [open] |
| S9 | Leading-order mechanism bias, ridge version and decay in $\lambda$, with explicit remainder constants. | exact remainder bounds |
| S10 | Bias of FWS's plug-in RSS with explicit remainder; a valid bias-aware interval under binding clipping with a public variance bound; closed form of the full-record sensitivity for blocks of dimension $\geq 2$ (S10c) and its exact enumeration for a block of dimension 1 (S10d); the full-record circuit's $B_{\mathrm{tot}}$, centre sensitivity and decoding offset (S10e). | exact; data-driven variance under clipping [open] |

Table 1: What is proved. Corrections to the specification are collected in Section 14; open steps in Section 15.

| $B_R$ | $B_O$ | $B_y$ | reg. | $\Delta$ exact | brute $d_O$=2 | $\Delta_{\neg yy}$ | $\Delta^{(1)}$ ($d_O$=1) | $\Delta_G$ | $\Delta_G^{(1)}$ |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 3.162278 | 3.162278 | 3.16228 | 2.82843 | 2.12132 | 2.00000 |
| 1 | 1 | 5 | 3 | 26.070598 | 26.070598 | 14.14214 | 26.05763 | 2.12132 | 2.00000 |
| 3 | 1 | 1 | 2 | 8.485281 | 8.485281 | 8.48528 | 8.48528 | 6.00000 | 6.00000 |
| 2.236 | 2.236 | 5 | 3 | 31.621815 | 31.621815 | 31.62182 | 31.62182 | 10.60596 | 9.99939 |
| 5.59 | 6.24 | 2.89 | 1 | 86.152245 | 86.152245 | 86.15224 | 78.53500 | 77.16183 | 69.76320 |
| 1 | 2 | 1 | 1 | 7.211103 | 7.211103 | 7.21110 | 5.65685 | 6.36396 | 4.53635 |
| 4 | 1 | 0.5 | 2 | 8.944272 | 8.944272 | 8.94427 | 8.94427 | 8.00000 | 8.00000 |
| 1 | 0.5 | 3 | 3 | 9.672258 | 9.672258 | 6.70820 | 9.67226 | 1.00000 | 1.00000 |
| 1.5 | 0.7 | 2.2 | 3 | 7.283296 | 7.283296 | 7.28330 | 7.28330 | 2.10000 | 2.10000 |
| 1 | 1 | 1.8 | 3 | 5.091169 | 5.091169 | 5.09117 | 5.09117 | 2.12132 | 2.00000 |
| 1 | 0.2 | 2 | 3 | 4.566806 | 4.566806 | 4.07922 | 4.56681 | 0.40000 | 0.40000 |
| 1 | 3 | 4 | 1 | 25.387005 | 25.387005 | 25.38700 | 25.29822 | 13.43503 | 9.51475 |

Table 2: Proposition S1: the exact value equals `delta_rep`, the brute-force triu maximum for $d_O = 2$ and $d_O = 3$ (the $d_O = 3$ column is identical and omitted), and the explicit configuration of Theorem 3.1(vii), in every case.

| shape | $2\sqrt{p}$ | $\mathbb{E}\lambda_{\max}$ | $\mathbb{E}\|\mathbf{E}\|_{\mathrm{op}}$ | $q_{.9975}(\lambda_{\max})$ | $r_1/\sigma_1$ | $\Pr[\lambda_{\max} \geq 2\sqrt{p}+2]$ |
|---|---|---|---|---|---|---|
| $p = 11$, $d_A = 5$ (protocol) | 6.633 | 4.976 | 5.383 | 7.43 | 11.53 | $2.5 \times 10^{-5}$ |
| $p = 11$, no mask, diag var 1 | 6.633 | 5.431 | 5.858 | 7.89 | 11.53 | $1.8 \times 10^{-4}$ |
| $p = 11$, GOE (diag var 2) | 6.633 | 5.645 | 6.148 | 8.29 | 11.53 | $9.0 \times 10^{-4}$ |
| $p = 2$, $d_A = 1$ | 2.828 | 0.965 | 1.368 | 3.60 | 7.72 | $9 \times 10^{-5}$ |
| $p = 20$, $d_A = 5$ | 8.944 | 7.702 | 8.096 | 10.00 | 13.84 | $4 \times 10^{-5}$ |
| $p = 50$, $d_A = 10$ | 14.142 | 13.095 | 13.446 | 15.09 | 19.04 | 0 |

Table 3: Lemma S2, Monte Carlo in units of $\sigma$; $r_1/\sigma_1$ at $\alpha_L = 0.0025$. The bound $e^{-t^2/4}$ at $t = 2$ is 0.37; the observed tails are far lighter, because the Lipschitz constant $\sqrt{2}\sigma$ is attained only by spread-out eigenvectors (empirical s.d. of $\lambda_{\max}$: $0.77\sigma$).

| scenario | $\vartheta_0$ | $\sigma_2 \operatorname{tr} \mathbf{P}_\lambda$ | $\mathbb{E}[\widetilde{\mathrm{RSS}}] - \mathrm{RSS}_\lambda$ (MC) | $b_{\mathrm{RSS}}$ | Corr mean (sd) | antithetic (sd) | $\max_j \lvert \mathrm{sd}_{\mathbf{V}_{\mathrm{mech}}}/\mathrm{sd} - 1 \rvert$ |
|---|---|---|---|---|---|---|---|
| off ($\lambda = 0$) | 48.7 | 0.017 | $-0.106 \pm 0.062$ | $-0.107$ | $-0.08$ (0.44) | $-0.107$ (0.004) | 0.009 |
| ridge $\mathbf{I}$ | 7.07 | 0.145 | $-5.19 \pm 0.27$ | $-5.29$ | $-5.42$ (1.90) | $-5.33$ (0.21) | 0.017 |
| ridge $\Pi_O$ | 8.05 | 0.111 | $-2.92 \pm 0.13$ | $-2.90$ | $-3.03$ (0.99) | $-2.94$ (0.12) | 0.015 |
| small $n$ | 8.17 | 0.183 | $-6.97 \pm 0.16$ | $-6.96$ | $-6.98$ (1.14) | $-7.04$ (0.23) | 0.018 |
| stress | 2.52 | 0.588 | $-23.61 \pm 0.17$ | $-22.49$ | $-26.48$ (1.98) | $-26.71$ (1.58) | 0.097 |

Table 4: Lemma S6 at fixed data and fixed $(\lambda, \Psi)$. "$\pm$" is the Monte Carlo standard error; "sd" is the s.d. across 200 independent releases.

| scenario | $n$ | $\sigma_2$ | med. $\lambda$ | $\Psi$ | refused | Wald cov. (range over $j$) | $\beta$ cov. | $\widetilde{\mathrm{RSS}} - \mathrm{RSS}_\lambda$ / after Corr |
|---|---|---|---|---|---|---|---|---|
| off | 4000 | 4 | 0 | – | 0 | 0.946–0.958 | 0.945–0.968 | $-0.14$ / $-0.05$ ($\pm 0.20$) |
| ridge $\mathbf{I}$ | 1500 | 20 | 477 | $\mathbf{I}$ | 0 | 0.948–0.960 | $\infty$ ($\hat{L} = 0$) | $-6.83$ / $-1.68$ ($\pm 0.86$) |
| ridge $\mathbf{I}$, $B^{\mathrm{pub}} = 1.5\|\boldsymbol{\beta}\|$ | 1500 | 20 | 477 | $\mathbf{I}$ | 0 | 0.948–0.960 | 1.000 | as above |
| ridge $\mathbf{I}$, private $\hat{B}_\beta$ | 1500 | 14 | 167 | $\mathbf{I}$ | 0 | 0.950–0.960 | 1.000$^\dagger$ | $-4.49$ / $-1.31$ ($\pm 0.66$) |
| ridge $\Pi_O$ | 3000 | 10 | 439 | $\Pi_O$ | 0 | 0.944–0.962 | $\infty$ ($\hat{L} = 0$) | $-3.03$ / $-0.07$ ($\pm 0.42$) |
| small $n$, $B^{\mathrm{pub}}$ | 200 | 20 | 1021 | $\mathbf{I}$ | 0 | 0.946–0.962 | 1.000 | $-7.43$ / $-0.41$ ($\pm 0.50$) |
| stress, $B^{\mathrm{pub}}$ | 60 | 20 | 319 | $\mathbf{I}$ | 0 | 0.960–0.974 | 1.000 | $-23.35$ / $+3.05$ ($\pm 0.57$) |

Table 5: Algorithm 6, 2000 replications each; Wald target $1 - \alpha_{\mathrm{CI}} = 0.955$, $\beta$ target $\geq 0.95$. In every scenario $\Pr[\Omega_L]$, $\Pr[\vartheta_0 \geq \vartheta^\star]$, $\Pr[\hat{L} \leq \lambda_{\min}(\mathbf{G})]$, $\Pr[U_{yy} \geq \mathbb{E}\mathbf{y}^\top\mathbf{y}]$ (both versions) and the agreement of closed-form $\lambda_O$ with bisection were all 1.000. "Stress" uses $\sigma_1 = 0.01\sigma_2$, so that $\vartheta_0 \approx \vartheta^\star$; it is outside the default design. $^\dagger$The private bound was finite in 96.1% of replications.

# Protocol v3.1: composition across runs

Proof of Proposition `prop:compose` for runs fixed in advance and for adaptively chosen runs under the ledger's filter, and a hockey-stick treatment of the smudging slack

Draft for review

**Abstract**

This note proves the composition statement of protocol v3.1 (Proposition `prop:compose` of the paper) and closes the check item "the adaptive hybrid of Proposition `prop:compose` is argued here but not yet written out in the proofs". We model all researchers as one adaptive coalition that chooses each run's parameters and inputs from the joint view of earlier runs, follows the protocol inside each run, may interleave and abort runs, and faces O's ledger. We prove that the joint view is $(\varepsilon_L, \delta_L)$-IND-CDP for adjacency on the base register, under each admission rule (Theorem 4.2). The proof has four parts: one statistical hybrid that removes every encryption error at once, a computational hybrid run by run, a self-contained Rényi filter for adaptively parameterized lattice Gaussian mechanisms, and the conversion. Part 2 measures the smudging step in hockey-stick divergence (Lemma 5.1, Theorem 5.2). With a reserved share $\xi_L = 10^{-3}$ of $\varepsilon_L$, the ledger's flooding requirement $\kappa_L$ at $(\varepsilon_L, \delta_L, \zeta_L) = (10, 10^{-6}, 0.1)$ falls from $48, 46, 44, 42, 40$ to $22, 21, 19, 17, 15$ for $\varepsilon_{\text{run}} = 0.5, 1, 2, 4, 8$, so it no longer binds against the floor $\kappa_{\min} = 40$.

**References to other documents.** This note is Part C of the proofs appendix (`protocol_v31_proofs.tex`). [CP] is `crypto_proofs.tex`, the proofs of the cryptographic and DP claims (Part A of the proofs appendix); result numbers are those of its current build, and labels are given in typewriter font. [P] is the paper, `paper/main.tex`. Notation follows [CP].

**Status (2026-10-25).** The paper, the implementation and [CP] now follow this note: the paper states Theorem 4.2 as Proposition `prop:compose` with base-register adjacency (`def:baseadj`); its ledger is the ledger of Definition 5.3 with $\xi_L = 10^{-3}$ and $\delta' = 2^{-100}$, charged in full at admission; its $\kappa_L$ is that of Proposition 5.5; and the floor is $2^{42}$ in the paper, in [CP] (Corollary 2.4, corrected) and in the implementation. The issues of Section 6 are resolved accordingly, except that rule (N) with adaptively chosen requests remains unproved (Remark 4.4).

**Summary of results.**

- **Theorem 4.2** replaces Proposition `prop:compose`. For every PPT coalition of researchers, with arbitrary collusion, adaptivity, interleaving and aborts, the joint view is simulated by an ideal experiment with slack $\mathbb{E}[\sum_{k \text{ admitted}} \gamma_k] + \text{negl}(\lambda)$. The ideal experiment is $\rho^*$-zCDP whenever the admitted runs satisfy $\sum_k \rho_k \le \rho^*$ pointwise. Under the filter rule the joint view is $(\varepsilon_L, \delta_L)$-IND-CDP for every adaptive coalition. Under the non-adaptive rule the same holds when the run requests are fixed in advance.
- **The adaptive case goes through as the paper's sketch claims**, with three precisions.

(i) All smudging steps of all runs are made in one statistical hybrid, before any computational step. The slack is then a single expectation under the real experiment, which the pointwise filter bounds. Taken step by step, the sketch's "expectation of $\gamma_k$" would be an expectation under a different hybrid for each step.

(ii) The $\delta_B$ part of a run's slack must be charged for *every planned release of every admitted run*, aborted or not. A researcher's decision to continue may depend on $s_R$ and on its own encryption errors, which enter $e_{\text{tot}}$. The simplest rule is to charge a run's full cost at admission, with no refund.

(iii) The RLWE assumptions are needed for adaptively chosen parameter sets (a finite menu suffices) and against non-uniform adversaries, since the reduction knows the data.

- **The Rényi filter** (Theorem 3.2) is proved here from scratch by the supermartingale argument. It covers lattice Gaussian steps whose dimension, centre and $\sigma$ depend on the history, together with refusals and other public steps.
- **The $\delta$ budget adds up exactly**: $\delta(\rho_L, \varepsilon_L) + (1 + e^{\varepsilon_L})\Gamma_L = (1 - \zeta_L)\delta_L + \zeta_L\delta_L = \delta_L$. Numerically the sum is $9.99999999999 \cdot 10^{-7}$. The paper's $\kappa_L$ values $48, 46, 44, 42, 40$ are reproduced.
- **Part 2.** A shifted lattice Gaussian with $\|\mathbf{e}\| \le B_{\text{tot}}$ and $\sigma \ge 2^{\kappa} B_{\text{tot}}/\sqrt{2\pi}$ is $(\varepsilon', \delta')$-indistinguishable from the unshifted one with $\varepsilon' \le 2\sqrt{\pi \ln(1/\delta')}\, 2^{-\kappa} + \pi 4^{-\kappa}$. This bound needs no smoothing condition. Composing with an $(\varepsilon', \delta')$ smudging step gives a three-budget ledger and a new rule for $\kappa_L$ (Section 5).
- **Issues in the paper** are collected in Section 6. None invalidates the stated guarantee once the ledger charges at admission. They include a constant in the smudging lemma: the factor $1 + 2^{-80}$ needs $\sigma_c \ge 2^{42}$, not $2^{20}$.

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# 1 Setting

## 1.1 The base register and its adjacency

The paper's adjacency ([P], Definition `def:adj`) is stated for the padded scope register of one run, with the cohort fixed. Runs on one base register may use different scopes, models and cohorts, so the joint statement needs an adjacency on the base register itself.

**Definition 1.1** (Base register; base adjacency)**.** A *base register* is $\mathfrak{D} = (\mathrm{id}_j, \mathbf{a}_j, \mathbf{x}_j, y_j)_{j \le n_\mathrm{b}}$. Here $\mathrm{id}_j$ is an identifier, $\mathbf{a}_j$ holds the public attributes that scopes may use, $\mathbf{x}_j$ holds all of $\mathsf{O}$'s feature columns and $y_j$ is the response. The *fixed part* of $\mathfrak{D}$ is $\mathrm{fix}(\mathfrak{D}) = (n_\mathrm{b}, (\mathrm{id}_j, \mathbf{a}_j)_j)$. Two base registers are adjacent, $\mathfrak{D} \sim_\mathrm{b} \mathfrak{D}'$, if $\mathrm{fix}(\mathfrak{D}) = \mathrm{fix}(\mathfrak{D}')$ and $(\mathbf{x}_j, y_j) \neq (\mathbf{x}'_j, y'_j)$ for exactly one $j$.

**Definition 1.2** (Dataset of a run)**.** A run's public parameters $P$ fix four things:

- a scope predicate $\Phi$ on public attributes;
- a row map $g_P$ from $(\mathbf{x}, y)$ to $\mathsf{O}$'s preprocessed and clipped columns, which selects the model's columns and applies public constants;
- a padded size $\bar{n}_O \ge |\{j : \Phi(\mathbf{a}_j)\}|$;
- the number of dummy rows.

Given dummy identifiers $\mathbf{d}$, the run's dataset is the scope register $D_P(\mathfrak{D}; \mathbf{d})$. It consists of the rows $(\mathrm{id}_j, g_P(\mathbf{x}_j, y_j))$ with $\Phi(\mathbf{a}_j)$, padded with rows $(\mathrm{d}_i, \mathbf{0}, 0)$ to $\bar{n}_O$ rows.

**Lemma 1.3** (Base adjacency induces scope adjacency)**.** *Let $\mathfrak{D} \sim_\mathrm{b} \mathfrak{D}'$. Then for every $P$ and every $\mathbf{d}$, the datasets $D_P(\mathfrak{D}; \mathbf{d})$ and $D_P(\mathfrak{D}'; \mathbf{d})$ either coincide or are adjacent in the sense of* [P]*, Definition* `def:adj`*. Adjacent here means that they have the same identifiers and the same scope, and differ in $(\mathbf{x}_O, y)$ of exactly one row.*

*Proof.* The selected rows depend on the public attributes, which are equal. The map $g_P$ acts row by row with public constants, so only the row $j$ in which $\mathfrak{D}$ and $\mathfrak{D}'$ differ can change, and only if $\Phi(\mathbf{a}_j)$ holds. The dummy rows are equal. □

The lemma needs row-wise preprocessing with public constants. A data-dependent transformation, such as standardizing $\mathsf{O}$'s columns by register means, would let one row change every row, and is excluded by Assumption 1.9(A4).

## 1.2 Runs, costs and the ledger

**Definition 1.4** (Run request; costs)**.** A *run request* is a pair $(P, \chi)$:

- $P \in \mathcal{P}_\lambda$ is a public parameter vector of the paper's Section `sec:params`. It comprises the scope, the model and its preprocessing, the bounds, the run budget $(\varepsilon_\mathrm{run}, \delta_\mathrm{run})$, the splits, $\zeta$, the policy for $\kappa$, $\delta_B$, and an HE parameter set $p \in \Pi_\lambda$ (ring dimension, modulus chain, distributions).
- $\chi$ is the researcher's private input: at most $n_R^{\max}$ cohort identifiers, with features $\mathbf{X}_R$.

Calibrate$(P)$ is deterministic. It either refuses or returns $\kappa$, $n_\mathrm{rel}$ and $(\sigma_{c,s}, \rho_s, B_{\mathrm{tot},s})_{s \le n_\mathrm{rel}}$. The run's *costs* are

$$\rho(P) = \sum_{s \le n_\mathrm{rel}} \rho_s, \qquad \gamma(P) = \sum_{s \le n_\mathrm{rel}} \gamma_s, \qquad \gamma_s = \delta_B + \gamma'(\kappa), \qquad \gamma'(\kappa) = 2^{-\kappa}(1 + 2^{-80}) + 2^{-97}.$$

The costs are public, because they are functions of $P$.

Throughout, $\delta(\rho,\varepsilon)$ is the tight conversion of Lemma 2.3, which is the function of [CP], Proposition 4.4 (`prop:conv`).

**Definition 1.5** (Ledger)**.** The ledger of a base register has public parameters $(\varepsilon_L, \delta_L, \zeta_L)$. It derives

$$\rho_L = \max\{\rho \geq 0 : \delta(\rho, \varepsilon_L) \leq (1-\zeta_L)\delta_L\}, \qquad \Gamma_L = \frac{\zeta_L \delta_L}{1+e^{\varepsilon_L}}.$$

Its state is $(\rho_\Sigma, \Gamma_\Sigma)$, initially $(0,0)$. It processes requests one at a time. Let a request have costs $(\rho_k, \gamma_k)$ and let CALIBRATE not refuse it. The ledger uses one of two rules, fixed when the ledger is created:

**(F)** *filter:* admit iff $\rho_\Sigma + \rho_k \leq \rho_L$ and $\Gamma_\Sigma + \gamma_k \leq \Gamma_L$;
**(N)** *non-adaptive:* admit iff $\delta(\rho_\Sigma + \rho_k, \varepsilon_L) + (1+e^{\varepsilon_L})(\Gamma_\Sigma + \gamma_k) \leq \delta_L$.

On admission the ledger sets $(\rho_\Sigma, \Gamma_\Sigma) \leftarrow (\rho_\Sigma + \rho_k, \Gamma_\Sigma + \gamma_k)$ at once. The full planned cost is charged before any message of the run that depends on a secret, and it is not refunded if the run is aborted.

**Remark 1.6** (Differences from the paper's ledger)**.** The paper's earlier text defined the state as the sum of the costs "of every release made" and admitted a run against its full cost. The implementation (`vfr/dp.py`, class `Ledger`, and `vfr/party_o.py`) checked a run's full cost at setup but charged each release only when it was made. This was a reservation-free scheme with implicit refunds on abort. Both now charge the full cost at admission, atomically (the implementation under an exclusive file lock), with no refund. Remark 4.3 shows which refunds are sound: those of $\rho_s$ and of $\gamma'(\kappa)$ for releases not made. The $\delta_B$ part of an aborted run's unmade releases must stay charged. Without reservation, two runs admitted against the same state can together exceed the cap. Definition 1.5 avoids both problems.

### 1.3 The multi-run experiment

**Definition 1.7** (The real experiment $\mathsf{Real}_{\mathcal{A}}(\mathfrak{D})$)**.** Let $\mathcal{A}$ be an interactive PPT machine, the *coalition*. It receives $1^\lambda$ and draws its random tape $\omega_{\mathcal{A}}$. Until it halts it repeats one of the following actions, each chosen as a function of $\omega_{\mathcal{A}}$ and of everything it has received so far.

(a) *Request.* $\mathcal{A}$ submits a run request $(P_k, \chi_k)$; requests are numbered in order of submission. $\mathsf{O}$ computes CALIBRATE$(P_k)$ and answers $\perp_{\text{cal}}$ if it refuses. Otherwise $\mathsf{O}$ submits the costs to the ledger and answers $\perp_{\text{led}}$ if the ledger refuses. Otherwise run $k$ is *admitted*, the ledger charges it, and the run becomes *open*. $\mathsf{O}$ draws fresh coins $\omega_{O,k}$, and the honest researcher machine $\mathsf{R}_k = \mathsf{R}(P_k, \chi_k; \omega_{R,k})$ gets fresh uniform coins $\omega_{R,k}$, which are given to $\mathcal{A}$. The coins $\omega_{O,k}$ comprise:
- $\mathsf{O}$'s key shares $s_O$ and their errors;
- the common random strings;
- the OPRF key, the shuffle $\tau_k$, the dummy identifiers and the PSI sanitization coins;
- for each release $s \leq n_{\text{rel}}$, the re-randomization coins $(u, e_0, e_1)$ and the noise $\eta \sim D_{\mathbb{Z}^N, \sigma_{c,s}}$.

(b) *Step.* $\mathcal{A}$ selects an open run. The next message of that run is computed by the honest party whose turn it is, either $\mathsf{R}_k$ or $\mathsf{O}(\mathfrak{D}, P_k; \omega_{O,k})$, and delivered. $\mathcal{A}$ receives every message that $\mathsf{R}_k$ sends or receives.

(c) *Abort.* $\mathcal{A}$ closes an open run, and no further message of that run is sent. In Algorithm 5 the researcher requests each release, so not requesting a release is an abort.

(d) *Halt.* $\mathcal{A}$ outputs a bit, which is the output of $\mathsf{Real}_{\mathcal{A}}(\mathfrak{D})$.

$\text{view}_{\mathcal{A}}(\mathfrak{D})$ consists of $\omega_{\mathcal{A}}$ and every answer, coin $\omega_{R,k}$ and message that $\mathcal{A}$ received.

The model makes the researchers semi-honest within each run: the researcher side of every run is the honest machine. Across runs they are fully adaptive in what they request, when, how they

interleave open runs, and when they abort. One machine stands for all researchers, so arbitrary collusion is covered. Non-colluding researchers are the special case in which $\mathcal{A}$'s actions for researcher $i$ depend only on $i$'s part of the view. The view of any subset of researchers is a function of $\text{view}_{\mathcal{A}}$.

**Definition 1.8** (IND-CDP of the multi-run protocol)**.** The protocol is $(\varepsilon, \delta)$-*IND-CDP for* $\sim_{\mathrm{b}}$ if, for every PPT coalition $\mathcal{A}$, there is a negligible $\nu$ such that for all $\mathfrak{D} \sim_{\mathrm{b}} \mathfrak{D}'$

$$\Pr[\mathsf{Real}_{\mathcal{A}}(\mathfrak{D}) = 1] \leq e^{\varepsilon} \Pr[\mathsf{Real}_{\mathcal{A}}(\mathfrak{D}') = 1] + \delta + \nu(\lambda).$$

$\mathcal{A}$'s output may be any PPT function of its view, so this is the IND-CDP of [P], Definition `def:simcdp`, applied to the joint view. The coalition's adaptive strategy is built into the experiment.

## 1.4 Assumptions

**Assumption 1.9.** (A1) *Honest data holder and honest, atomic ledger.* $\mathsf{O}$ follows the protocol. Its ledger behaves as in Definition 1.5, with one rule. $\mathsf{O}$ produces decryption shares only within admitted runs. The answers $\perp_{\text{cal}}$ and $\perp_{\text{led}}$ depend only on $P_k$ and the ledger state. $\mathsf{O}$ aborts a run only on events determined by $\text{fix}(\mathfrak{D})$, $P_k$, its coins other than ($s_O$, its errors, $u, e_0, e_1, \eta$), and the messages it has received. An overflow of the PSI tables is an example of such an event.

(A2) *Semi-honest researchers with arbitrary collusion*, as in Definition 1.7. In particular each $\mathsf{R}_k$ clips its features, and its coins are uniform.

(A3) *Fresh coins.* Each admitted run draws $\omega_{O,k}$ and $\omega_{R,k}$ independently of each other and of everything before, with the protocol's distributions. The CKKS key material of $\mathsf{O}$ and $\mathsf{R}$ ($s_O$, $s_R$, their errors and the common random strings) is fresh in every run. $(u, e_0, e_1, \eta)$ is fresh in every release, and $\eta$ is *exactly* $D_{\mathbb{Z}^N, \sigma_{c,s}}$.

(A4) *Static register, public scopes, row-wise public preprocessing.* $\mathfrak{D}$ is the same in all runs. Every scope uses public attributes only, and every $g_P$ is row-wise with public constants. Lemma 1.3 therefore holds.

(A5) *Per-run hypotheses.* Every $P \in \mathcal{P}_\lambda$ that Calibrate does not refuse satisfies the hypotheses of [CP], Theorem 4.11 (`thm:output`), for every release $s$:

- (M1)–(M5);
- $\sigma_{c,s} \geq \max\{2^{\kappa} B_{\text{tot},s}/\sqrt{2\pi},\ 2^{42}\}$ (see Remark 2.6 for the floor);
- $\rho_s \geq \widehat{\Delta}^2_{\text{enc},s}/(2\sigma^2_{c,s})$, where $\widehat{\Delta}_{\text{enc},s}$ bounds the sensitivity of the centre $f_s$ under scope adjacency, for every arrangement and every researcher input in the clipped domain ([CP], Corollary 3.13, `cor:sensZ`);
- $B_{\text{tot},s}$ satisfies [CP], Theorem 3.12 (`thm:01`): $\Pr[\|e_{\text{tot},s}\|_2 > B_{\text{tot},s}] \leq \delta_B$ for every dataset in the clipped domain, every researcher input and every value of the public randomness. The probability is over the run's key shares, errors and fresh coins.

(A6) *RLWE for every parameter set in a finite menu.* [CP], Assumptions 4.9 (`as:rlweu`) and 4.10 (`as:rlweO`) hold for every $p \in \Pi_\lambda$, and $|\Pi_\lambda| \leq \text{poly}(\lambda)$. They hold against non-uniform adversaries: for every polynomial $t$, the best advantages $\text{Adv}_{1,p}(\lambda, t(\lambda))$ and $\text{Adv}_{2,p}(\lambda, t(\lambda))$ of circuits of size $t$ are negligible.

(A7) *Efficiency.* Calibrate and the ledger tests run in deterministic polynomial time. The centre $f_s(D; \pi)$ is computable in time polynomial in $\lambda$ and $|\mathfrak{D}|$. The exact sampler runs in expected polynomial time; a reduction may cut it off at a polynomial time bound, which changes its output with probability $2^{-\lambda}$, so it suffices.

(A8) *PPT coalition.* $\mathcal{A}$ runs in time $t_{\mathcal{A}}(\lambda) \leq \text{poly}(\lambda)$, so it makes at most $t_{\mathcal{A}}(\lambda)$ requests and actions.

(A3) is used only for the CKKS key material and for the release coins. Fresh OPRF keys and shuffles matter for cohort and linkage privacy (G2, G4) across runs, not for Theorem 4.2: the theorem never replaces the Link phase, and the sensitivity bound holds for every arrangement.

## 2 Divergences on countable spaces

Every object below is a probability mass function (pmf) on a countable set. This covers every transcript of the protocol, which consists of bit strings, ring elements and lattice points.

**Definition 2.1** (Divergences)**.** Let $P, Q$ be pmfs on a countable set $\mathcal{Y}$. Let $\phi_a(p,q) = p^a q^{1-a}$ for $q > 0$, with $\phi_a(0,0) = 0$ and $\phi_a(p,0) = \infty$ for $p > 0$.

- For $a > 1$, $\mathsf{M}_a(P\|Q) = \sum_y \phi_a(P(y), Q(y))$, and the Rényi divergence is $D_a(P\|Q) = \frac{1}{a-1} \ln \mathsf{M}_a(P\|Q)$.
- For $\varepsilon \geq 0$, the hockey-stick divergence is

$$\mathsf{E}_\varepsilon(P\|Q) = \sum_y (P(y) - e^\varepsilon Q(y))_+ = \sup_{\varphi:\mathcal{Y}\to[0,1]} \big(\mathbb{E}_P\varphi - e^\varepsilon \mathbb{E}_Q\varphi\big).$$

- $\mathrm{SD} = \mathsf{E}_0$ is the statistical distance.

$P$ and $Q$ are $(\varepsilon, \delta)$-*indistinguishable* if $\mathsf{E}_\varepsilon(P\|Q) \leq \delta$ and $\mathsf{E}_\varepsilon(Q\|P) \leq \delta$. A finite $D_a(P\|Q)$ implies $P \ll Q$.

The supremum form of $\mathsf{E}_\varepsilon$ is attained at $\varphi = \mathbf{1}\{P > e^\varepsilon Q\}$. It covers randomized tests: a distinguisher $\mathcal{B}$ with coins defines $\varphi(y) = \Pr[\mathcal{B}(y) = 1] \in [0,1]$.

**Lemma 2.2** (Post-processing; mixtures with common weights)**.** *Let $K$ be a Markov kernel from $\mathcal{Y}$ to a countable $\mathcal{Z}$, and write $PK(z) = \sum_y P(y)K(z \mid y)$. Then $\mathsf{M}_a(PK\|QK) \leq \mathsf{M}_a(P\|Q)$ and $\mathsf{E}_\varepsilon(PK\|QK) \leq \mathsf{E}_\varepsilon(P\|Q)$. In particular, for pmfs $\mu$ on a countable index set and families $(P_r)$, $(Q_r)$,*

$$\mathsf{M}_a\Big(\textstyle\sum_r \mu_r P_r \,\Big\|\, \sum_r \mu_r Q_r\Big) \leq \sum_r \mu_r \mathsf{M}_a(P_r\|Q_r), \qquad \mathsf{E}_\varepsilon\Big(\textstyle\sum_r \mu_r P_r \,\Big\|\, \sum_r \mu_r Q_r\Big) \leq \sum_r \mu_r \mathsf{E}_\varepsilon(P_r\|Q_r).$$

*Proof.* Both $\phi_a$ and $\psi_\varepsilon(p,q) = (p - e^\varepsilon q)_+$ are convex, lower semicontinuous and positively homogeneous of degree one on $[0,\infty)^2$. Here $\phi_a$ is the closed perspective of $x \mapsto x^a$. Such a function $f$ is subadditive, since $f(x + x') = 2f(\frac{x+x'}{2}) \leq f(x) + f(x')$. By lower semicontinuity this extends to countable sums: $f(\sum_i x_i) \leq \liminf_n \sum_{i\leq n} f(x_i) \leq \sum_i f(x_i)$. Hence

$$\mathsf{M}_a(PK\|QK) = \sum_z \phi_a\Big(\sum_y K(z \mid y)(P(y), Q(y))\Big) \leq \sum_z \sum_y K(z \mid y)\, \phi_a(P(y), Q(y)) = \mathsf{M}_a(P\|Q),$$

and likewise for $\psi_\varepsilon$. For the mixture statement, apply this to the joint pmfs $\mu_r P_r(y)$ and $\mu_r Q_r(y)$ of $(r, y)$, whose $\mathsf{M}_a$ and $\mathsf{E}_\varepsilon$ equal the right-hand sides, and to the projection $(r, y) \mapsto y$. □

**Lemma 2.3** (Conversion; [2], [CP], Proposition 4.4)**.** *Suppose $D_a(P\|Q) \leq a\rho$ for every $a > 1$. Then for every $\varepsilon \geq 0$,*

$$\mathsf{E}_\varepsilon(P\|Q) \leq \delta(\rho,\varepsilon) = \inf_{a>1} e^{(a-1)(a\rho-\varepsilon)} \frac{(a-1)^{a-1}}{a^a}.$$

*Moreover $\delta(\rho,\varepsilon) \leq e^{-(\varepsilon-\rho)^2/(4\rho)}$ for $\varepsilon \geq \rho$. Hence $\mathsf{E}_\varepsilon(P\|Q) \leq \delta'$ whenever $\varepsilon \geq \rho + 2\sqrt{\rho \ln(1/\delta')}$. The function $\delta(\cdot,\varepsilon)$ is continuous, and strictly increasing on the set where it is below 1; for $\varepsilon > 0$, $\delta(0,\varepsilon) = 0$.*

*Proof.* For $x \ge 0$ and $a > 1$ we have $(x - e^{\varepsilon})_+ \le c_a x^a$, where

$$c_a = \sup_{x>0} \frac{(x-e^{\varepsilon})_+}{x^a} = e^{-(a-1)\varepsilon} \frac{(a-1)^{a-1}}{a^a}.$$

The supremum is attained at $x = ae^{\varepsilon}/(a-1)$. Since $P \ll Q$,

$$\mathsf{E}_{\varepsilon}(P\|Q) = \sum_{Q(y)>0} Q(y)\Big(\frac{P(y)}{Q(y)} - e^{\varepsilon}\Big)_+ \le c_a \mathsf{M}_a(P\|Q) \le c_a e^{(a-1)a\rho}.$$

Take the infimum over $a$. For the simple form, note that $(a-1)^{a-1}/a^a = \frac{1}{a}(1-\frac{1}{a})^{a-1} \le 1$, and take $a = (\varepsilon+\rho)/(2\rho)$, which exceeds 1 when $\varepsilon > \rho$. The exponent is then $(a-1)(a\rho-\varepsilon) = -(\varepsilon-\rho)^2/(4\rho)$.

Continuity is [CP], Proposition 4.4. For strict monotonicity, let $\rho < \rho'$ with $\delta(\rho',\varepsilon) < 1$. The objective is continuous in $a$, tends to 1 as $a \to 1$ and to $+\infty$ as $a \to \infty$, so its infimum $\delta(\rho',\varepsilon) < 1$ is attained at some $a^*$. Then $\delta(\rho,\varepsilon) \le e^{(a^*-1)(a^*\rho-\varepsilon)}c' < e^{(a^*-1)(a^*\rho'-\varepsilon)}c' = \delta(\rho',\varepsilon)$, where $c' = (a^*-1)^{a^*-1}/a^{*a^*}$. □

**Lemma 2.4** (Integer shifts of the lattice Gaussian; [CP], Lemma 2.2)**.** *Let $\eta \sim D_{\mathbb{Z}^n,\sigma}$ and $\mathbf{e} \in \mathbb{Z}^n$. Then $D_a(\mathbf{e}+\eta\|\eta) \le a\|\mathbf{e}\|_2^2/(2\sigma^2)$ for every $a > 1$.*

*Proof.* Let $Z = \rho_\sigma(\mathbb{Z}^n)$. Since $\mathbf{e}$ is integral, $\mathbf{e}+\eta$ has pmf $\rho_\sigma(y-\mathbf{e})/Z$ on $\mathbb{Z}^n$. By the identity $a\|y-\mathbf{e}\|^2 + (1-a)\|y\|^2 = \|y-a\mathbf{e}\|^2 + a(1-a)\|\mathbf{e}\|^2$,

$$\mathsf{M}_a = e^{a(a-1)\|\mathbf{e}\|^2/(2\sigma^2)} \frac{\rho_\sigma(\mathbb{Z}^n - a\mathbf{e})}{Z} \le e^{a(a-1)\|\mathbf{e}\|^2/(2\sigma^2)}.$$

The inequality is [CP], Lemma 2.1 (`lem:gsum`), proved there by Poisson summation. □

The same bound holds for $D_a(\eta\|\mathbf{e}+\eta)$: shift both laws by $-\mathbf{e}$, which is a bijection of $\mathbb{Z}^n$.

**Lemma 2.5** (Smudging in statistical distance; [CP], Lemma 2.3(b) and Corollary 2.4)**.** *Let $\eta \sim D_{\mathbb{Z}^N,\sigma}$ with $N \le 2^{20}$, let $\mathbf{e} \in \mathbb{Z}^N$ with $\|\mathbf{e}\|_2 \le B$, and let $\sigma \ge \max\{2^\kappa B/\sqrt{2\pi}, 2^{42}\}$. Then $\mathrm{SD}(\mathbf{e}+\eta,\eta) \le \gamma'(\kappa) = 2^{-\kappa}(1+2^{-80}) + 2^{-97}$.*

*Proof.* [CP], Lemma 2.3(b), with smoothing error $\epsilon = 2^{-100}$ and $\sigma_2 = 2.1$, gives

$$\mathrm{SD} \le \frac{\|\mathbf{e}\|}{\sqrt{2\pi}\,\sigma_1} + \frac{4\epsilon}{1-\epsilon} \le 2^{-\kappa}\frac{\sigma}{\sigma_1} + 2^{-97.99},$$

where $\sigma_1 = \sqrt{\sigma^2 - \sigma_2^2}$. The choice $\sigma_2 = 2.1$ is admissible for $N \le 2^{20}$, because by [4, Lemma 3.3]

$$\frac{\eta_\epsilon(\mathbb{Z}^N)}{\sqrt{2\pi}} \le \frac{\sqrt{\ln(2N(1+2^{100}))/\pi}}{\sqrt{2\pi}} \le 2.07.$$

Finally, $\sigma/\sigma_1 = (1-\sigma_2^2/\sigma^2)^{-1/2} \le 1 + \sigma_2^2/\sigma^2 \le 1 + 4.41\cdot 2^{-84} < 1 + 2^{-81}$. □

**Remark 2.6** (The floor on $\sigma_c$)**.** [CP] and [P] stated the bound $\gamma'(\kappa)$ under $\sigma_c \ge \max\{2^\kappa B_{\mathrm{tot}}/\sqrt{2\pi}, 2^{20}\}$; both now use $2^{42}$. At $\sigma = 2^{20}$, however, $\sigma/\sigma_1$ is $1 + 2^{-38.9}$, and the proof's bound $1 + \sigma_2^2/\sigma^2$ is $1 + 2^{-37.9}$. Either way the factor is not $1 + 2^{-80}$. Correspondingly, the last step of [CP], Corollary 2.4 ($2^{-\kappa}\sigma/\sigma_1 + 2^{-97.9} \le 2^{-\kappa} + 2^{-97}$ "for every $\sigma \ge 2^{20}$") fails for $\kappa < 60$. The factor $1 + 2^{-80}$ needs $\sigma \ge 2.1 \cdot 2^{40} \approx 2^{41.1}$. This has no practical effect: at every evaluation setting $\kappa \ge 40$ and $B_{\mathrm{tot}} \ge 2^{17}$, so $\sigma_c \ge 2^\kappa B_{\mathrm{tot}}/\sqrt{2\pi} \ge 2^{55} \gg 2^{42}$. The fix is to write $2^{42}$ for $2^{20}$, or $1 + 2^{-37}$ for $1 + 2^{-80}$. Nothing below depends on the exact form of $\gamma'(\kappa)$, only on Lemma 2.5 holding for it.

## 3 Adaptive processes: a Rényi filter and two hybrid lemmas

**Definition 3.1** (Adaptive process)**.** Let $\mathcal{Y}$ be countable and $M \in \mathbb{N}$. An *adaptive process* $P$ of horizon $M$ on $\mathcal{Y}$ is a family of kernels: for each $t \le M$ and each history $h \in \mathcal{Y}^{t-1}$, a pmf $P_t(\cdot \mid h)$ on $\mathcal{Y}$. Its law is $P(y) = \prod_{t \le M} P_t(y_t \mid y_{<t})$, also written $P$. A process that stops early outputs a fixed symbol $\perp$ with probability one from then on. A family $c = (c_t)$ of functions $c_t : \mathcal{Y}^{t-1} \to [0, \infty)$ is *predictable*: the value at step $t$ depends only on the history before $t$.

**Theorem 3.2** (Rényi filter for zCDP)**.** *Let $P$ and $Q$ be adaptive processes of horizon $M$ on $\mathcal{Y}$, and let $c$ be predictable. Suppose*

*(i)* $D_a\big(P_t(\cdot \mid h) \| Q_t(\cdot \mid h)\big) \le a\, c_t(h)$ *for all* $t \le M$*, all* $h \in \mathcal{Y}^{t-1}$ *and all* $a > 1$*;*

*(ii)* $\sum_{t \le M} c_t(y_{<t}) \le \rho^*$ *for every* $y \in \mathcal{Y}^M$*.*

*Then* $D_a(P\|Q) \le a\rho^*$ *for every* $a > 1$*.*

*Proof.* Fix $a > 1$ and abbreviate $P_t(y_t \mid y_{<t})$ to $P_t$, and similarly $Q_t$ and $c_t$. By (i), $P_t(\cdot \mid h) \ll Q_t(\cdot \mid h)$ for every $h$, so the products below contain no term $\phi_a(p, 0)$ with $p > 0$. For $0 \le t \le M$ put

$$S_t = \sum_{y_{\le t} \in \mathcal{Y}^t} \prod_{j \le t} \phi_a(P_j, Q_j)\, e^{-a(a-1)c_j}, \qquad S_0 = 1.$$

Sum first over $y_t$, with $y_{<t}$ fixed. Since $c_t$ depends only on $y_{<t}$,

$$S_t = \sum_{y_{<t}} \Big( \prod_{j<t} \phi_a(P_j, Q_j) e^{-a(a-1)c_j} \Big)\, e^{-a(a-1)c_t}\, \mathsf{M}_a\big(P_t(\cdot \mid y_{<t}) \| Q_t(\cdot \mid y_{<t})\big) \le S_{t-1},$$

because $\mathsf{M}_a(P_t\|Q_t) = e^{(a-1)D_a(P_t\|Q_t)} \le e^{a(a-1)c_t}$ by (i). Hence $S_M \le 1$. The function $\phi_a$ is multiplicative, so $\phi_a(P(y), Q(y)) = \prod_t \phi_a(P_t, Q_t)$, and by (ii)

$$\mathsf{M}_a(P\|Q) = \sum_y \prod_t \phi_a(P_t, Q_t) \le e^{a(a-1)\rho^*} \sum_y \prod_t \phi_a(P_t, Q_t) e^{-a(a-1)c_t} = e^{a(a-1)\rho^*} S_M \le e^{a(a-1)\rho^*}.$$

Take $\frac{1}{a-1} \ln$. □

**Remark 3.3** (What the proof is, and attribution)**.** $S_t$ is $\mathbb{E}_Q[Z_t]$ for $Z_t = \prod_{j \le t} (P_j/Q_j)^a e^{-a(a-1)c_j}$, and the inequality $S_t \le S_{t-1}$ is the $Q$-supermartingale property of $Z_t$. This is the standard argument for Rényi filters. Condition (ii) holds for every history, which is exactly what a filter guarantees. A filter stops the process, or makes it output $\perp$ at cost 0, as soon as the next cost would exceed the remaining budget. That decision depends only on the accumulated public costs, so it is a stopping time.

The horizon is finite because the coalition is PPT (Assumption 1.9(A8)), so no optional-stopping theorem is needed. An unbounded horizon would need a passage to the limit $M \to \infty$, which is not needed here.

The cost $a\, c_t$ is linear in $a$ with one predictable $c_t$ for all orders, so one stopping rule serves every order and the conclusion is zCDP. Privacy filters were introduced by Rogers, Roth, Ullman and Vadhan [6]. The Rényi version, with adaptively chosen per-step Rényi costs and a fixed total, is due to Feldman and Zrnic [3], who prove a more general individual form [*citation detail unverified:* their main theorem is quoted as Theorem 1.1 in the arXiv version; numbering in the NeurIPS version not checked]. Whitehouse et al. [7] give fully adaptive filters for approximate DP and approximate zCDP. Nothing below depends on the exact statements of these papers.

**Corollary 3.4** (Steps of the protocol)**.** *The following steps satisfy condition (i) of Theorem 3.2 for the processes $P$ (dataset $\mathfrak{D}$) and $Q$ (dataset $\mathfrak{D}'$).*

*(a) A* public step*, whose kernel is the same under $\mathfrak{D}$ and $\mathfrak{D}'$, with $c_t = 0$. This includes deterministic steps (a function of $h$), refusals, and draws of coins that do not depend on $(\mathbf{x}_j, y_j)_j$.*

*(b) A* lattice Gaussian step*, $P_t(\cdot \mid h) = \mathrm{law}(F(h) + \eta)$ and $Q_t(\cdot \mid h) = \mathrm{law}(F'(h) + \eta)$, with $\eta \sim D_{\mathbb{Z}^{n(h)},\sigma(h)}$, $F(h) - F'(h) \in \mathbb{Z}^{n(h)}$ and $\|F(h) - F'(h)\|_2 \le \Delta(h)$, with $c_t(h) = \Delta(h)^2/(2\sigma(h)^2)$. The dimension, the centre, the bound and $\sigma$ may all depend on the history.*

*Deterministic post-processing of the transcript preserves the conclusion of Theorem 3.2 (Lemma 2.2).*

*Proof.* (a) is immediate. For (b), $D_a$ is invariant under the bijection $y \mapsto y - F'(h)$ of $\mathbb{Z}^{n(h)}$; apply Lemma 2.4 with $\mathbf{e} = F(h) - F'(h)$. □

**Lemma 3.5** (Statistical hybrid lemma with predictable distances)**.** *Let $P$ and $Q$ be adaptive processes of horizon $M$ on $\mathcal{Y}$, and let $\beta$ be predictable with $\mathrm{SD}(P_t(\cdot \mid h), Q_t(\cdot \mid h)) \le \beta_t(h)$ for all $t$ and $h$. Then*

$$\mathrm{SD}(P, Q) \le \mathbb{E}_{Y\sim P}\Big[\sum_{t\le M} \beta_t(Y_{<t})\Big].$$

*Proof.* For $0 \le t \le M$, let $H^{(t)}$ be the process with kernels $P_j$ for $j \le t$ and $Q_j$ for $j > t$, so that $H^{(M)} = P$ and $H^{(0)} = Q$. The processes $H^{(t)}$ and $H^{(t-1)}$ have the same law of $Y_{<t}$, namely $P$'s, and the same kernels after step $t$; they differ only in the kernel at step $t$. By Lemma 2.2 (a mixture over $y_{<t}$ with common weights, followed by the common continuation kernel),

$$\mathrm{SD}(H^{(t)}, H^{(t-1)}) \le \sum_{y_{<t}} P(y_{<t})\, \mathrm{SD}\big(P_t(\cdot \mid y_{<t}), Q_t(\cdot \mid y_{<t})\big) \le \mathbb{E}_P[\beta_t(Y_{<t})].$$

Sum over $t$ and use the triangle inequality. □

The point of the lemma is that every term is an expectation under the *same* law $P$, because each hybrid keeps $P$'s kernels on the prefix. Applied with $P$ the real experiment, it gives a slack that is one expectation under the real experiment.

**Lemma 3.6** (Adaptive composition of hockey-stick closeness)**.** *Let $P$ and $Q$ be adaptive processes of horizon $M$ on $\mathcal{Y}$. Let $\varepsilon_t \ge 0$ and $\delta_t \ge 0$ be predictable with $\mathsf{E}_{\varepsilon_t(h)}\big(P_t(\cdot \mid h) \| Q_t(\cdot \mid h)\big) \le \delta_t(h)$ for all $t$ and $h$, and suppose $\sum_t \varepsilon_t(y_{<t}) \le E$ for every $y$. Then*

$$\mathsf{E}_E(P\|Q) \le \mathbb{E}_{Y\sim P}\Big[\sum_{t\le M} \delta_t(Y_{<t})\Big].$$

*Proof.* Put $P'_t(y_t \mid h) = \min\{P_t(y_t \mid h), e^{\varepsilon_t(h)} Q_t(y_t \mid h)\}$. Then $\sum_{y_t}(P_t - P'_t)(y_t \mid h) = \mathsf{E}_{\varepsilon_t(h)}(P_t\|Q_t) \le \delta_t(h)$. Let $P'(y) = \prod_t P'_t(y_t \mid y_{<t})$. Then $P'(y) \le e^{\sum_t \varepsilon_t} Q(y) \le e^E Q(y)$. Telescoping, with $0 \le P'_j \le P_j$,

$$P(y) - P'(y) = \sum_{t\le M} \Big(\prod_{j<t} P'_j\Big)(P_t - P'_t)\Big(\prod_{j>t} P_j\Big) \ \ge 0.$$

Summing over $y$, the factor $\prod_{j>t} P_j$ sums to 1 over $y_{>t}$. Bounding $P'_j \le P_j$ for $j < t$ gives

$$\sum_y (P - P')(y) \le \sum_t \sum_{y_{<t}} P(y_{<t})\, \delta_t(y_{<t}) = \mathbb{E}_P\Big[\sum_t \delta_t\Big].$$

For any $\varphi : \mathcal{Y} \to [0,1]$, $\mathbb{E}_P\varphi \le \mathbb{E}_{P'}\varphi + \sum_y (P - P')(y) \le e^E \mathbb{E}_Q\varphi + \mathbb{E}_P[\sum_t \delta_t]$. □

The $\varepsilon_t$ must sum to at most $E$ pointwise, because they enter as a factor $e^E$. The $\delta_t$ enter only through their expectation under $P$.

**Lemma 3.7** (Adaptively chosen parameter sets)**.** *For each $p$ in a finite set $\Pi$, let $G_p$ be a decision problem: a challenger draws $b \in \{0,1\}$ and hands out a challenge $X_p^b$. Let $\mathcal{B}$ be an adversary that first computes, possibly at random, a choice $p \in \Pi$, then receives $X_p^b$, and outputs a bit. Let $\mathcal{B}_p$ run $\mathcal{B}$ until it chooses some $p'$, output 0 if $p' \neq p$, and otherwise forward its own challenge to $\mathcal{B}$ and output $\mathcal{B}$'s bit. Then $\mathrm{Adv}(\mathcal{B}) \leq \sum_{p\in\Pi} \mathrm{Adv}_p(\mathcal{B}_p)$, and each $\mathcal{B}_p$ runs in the time of $\mathcal{B}$.*

*Proof.* $\mathcal{B}$'s choice is made before it sees the challenge, so it is independent of $b$. Hence $\Pr[\mathcal{B}=1 \mid b] = \sum_p \Pr[\mathcal{B} \text{ chooses } p]\Pr[\mathcal{B}=1 \mid p,b] = \sum_p \Pr[\mathcal{B}_p = 1 \mid b]$. Take the difference over $b$ and apply the triangle inequality. □

# 4 Composition across runs

## 4.1 The ideal experiment

**Definition 4.1** (The ideal experiment $\mathsf{Ideal}_{\mathcal{A}}(\mathfrak{D})$)**.** $\mathsf{Ideal}_{\mathcal{A}}(\mathfrak{D})$ is $\mathsf{Real}_{\mathcal{A}}(\mathfrak{D})$ with the same coalition, CALIBRATE, ledger, refusals and honest machines $\mathsf{R}_k$, with one change: in every admitted run $k$, $\mathsf{O}$ is replaced by the ideal party $\tilde{\mathsf{O}}_k$. It sends its messages at the same points of the protocol as $\mathsf{O}$.

(I1) It sends the public parameters, as $\mathsf{O}$ does.

(I2) It replaces $\mathsf{O}$'s key-generation shares (the public-key share and the rotation-key shares) by independent uniform elements of the corresponding rings.

(I3) It runs $\mathsf{O}$'s Link phase (OPRF, labeled PSI and SANITIZEBFV) honestly on the identifiers of $D_{P_k}(\mathfrak{D};\mathbf{d}_k)$, with fresh coins (OPRF key, $\tau_k$, $\mathbf{d}_k$, sanitization coins). This uses only $\mathrm{fix}(\mathfrak{D})$ and $P_k$.

(I4) It receives $\mathsf{R}_k$'s uploads and ignores them; it performs no aggregation.

(I5) At each release $s$ that $\mathcal{A}$ requests, it draws $c^* \leftarrow \mathrm{U}(\mathcal{R}_{q_0})$ and $\eta \sim D_{\mathbb{Z}^N,\sigma_{c,s}}$, sets

$$w_{k,s} = f_{k,s}(\mathfrak{D}) + \eta, \qquad f_{k,s}(\mathfrak{D}) = f_s\big(D_{P_k}(\mathfrak{D};\mathbf{d}_k);\pi_k,\chi_k\big) \in \mathbb{Z}^N,$$

and sends $(c^*,\ w_{k,s} - c^* s_R \bmod q_0)$. Here $f_s$ is the centre of [CP], Definition 3.7 (`def:centre`), $\pi_k$ is the arrangement defined by $\tau_k$, and $s_R$ is read from $\omega_{R,k}$.

$\tilde{\mathsf{O}}_k$ uses $(\mathbf{x}_j, y_j)_j$ only through the values $w_{k,s}$. Run by run, $\tilde{\mathsf{O}}_k$ is the simulator of [CP], Theorem 4.11, applied to that theorem's mechanism $\mathcal{M}_k = (\omega_{R,k}, \Lambda_k, (f_{k,s} + \eta_{k,s})_s)$.

**Theorem 4.2** (Composition across runs; replaces Proposition `prop:compose`)**.** *Assume Assumption 1.9, and let $\mathcal{A}$ be a PPT coalition. Let $n_{\mathrm{rel}} \leq n_{\max}$ for every $P \in \mathcal{P}_\lambda$. Let $t_{\mathcal{B}}(\lambda)$ be the polynomial running time of the reductions in Step 3 of the proof. Put*

$$\nu_{\mathcal{A}}(\lambda) = t_{\mathcal{A}}(\lambda) \sum_{p\in\Pi_\lambda} \big(n_{\max}\mathrm{Adv}_{1,p}(\lambda, t_{\mathcal{B}}(\lambda)) + \mathrm{Adv}_{2,p}(\lambda, t_{\mathcal{B}}(\lambda))\big),$$

*which is negligible.*

*(a)* Simulation. *For every $\mathfrak{D}$,*

$$\big|\Pr[\mathsf{Real}_{\mathcal{A}}(\mathfrak{D})=1] - \Pr[\mathsf{Ideal}_{\mathcal{A}}(\mathfrak{D})=1]\big| \leq \Gamma_{\mathcal{A}}(\mathfrak{D}) + \nu_{\mathcal{A}}(\lambda),$$

*where $\Gamma_{\mathcal{A}}(\mathfrak{D}) = \mathbb{E}_{\mathsf{Real}_{\mathcal{A}}(\mathfrak{D})}\big[\sum_{k \text{ admitted}} \gamma_k\big]$.*

*(b)* Ideal composition. *Suppose that in every execution the admitted runs satisfy* $\sum_k \rho_k \le \rho^*$*, for a constant* $\rho^*$*. Then* $D_a\big(\mathrm{view}_{\mathcal{A}}^{\mathsf{Ideal}}(\mathfrak{D}) \| \mathrm{view}_{\mathcal{A}}^{\mathsf{Ideal}}(\mathfrak{D}')\big) \le a\rho^*$ *for all* $a > 1$ *and* $\mathfrak{D} \sim_{\mathrm{b}} \mathfrak{D}'$*. Hence, for every* $\varepsilon \ge 0$*,*

$$\Pr[\mathsf{Ideal}_{\mathcal{A}}(\mathfrak{D}) = 1] \le e^{\varepsilon} \Pr[\mathsf{Ideal}_{\mathcal{A}}(\mathfrak{D}') = 1] + \delta(\rho^*, \varepsilon).$$

*(c)* Accounting. *Suppose in addition that* $\sum_k \gamma_k \le \Gamma^*$ *over the admitted runs in every execution. Then for every* $\varepsilon \ge 0$ *and* $\mathfrak{D} \sim_{\mathrm{b}} \mathfrak{D}'$*,*

$$\Pr[\mathsf{Real}_{\mathcal{A}}(\mathfrak{D}) = 1] \le e^{\varepsilon} \Pr[\mathsf{Real}_{\mathcal{A}}(\mathfrak{D}') = 1] + \delta(\rho^*, \varepsilon) + (1 + e^{\varepsilon})\Gamma^* + (1 + e^{\varepsilon})\nu_{\mathcal{A}}(\lambda).$$

*(d)* Rule (F), any adaptive coalition. *Under the filter rule, (b) and (c) hold with* $(\rho^*, \Gamma^*) = (\rho_L, \Gamma_L)$*, and the protocol is* $(\varepsilon_L, \delta_L)$*-IND-CDP for* $\sim_{\mathrm{b}}$*. The* $\delta$ *budget is used exactly:* $\delta(\rho_L, \varepsilon_L) + (1 + e^{\varepsilon_L})\Gamma_L = \delta_L$*.*

*(e)* Rule (N), requests fixed in advance. *Suppose the ledger uses the non-adaptive rule and* $\mathcal{A}$*'s requests are a fixed list* $(P_k, \chi_k)_{k \le K}$*, submitted in this order and chosen independently of* $\omega_{\mathcal{A}}$ *and of every message.* $\mathcal{A}$ *may still schedule, abort and stop adaptively. Let* $(\rho^{\mathrm{fin}}, \Gamma^{\mathrm{fin}})$ *be the ledger's final state when the whole list is submitted. Then (b) and (c) hold with* $(\rho^*, \Gamma^*) = (\rho^{\mathrm{fin}}, \Gamma^{\mathrm{fin}})$*,* $\delta(\rho^{\mathrm{fin}}, \varepsilon_L) + (1 + e^{\varepsilon_L})\Gamma^{\mathrm{fin}} \le \delta_L$*, and the protocol is* $(\varepsilon_L, \delta_L)$*-IND-CDP for* $\sim_{\mathrm{b}}$ *against such coalitions.*

Parts (a) and (b) together say that the joint view is SIM-CDP with an *interactive* mechanism: $\mathcal{A}$ interacting with the parties $\tilde{\mathsf{O}}_k$ is a post-processing of an adaptive $\rho^*$-zCDP mechanism, with statistical slack $\Gamma_{\mathcal{A}}(\mathfrak{D}) \le \Gamma^*$.

## 4.2 Proof of Theorem 4.2

**Step 0: the experiments as adaptive processes.** Make three changes of presentation, none of which changes the law of $\mathrm{view}_{\mathcal{A}}$.

(i) *Draw a run's coins at admission.* When run $k$ is admitted, draw $\omega_{R,k}$ and all of $\omega_{O,k}$ except the noises $\eta_{k,s}$. Each $\eta_{k,s}$ is drawn at release $s$. By (A3) these coins are independent of everything earlier, and drawing them earlier changes nothing.

(ii) *Rewrite the share* as in hybrid H1 of [CP], Theorem 4.11. $\mathsf{O}$'s share $h_O = c'_0 + c'_1 s_O + \eta$ equals $v - c'_1 s_R$ with $v = c'_0 + c'_1 s + \eta$, and by [CP], Proposition 3.8 (`prop:decomp`),

$$v \equiv f_{k,s}(\mathfrak{D}) + e_{k,s} + \eta \pmod{q_0}, \qquad e_{k,s} = e_{\mathrm{tot},s} \text{ of run } k \in \mathbb{Z}^N.$$

(iii) *Honest completions.* The honest machine $\mathsf{R}_k$ depends only on $(P_k, \chi_k, \omega_{R,k})$ and on $\mathsf{O}$'s messages in run $k$, and those depend only on $(\mathfrak{D}, P_k, \omega_{O,k})$ and $\mathsf{R}_k$'s messages. So each admitted run has an *honest completion*: the transcript it would have if $\mathcal{A}$ never aborted it. The honest completion is a function of $(\mathfrak{D}, P_k, \chi_k, \omega_{O,k}, \omega_{R,k})$ and does not depend on other runs or on the scheduling. Let $e_{k,s}$ be the error of release $s$ in the honest completion, for every $s \le n_{\mathrm{rel},k}$. When release $s$ is made, it is the error of the actual release. It does not involve any $\eta$, because $M_1$, $M_2$ (and $M_y$) are computed before any release. By (A5) and (A3), for every history up to the admission of run $k$,

$$\Pr\big[\|e_{k,s}\|_2 > B_{\mathrm{tot},k,s} \;\big|\; \text{history up to the admission of run } k\big] \le \delta_{B,k}. \tag{1}$$

Every experiment below is then an adaptive process on a countable alphabet. Its steps are the draw of $\omega_{\mathcal{A}}$, the actions of $\mathcal{A}$ (deterministic given the history), $\mathsf{O}$'s answers, the draws of coins at admission, and the messages of open runs. The horizon is $M \le \mathrm{poly}(\lambda)$, by (A8). $\mathcal{A}$'s output is a function of the process. The ledger's state is a function of the history, and every refusal is a deterministic function of the history.

**Step 1: one statistical hybrid for all runs.** Let $\mathsf{Hyb}^{\mathrm{st}}_{\mathcal{A}}(\mathfrak{D})$ be $\mathsf{Real}_{\mathcal{A}}(\mathfrak{D})$ with $v$ replaced by $f_{k,s}(\mathfrak{D})+\eta$ in every release that is made. The share sent is then $f_{k,s}(\mathfrak{D})+\eta-c'_1 s_R \bmod q_0$. This is H2 of [CP], Theorem 4.11, done in every run at once.

The two processes have the same kernels at every step except the release steps. At a release step $(k,s)$ with history $h$, $c'_1$, $s_R$, $f_{k,s}(\mathfrak{D})$ and $e_{k,s}$ are functions of $h$, and $\eta$ is fresh. The real kernel is the law of $(c'_1,\ f_{k,s}+e_{k,s}+\eta-c'_1 s_R \bmod q_0)$, and the hybrid kernel is the same with $e_{k,s}$ removed. By Lemma 2.2, which covers the shift and the reduction modulo $q_0$, and Lemma 2.5, their distance is at most

$$\beta_{(k,s)}(h)=\mathrm{SD}(e_{k,s}+\eta,\eta)\le \mathbf{1}\{\|e_{k,s}\|>B_{\mathrm{tot},k,s}\}+\gamma'(\kappa_k).$$

By Lemma 3.5, with the real experiment as $P$,

$$\begin{aligned}\mathrm{SD}\big(\mathsf{Real}_{\mathcal{A}}(\mathfrak{D}),\mathsf{Hyb}^{\mathrm{st}}_{\mathcal{A}}(\mathfrak{D})\big)&\le \mathbb{E}\Big[\sum_{(k,s)\ \mathrm{made}}\mathbf{1}\{\|e_{k,s}\|>B_{\mathrm{tot},k,s}\}\Big]+\mathbb{E}\Big[\sum_{(k,s)\ \mathrm{made}}\gamma'(\kappa_k)\Big]\\&\le\sum_k\sum_{s\le n_{\mathrm{rel},k}}\mathbb{E}\big[\mathbf{1}\{k\ \mathrm{adm}\}\,\mathbf{1}\{\|e_{k,s}\|>B_{\mathrm{tot},k,s}\}\big]+\mathbb{E}\Big[\sum_{k\ \mathrm{adm}}n_{\mathrm{rel},k}\gamma'(\kappa_k)\Big]\\&\le\mathbb{E}\Big[\sum_{k\ \mathrm{adm}}n_{\mathrm{rel},k}\big(\delta_{B,k}+\gamma'(\kappa_k)\big)\Big]=\Gamma_{\mathcal{A}}(\mathfrak{D}).\end{aligned}$$

Expectations are under $\mathsf{Real}_{\mathcal{A}}(\mathfrak{D})$. The second line uses $\{(k,s)\text{ made}\}\subseteq\{k\text{ adm}\}$. The third line conditions on the history up to the admission of run $k$, of which $\{k\text{ adm}\}$ is a function, and applies (1).

*Why every planned release is charged $\delta_B$.* The only usable bound on $\mathbf{1}\{\text{made}\}\mathbf{1}\{\text{bad}\}$ is $\mathbf{1}\{\text{adm}\}\mathbf{1}\{\text{bad}\}$. $\mathcal{A}$'s decision to request release $s$ may depend on its view, which contains $s_R$ and $\mathsf{R}$'s encryption errors. Both enter $e_{k,s}$, through $T_1$, $T_2$ and $T_6$ of [CP], Proposition 3.8. So $\Pr[\text{made}\wedge\text{bad}]$ need not be at most $\delta_B\Pr[\text{made}]$. The term $\gamma'(\kappa_k)$, by contrast, bounds the distance for every value of $e_{k,s}$ of norm at most $B_{\mathrm{tot}}$, so it is charged only for releases made.

*Expectation versus pointwise bound.* Step 1 needs only the expectation $\Gamma_{\mathcal{A}}(\mathfrak{D})$ under the real experiment, for $\mathfrak{D}$ and for $\mathfrak{D}'$. Lemma 3.5 makes it a single expectation; doing the smudging steps one at a time, interleaved with computational steps, would give expectations under different hybrids. The pointwise bound $\sum_{\mathrm{adm}}\gamma_k\le\Gamma^*$ that the ledger enforces implies $\Gamma_{\mathcal{A}}(\mathfrak{D})\le\Gamma^*$ for every $\mathfrak{D}$ and every $\mathcal{A}$. The ledger cannot compute expectations, which depend on $\mathfrak{D}$ and $\mathcal{A}$, so the pointwise rule is the implementable one. The zCDP part (Step 4) genuinely needs the pointwise bound on $\sum\rho_k$.

**Step 2: the order of hybrids.** Order the admitted runs by admission time, $k=1,2,\dots,T$ with $T\le t_{\mathcal{A}}(\lambda)$. For $1\le k\le t_{\mathcal{A}}(\lambda)$ and $0\le j\le n_{\max}+1$, let $\mathsf{Hyb}_{k,j}$ be the following experiment.

- Admitted runs $k'<k$ are played by the ideal parties $\tilde{\mathsf{O}}_{k'}$.
- Run $k$ is played as in $\mathsf{Hyb}^{\mathrm{st}}$, except that $c'_{1,k,s}$ is uniform in $\mathcal{R}_{q_0}$ for $s\le\min(j,n_{\mathrm{rel},k})$, and, if $j=n_{\max}+1$, $\mathsf{O}$'s key-generation shares are uniform.
- Runs $k'>k$ are played as in $\mathsf{Hyb}^{\mathrm{st}}$.

Then $\mathsf{Hyb}_{1,0}=\mathsf{Hyb}^{\mathrm{st}}$, $\mathsf{Hyb}_{k,n_{\max}+1}=\mathsf{Hyb}_{k+1,0}$, and $\mathsf{Hyb}_{t_{\mathcal{A}},n_{\max}+1}=\mathsf{Ideal}_{\mathcal{A}}(\mathfrak{D})$.

The middle identity holds because, in $\mathsf{Hyb}_{k,n_{\max}+1}$, run $k$ is already ideal. Every share sent is $f_{k,s}+\eta-c^*s_R$ with $c^*$ uniform, and the key shares are uniform. The Link phase is $\mathsf{O}$'s honest code. The ciphertexts $M_s$ no longer enter the view, so neither do the conversion, the aggregation and the key switching that produce them. This is exactly $\tilde{\mathsf{O}}_k$. If fewer than $k$ runs are admitted, $\mathsf{Hyb}_{k,j}$ does not depend on $j$.

**Step 3: the computational steps.** *Re-randomizer steps,* $1 \le j \le n_{\max}$. The reduction $\mathcal{B}_{k,j}$ plays the adaptive-parameter game of Lemma 3.7 for [CP], Assumption 4.9. It has $\mathfrak{D}$ as non-uniform advice. It draws $\omega_{\mathcal{A}}$ and runs $\mathcal{A}$, playing every other party and the ledger itself, exactly as in $\mathsf{Hyb}_{k,j-1}$.

- *Runs admitted before run k* are played by $\tilde{\mathsf{O}}_{k'}$. $\mathcal{B}_{k,j}$ can do this: the Link phase needs $\mathrm{fix}(\mathfrak{D})$; a release needs $f_{k',s}(\mathfrak{D}) + \eta$, computable in polynomial time by (A7); the key shares and $c^*$ are uniform.
- *Run k.* When run $k$ is admitted with parameter set $p_k$, $\mathcal{B}_{k,j}$ announces $p_k$ and receives a challenge $(a^*, b^*)$ over $\mathcal{R}_{q_0}$. It draws all of run $k$'s coins itself, with two exceptions. First, the public-key CRS $a$ is a uniform lift of $a^*$: $a \equiv a^* \pmod{q_0}$, uniform modulo the full modulus. Second, $(u_{k,j}, e_{1,k,j})$ is not drawn.
  - It sets $c'_{1,k,s}$ uniform for $s < j$.
  - It sets $c'_{1,k,j} = c_{1,k,j} + b^* \bmod q_0$. Here $c_{1,k,j}$ is computable from the public randomness and the data, by [CP], Lemma 3.6 (`lem:c1indep`).
  - It sets $c'_{1,k,s} = c_{1,k,s} + a u_{k,s} + e_{1,k,s}$ for $s > j$, which it can compute since it knows $a$.

  Each release that is made sends $(c'_{1,k,s},\ f_{k,s}(\mathfrak{D}) + \eta - c'_{1,k,s} s_R \bmod q_0)$. In $\mathsf{Hyb}^{\mathrm{st}}$ nothing else in run $k$ uses $(u_{k,j}, e_{1,k,j})$: $c'_0$ is never sent, and it is no longer used, because $v$ no longer contains $e_{k,j}$, whose term $T_6$ was the only other occurrence. If $j > n_{\mathrm{rel},k}$, $\mathcal{B}_{k,j}$ ignores the challenge.
- *Runs admitted after run k* are played as in $\mathsf{Hyb}^{\mathrm{st}}$. $\mathcal{B}_{k,j}$ draws their keys and coins and computes everything from $\mathfrak{D}$.
- *Public computations.* $\mathcal{B}_{k,j}$ computes Calibrate, the ledger and every refusal from public data (A7). It obtains the adaptively chosen requests by running $\mathcal{A}$ on the view it has generated, which has exactly the law of $\mathcal{A}$'s view in the hybrid. This is why the adaptively chosen parameters are efficiently computable by the reduction: they are the outputs of a PPT machine on an input the reduction has produced.
- $\mathcal{B}_{k,j}$ outputs $\mathcal{A}$'s output.

If $b^* = a^* u + e_1$ with fresh $u \leftarrow \chi_{\mathrm{key}}$ and $e_1 \leftarrow \chi_{\mathrm{err}}$, the simulation has the law of $\mathsf{Hyb}_{k,j-1}$. If $b^*$ is uniform, $c'_{1,k,j}$ is uniform and the simulation has the law of $\mathsf{Hyb}_{k,j}$. If run $k$ is never admitted, or its release $j$ is never made, the two hybrids coincide, and $\mathcal{B}_{k,j}$'s output does not depend on the challenge.

$\mathcal{B}_{k,j}$ runs in time $t_{\mathcal{B}}(\lambda) = O(t_{\mathcal{A}}(\lambda)) + t_{\mathcal{A}}(\lambda) \cdot \mathrm{poly}(\lambda)$, because it plays at most $t_{\mathcal{A}}$ runs of polynomial cost. By Lemma 3.7,

$$\left|\Pr[\mathsf{Hyb}_{k,j-1} = 1] - \Pr[\mathsf{Hyb}_{k,j} = 1]\right| \le \sum_{p \in \Pi_\lambda} \mathrm{Adv}_{1,p}(\lambda, t_{\mathcal{B}}(\lambda)).$$

*Key-share step,* $j = n_{\max} + 1$. $\mathcal{B}_{k,n_{\max}+1}$ plays the game for [CP], Assumption 4.10, in the same way. It embeds the challenge as $\mathsf{O}$'s key-generation shares in run $k$, uses the challenge's $a$ and $a_{g,j}$ as run $k$'s common random strings, and computes the joint keys from the challenge and $\mathsf{R}_k$'s shares; it knows $\omega_{R,k}$. In $\mathsf{Hyb}_{k,n_{\max}}$ the view depends on run $k$'s $(s_O, e_O^{\mathrm{pk}}, e_O^{\mathrm{rk}})$ only through these shares:

- every release share is $f + \eta - c'_1 s_R$ with $c'_1$ uniform;
- $c'_0$ is unused;
- the conversion $c_0 - a s_O$, the aggregation and the key switching produce only the ciphertexts $M_s$, which no longer enter the view.

So $\mathcal{B}_{k,n_{\max}+1}$ never needs $s_O$. A real challenge gives $\mathsf{Hyb}_{k,n_{\max}}$, and a uniform one gives $\mathsf{Hyb}_{k,n_{\max}+1}$. The loss is at most $\sum_p \mathrm{Adv}_{2,p}(\lambda, t_{\mathcal{B}}(\lambda))$.

*Number of runs and total loss.* There are at most $t_{\mathcal{A}}(\lambda)(n_{\max} + 1)$ steps, by (A8), so

$$\left|\Pr[\mathsf{Hyb}^{\mathrm{st}} = 1] - \Pr[\mathsf{Ideal}_{\mathcal{A}}(\mathfrak{D}) = 1]\right| \le \nu_{\mathcal{A}}(\lambda).$$

$\nu_{\mathcal{A}}$ is negligible by (A6), since $t_{\mathcal{A}}$, $n_{\max}$ and $|\Pi_\lambda|$ are polynomial and $t_{\mathcal{B}}$ is polynomial. Together with Step 1 this proves (a).

**Step 4: the ideal experiment is $\rho^*$-zCDP.** Write $\mathsf{Ideal}_{\mathcal{A}}(\mathfrak{D})$ as an adaptive process $Y$ with three kinds of random step.

(a) The draw of $\omega_{\mathcal{A}}$.

(b) For each admitted run $k$, one step that draws $r_k$. It consists of $\omega_{R,k}$, $\tilde{\mathsf{O}}_k$'s Link coins (OPRF key, $\tau_k$, $\mathbf{d}_k$, sanitization coins), its uniform key shares and $(c^*_{k,s})_{s\le n_{\mathrm{rel},k}}$.

(c) For each release $(k,s)$ that $\mathcal{A}$ requests, one step that outputs $w_{k,s}\in\mathbb{Z}^N$.

Everything else is a deterministic function of the history: $\mathcal{A}$'s actions, the refusals, the ledger state, the PSI messages, and the release messages $(c^*, w - c^* s_R \bmod q_0)$. The transcript $Y$ contains $\tau_k$ even though $\mathcal{A}$ does not see it. $\mathcal{A}$'s view is a deterministic function of $Y$.

Compare $Y$ under $\mathfrak{D}$ and $\mathfrak{D}'$, with $\mathfrak{D}\sim_{\mathrm{b}}\mathfrak{D}'$. Steps (a) and (b) have kernels that do not depend on $(\mathbf{x}_j,y_j)_j$, so they are public steps with cost 0. A step (c) with history $h$ is a lattice Gaussian step in the sense of Corollary 3.4(b), with $F(h)=f_{k,s}(\mathfrak{D})$ and $F'(h)=f_{k,s}(\mathfrak{D}')$. Its dimension $N$ and $\sigma_{c,k,s}$ are given by $P_k$, and $\pi_k$ and $\chi_k$ are coordinates of $h$. By Lemma 1.3, the two scope registers coincide or are adjacent, so by (A5) $\|F(h)-F'(h)\|_2\le\widehat{\Delta}_{\mathrm{enc},k,s}$ for every arrangement and every researcher input. Both centres are integral, so the step's cost is $c(h)=\widehat{\Delta}^2_{\mathrm{enc},k,s}/(2\sigma^2_{c,k,s})\le\rho_{k,s}$.

For every history, $\sum_t c_t\le\sum_{k\ \mathrm{adm}}\sum_s\rho_{k,s}=\sum_{k\ \mathrm{adm}}\rho_k\le\rho^*$. This holds for every history, not only almost surely, because the ledger state is a function of the history and enforces the bound. Theorem 3.2 gives $D_a(Y(\mathfrak{D})\|Y(\mathfrak{D}'))\le a\rho^*$, and Lemma 2.2 transfers this to $\mathcal{A}$'s view and output. Lemma 2.3 gives the hockey-stick bound. The roles of $\mathfrak{D}$ and $\mathfrak{D}'$ are symmetric. This proves (b).

*Refusals and the gate.* The refusals are deterministic functions of the history, that is, public steps of cost 0. The stage-2 gate is part of $\mathcal{A}$'s computation. The theorem bounds the unconditional joint law of the view, of which the refusal pattern and the gate's outcome are coordinates. We never condition on them. A guarantee conditional on an event such as "run $k$ was admitted" or "the gate passed" does not follow and is not claimed.

**Step 5: conversion and the budget.** Write $\Gamma=\Gamma_{\mathcal{A}}(\mathfrak{D})$, $\Gamma'=\Gamma_{\mathcal{A}}(\mathfrak{D}')$ and $\nu=\nu_{\mathcal{A}}(\lambda)$. By (a) for $\mathfrak{D}$, then (b), then (a) for $\mathfrak{D}'$,

$$\begin{aligned}\Pr[\mathsf{Real}(\mathfrak{D})=1]&\le\Pr[\mathsf{Ideal}(\mathfrak{D})=1]+\Gamma+\nu\le e^{\varepsilon}\Pr[\mathsf{Ideal}(\mathfrak{D}')=1]+\delta(\rho^*,\varepsilon)+\Gamma+\nu\\&\le e^{\varepsilon}\big(\Pr[\mathsf{Real}(\mathfrak{D}')=1]+\Gamma'+\nu\big)+\delta(\rho^*,\varepsilon)+\Gamma+\nu.\end{aligned}$$

With $\Gamma,\Gamma'\le\Gamma^*$ this is (c). This is [P], Lemma `lem:sim2ind`, with the factor $1+e^{\varepsilon}$ at the final $\varepsilon$.

(d) Under rule (F), every admission keeps both sums within their caps, and the sums are the ledger's state. So in every execution $\sum_{\mathrm{adm}}\rho_k\le\rho_L$ and $\sum_{\mathrm{adm}}\gamma_k\le\Gamma_L$. Take $\varepsilon=\varepsilon_L$. By the definition of $\rho_L$ and the continuity and strict monotonicity of $\delta(\cdot,\varepsilon_L)$ (Lemma 2.3), $\delta(\rho_L,\varepsilon_L)=(1-\zeta_L)\delta_L$. Also $(1+e^{\varepsilon_L})\Gamma_L=\zeta_L\delta_L$, so the sum is $\delta_L$ exactly. The term $(1+e^{\varepsilon_L})\nu$ is negligible because $\varepsilon_L$ is a constant. At $(10,10^{-6},0.1)$ the sum evaluates to $9.99999999999\cdot10^{-7}$ (script `scripts/kappa_numbers.py`).

(e) The requests are fixed, so their costs are fixed. The full cost is charged at admission, so the ledger's decisions depend only on the cost sequence. The set $\mathcal{K}$ of runs admitted when the whole list is submitted is therefore deterministic. In every execution the admitted runs form a subset of $\mathcal{K}$, since $\mathcal{A}$ may stop early, so the sums are at most $(\rho^{\mathrm{fin}},\Gamma^{\mathrm{fin}})$. If $\mathcal{K}\neq\emptyset$, the test at the last admission in $\mathcal{K}$ reads exactly $\delta(\rho^{\mathrm{fin}},\varepsilon_L)+(1+e^{\varepsilon_L})\Gamma^{\mathrm{fin}}\le\delta_L$. If $\mathcal{K}=\emptyset$, no run is admitted and the view does not depend on $(\mathbf{x}_j,y_j)_j$ at all. □

### 4.3 Corner cases

**Refused runs.** A run refused by CALIBRATE or by the ledger produces one message, $\perp_{\mathrm{cal}}$ or $\perp_{\mathrm{led}}$. That message is a function of $P_k$ and the ledger state, both public. No key is generated and no share is sent. In the proof such a run is a public step of cost 0 (Step 4). The refusal pattern can depend on earlier outputs, through $\mathcal{A}$'s requests, but it is part of the view, which the theorem covers unconditionally.

**The stage-2 gate.** A refused gate means that $\mathsf{R}$ withholds its estimate after the stage-2 release has been made and charged. This is post-processing of $\mathcal{A}$'s view and changes nothing. The paper's sentence "a refused stage-2 gate still spends the budget" is therefore correct and necessary.

**Aborted runs.** $\mathcal{A}$ may abort any open run at any time, adaptively. The proof handles this through the honest completion (Step 0(iii)): the ideal experiment releases only the prefix that $\mathcal{A}$ requests, and the costs of unmade releases are simply not used in Step 4. An abort by $\mathsf{O}$ is covered by (A1). Such an abort depends only on public and data-independent quantities, such as a PSI table overflow, which depends on the identifiers and the OPRF key. The accounting must keep the $\delta_B$ part of every planned release of an admitted run (Step 1 and Remark 4.3).

**The PSI transcript.** The identifiers are part of $\mathrm{fix}(\mathfrak{D})$, so the Link phase of every run is identical under $\mathfrak{D}$ and $\mathfrak{D}'$. The ideal experiment runs $\mathsf{O}$'s real Link code (I3), so no PSI security is used for this theorem. What the Link phase discloses $(I,\ \pi|_I,\ n)$ is outside the DP guarantee, and over many runs this disclosure accumulates: [P], Remark `rem:revealed`, and G4. The OPRF quota limits it as a matter of policy, not as part of this theorem.

**Different researchers' cohorts.** The cohorts are inputs $\chi_k$ of the coalition, chosen adaptively, possibly overlapping, and not protected. Step 4 compares the two datasets for a *fixed* history, which fixes $\chi_k$. The sensitivity bound holds for every cohort in the clipped domain. So the cohorts may depend on earlier outputs in arbitrary ways. A row matched in many runs is charged in every run, through $\sum \rho_k$.

**One base register, several scopes and models.** By Lemma 1.3, base adjacency induces adjacency or equality of every run's scope register. This holds for every scope, model, column subset and public row-wise preprocessing. The ledger must therefore be shared across all scopes and models of a base register, as the paper specifies; a ledger per scope would not give Theorem 4.2. The guarantee is per base register and per row. If $\mathsf{O}$ holds several base registers that contain the same people, protecting a person across them needs one ledger over their union, which is outside this theorem.

**Concurrency.** Several runs may be open at once and interleaved arbitrarily (Definition 1.7(b)). The proof indexes hybrids by admission order and orders the statistical and filter steps by global time. Both are insensitive to interleaving, provided admission is atomic and charges the full cost.

**The number of runs.** By (A8) the coalition makes at most $t_{\mathcal{A}}(\lambda)$ requests, which bounds the number of computational hybrids. Under rule (F) the ledger gives a concrete bound as well. Every admitted run has $\gamma_k \geq n_{\mathrm{rel},k}(\delta_{B,k} + 2^{-97})$, so at most $\Gamma_L / \min_k \gamma_k$ runs are admitted. At $(10, 10^{-6}, 0.1)$ with $n_{\mathrm{rel}} = 2$ and $\delta_B = 2^{-50}$ this is 2555. In $\nu_{\mathcal{A}}$ one may therefore replace $t_{\mathcal{A}}$ by this

constant, which makes the computational loss independent of the coalition's running time. Refused requests involve no key and add nothing.

**Not covered.** The following are outside Theorem 4.2:

- a register that changes between runs;
- researchers who deviate within a run, for example by uploading noisier ciphertexts, which would void $B_{\mathrm{tot}}$;
- a dishonest or non-atomic ledger;
- side channels such as the time $\mathsf{O}$ takes to refuse.

**Remark 4.3** (Which refunds are sound)**.** Suppose the ledger reserves the full cost at admission and refunds part of it when a run ends early. The proof uses only two bounds. Step 4 uses $\sum_{\mathrm{made}} \rho_{k,s} \le \rho^*$. Step 1 uses $\sum_{\mathrm{made}} \gamma'(\kappa_k) + \sum_{\mathrm{adm}} n_{\mathrm{rel},k}\delta_{B,k} \le \Gamma^*$ in expectation. Refunding $\rho_{k,s}$ and $\gamma'(\kappa_k)$ of the releases not made is therefore sound. Refunding their $\delta_{B,k}$ is not covered. The paper's state, "the sum of the costs of every release made", refunds all three. The simplest correct rule charges the whole run at admission. Since $\delta_B \le \gamma_s$, the difference is a fraction $\delta_B/\gamma_s \approx \delta_B 2^{\kappa}$ of the slack of unmade releases: about 1.5% at $\kappa = 44$.

**Remark 4.4** (Rule (N) with adaptive requests)**.** With adaptively chosen requests, the final state $(\rho_\Sigma, \Gamma_\Sigma)$ of rule (N) is random. Each trajectory satisfies the rule's inequality, but the trade-off between $\rho$ and $\Gamma$ varies from one trajectory to another. Theorem 4.2(c) then applies only with $\rho^* = \sup \rho_\Sigma$ and $\Gamma^* = \sup \Gamma_\Sigma$ over executions. If some trajectories spend almost all of $\delta_L$ on $\rho$ and others on $\Gamma$, this can approach $2\delta_L$. We do not claim that rule (N) fails adaptively, only that the proof does not cover it. The paper's prescription is consistent with this: use the filter unless every run is fixed in advance.

**Remark 4.5** (Comparison with the paper's proof sketch)**.** The sketch's three parts (simulation run by run, ideal composition by a Rényi filter, conversion at $\varepsilon_L$) are those of the proof above, with four differences:

(i) the smudging steps of all runs are done together, first, so that the slack is a single expectation under $\mathsf{Real}$ (Lemma 3.5);
(ii) the slack counts $\delta_B$ for all planned releases of admitted runs;
(iii) the reductions need RLWE for adaptively chosen parameter sets and against non-uniform adversaries;
(iv) the Rényi filter is proved here (Theorem 3.2), so the citation of [3] becomes an attribution.

# 5 Part 2: the smudging step in hockey-stick divergence

## 5.1 The shifted lattice Gaussian

**Lemma 5.1** (Hockey-stick smudging, explicit constants)**.** *Let $\eta \sim D_{\mathbb{Z}^N,\sigma}$ and $\mathbf{e} \in \mathbb{Z}^N$ with $\|\mathbf{e}\|_2 \le B$ and $\sigma \ge 2^{\kappa}B/\sqrt{2\pi}$. For $\delta' \in (0,1)$ let*

$$\varepsilon'(\kappa,\delta') = \min\{\varepsilon \ge 0 : \delta(\pi 4^{-\kappa}, \varepsilon) \le \delta'\} \le \pi 4^{-\kappa} + 2\sqrt{\pi \ln(1/\delta')}\, 2^{-\kappa}.$$

*(a) $\mathbf{e} + \eta$ and $\eta$ are $(\varepsilon'(\kappa,\delta'),\delta')$-indistinguishable.*
*(b)* Side information. *Let $(Z,\mathbf{e})$ have any joint law with $\mathbf{e} \in \mathbb{Z}^N$, and let $\eta$ be independent of $(Z,\mathbf{e})$. Then both $\mathsf{E}_{\varepsilon'}\big((Z,\mathbf{e}+\eta)\|(Z,\eta)\big)$ and $\mathsf{E}_{\varepsilon'}\big((Z,\eta)\|(Z,\mathbf{e}+\eta)\big)$ are at most $\delta' + \Pr[\|\mathbf{e}\| > B]$.*

*No smoothing condition and no floor on $\sigma$ are needed, and $N$ is arbitrary. For $\delta' = 2^{-100}$, $\varepsilon' \le 29.52 \cdot 2^{-\kappa} + \pi 4^{-\kappa}$; at $\kappa = 40$ this is $2.7 \cdot 10^{-11}$, and the tight value is $2.0 \cdot 10^{-11}$.*

*Proof.* (a) Let $\rho' = \|\mathbf{e}\|^2/(2\sigma^2) \le B^2 \cdot 2\pi/(2 \cdot 4^\kappa B^2) = \pi 4^{-\kappa}$. By Lemma 2.4, and the remark after it, $D_a(\mathbf{e}+\eta\|\eta) \le a\rho'$ and $D_a(\eta\|\mathbf{e}+\eta) \le a\rho'$ for all $a > 1$. By Lemma 2.3 and the monotonicity of $\delta(\cdot,\varepsilon)$, both hockey-stick divergences at $\varepsilon'(\kappa,\delta')$ are at most $\delta(\rho',\varepsilon') \le \delta(\pi 4^{-\kappa},\varepsilon') \le \delta'$. The closed-form bound on $\varepsilon'$ is the simple form in Lemma 2.3.

(b) Let $\mu$ be the law of $(Z,\mathbf{e})$. Both laws are mixtures over $(z,\mathbf{e})$ with the common weights $\mu(z,\mathbf{e})$: $(Z,\mathbf{e}+\eta)$ mixes $\delta_z \otimes \mathrm{law}(\mathbf{e}+\eta)$, and $(Z,\eta)$ mixes $\delta_z \otimes \mathrm{law}(\eta)$. By Lemma 2.2, each divergence is at most $\sum \mu(z,\mathbf{e})\mathsf{E}_{\varepsilon'}(\cdot\|\cdot)$. Each term is at most $\delta'$ when $\|\mathbf{e}\| \le B$, by (a), and at most 1 otherwise. □

Table 1: Exact check of Lemma 5.1 in one dimension, worst case $e = B$, $\sigma = 2^\kappa B/\sqrt{2\pi}$. "Exact $\varepsilon$" is the smallest $\varepsilon$ whose exact hockey-stick divergence is at most $\delta'$, summed over $\pm 40\sigma$. The last column is the exact divergence at the tight bound, in both directions. The Rényi bound of Lemma 2.4 was also checked exactly ($\sigma \in \{3,10,50\}$, $a \in \{1.5,2,8\}$). Script `scripts/kappa_numbers.py`.

| $\kappa$ | $B$ | $\sigma$ | $\delta'$ | $\varepsilon'$ tight | $\varepsilon'$ simple | exact $\varepsilon$ | $\mathsf{E}$ at tight bound |
|---|---|---|---|---|---|---|---|
| 6 | 50 | 1276.6 | $10^{-6}$ | 0.15886 | 0.20664 | 0.14574 | $2.36 \cdot 10^{-7}$ |
| 6 | 50 | 1276.6 | $10^{-9}$ | 0.21364 | 0.25291 | 0.20188 | $1.85 \cdot 10^{-10}$ |
| 8 | 40 | 4085.2 | $10^{-6}$ | 0.036278 | 0.051517 | 0.032917 | $2.53 \cdot 10^{-7}$ |
| 8 | 40 | 4085.2 | $10^{-9}$ | 0.050799 | 0.063085 | 0.047807 | $1.93 \cdot 10^{-10}$ |
| 10 | 30 | 12255.5 | $10^{-6}$ | 0.0081637 | 0.012870 | 0.0072998 | $2.74 \cdot 10^{-7}$ |
| 10 | 30 | 12255.5 | $10^{-9}$ | 0.012044 | 0.015762 | 0.011281 | $2.02 \cdot 10^{-10}$ |

### 5.2 Composition with an $(\varepsilon',\delta')$ smudging step

For an admitted run $k$, let $\delta'_k \in (0,1)$ be a public constant (a protocol constant such as $2^{-100}$, or part of $P_k$), and put

$$\varepsilon'_k = n_{\mathrm{rel},k}\,\varepsilon'(\kappa_k,\delta'_k), \qquad \hat{\gamma}_k = n_{\mathrm{rel},k}(\delta_{B,k} + \delta'_k).$$

**Theorem 5.2** (Composition with the hockey-stick slack)**.** *Assume Assumption 1.9; the floor $2^{42}$ in (A5) is not needed here. Let $\mathcal{A}$ be a PPT coalition, and suppose that in every execution the admitted runs satisfy $\sum_k \rho_k \le \rho^*$, $\sum_k \varepsilon'_k \le E^*$ and $\sum_k \hat{\gamma}_k \le \Gamma^*$. Then for every $\varepsilon \ge 0$ and $\mathfrak{D} \sim_{\mathrm{b}} \mathfrak{D}'$,*

$$\Pr[\mathsf{Real}_{\mathcal{A}}(\mathfrak{D}) = 1] \le e^{\varepsilon+2E^*}\Pr[\mathsf{Real}_{\mathcal{A}}(\mathfrak{D}') = 1] + e^{E^*}\delta(\rho^*,\varepsilon) + \big(1 + e^{\varepsilon+E^*}\big)\Gamma^* + \mathrm{negl}(\lambda).$$

*Proof. Step 1′, replacing Step 1.* Apply Lemma 3.6 twice: once to $(P,Q) = (\mathsf{Real},\mathsf{Hyb}^{\mathrm{st}})$ and once to $(\mathsf{Hyb}^{\mathrm{st}},\mathsf{Real})$, both under $\mathfrak{D}$. At a release step $(k,s)$ with history $h$, take $\varepsilon_t(h) = \varepsilon'(\kappa_k,\delta'_k)$ and $\delta_t(h) = \delta'_k + \mathbf{1}\{\|e_{k,s}(h)\| > B_{\mathrm{tot},k,s}\}$; at every other step take $(0,0)$. The two kernels at a release step differ by the shift $e_{k,s}(h)$ of $f + \eta - c'_1 s_R$. By Lemma 2.2 and Lemma 5.1(a), their hockey-stick divergence at $\varepsilon'$ is at most $\delta'_k$ in both directions when $\|e_{k,s}(h)\| \le B_{\mathrm{tot},k,s}$, and at most 1 trivially otherwise.

In every execution $\sum_t \varepsilon_t \le \sum_{\mathrm{adm}} \varepsilon'_k \le E^*$. As in Step 1, $\mathbb{E}_P[\sum_t \delta_t] \le \mathbb{E}[\sum_{\mathrm{adm}} \hat{\gamma}_k] \le \Gamma^*$. This holds for $P = \mathsf{Real}$, and also for $P = \mathsf{Hyb}^{\mathrm{st}}$, because there the coins of each run are drawn as in $\mathsf{Real}$ and (1) holds unchanged. Hence for every test $\varphi$, in particular $\mathcal{A}$'s output,

$$\mathbb{E}_{\mathsf{Real}(\mathfrak{D})}\varphi \le e^{E^*}\mathbb{E}_{\mathsf{Hyb}^{\mathrm{st}}(\mathfrak{D})}\varphi + \Gamma^*, \qquad \mathbb{E}_{\mathsf{Hyb}^{\mathrm{st}}(\mathfrak{D})}\varphi \le e^{E^*}\mathbb{E}_{\mathsf{Real}(\mathfrak{D})}\varphi + \Gamma^*,$$

and likewise for $\mathfrak{D}'$.

*Steps 2–4* are unchanged: $|\Pr[\mathsf{Hyb}^{\mathrm{st}}(\mathfrak{D}) = 1] - \Pr[\mathsf{Ideal}(\mathfrak{D}) = 1]| \le \nu$, and the ideal experiment is $\rho^*$-zCDP.

*Chain.* Write $p_X(\mathfrak{D}) = \Pr[X(\mathfrak{D}) = 1]$. Then

$$\begin{aligned} p_{\mathsf{Real}}(\mathfrak{D}) &\le e^{E^*} p_{\mathsf{Hyb}^{\mathrm{st}}}(\mathfrak{D}) + \Gamma^* \le e^{E^*}\big(p_{\mathsf{Ideal}}(\mathfrak{D}) + \nu\big) + \Gamma^* \\ &\le e^{E^*}\big(e^{\varepsilon} p_{\mathsf{Ideal}}(\mathfrak{D}') + \delta(\rho^*, \varepsilon) + \nu\big) + \Gamma^* \\ &\le e^{E^*}\big(e^{\varepsilon}(p_{\mathsf{Hyb}^{\mathrm{st}}}(\mathfrak{D}') + \nu) + \delta(\rho^*, \varepsilon) + \nu\big) + \Gamma^* \\ &\le e^{E^*}\big(e^{\varepsilon}(e^{E^*} p_{\mathsf{Real}}(\mathfrak{D}') + \Gamma^* + \nu) + \delta(\rho^*, \varepsilon) + \nu\big) + \Gamma^*, \end{aligned}$$

which is the claim, with $\mathrm{negl} = (e^{E^*} + e^{\varepsilon + E^*})\nu$. □

**How computational and $(\varepsilon', \delta')$ steps are mixed.** The hockey-stick step is information-theoretic and holds for every $[0,1]$-valued test, so in particular for the event that a PPT distinguisher outputs 1. It is applied only between $\mathsf{Real}$ and $\mathsf{Hyb}^{\mathrm{st}}$, before any computational replacement, where $e_{k,s}$ is defined and Lemma 5.1 applies. The computational steps are applied between $\mathsf{Hyb}^{\mathrm{st}}$ and $\mathsf{Ideal}$, and they are additive. In the chain the multiplicative factors $e^{E^*}$ and $e^{\varepsilon}$ multiply the negligible $\nu$, which stays negligible because $\varepsilon$ and $E^*$ are constants. No hockey-stick bound is ever applied across a computational step. That would need a separate argument, and it is not needed here.

### 5.3 A ledger with a hockey-stick slack, and the new $\kappa_L$

**Definition 5.3** (Ledger with an $\varepsilon$-share for smudging)**.** The parameters are $(\varepsilon_L, \delta_L, \zeta_L)$, a share $\xi_L \in (0,1)$ of $\varepsilon_L$ reserved for smudging, and a per-release $\delta'$, a protocol constant such as $2^{-100}$. Put

$$E'_L = \frac{\xi_L \varepsilon_L}{2}, \qquad \rho_L = \max\big\{\rho : \delta\big(\rho, (1-\xi_L)\varepsilon_L\big) \le e^{-E'_L}(1-\zeta_L)\delta_L\big\}, \qquad \Gamma'_L = \frac{\zeta_L \delta_L}{1 + e^{\varepsilon_L - E'_L}}.$$

The state is $(\rho_\Sigma, E_\Sigma, \Gamma_\Sigma)$. Rule (F′) admits a run iff $\rho_\Sigma + \rho_k \le \rho_L$, $E_\Sigma + \varepsilon'_k \le E'_L$ and $\Gamma_\Sigma + \hat{\gamma}_k \le \Gamma'_L$, and charges all three in full at admission.

**Corollary 5.4.** *Under rule (F′) the protocol is $(\varepsilon_L, \delta_L)$-IND-CDP for $\sim_{\mathrm{b}}$, against every PPT coalition.*

*Proof.* Apply Theorem 5.2 with $\varepsilon = (1-\xi_L)\varepsilon_L$, $(\rho^*, E^*, \Gamma^*) = (\rho_L, E'_L, \Gamma'_L)$. Then:

- $\varepsilon + 2E'_L = \varepsilon_L$;
- $e^{E'_L}\delta(\rho_L, \varepsilon) \le (1-\zeta_L)\delta_L$;
- $\varepsilon + E'_L = \varepsilon_L - E'_L$, so $(1 + e^{\varepsilon + E'_L})\Gamma'_L = \zeta_L \delta_L$.

□

For runs fixed in advance, the analogue of rule (N) admits a run iff

$$e^{E_\Sigma}\delta(\rho_\Sigma, \varepsilon_L - 2E_\Sigma) + (1 + e^{\varepsilon_L - E_\Sigma})\Gamma_\Sigma \le \delta_L,$$

with the sums taken after adding the run. Correctness follows from Theorem 5.2 as in Theorem 4.2(e).

**Proposition 5.5** (The new $\kappa_L$ rule)**.** *Let $\tilde{\rho}_{\mathrm{run}}$ be as in* [P]*, equation* `eq:kappa`*: the budget at $\delta_{\mathrm{DP}} = (1-\zeta)\delta_{\mathrm{run}}$, a lower bound on $\rho_{\mathrm{run}}$ once $\kappa \ge \kappa_{\mathrm{run}}$. Let $\rho_L$ and $E'_L$ be as in Definition 5.3. Define*

$$\kappa_L = \min\Big\{\kappa \in \mathbb{Z} :\ n_{\mathrm{rel}}\, \varepsilon'(\kappa, \delta') \le \frac{\tilde{\rho}_{\mathrm{run}}}{\rho_L} E'_L\Big\}.$$

*Suppose every run on the register uses $\kappa_k \ge \kappa_L$. Then the condition $E_\Sigma + \varepsilon'_k \le E'_L$ holds whenever $\rho_\Sigma + \rho_k \le \rho_L$. The slack condition of rule (F′) holds whenever the zCDP condition does, provided $\delta_B$ satisfies the admissibility constraint $n_{\mathrm{rel}}(\delta_B + \delta') \le (\tilde{\rho}_{\mathrm{run}}/\rho_L)\Gamma'_L$, which involves $\delta_B$ alone and not $\kappa$.*

*Proof.* $\varepsilon'_k \le (\tilde\rho_{\mathrm{run},k}/\rho_L)E'_L \le (\rho_k/\rho_L)E'_L$, so $\sum_k \varepsilon'_k \le (\sum_k \rho_k/\rho_L)E'_L \le E'_L$. The same argument gives the bound on $\sum_k \hat\gamma_k$. □

Table 2: The ledger's flooding requirement $\kappa_L$ at $(\varepsilon_L, \delta_L, \zeta_L) = (10, 10^{-6}, 0.1)$, with $n_{\mathrm{rel}} = 2$, $\delta_B = 2^{-50}$, $\delta_{\mathrm{run}} = 10^{-9}$ and $\zeta = 0.1$. "SD" is the paper's rule: the smallest $\kappa$ with $\gamma \le (\tilde\rho_{\mathrm{run}}/\rho_L)\Gamma_L$. The hockey-stick columns use Proposition 5.5 with $\delta' = 2^{-100}$ and the tight $\varepsilon'$; the simple closed form gives the same values except 23 instead of 22 at $\xi_L = 10^{-3}$, $\varepsilon_{\mathrm{run}} = 0.5$, and 14 instead of 13 at $\xi_L = 10^{-2}$, $\varepsilon_{\mathrm{run}} = 4$. "Loss in $\rho_L$" is the relative decrease of $\rho_L$ from reserving $\xi_L$. The $\delta_B$ admissibility constraint of Proposition 5.5 holds in every cell, with a margin of 6.6× at $\varepsilon_{\mathrm{run}} = 0.5$ up to 1300× at $\varepsilon_{\mathrm{run}} = 8$.

| | | $\kappa_L$ for $\varepsilon_{\mathrm{run}} =$ | | | | |
|---|---|---|---|---|---|---|
| rule | loss in $\rho_L$ | 0.5 | 1 | 2 | 4 | 8 |
| SD (paper), $\rho_L = 1.52971$ | – | 48 | 46 | 44 | 42 | 40 |
| hockey-stick, $\xi_L = 10^{-2}$ $(E'_L = 0.05)$ | 1.98% | 19 | 17 | 15 | 13 | 12 |
| hockey-stick, $\xi_L = 10^{-3}$ $(E'_L = 0.005)$ | 0.198% | 22 | 21 | 19 | 17 | 15 |
| hockey-stick, $\xi_L = 10^{-4}$ $(E'_L = 0.0005)$ | 0.020% | 26 | 24 | 22 | 20 | 18 |

**Effect on $\kappa$.** With $\kappa = \max\{\kappa_{\min}, \kappa_{\mathrm{run}}, \kappa_L\}$, $\kappa_{\min} = 40$ and the paper's $\kappa_{\mathrm{run}} = 36, 37, 38, 41, 46$, the ledger cap $(10, 10^{-6})$ now gives $\kappa = 40, 40, 40, 41, 46$, compared with $48, 46, 44, 42, 46$ under the statistical-distance rule. This is the $\kappa$ of a single-run ledger, so the right-hand columns of [P], Table `tab:kappa`, become equal to its single-run columns. The run budget is unchanged, so $\log_2 \Omega_f$ at $\bar n_O = 10^6$ becomes 61.78, 62.75, 63.69, 65.66 and 71.94, compared with 69.88, 68.78, 67.72, 66.66 and 71.94. That saves 8.1, 6.0, 4.0, 1.0 and 0 bits of scale and of $q_0$. The cost is a relative decrease of 0.2% in $\rho_L$ at $\xi_L = 10^{-3}$, and one more ledger counter.

**Remark 5.6** (The per-run statement)**.** The same lemma applies to a single run ([CP], Theorem 4.11), with the share $\xi$ taken from $\varepsilon_{\mathrm{run}}$. At $\xi = 10^{-3}$ and $\delta' = 2^{-100}$ the per-run requirement becomes $\kappa_{\mathrm{run}} = 18, 17, 16, 15, 14$, and the slack is at most 0.53% of $\delta_{\mathrm{run}}$. Then $\kappa = \kappa_{\min} = 40$ for every row. This changes Theorem `thm:output`'s statement, so it is not part of the proposed paper text.

**Remark 5.7** (A Rényi-aggregated variant)**.** One can also account the smudging steps in Rényi divergence. Truncate the error to the event $\|e_{k,s}\| \le B$, which costs $\sum_{\mathrm{adm}} n_{\mathrm{rel}}\delta_B$ in statistical distance. The truncated real process is then Rényi-close to $\mathsf{Hyb}^{\mathrm{st}}$ with cost $\pi 4^{-\kappa_k}$ per release, and these costs compose by Theorem 3.2. Converting once at $(E'_L, \delta'')$ makes the smudging $\varepsilon$ grow like the square root of the number of releases rather than linearly. At $\xi_L = 10^{-3}$ and $\delta'' = 2^{-100}$ this gives $\kappa_L = 18, 17, 16, 15, 14$. Since $\kappa_{\min}$ binds either way, the simpler per-release form is proposed.

**Remark 5.8** ($\delta_B$ is not helped)**.** $\delta_B$ remains multiplied by $1 + e^{\varepsilon_L - E'_L}$. At $\varepsilon_L = 20$, $n_{\mathrm{rel}}(1 + e^{20})\delta_B \approx 8.6 \cdot 10^{-7}$ exceeds $\zeta_L\delta_L = 10^{-7}$ for $\delta_L = 10^{-6}$, so the paper's remark that $\delta_B$ must then be lowered stands.

**Remark 5.9** (The budgeted sampler's failure bound)**.** Each release draws its noise with the fixed-work sampler of Part A, Section 5.5, whose failure probability is at most $\xi_T$ (Part A, Lemma 5.17); outside the failure event the release's noise is exact and its timing independent of the noise and of the data (Part A, Lemma 5.20, under the cost assumption (W)). Replacing every planned release of every admitted run by its ideal counterpart is a statistical hybrid made before any other, like the truncation that costs $\delta_B$; so $\xi_T$ enters the slack exactly as $\delta_B$ does, and the implementation charges

$\hat{\gamma}_k = n_{\mathrm{rel}}(\delta_B + \delta' + \xi_T)$ in rule (F′) (and $n_{\mathrm{rel}}\xi_T$ more in the statistical-distance rule). At $\xi_T = 2^{-80}$ this is negligible beside $\delta_B = 2^{-50}$.

# 6 Issues found in the paper's current text

(I1) *The sketch's slack.* "Step $k$ costs the expectation of $\gamma_k$ … slack $\Gamma = \mathbb{E}\sum_k \gamma_k$." When the smudging steps are interleaved with the computational steps run by run, each expectation is taken under a different hybrid, so the sum is not a single expectation. The fix is to do all smudging first (Lemma 3.5); the conclusion stands.

(I2) *Aborts and the $\delta_B$ part.* The ledger state "of every release made", together with admission against the full cost, refunds unmade releases. The $\delta_B$ part of an admitted run must stay charged for every planned release (Step 1; Remark 4.3). The fix is to charge a run's full cost at admission. The numerical effect is tiny, but the statement as written is not covered by the proof.

(I3) *Atomicity.* The implementation checks a run's full cost at setup but charges at each release, so two concurrently admitted runs can together exceed the cap. Admission must reserve the full cost atomically.

(I4) *Adjacency of the joint statement.* G3 and Proposition `prop:compose` speak of the joint view over a base register, but [P], Definition `def:adj`, defines adjacency on one scope register with the cohort fixed. A base-register adjacency is needed (Definition 1.1). So are the facts that it induces scope adjacency (Lemma 1.3, which needs public scopes and row-wise public preprocessing) and that cohorts may be chosen adaptively.

(I5) *RLWE for adaptively chosen parameters.* The run's HE parameter set may depend on earlier outputs. The reduction needs the assumption for a finite menu of parameter sets, or an adaptive-parameter form; it loses a factor $|\Pi_\lambda|$ (Lemma 3.7). It also needs security against non-uniform adversaries, because the reduction knows the data.

(I6) *The smudging constant.* The factor $1 + 2^{-80}$ in $\gamma_s$ needs $\sigma_c \geq 2^{41.1}$, not $2^{20}$ (Remark 2.6). With $2^{20}$ the factor is $1 + 2^{-37.9}$, and [CP], Corollary 2.4's last inequality fails for $\kappa < 60$. This affects [P], Lemma `lem:smudge` and Theorem `thm:output`(i), and [CP], Corollary 2.4. It has no numerical effect at any admissible setting.

(I7) *Wording of the $\kappa_L$ guarantee.* "$\kappa_L$ makes the ledger run out of zCDP budget before it runs out of slack, whatever the order and the sizes of the runs." This holds only if every run on the register uses $\kappa \geq \kappa_L$ computed with the same $(\rho_L, \Gamma_L)$. A run with a fixed, smaller $\kappa$ can exhaust the slack first. Privacy is unaffected, because the filter checks both conditions.

(I8) *Rule (N).* "Runs fixed in advance" should say that the requests (parameters and inputs) are fixed before the first run and do not depend on any output. Rule (N) is not shown to be valid for adaptive requests (Remark 4.4).

(I9) *Implementation details.* The filter test in `Ledger.admits` allows a relative excess of $10^{-12}$ over the caps. `rho_tight` returns the root of a bracket, which may lie slightly above the true $\rho_L$. A rigorous ledger should round $\rho_L$ and $\Gamma_L$ down and test without tolerance. The effect on $\delta$ is a relative $10^{-11}$.

No inequality in the budget is reversed. The definitions of $\rho_L$ and $\Gamma_L$ make the $\delta$ budget add up exactly (Theorem 4.2(d)). The paper's $\kappa_L$ values $48, 46, 44, 42, 40$ and $\kappa_{\text{run}}$ values $36, 37, 38, 41, 46$ are reproduced exactly.

# 7 What is used and what is proved here

**Proved here, self-contained.**

- Lemmas 2.2 (post-processing, common mixtures) and 2.3 (conversion), with proofs.
- Theorem 3.2 (Rényi filter for zCDP) and Corollary 3.4.
- Lemma 3.5 (statistical hybrid lemma), Lemma 3.6 (adaptive composition of hockey-stick closeness) and Lemma 3.7 (adaptive parameter choice).
- Lemma 1.3, Theorem 4.2, Lemma 5.1, Theorem 5.2, Corollary 5.4 and Proposition 5.5.
- All numbers, in `scripts/kappa_numbers.py`, with output in `scripts/kappa_numbers.out`. The script also computes exactly a two-step adaptive composition of one-dimensional lattice Gaussians, in which the second step's $\sigma$ depends on the first output. With the filter, $D_a \leq a\rho^*$ holds at every tested order; a rule that overspends on some branches violates it.

**Used from [CP] (proved there, not re-verified here).**

- Lemma 2.1 (Gaussian sums maximal at the lattice) and Lemma 2.2 (integer shifts), restated with proof as Lemma 2.4.
- Lemma 2.3(b) (smudging), with the constant corrected in Remark 2.6.
- Lemma 3.6 ($c_1$ independent of secrets), Proposition 3.8 (error decomposition), Theorem 3.12 (O1) and Corollary 3.13 (sensitivity of the centre).
- The per-run hybrids H1–H4 of Theorem 4.11, which are re-run here inside the multi-run experiment.

**External results.**

- [4, Lemma 3.3], the bound on the smoothing parameter of $\mathbb{Z}^N$, through Lemma 2.5.
- Decision-RLWE (Assumption 4.9) and RLWE with circular security for $\mathsf{O}$'s key shares (Assumption 4.10), as assumptions.
- The definitions of zCDP [1] and of SIM-/IND-CDP [5].
- [6, 3, 7] are cited for attribution only. No step depends on their exact statements, which were not checked against the full texts.
- The CKS conversion [2] is reproved in Lemma 2.3.

# References


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